

FOUNDATIONS OF DIFFUSION LANGUAGE MODELS

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Diffusion models have recently emerged as a powerful alternative to autoregressive (AR) models for generative modeling, with strong results in continuous domains such as images and video. In the discrete setting of language, however, diffusion models still lag behind AR approaches in both likelihood and sampling efficiency. This thesis investigates the foundations of diffusion language models and asks whether they can be made competitive with traditional AR language models.

First, we revisit the role of the forward noising process. Contrary to the prevailing theory that likelihood is invariant to the noise schedule, we prove that this invariance holds only for univariate schedules. We introduce context-adaptive, multivariate noise processes that are learned from data and show that they strictly improve likelihood and sample quality.

Second, we propose a simple and unified framework for discrete diffusion that encompasses masked and Uniform-state processes. Within this framework, we derive Rao–Blackwellized variational bounds that are tighter and exhibit lower variance than existing ELBO formulations. Using these bounds, our MDLM models achieve state-of-the-art diffusion perplexities, approach AR perplexities on standard language benchmarks, and surpass AR models on several likelihood evaluations, while enabling efficient conversion of BERT-style encoders into generative models.

Third, we establish a discrete–continuous duality that links Uniform-state diffusion in discrete space with Gaussian diffusion in continuous space. This

connection allows us to transfer efficient parameterizations, improved training objectives, and distillation-based fast samplers to the discrete domain. Leveraging this, we obtain a twofold speedup in training convergence and two orders of magnitude faster sampling, and we introduce the first few-step discrete diffusion samplers via discrete consistency distillation.

Collectively, these contributions show that diffusion language models can substantially close the gap to AR models in both quality and speed, while providing a principled foundation for future advances in discrete diffusion, fast sampling, and controllable language generation.

BIOGRAPHICAL SKETCH

Subham grew up in the small town of Cuttack, Odisha. From an early age, he was deeply fascinated by mathematics and physics. He pursued his undergraduate studies at IIT Kharagpur, where he majored in Electrical Engineering. His interest in both digital and analog circuits sparked a curiosity about the interplay between discrete and continuous entities in physical and mathematical contexts, laying the foundation for his future research.

After graduation, he spent two formative years at Google Research, Mountain View, where his work focused on explaining the decisions made by black-box AI models (neural networks).

In early 2022, Subham began his Ph.D., where he was introduced to generative modeling. Building on his passion for discrete structures, his research centered on developing generative algorithms for modeling such structures. During his doctoral studies, he led several foundational developments that helped shape the emerging field of Diffusion Language Models. His work has been adopted at an industrial scale by organizations including Google, NVIDIA, and ByteDance, with applications spanning language generation and drug discovery.

After completing his Ph.D., Subham will join the Institute of Foundation Models at the Mohamed Bin Zayed University of Artificial Intelligence in Sunnyvale, California as a Sr. Research Scientist, where he will lead a research group focusing on discrete diffusion.

To my family,
Papa, Mama, and Twinkle.

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CHAPTER 1

INTRODUCTION

Diffusion models, inspired by the physics of heat diffusion, have emerged as powerful tools for generative modeling. They can synthesize realistic, high-quality images [Sohl-Dickstein et al., 2015, Ho et al., 2020, Rombach et al., 2021], audio [Kong et al., 2021, Liu et al., 2023b], and videos [Ho et al., 2022, Wu et al., 2023, Esser et al., 2023, Blattmann et al., 2023]. More recently, they have also received significant attention for generating discrete data such as text [Austin et al., 2021, Li et al., 2021, Lou et al., 2024], biological sequences [Avdeyev et al., 2023, Rastogi and Schiff, 2023], and graphs [Sun and Yang, 2023, Vignac et al., 2022].

In the context of language modeling, however, discrete diffusion methods lag behind autoregressive (AR) models [Austin et al., 2021, Gulrajani and Hashimoto, 2024, He et al., 2022, Lou et al., 2024]. Unlike AR approaches, diffusion-based methods are not bound to generate tokens sequentially, and thus offer the potential for improved long-range reasoning, controllable generation, and faster sampling. These advantages make discrete diffusion a compelling alternative to AR models.

Yet, diffusion language models face two key limitations:

Generation quality Perplexity—a monotone transformation of log-likelihood—has long been the standard metric for language models, as it strongly correlates with downstream performance [Hoffmann et al., 2022, Xia et al., 2022]. Discrete diffusion models still exhibit a noticeable likelihood gap on challenging language modeling tasks [Lou et al., 2024]. While many likelihood-improvement methods exist for Gaussian diffusion [Kingma et al., 2021, Zheng et al., 2023a], these typically rely on variance-minimization techniques that do not readily extend to the discrete domain.

For example, the ELBO invariance property exploited by [Kingma et al., 2021] is unknown in the discrete case, and the efficient denoising parameterizations proposed in [Zheng et al., 2023a] have no clear analogue. Moreover, discrete diffusion training objectives tend to be numerically unstable, often requiring high-precision training.

Inference speed. Although diffusion models support parallel generation, each denoising step is significantly more computationally expensive than a single AR decoding step. As a result, discrete diffusion models remain much slower in practice, limiting their applicability in latency-sensitive scenarios such as conversational systems. Efficient samplers [Karras et al., 2022] and single-step distillation methods [Song et al., 2023, Song and Dhariwal, 2023, Yin et al., 2024], which have proven effective in Gaussian diffusion, are not yet available for the discrete case.

These challenges motivate a central question: **Can diffusion language models be made competitive with traditional autoregressive approaches?**

1.1 Contributions

This dissertation addresses the above question by presenting three methods to improve the quality and speed of diffusion language models.

Adaptive Noise Processes To improve likelihood, we propose learning the noising process directly from data. **We disprove the prevailing theory that the log-likelihood is invariant to the choice of noise schedule** [Kingma et al., 2021] by showing that if the noise schedule is multivariate and conditioned on the input, diffusion models achieve better (lower) likelihoods.

Simple Discrete Diffusion Models We introduce a simplified framework for discrete diffusion that captures Masked and Uniform-State diffusion processes— the two most popular discrete diffusion processes proposed in [Austin et al., 2021]. We derive Rao-Blackwellized likelihood bounds that are provably tighter and exhibit lower variance than existing ELBO formulations [Austin et al., 2021, Campbell et al., 2022, Lou et al., 2024]. Empirically, **these models approach AR perplexities on standard datasets and surpass AR models on several natural language likelihood benchmarks for the first time.**

The Discrete-Continuous Diffusion Duality Finally, we establish a theoretical connection between discrete space and continuous space (Gaussian) diffusion models. This link enables the transfer of key Gaussian diffusion techniques—such as efficient parameterizations for the denoising model [Ho et al., 2020, Salimans and Ho, 2022b, Zheng et al., 2023a], faster training methods [Kingma et al., 2021, Zheng et al., 2023a], and distillation strategies [Song et al., 2023, Song and Dhariwal, 2023, Yin et al., 2024]—to the discrete domain. Leveraging this duality, we improve the likelihood of Uniform-State diffusion by proposing a curriculum learning strategy, and **introduce the first few-step discrete diffusion samplers through discrete consistency distillation.**

1.2 Dissertation Structure

The remainder of this dissertation is organized as follows:

- **Chapter 3** introduces *Diffusion Models with Learned Adaptive Noise*, where we prove that input-conditioned noise schedules can improve likelihood.

- [Chapter 4](#) presents *Simple Discrete Diffusion Models*, a unified training framework with Rao-Blackwellized bounds that, for the first time, enable diffusion language models to outperform AR models on downstream benchmarks.
- [Chapter 5](#) presents *The Diffusion Duality*, establishing the connection between the diffusion processes in the discrete space and the continuous space. This perspective leads to new training techniques and the first demonstration of few-step discrete diffusion generation.

1.3 Summary of Publications

The contents of [Chapter 3](#) consist of research ideas and results taken from [Sahoo et al. \[2024b\]](#):

[Subham Sekhar Sahoo](#), Aaron Gokaslan, Christopher De Sa, and Volodymyr Kuleshov. **Diffusion models with learned adaptive noise**. In The Thirty-eighth Annual Conference on Neural Information Processing Systems, 2024.

The contents of [Chapter 4](#) consist of research ideas and results taken from [Sahoo et al. \[2024a\]](#):

[Subham Sekhar Sahoo](#), Marianne Arriola, Aaron Gokaslan, Edgar Mariano Marroquin, Alexander M Rush, Yair Schiff, Justin T Chiu, and Volodymyr Kuleshov. **Simple and effective masked diffusion language models**. In The Thirty-eighth Annual Conference on Neural Information Processing Systems, 2024.

The contents of [Chapter 5](#) consist of research ideas and results taken from [Sahoo et al. \[2025\]](#):

Subham Sekhar Sahoo, Justin Deschenaux, Aaron Gokaslan, Guanghan Wang, Justin T Chiu, and Volodymyr Kuleshov. **The diffusion duality**. In Forty- second International Conference on Machine Learning, 2025.

Non-thesis research: During my Ph.D., I have also contributed to the following research publications, which are excluded from the thesis.

Subham Sekhar Sahoo, Anselm Paulus, Marin Vlastelica, Vít Musil, Volodymyr Kuleshov, and Georg Martius. **Backpropagation through combinatorial algorithms: Identity with projection works**. In The Eleventh International Conference on Learning Representations, 2023.

Phillip Si, Zeyi Chen, Subham Sekhar Sahoo, Yair Schiff, and Volodymyr Kuleshov. **Semi-autoregressive energy flows: exploring likelihood-free training of normalizing flows**. In International Conference on Machine Learning, pages 31732–31753. PMLR, 2023

Yair Schiff*, Subham Sekhar Sahoo*, Hao Phung*, Guanghan Wang*, Sam Boshar, Hugo Dalla-torre, Bernardo P de Almeida, Alexander M Rush, Thomas PIERROT, and Volodymyr Kuleshov. **Simple guidance mechanisms for discrete diffusion models**. In The Thirteenth International Conference on Learning Representations, 2025.

Marianne Arriola, Aaron Gokaslan, Zhihan Yang, Zhix- uan Qi, Jiaqi Han, Justin T Chiu, Subham Sekhar Sahoo, and Volodymyr Kuleshov. **Block diffusion: Interpolating between autoregressive and diffusion language models**. In The Thirteenth International Conference on Learning Representations, 2025.

Justin Deschenaux, Caglar Gulchere, and Subham Sekhar Sahoo. **The Diffusion Duality, Chapter II: Ψ -Samplers and Efficient Curriculum**, In The Eleventh International Conference on Learning Representations, 2026.

Subham Sekhar Sahoo, Zhihan Yang, Yash Akhauri, Johnna Liu, Deepansha Singh, Zhoujun Cheng, Zhengzhong Liu, Eric Xing, John Thickstun, and Arash Vahdat. **Esoteric language models.** arXiv, 2026.

CHAPTER 2

BACKGROUND

The contents of this thesis are centered around probabilistic machine learning. We assume the reader is familiar with the basics of probability, calculus, and linear algebra. The reader should have an understanding of the core concepts of machine learning and Bayesian statistics [Murphy, 2012, Bishop, 2007, Gelman et al., 2013]. Our contributions make heavy use of the deep learning modeling paradigm [Schmidhuber, 2015, Goodfellow et al., 2016] and gradient-based learning through automatic differentiation [Baydin et al., 2015, Paszke et al., 2019].

2.1 Problem Statement

Let $\{\mathbf{x}_{(1)}, \mathbf{x}_{(2)}, \dots, \mathbf{x}_{(N)}\}$ be a dataset of size N . A standard assumption in statistics and machine learning is that the data points are independent and identically distributed (i.i.d.) draws from some unknown data-generating distribution p_{data} :

$$\{\mathbf{x}_{(1)}, \mathbf{x}_{(2)}, \dots, \mathbf{x}_{(N)}\} \stackrel{\text{iid}}{\sim} p_{\text{data}}(\mathbf{x}) \quad (2.1)$$

For brevity, we use p to denote both a probability distribution and its density or mass function, with the meaning clear from context. If the true data distribution $p_{\text{data}}(\mathbf{x})$ were known, we could sample from it to generate arbitrarily many new observations.

The aim of generative modeling is to approximate this unknown distribution $p_{\text{data}}(\mathbf{x})$ using a parameterized model, so that we can both generate realistic synthetic data and, when possible, evaluate probabilities of arbitrary configurations. To do this, we introduce a family of models $p_{\text{data}}(\mathbf{x})$, where $\theta \in \Theta$ denotes the

parameters and Θ is the parameter space. Learning consists of finding parameters $\theta^* \in \Theta$ such that

$$p_{\theta^*}(\mathbf{x}) \approx p_{\text{data}}(\mathbf{x}). \quad (2.2)$$

When the model distribution $p_{\theta^*}(\mathbf{x})$ is close to $p_{\text{data}}(\mathbf{x})$, we can use $p_{\theta^*}(\mathbf{x})$ as a surrogate for $p_{\text{data}}(\mathbf{x})$ for sampling and probability evaluation. In the machine learning literature, such a model $p_{\theta^*}(\mathbf{x})$ is typically referred to as a generative model.

2.2 Learning A Generative Model

To learn θ , we introduce a discrepancy measure $D(\cdot\|\cdot)$ between the true distribution $p_{\text{data}}(\mathbf{x})$ and the model $p_{\theta}(\mathbf{x})$, and then solve

$$\theta^* = \arg \min_{\theta \in \Theta} D(p_{\text{data}}\|p_{\theta}). \quad (2.3)$$

Common choices for D include f -divergences [Csiszár, 1964, Ali and Silvey, 1966] and integral probability metrics [Müller, 1997]. Given a convex, lower semi-continuous function $f: [0, \infty) \rightarrow \mathcal{R}$ with $f(0) = 1$, the corresponding f -divergence between p_{data} and p_{θ} is defined as

$$D_f(p_{\text{data}}\|p_{\theta}) = \int p_{\theta}(\mathbf{x}) f\left(\frac{p_{\text{data}}(\mathbf{x})}{p_{\theta}(\mathbf{x})}\right) d\mathbf{x}. \quad (2.4)$$

In practice, we only have access to samples $\{\mathbf{x}_{(1)}, \dots, \mathbf{x}_{(N)}\} \stackrel{\text{iid}}{\sim} p_{\text{data}}(\mathbf{x})$, so $D(\cdot\|\cdot)$ must be approximated from data.

A notable example arises by setting $f(u) = u \log u$ in Eq. 2.4, which yields the

Kullback–Leibler (KL) divergence [Kullback and Leibler, 1951]:

$$\begin{aligned}
& D_{\text{KL}}(p_{\text{data}} \| p_{\theta}) \\
&= \int p_{\text{data}} \log \left(\frac{p_{\text{data}}}{p_{\theta}} \right) d\mathbf{x} \\
&= \mathbb{E}_{p_{\text{data}}} \left[\log \left(\frac{p_{\text{data}}}{p_{\theta}} \right) \right] \\
&\approx \frac{1}{N} \sum_{i=1}^N \log \left(\frac{p_{\text{data}}(\mathbf{x}_{(i)})}{p_{\theta}(\mathbf{x}_{(i)})} \right) \\
&= \frac{1}{N} \sum_{i=1}^N \log (p_{\text{data}}(\mathbf{x}_{(i)})) - \frac{1}{N} \sum_{i=1}^N \log (p_{\theta}(\mathbf{x}_{(i)})) \\
&\because \frac{1}{N} \sum_{i=1}^N \log (p_{\text{data}}(\mathbf{x}_{(i)})) \text{ doesn't depend on model parameters } \theta; \text{ we get,} \\
&= -\frac{1}{N} \sum_{i=1}^N \log (p_{\theta}(\mathbf{x}_{(i)})) + \text{const.} \tag{2.5}
\end{aligned}$$

where the term $\frac{1}{N} \sum_{i=1}^N \log p_{\text{data}}(\mathbf{x}_i)$ does not depend on θ . Equation 2.5 shows that minimizing $D_{\text{KL}}(p_{\text{data}} \| p_{\theta})$ is equivalent to maximizing the average log-likelihood under the model:

$$\theta^* = \arg \max_{\theta} \frac{1}{N} \sum_{i=1}^N \log p_{\theta}(\mathbf{x}_i), \tag{2.6}$$

which is the classical maximum likelihood estimation (MLE) principle for learning probabilistic models.

2.3 Existing Methods

Below we give an introduction to several predominant families of generative models in the literature. We classify them into three categories:

Notation We represent scalar discrete random variables that can take K values as ‘one-hot’ column vectors and define $\mathcal{V} \in \{\mathbf{x} \in \{0, 1\}^K : \sum_{i=1}^K \mathbf{x}_i = 1\}$ as the set

of all such vectors. In the context of language modeling, K is the vocabulary size and $\mathcal{V}$ is the vocabulary. Let $\mathbf{m} \in \mathcal{V}$ be a special mask vector such that its K -th entry is one, i.e., $\mathbf{m}_K = 1$. Define $\text{Cat}(\cdot; \boldsymbol{\pi})$ as the categorical distribution over K classes with probabilities given by $\boldsymbol{\pi} \in \Delta^K$, where Δ^K denotes the K -simplex. Additionally, let $\langle \mathbf{a}, \mathbf{b} \rangle$ denote the dot product between vectors $\mathbf{a}$ and $\mathbf{b}$. We use parentheses $()$ to denote ordered sets (tuples) and curly brackets $\{\}$ to denote unordered sets. $|A|$ denotes the cardinality of the set A . We use $\mathbf{x}^{1:L} \in \mathcal{V}^L$ and $[\mathbf{x}^\ell]_{\ell=1}^L \in \mathcal{V}^L$ to denote sequences of length L .

2.3.1 Autoregressive Model

Given a sequence $\mathbf{x} \in \mathcal{V}^L \sim q_{\text{data}}$, AR models define the following factorization of the joint distribution:

$$\log p_\theta(\mathbf{x}) = \sum_{\ell=1}^L \log p_\theta(\mathbf{x}^\ell \mid \mathbf{x}^{<\ell}),$$

where the model p_θ is usually parameterized by a causal transformer [Vaswani et al., 2017] model. Sampling takes L steps or NFEs but each is computationally efficient due to KV caching. AR models achieve the best likelihood and generation quality.

2.3.2 Discrete Diffusion Models

Consider the clean data $\mathbf{x} \in \mathcal{V}$ drawn from the data distribution q_{data} . Applications of diffusion modeling to discrete data can be broken into two broad categories. First are works that embed discrete structures in continuous space and then perform the Gaussian diffusion defined above on these continuous representations [Chen et al., 2022, Dieleman et al., 2022, Gulrajani and Hashimoto, 2024, Han et al.,

2022, Li et al., 2022, Lovelace et al., 2024, Strudel et al., 2022]. In contrast we focus on diffusion process directly on discrete structures. In the discrete diffusion framework [Sohl-Dickstein et al., 2015, Austin et al., 2021] the complex data distribution q_{data} is mapped to a simple distribution through a sequence of Markov states. The forward process $(q_t)_{t \in [0,1]}$ smoothly transitions from q_{data} to a prior distribution $\text{Cat}(\cdot; \boldsymbol{\pi})$, by introducing latent variables $\mathbf{z}_t \in \mathcal{V}$ whose marginals conditioned on $\mathbf{x}$ at time t are given by:

$$q_t(\cdot | \mathbf{x}; \boldsymbol{\alpha}_t) = \text{Cat}(\cdot; \bar{Q}_t \mathbf{x}) = \text{Cat}(\mathbf{z}_t; Q_t \cdot Q_{t-\frac{1}{T}} \cdots Q_{\frac{1}{T}} \cdot Q_0 \mathbf{x}), \quad (2.7)$$

where the diffusion parameter $\boldsymbol{\alpha}_t \in [0, 1]$ is a strictly decreasing function in t , with $\boldsymbol{\alpha}_{t=0} \approx 1$ and $\boldsymbol{\alpha}_{t=1} \approx 0$, and $Q_t \in \mathbb{R}^{K \times K}$ is the state transition matrix with $\bar{Q}_{t=1} = \boldsymbol{\pi}$. A discrete diffusion process is characterized by the time evolution of marginals follows a linear ordinary differential equation [Anderson, 2012]:

$$\frac{d}{dt} q_t = Q_t q_t. \quad (2.8)$$

Austin et al. [2021] introduce two noising frameworks: Masked Diffusion Models (MDMs), which use a masked token prior $\boldsymbol{\pi} = \mathbf{m}$ with $\mathbf{m} \in \mathcal{V}$ as a special mask token, and Uniform State Diffusion Models (USDMs), which uses a uniform prior over $\mathcal{V}$, i.e., $\boldsymbol{\pi} = \mathbf{1}/K$. These frameworks differ in their forward corruption dynamics. In MDMs, the clean data $\mathbf{x}$ either stays unchanged or transitions to the mask token $\mathbf{m}$, after which it remains masked for the rest of the process. In contrast, USDMs allow each token to either stay the same or transition uniformly to any other token in $\mathcal{V}$, with the transition probability determined by the diffusion timestep (see Fig. C.1 for examples). These forward dynamics impact the reverse generation process: USDMs permit continual token updates, while MDMs fix tokens once unmasked. To mitigate this limitation, predictor-corrector methods have been proposed for MDMs [Campbell et al., 2022, Gat et al., 2024, Wang et al., 2025],

but at the cost of added computation. In contrast, USDMs naturally exhibit a self-correcting property absent in AR and MDM approaches.

We fit p_θ by minimizing the evidence lower bound (ELBO) on the marginal log-likelihood given by:

$$\log p_\theta(\mathbf{x}_0) = \text{ELBO}(p_\theta, q) + D_{\text{KL}}[q(\mathbf{x}_{0:1}|\mathbf{x})\|p_\theta(\mathbf{x}_{0:1}|\mathbf{x}_0)] \geq \text{ELBO}(p_\theta, q) \quad (2.9)$$

where the Negative Evidence Lower Bound (NELBO) equals [Sohl-Dickstein et al., 2015]:

$$\begin{aligned} -\log p_\theta(\mathbf{x}) &\leq \text{NELBO}(p_\theta, q) \\ &= \mathbb{E}_q \left[\underbrace{-\log p_\theta(\mathbf{x}|\mathbf{z}_{t(0)})}_{\mathcal{L}_{\text{recons}}} + \underbrace{\sum_{i=1}^T D_{\text{KL}}[q(\mathbf{z}_{s(i)}|\mathbf{z}_{t(i)}, \mathbf{x})\|p_\theta(\mathbf{z}_{s(i)}|\mathbf{z}_{t(i)})]}_{\mathcal{L}_{\text{diffusion}}} \right] \\ &\quad + \underbrace{D_{\text{KL}}[q(\mathbf{z}_1|\mathbf{x})\|p_\theta(\mathbf{z}_1)]}_{\mathcal{L}_{\text{prior}}} \end{aligned} \quad (2.10)$$

where $s(i) = (i - 1)/T$ and $t(i) = i/T$.

2.3.3 Gaussian Diffusion Models

In contrast to discrete diffusion processes, Gaussian diffusion maps the data distribution q_{data} to a simple prior distribution usually a Normal distribution $\mathcal{N}(0, \mathbf{I}_K)$ via a sequence of noisy latent variables $\mathbf{w}_t \in \mathbb{R}^K$ for $t \in [0, 1]$ by progressively adding Gaussian noise of increasing magnitude [Sohl-Dickstein et al., 2015, Ho et al., 2020, Song et al., 2020] to the clean data $\mathbf{x} \in \mathbb{R}^K$. The marginal distribution of each latent is defined by

$$\tilde{q}_t(\cdot|\mathbf{x}; \tilde{\alpha}_t) = \mathcal{N}(\tilde{\alpha}_t \mathbf{x}, (1 - \tilde{\alpha}_t^2) \mathbf{I}_K), \quad (2.11)$$

where the diffusion parameters $\alpha_t, \sigma_t \in \mathbb{R}^+$ implicitly define a noise schedule as a function of t , such that $\nu(t) = \alpha_t^2 / \sigma_t^2$ is a monotonically decreasing function in t . For $\tilde{\alpha}_{t=0} = 1$ and $\tilde{\alpha}_{t=1} = 0$, the NELBO Eq. 2.10 for such a process is given as [Kingma et al., 2021]:

$$\text{NELBO}(\tilde{q}, p_\theta; \mathbf{x}) = -\mathbb{E}_{t \sim \mathcal{U}[0,1], \tilde{q}_t(\mathbf{w}_t | \mathbf{x}; \tilde{\alpha}_t)} \nu'(t) \|\mathbf{x} - \mathbf{x}_\theta(\mathbf{w}_t, t)\|_2^2 \quad (2.12)$$

where $\nu'(t)$ is the time derivative of the signal-to-noise ratio $\nu(t) = \tilde{\alpha}_t^2 / (1 - \tilde{\alpha}_t^2)$ for the Gaussian diffusion process.

CHAPTER 3

DIFFUSION MODELS WITH LEARNED ADAPTIVE NOISE

3.1 Introduction

Central to diffusion algorithms is the diffusion process, a gradual mapping of clean images into white noise. The reverse of this mapping defines the data-generating process we seek to learn—hence, its choice can significantly impact performance [Kingma and Gao, 2023]. The conventional approach involves adopting a diffusion process derived from the laws of thermodynamics, which, albeit simple and principled, may be suboptimal due to its lack of adaptability to the dataset.

In this chapter, we investigate whether the notion of diffusion can be instead *learned from data*. Our motivating goal is to perform accurate log-likelihood estimation and probabilistic modelling, and we take an approach grounded in Bayesian inference [Kingma and Welling, 2013]. We view the diffusion process as an approximate variational posterior: learning this process induces a tighter lower bound (ELBO) on the marginal likelihood of the data. Although previous work argued that the ELBO objective of a diffusion model is invariant to the choice of diffusion process [Kingma et al., 2021, Kingma and Gao, 2023], we show that this claim is only true for the simplest types of univariate Gaussian noise: we identify a broader class of noising processes whose optimization yields significant performance gains.

Specifically, we propose a new diffusion process, multivariate learned adaptive noise (MuLAN), which augments classical diffusion models [Sohl-Dickstein et al., 2015, Kingma et al., 2021] with three innovations: a per-pixel polynomial noise

schedule, an adaptive input-conditional noising process, and auxiliary latent variables. In practice, this method learns the schedule by which Gaussian noise is applied to different parts of an image, and allows tuning this noise schedule to the each image instance.

Our learned diffusion process yields improved log-likelihood estimates on two standard image datasets, CIFAR-10 and ImageNet, and it achieves state-of-the-art performance with less than half the training time of previous methods. The approach requires no changes to the underlying UNet architecture, making it readily compatible with most existing diffusion algorithms. Building on these advantages, we first show that the ELBO of a diffusion model is not invariant to the choice of noise process for many common forms of noise, challenging a widely held assumption in the field. We then introduce MULAN, a learned noise process that adaptively applies multivariate Gaussian noise at different rates across an image, conditioned on arbitrary context, including the image itself. Finally, we demonstrate empirically that learning the diffusion process accelerates training, matches prior state-of-the-art models using only half the compute, and sets a new state-of-the-art in density estimation on CIFAR-10 and ImageNet.

This chapter was previously published as [Sahoo et al. \[2024b\]](#).

3.2 Diffusion Models With Multivariate Learned Adaptive Noise

In most works, the noise schedule, as defined by $\nu(t)$, is either fixed or treated as a hyper-parameter [[Ho et al., 2020](#), [Chen, 2023](#), [Hoogeboom et al., 2023](#)]. [Chen](#)

[2023], Hooigeboom et al. [2023] show that the noise schedule can have a significant impact on sample quality. Kingma et al. [2021] consider learning $\nu(t)$, but argue that the KL divergence terms in the ELBO are invariant to the choice of function ν , except for the initial values $\nu(0), \nu(1)$, and they set these values to hand-specified constants in their experiments. They only consider learning ν for the purpose of minimizing the variance of the gradient of the ELBO. In this work, we show that the ELBO is not invariant to more complex forward processes.

Here, we introduce a new diffusion process, multivariate learned adaptive noise (MuLAN), which introduces three innovations: a per-pixel polynomial noise schedule, a conditional noising process, and auxiliary-variable reverse diffusion. We describe these below.

3.2.1 Why Learned Diffusion?

Our goal is to perform accurate density estimation and probabilistic modelling, and we take an approach grounded in Bayesian inference [Kingma and Welling, 2013]. Notice that the gap between the evidence lower bound $\text{ELBO}(p, q)$ and the marginal log-likelihood (MLL) in Eq. 2.9 is precisely the KL divergence $D_{\text{KL}}[q(\mathbf{w}_{t(1):t(T)}|\mathbf{x})||p_{\theta}(\mathbf{w}_{t(1):t(T)}|\mathbf{x})]$ between the diffusion process q over the latents $\mathbf{w}_t$ and the true posterior of the diffusion model. The diffusion process plays the role of a variational posterior q in $\text{ELBO}(p, q)$; optimizing q thus tightens the gap (MLL – ELBO).

This observation suggests that the ELBO can be made tighter by choosing a diffusion processes q that is closer to the true posterior $p_{\theta}(\mathbf{w}_{t(1):t(T)}|\mathbf{x})$. In fact, the key idea of variational inference is to optimize $\max_{q \in \mathcal{Q}} \text{ELBO}(p, q)$ over a family of

approximate posteriors $\mathcal{Q}$ to induce a tighter ELBO [Kingma and Welling, 2013]. Most diffusion algorithms, however optimize $\max_{p \in \mathcal{P}} \text{ELBO}(p, q)$ within some family $\mathcal{P}$ with a fixed q . Our work seeks to jointly optimize $\max_{p \in \mathcal{P}, q \in \mathcal{Q}} \text{ELBO}(p, q)$; we will show in our experiments that this improves the likelihood estimation.

The task of log-likelihood estimation is directly motivated by applied problems such as data compression [MacKay, 2003]. In that domain, arithmetic coding techniques can take a generative model and produce a compression algorithm that provably achieves a compression rate (in bits per dimension) that equals the model’s log-likelihood [Cover and Thomas, 2005]. Other applications of log-likelihood estimation include adversarial example detection [Song et al., 2017], semi-supervised learning [Dai et al., 2017], and others.

As described in Chapter 1, the central goal of this thesis is to improve the generation quality of language models, and log-likelihood is the metric that most directly captures this objective. Although this chapter evaluates our proposed method on images, our focus is to improve log-likelihood and therefore benefit language generation. The visual appeal of generated images (as measured by e.g., FID) correlates imperfectly with log-likelihood. We focus here on pushing the state-of-the-art in log-likelihood estimation, and while we report FID for completeness, we defer sample quality optimization to future work.

3.2.2 A Forward Diffusion Process With Multivariate Adaptive Noise

Next, our plan is to define a family of approximate posteriors $\mathcal{Q}$, as well as a family suitably matching reverse processes $\mathcal{P}$, such that the optimization problem

$\max_{p \in \mathcal{P}, q \in \mathcal{Q}} \text{ELBO}(p, q)$ is tractable and does not suffer from the aforementioned invariance to the choice of q . This subsection focuses on defining $\mathcal{Q}$; the next sections will show how to parameterize and train a reverse model $p \in \mathcal{P}$.

Notation. Given two vectors $\mathbf{a}$ and $\mathbf{b}$, we use the notation $\mathbf{a}\mathbf{b}$ to represent the Hadamard product (element-wise multiplication). Additionally, we denote element-wise division of $\mathbf{a}$ by $\mathbf{b}$ as $\mathbf{a} / \mathbf{b}$. We denote the mapping $\text{diag}(\cdot)$ that takes a vector as input and produces a diagonal matrix as output.

Multivariate Gaussian Noise Schedule

Intuitively, a multivariate noise schedule injects noise at different rates for each pixel of an input image. This enables adapting the diffusion process to spatial variations within the image. We will also see that this change is sufficient to make the ELBO no longer invariant in q .

Formally, we define a forward diffusion process with a multivariate noise schedule q via the marginal for each latent noise variable $\mathbf{w}_t$ for $t \in [0, 1]$, where the marginal is given by:

$$q(\mathbf{w}_t | \mathbf{x}) = \mathcal{N}(\mathbf{w}_t; \boldsymbol{\alpha}_t \mathbf{x}, \text{diag}(\boldsymbol{\sigma}_t^2)), \quad (3.1)$$

where $\mathbf{w}_t, \mathbf{x} \in \mathbb{R}^d$, $\boldsymbol{\alpha}_t, \boldsymbol{\sigma}_t \in \mathbb{R}_+^d$ and d is the dimensionality of the input data. The $\boldsymbol{\alpha}_t, \boldsymbol{\sigma}_t$ denote varying amounts of signal associated with each component (i.e., each pixel) of $\mathbf{x}$ as a function of time $t(i)$. We define the multivariate signal-to-noise ratio as $\boldsymbol{\nu}(t) = \boldsymbol{\alpha}_t^2 / \boldsymbol{\sigma}_t^2$ and choose $\boldsymbol{\alpha}_t, \boldsymbol{\sigma}_t$ so that $\boldsymbol{\nu}(t)$ decreases monotonically in t along all dimensions and is differentiable in $t \in [0, 1]$. Let $\boldsymbol{\alpha}_{t|s} = \boldsymbol{\alpha}_t / \boldsymbol{\alpha}_s$ and $\boldsymbol{\sigma}_{t|s}^2 = \boldsymbol{\sigma}_t^2 - \boldsymbol{\alpha}_{t|s}^2 / \boldsymbol{\sigma}_s^2$ with all operations applied elementwise.

These marginals induce transition kernels between steps $s < t$ given by

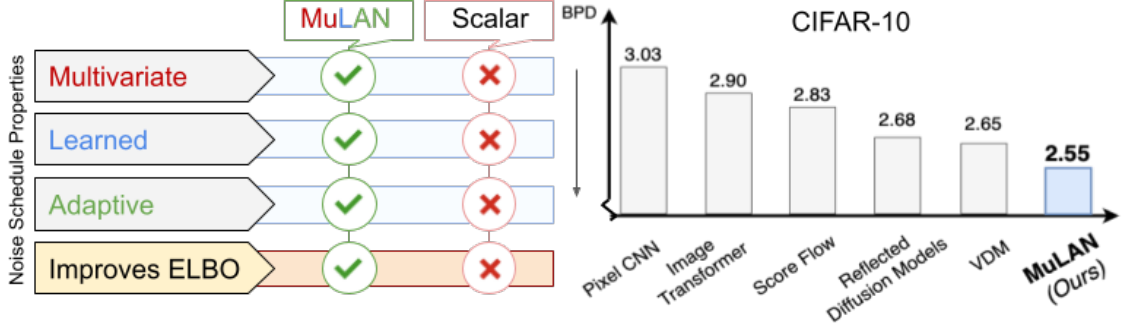


Figure 3.1: (Left) Comparison of noise schedule properties: Multivariate Learned Adaptive Noise schedule (MuLAN) (ours) versus a typical scalar noise schedule. Unlike scalar noise schedules, MuLAN’s multivariate and input-adaptive properties improve likelihood. (Right) Likelihood in bits-per-dimension (BPD) on CIFAR-10 without data augmentation.

(Suppl. A.7):

$$q(\mathbf{w}_s | \mathbf{w}_t, \mathbf{x}) = \mathcal{N} \left(\mathbf{w}_s; \boldsymbol{\mu}_q = \frac{\alpha_{t|s} \sigma_s^2}{\sigma_t^2} \mathbf{w}_t + \frac{\sigma_{t|s}^2 \alpha_s}{\sigma_t^2} \mathbf{x}, \Sigma_q = \text{diag} \left(\frac{\sigma_s^2 \sigma_{t|s}^2}{\sigma_t^2} \right) \right). \quad (3.2)$$

In Sec. 3.2.5, we argue that this class of diffusion process $\mathcal{Q}$ induces an ELBO that is not invariant to $q \in \mathcal{Q}$. The ELBO consists of a line integral along the diffusion trajectory specified by $\boldsymbol{\nu}(t)$. A line integrand is almost always path-dependent, unless its integral corresponds to a conservative force field, which is rarely the case for a diffusion process [Spinney and Ford, 2012]. See Sec. 3.2.5 for details.

Adaptive Noise Schedule Conditioned On Context

Next, we extend the diffusion process to support context-adaptive noise. This enables injecting noise in a way that is dependent on the features of an image. Formally, suppose we have access to a context variable $\mathbf{c} \in \mathbb{R}^m$ which encapsulates high-level information regarding $\mathbf{x}$. Examples of $\mathbf{c}$ could be a class label, a vector of attributes (e.g., features characterizing a human face), or even the input $\mathbf{x}$ itself.

We define the marginal of the latent $\mathbf{w}_t$ in the forward process as $q(\mathbf{w}_t|\mathbf{x}, \mathbf{c}) = \mathcal{N}(\mathbf{w}_t; \boldsymbol{\alpha}_t(\mathbf{c})\mathbf{x}, \boldsymbol{\sigma}_t^2(\mathbf{c}))$; the reverse process can be similarly derived (Suppl. A.7) as:

$$\begin{aligned} & q(\mathbf{w}_s|\mathbf{w}_t, \mathbf{x}, \mathbf{c}) \\ &= \mathcal{N}\left(\boldsymbol{\mu}_q = \frac{\boldsymbol{\alpha}_{t|s}(\mathbf{c})\boldsymbol{\sigma}_s^2(\mathbf{c})}{\boldsymbol{\sigma}_t^2(\mathbf{c})}\mathbf{w}_t + \frac{\boldsymbol{\sigma}_{t|s}^2(\mathbf{c})\boldsymbol{\alpha}_s(\mathbf{c})}{\boldsymbol{\sigma}_t^2(\mathbf{c})}\mathbf{x}, \boldsymbol{\Sigma}_q = \text{diag}\left(\frac{\boldsymbol{\sigma}_s^2(\mathbf{c})\boldsymbol{\sigma}_{t|s}^2(\mathbf{c})}{\boldsymbol{\sigma}_t^2(\mathbf{c})}\right)\right), \end{aligned} \quad (3.3)$$

where the diffusion parameters $\boldsymbol{\alpha}_t, \boldsymbol{\sigma}_t$ are now conditioned on $\mathbf{c}$ via a neural network.

Specifically, we parameterize the diffusion parameters $\boldsymbol{\alpha}_t(\mathbf{c}), \boldsymbol{\sigma}_t(\mathbf{c}), \boldsymbol{\nu}(t, \mathbf{c})$ as $\boldsymbol{\alpha}_t^2(\mathbf{c}) = \text{sigmoid}(-\boldsymbol{\gamma}_\phi(\mathbf{c}, t))$, $\boldsymbol{\sigma}_t^2(\mathbf{c}) = \text{sigmoid}(\boldsymbol{\gamma}_\phi(\mathbf{c}, t))$, and $\boldsymbol{\nu}(\mathbf{c}, t) = \exp(-\boldsymbol{\gamma}_\phi(\mathbf{c}, t))$. Here, $\boldsymbol{\gamma}_\phi(\mathbf{c}, t) : \mathbb{R}^m \times [0, 1] \rightarrow [\gamma_{\min}, \gamma_{\max}]^d$ is a neural network with the property that $\boldsymbol{\gamma}_\phi(\mathbf{c}, t)$ is monotonic in t . Following Kingma et al. [2021], Zheng et al. [2023a], we set $\gamma_{\min} = -13.30$, $\gamma_{\max} = 5.0$.

We explore various parameterizations for $\boldsymbol{\gamma}_\phi(\mathbf{c}, t)$. These schedules are designed in a manner that guarantees $\boldsymbol{\gamma}_\phi(\mathbf{c}, 0) = \gamma_{\min}\mathbf{1}_d$ and $\boldsymbol{\gamma}_\phi(\mathbf{c}, 1) = \gamma_{\max}\mathbf{1}_d$. Below, we list these parameterizations. The polynomial parameterization is novel to our work and yields significant performance gains.

Monotonic Neural Network [Kingma et al., 2021]. We use the monotonic neural network $\boldsymbol{\gamma}_{\text{vdm}}(t)$, proposed in VDM to express $\boldsymbol{\gamma}$ as a function of t such that $\boldsymbol{\gamma}_{\text{vdm}}(t) : [0, 1] \rightarrow [\gamma_{\min}, \gamma_{\max}]^d$. Then we use FiLM conditioning [Perez et al., 2018] in the intermediate layers of this network via a neural network that maps $\mathbf{z}$. The activations of the FiLM layer are constrained to be positive.

Polynomial. (Ours) We express $\boldsymbol{\gamma}_\phi(\mathbf{c}, t)$ as a monotonic degree 5 polynomial in t . Details about the exact functional form of this polynomial and its implementation can be found in Suppl. A.5.2.

3.2.3 Auxiliary-Variable Reverse Diffusion Processes

In principle, we can fit a normal diffusion model in conjunction with our proposed forward diffusion process. However, variational inference suggests that the variational and the true posterior ought to have the same dependency structure: that is the only way for the KL divergence between these two distributions to be zero. Thus, we introduce a class of approximate reverse processes $\mathcal{P}$ that match the structure of $\mathcal{Q}$ and that are naturally suitable for joint optimization $\max_{p \in \mathcal{P}, q \in \mathcal{Q}} \text{ELBO}(p, q)$.

Formally, we define a diffusion model where the reverse diffusion process is conditioned on the context $\mathbf{c}$. Specifically, given any discretization of $t \in [0, 1]$ into T time steps as in [Chapter 2](#), we introduce a context-conditional diffusion model $p_\theta(\mathbf{w}_{0:1}|\mathbf{c})$ that factorizes as the Markov chain

$$p_\theta(\mathbf{w}_{0:1}|\mathbf{c}) = p_\theta(\mathbf{w}_1|\mathbf{c}) \prod_{i=1}^T p_\theta(\mathbf{w}_{s(i)}|\mathbf{w}_{t(i)}, \mathbf{c}). \quad (3.4)$$

Given that the true reverse process is a Gaussian as specified in [Eq. 3.3](#), the ideal p_θ matches this parameterization (the proof mirrors that of regular diffusion models; [Suppl. A.4](#)), which yields

$$\begin{aligned} p_\theta(\mathbf{w}_s|\mathbf{c}, \mathbf{w}_t) \\ = \mathcal{N}\left(\boldsymbol{\mu}_p = \frac{\boldsymbol{\alpha}_{t|s}(\mathbf{c})\boldsymbol{\sigma}_s^2(\mathbf{c})}{\boldsymbol{\sigma}_t^2(\mathbf{c})}\mathbf{w}_t + \frac{\boldsymbol{\sigma}_{t|s}^2(\mathbf{c})\boldsymbol{\alpha}_s(\mathbf{c})}{\boldsymbol{\sigma}_t^2(\mathbf{c})}\mathbf{x}_\theta(\mathbf{w}_t, t), \boldsymbol{\Sigma}_p = \text{diag}\left(\boldsymbol{\sigma}_s^2(\mathbf{c})\boldsymbol{\sigma}_{t|s}^2(\mathbf{c})/\boldsymbol{\sigma}_t^2(\mathbf{c})\right)\right), \end{aligned} \quad (3.5)$$

where $\mathbf{x}_\theta(\mathbf{w}_t, t)$, is a neural network that approximates $\mathbf{x}$. Instead of parameterizing $\mathbf{x}_\theta(\mathbf{w}_t, t)$ directly using a neural network, we consider two other parameterizations. One is the noise parameterization [[Ho et al., 2020](#)] where $\boldsymbol{\epsilon}_\theta(\mathbf{w}_t, \mathbf{c}, t)$ is the denoising model which is parameterized as $\boldsymbol{\epsilon}_\theta(\mathbf{w}_t, t) = (\mathbf{w}_t - \boldsymbol{\alpha}_t(\mathbf{c})\mathbf{x}_\theta(\mathbf{w}_t, t, \mathbf{c}))/\boldsymbol{\sigma}_t(\mathbf{c})$; see [Suppl. A.5.1](#) and the other is v-parameterization [[Salimans and Ho, 2022a](#)] where $\mathbf{v}_\theta(\mathbf{w}_t, \mathbf{c}, t)$ is a neural network that models $\mathbf{v}_\theta(\mathbf{w}_t, \mathbf{c}, t) = (\boldsymbol{\alpha}_t(\mathbf{c})\mathbf{w}_t -$

$\mathbf{x}_\theta(\mathbf{w}_t, \mathbf{c}, t))/\sigma_t(\mathbf{c})$; see Suppl. A.5.1.

Challenges in Conditioning on Context

Note that the model $p_\theta(\mathbf{w}_{0:1}|\mathbf{c})$ implicitly assumes the availability of $\mathbf{c}$ at generation time. Sometimes, this context may be available, such as when we condition on a label. We may then fit a conditional diffusion process with a standard diffusion objective $\mathbb{E}_{\mathbf{x},c}[\text{ELBO}(\mathbf{x}, p_\theta(\mathbf{w}_{0:1}|\mathbf{c}), q_\phi(\mathbf{w}_{0:1}|\mathbf{c}))]$, in which both the forward and the backward processes are conditioned on $\mathbf{c}$ (see Sec. 3.2.4).

When $\mathbf{c}$ is not known at generation time, we may fit a model p_θ that does not condition on $\mathbf{c}$. Unfortunately, this also forces us to define $p_\theta(\mathbf{w}_s|\mathbf{w}_t) = \mathcal{N}(\boldsymbol{\mu}_p(\mathbf{w}_t, t), \boldsymbol{\Sigma}_p(\mathbf{w}_t, t))$ where $\boldsymbol{\mu}_p(\mathbf{w}_t, t), \boldsymbol{\Sigma}_p(\mathbf{w}_t, t)$ is parameterized directly by a neural network. We can no longer use a noise parameterization $\boldsymbol{\epsilon}_\theta(\mathbf{w}_t, t) = (\mathbf{w}_t - \boldsymbol{\alpha}_t(\mathbf{c})\mathbf{x}_\theta(\mathbf{w}_t, t, \mathbf{c}))/\sigma_t(\mathbf{c})$ because it requires us to compute $\boldsymbol{\alpha}_t(\mathbf{c})$ and $\sigma_t(\mathbf{c})$, which we do not know. Since noise parameterization plays a key role in the sample quality of diffusion models [Ho et al., 2020], this approach limits performance.

Conditioning Noise on an Auxiliary Latent Variable

We propose an alternative strategy for learning conditional forward and reverse processes p, q that feature the same structure and hence support efficient noise parameterization. Our approach is based on the introduction of auxiliary variables [Wang et al., 2023], which lift the distribution p_θ into an augmented latent space. Experiments (Suppl. A.4.3) and theory (Suppl. A.4) confirm that this approach performs better than parameterizing $\mathbf{c}$ using a neural network, $\mathbf{c}_\theta(\mathbf{w}_t, t)$.

Specifically, we introduce an auxiliary latent variable $\mathbf{z} \in \mathbb{R}^m$ and define a lifted

$p_\theta(\mathbf{x}, \mathbf{z}) = p_\theta(\mathbf{x}|\mathbf{z})p_\theta(\mathbf{z})$, where $p_\theta(\mathbf{x}|\mathbf{z})$ is the conditional diffusion model from Eq. 3.4 (with context $\mathbf{c}$ set to $\mathbf{z}$) and $p_\theta(\mathbf{z})$ is a simple prior (e.g., unit Gaussian or fully factored Bernoulli). The latents $\mathbf{z}$ can be interpreted as a high-level semantic representation of $\mathbf{x}$ that conditions both the forward and the reverse processes. Unlike $\mathbf{w}_{0:1}$, the $\mathbf{z}$ are not constrained to have a particular dimension and can be a low-dimensional vector of latent factors of variation. They can be continuous or discrete. The learning objective for the lifted p_θ is given by:

$$\log p_\theta(\mathbf{x}) \geq \mathbb{E}_{q_\phi(\mathbf{z}|\mathbf{x})}[\log p_\theta(\mathbf{x}|\mathbf{z})] - \text{D}_{\text{KL}}(q_\phi(\mathbf{z}|\mathbf{x})||p_\theta(\mathbf{z})) \quad (3.6)$$

$$\geq \mathbb{E}_{q_\phi(\mathbf{z}|\mathbf{x})}\text{ELBO}(p_\theta(\mathbf{w}_{0:1}|\mathbf{z}), q_\phi(\mathbf{w}_{0:1}|\mathbf{z})) - \text{D}_{\text{KL}}(q_\phi(\mathbf{z}|\mathbf{x})||p_\theta(\mathbf{z})), \quad (3.7)$$

where $\text{ELBO}(p_\theta(\mathbf{w}_{0:1}|\mathbf{z}), q_\phi(\mathbf{w}_{0:1}|\mathbf{z}))$ denotes the variational lower bound (VLB) of a diffusion model (defined in Eq. 2.9) with a forward process $q_\phi(\mathbf{w}_{0:1}|\mathbf{z})$ (defined in Eq. 3.3 and Sec. 3.2.2) and an approximate reverse process $p_\theta(\mathbf{w}_{0:1}|\mathbf{z})$ (defined in Eq. 3.4), both conditioned on $\mathbf{z}$. The distribution $q_\phi(\mathbf{z}|\mathbf{x})$ is an approximate posterior for $\mathbf{z}$ parameterized by a neural network with parameters ϕ .

Crucially, note that in the learning objective (Eq. 3.7), the context, which in this case is $\mathbf{z}$, is available at training time in both the forward and reverse processes. At generation time, we can still obtain a valid context vector by sampling an auxiliary latent from $p_\theta(\mathbf{z})$. Thus, this approach addresses the aforementioned challenges and enables us to use the noise parameterization in Eq. 3.5.

Although we apply Jensen’s inequality twice to get (3.7), this also enables us to learn the noise process, which significantly offsets any potential increase in ELBO gap reduction and improves $\text{ELBO}(p_\theta(\mathbf{w}_{0:1}|\mathbf{z}), q_\phi(\mathbf{w}_{0:1}|\mathbf{z}))$ by optimizing over a more expressive class of posteriors. This claim is empirically validated in Table 3.2.

3.2.4 Variational Lower Bound

Next, we derive a precise formula for the learning objective (3.7) of the auxiliary-variable diffusion model. Using the objective of a diffusion model in (2.9) we can write (3.7) as the sum of four terms:

$$\log p_\theta(\mathbf{x}) \geq \mathbb{E}_{q_\phi}[\mathcal{L}_{\text{recons}} + \mathcal{L}_{\text{diffusion}} + \mathcal{L}_{\text{prior}} + \mathcal{L}_{\text{latent}}], \quad (3.8)$$

The reconstruction loss, $\mathcal{L}_{\text{recons}}$, can be (stochastically and differentiably) estimated using standard techniques; see [Kingma and Welling, 2013], $\mathcal{L}_{\text{prior}} = -\text{D}_{\text{KL}}[q_\phi(\mathbf{w}_1|\mathbf{x}, \mathbf{z})||p_\theta(\mathbf{w}_1)]$ is the diffusion prior term, $\mathcal{L}_{\text{latent}} = -\text{D}_{\text{KL}}[q_\phi(\mathbf{z}|\mathbf{x})||p_\theta(\mathbf{z})]$ is the latent prior term, and $\mathcal{L}_{\text{diffusion}}$ is the diffusion loss term, which we examine below. The complete derivation is given in Suppl. A.5.3.

Diffusion Loss

Discrete-Time Diffusion. We start by defining p_θ in discrete time, and as in Chapter 2, we let $T > 0$ be the number of total time steps and define $t(i) = i/T$ and $s(i) = (i - 1)/T$ as indexing variables over the time steps. We also use $\mathbf{w}_{0:1}$ to denote the subset of variables associated with these timesteps. Starting with the expression in Eq. 2.9 and following the steps in Suppl. A.5, we can write $\mathcal{L}_{\text{diffusion}}$ as:

$$\begin{aligned} \mathcal{L}_{\text{diffusion}} &= - \sum_{i=2}^T \text{D}_{\text{KL}}[q_\phi(\mathbf{w}_{s(i)}|\mathbf{w}_{t(i)}, \mathbf{x}, \mathbf{z})||p_\theta(\mathbf{w}_{s(i)}|\mathbf{w}_{t(i)}, \mathbf{z})] \\ &= \frac{1}{2} \sum_{i=2}^T [(\boldsymbol{\epsilon}_t - \boldsymbol{\epsilon}_\theta(\mathbf{w}_t, \mathbf{z}, t(i)))^\top \text{diag}(\boldsymbol{\gamma}(\mathbf{z}, s(i)) - \boldsymbol{\gamma}(\mathbf{z}, t(i))) (\boldsymbol{\epsilon}_t - \boldsymbol{\epsilon}_\theta(\mathbf{w}_t, \mathbf{z}, t(i)))] \end{aligned} \quad (3.9)$$

Continuous-Time Diffusion. We can also consider the limit of the above objective as we take an infinitesimally small partition of $t \in [0, 1]$, which corresponds to the limit when $T \rightarrow \infty$. In Suppl. A.5 we show that taking this limit of Eq. 3.9 yields the continuous-time diffusion loss:

$$\mathcal{L}_{\text{diffusion}} = -\frac{1}{2} \mathbb{E}_{t \sim [0,1]} [(\boldsymbol{\epsilon}_t - \boldsymbol{\epsilon}_\theta(\mathbf{w}_t, \mathbf{z}, t))^\top \text{diag}(\nabla_t \boldsymbol{\gamma}(\mathbf{z}, t)) (\boldsymbol{\epsilon}_t - \boldsymbol{\epsilon}_\theta(\mathbf{w}_t, \mathbf{z}, t))] \quad (3.10)$$

where $\nabla_t \boldsymbol{\gamma}(\mathbf{z}, t) \in \mathbb{R}^d$ denotes the Jacobian of $\boldsymbol{\gamma}(\mathbf{z}, t)$ with respect to the scalar t . We observe that the limit of $T \rightarrow \infty$ yields improved performance, matching the existing theoretical argument by Kingma et al. [2021].

Auxiliary latent loss

We try two different kinds of priors for $p_\theta(\mathbf{z})$: discrete ($\mathbf{z} \in \{0, 1\}^m$) and continuous ($\mathbf{z} \in \mathbb{R}^m$).

Continuous Auxiliary Latents. In the case where $\mathbf{z}$ is continuous, we select $p_\theta(\mathbf{z})$ as $\mathcal{N}(\mathbf{0}, \mathbf{I}_m)$. This leads to the following KL loss term:

$$\text{D}_{\text{KL}}(q_\phi(\mathbf{z}|\mathbf{x})||p_\theta(\mathbf{z})) = \frac{1}{2}(\boldsymbol{\mu}^\top(\mathbf{x})\boldsymbol{\mu}(\mathbf{x})) + \text{tr}(\boldsymbol{\Sigma}^2(\mathbf{x}) - \mathbf{I}_m) - \log |\boldsymbol{\Sigma}^2(\mathbf{x})|. \quad (3.11)$$

Discrete Auxiliary Latents. In the case where $\mathbf{z}$ is discrete, we select $p_\theta(\mathbf{z})$ as a uniform distribution. Let $\mathbf{z} \in \{0, 1\}^m$ be a k -hot vector sampled from a discrete Exponential Family distribution $p_\theta(\mathbf{z}; \theta)$ with logits θ . Niepert et al. [2021] show that $\mathbf{z} \sim p_\theta(\mathbf{z}; \theta)$ is equivalent to $\mathbf{z} = \arg \max_{y \in Y} \langle \theta + \epsilon_g, y \rangle$ where ϵ_g denotes the sum of gamma distribution Suppl. A.6, Y denotes the set of all k -hot vectors of some fixed length m . For $k > 1$, To differentiate through the arg max we use a relaxed estimator, Identity, as proposed by Sahoo et al. [2023]. This leads to the

following KL loss term:

$$D_{\text{KL}}(q_\phi(\mathbf{z}|\mathbf{x})||p_\theta(\mathbf{z})) = -\sum_{i=1}^m q_\phi(\mathbf{z}|\mathbf{x})_i (\log q_\phi(\mathbf{z}|\mathbf{x})_i + \log m). \quad (3.12)$$

3.2.5 The Variational Lower Bound as a Line Integral Over The Noise Schedule

Having defined our loss, we now return to the question of whether it is invariant to the choice of diffusion process. Notice that we may rewrite Eq. 3.10 in the following vectorized form:

$$\mathcal{L}_{\text{diffusion}} = -\frac{1}{2} \int_0^1 (\mathbf{x} - \mathbf{x}_\theta(\mathbf{w}_t, \mathbf{z}, t))^2 \cdot \nabla_t \boldsymbol{\nu}(\mathbf{z}, t) dt \quad (3.13)$$

where the square is applied elementwise. We seek to rewrite (3.13) as a line integral $\int_a^b \mathbf{f}(\mathbf{r}(t)) \cdot \frac{d}{dt} \mathbf{r}(t) dt$ for some vector field $\mathbf{f}$ and trajectory $\mathbf{r}(t)$. Recall that $\boldsymbol{\nu}(\mathbf{z}, t)$ is monotonically decreasing in each coordinate as a function of t ; hence, it is invertible on its image, and we can write $t = \boldsymbol{\nu}_z^{-1}(\boldsymbol{\nu}(\mathbf{z}, t))$ for some $\boldsymbol{\nu}_z^{-1}$. Let $\bar{\mathbf{x}}_\theta(\mathbf{w}_{\boldsymbol{\nu}(\mathbf{z}, t)}, \mathbf{z}, \boldsymbol{\nu}(\mathbf{z}, t)) \equiv \mathbf{x}_\theta(\mathbf{w}_{\boldsymbol{\nu}_z^{-1}(\boldsymbol{\nu}(\mathbf{z}, t))}, \mathbf{z}, \boldsymbol{\nu}_z^{-1}(\boldsymbol{\nu}(\mathbf{z}, t)))$ and note that for all t , we can write $\mathbf{w}_t$ as $\mathbf{w}_{\boldsymbol{\nu}(\mathbf{z}, t)}$; see Eq. A.18, and have $\bar{\mathbf{x}}_\theta(\mathbf{w}_{\boldsymbol{\nu}(\mathbf{z}, t)}, \mathbf{z}, \boldsymbol{\nu}(\mathbf{z}, t)) \equiv \mathbf{x}_\theta(\mathbf{w}_t, \mathbf{z}, t)$. We can then write the integral in (3.13) as

$$\int_0^1 (\mathbf{x} - \bar{\mathbf{x}}_\theta(\mathbf{w}_{\boldsymbol{\nu}(\mathbf{z}, t)}, \mathbf{z}, \boldsymbol{\nu}(\mathbf{z}, t)))^2 \cdot \frac{d}{dt} \boldsymbol{\nu}(\mathbf{z}, t) dt,$$

which is a line integral with $\mathbf{f}(\mathbf{r}(t)) \equiv (\mathbf{x} - \bar{\mathbf{x}}_\theta(\mathbf{w}_{\boldsymbol{\nu}(\mathbf{z}, t)}, \mathbf{z}, \boldsymbol{\nu}(\mathbf{z}, t)))^2$ and $\mathbf{r}(t) \equiv \boldsymbol{\nu}(\mathbf{z}, t)$.

Intuitive explanation. Imagine piloting a plane across a region with cyclones and strong winds, as shown in Fig. A.2. Plotting a direct, straight-line course through these adverse weather conditions requires more fuel and effort due to

increased resistance. By navigating around the cyclones and winds, however, the plane reaches its destination with less energy, even if the route is longer.

This intuition translates into mathematical and physical terms. The plane’s trajectory is denoted by $\mathbf{r}(t) \in \mathbb{R}_+^n$, while the forces acting on it are represented by $\mathbf{f}(\mathbf{r}(t)) \in \mathbb{R}^n$. The work required to navigate is given by $\int_0^1 \mathbf{f}(\mathbf{r}(t)) \cdot \frac{d}{dt}\mathbf{r}(t), dt$. Here, the work depends on the trajectory because $\mathbf{f}(\mathbf{r}(t))$ is not a conservative field.

This concept also applies to the diffusion NELBO. From Eq. 3.13, it’s clear that the trajectory $\mathbf{r}(t)$ is parameterized by the noise schedule $\boldsymbol{\nu}(\mathbf{z}, t)$, which is influenced by complex forces, $\mathbf{f}$ (analogous to weather patterns), represented by the dimension-wise reconstruction error of the denoising model, $(\mathbf{x} - \mathbf{x}_\theta(\mathbf{w}_t, \mathbf{z}, t))^2$. Thus, the diffusion loss, $\mathcal{L}_{\text{diffusion}}$, can be interpreted as the work done along the trajectory $\boldsymbol{\nu}(\mathbf{z}, t)$ in the presence of these vector field forces $\mathbf{f}$. By learning the noise schedule, we can avoid “high-resistance” paths (those where the loss accumulates rapidly), thereby minimizing the overall “energy” expended, as measured by the NELBO. Since the diffusion process corresponds to non-conservative force fields, as noted in Spinney and Ford [2012], different noise schedules should yield different NELBOs—a result supported by our empirical findings. In Suppl. A.5.5, we show that variational diffusion models are limited to linear trajectories $\boldsymbol{\nu}(t)$, rendering their objective invariant to the noise schedule. In contrast, our approach learns a multivariate $\boldsymbol{\nu}$, enabling paths that achieve a better ELBO.

3.3 Experiments

This section reports experiments on the CIFAR-10 [Krizhevsky et al., 2009] and ImageNet-32 [Van Den Oord et al., 2016] datasets. We don’t employ data augmen-

tation and we use the same architecture and settings as in the VDM model [Kingma et al., 2021]. The encoder, $q_\phi(\mathbf{z}|\mathbf{x})$, is modeled using a sequence of 4 ResNet blocks which is much smaller than the denoising network that uses 32 such blocks (i.e., we increase parameter count by only about 10%); the noise schedule γ_ϕ is modeled using a two-layer MLP. In all our experiments, we use discrete auxiliary latents with $m = 50$ and $k = 15$. A detailed description can be found in Suppl. A.7.

3.3.1 Training Speed

In these experiments, we replace VDM’s noise process with MuLAN. On CIFAR-10, MuLAN attains VDM’s likelihood score of 2.65 in just 2M steps, compared to VDM’s 10M steps 3.1). When trained on 4 V100 GPUs, VDM achieves a training rate of 2.6 steps/second, while MuLAN trains slightly slower at 2.24 steps/second due to the inclusion of an additional encoder network. However, despite this slower training pace, VDM requires 30 days to reach a BPD of 2.65, whereas Mulan achieves the same BPD within a significantly shorter timeframe of 10 days. On ImageNet-32, VDM integrated with MuLAN reaches a likelihood of 3.71 in half the time, achieving this score in 1M steps versus the 2M steps required by VDM.

3.3.2 Likelihood Estimation

In Table 3.2, we also compare MuLAN with other recent methods on CIFAR-10 and ImageNet-32. MuLAN was trained using v-parameterization for 8M steps on CIFAR-10 and 2M steps on Imagenet-32. During inference, we extract the underlying probability flow ODE and use it to estimate the log-likelihood;

see Suppl. A.9.2. Our algorithm **establishes a new state-of-the-art in density estimation** on both ImageNet-32 and CIFAR-10. In Table A.5, we also compute variational lower bounds (VLBs) of ≤ 2.59 and ≤ 3.71 on CIFAR-10 and ImageNet, respectively. Each bound improves over published results (Table 3.2); our true NLLs (via flow ODEs) are even lower.

Table 3.1: Likelihood in bits per dimension (BPD) based on the Variational Lower Bound (VLB) estimate (Suppl. A.9.1), sample quality (FID scores) and number of function evaluations (NFE) on CIFAR-10, for vanilla VDM and VDM when endowed with MuLAN. FID and NFE were computed for 10k samples generated using an adaptive-step ODE solver. Both methods use noise parameterization (Suppl. A.5.1).

Model	Steps	CIFAR-10				ImageNet			
		VLB ($\downarrow$)	FID ($\downarrow$)	NFE ($\downarrow$)		Steps	VLB ($\downarrow$)	FID ($\downarrow$)	NFE ($\downarrow$)
VDM [Kingma et al., 2021]	10M	2.65	23.91	56		2M	3.72	14.26	56
+ MuLAN	2M	2.65	18.54	55		1M	3.72	15.00	62
+ MuLAN	10M	2.60	17.62	50		2M	3.71	13.19	62

3.3.3 Alternative Learned Diffusion Methods

Concurrent work that seeks to improve log-likelihood estimation by learning the forward diffusion process includes Neural Diffusion Models (NDMs) [Bartosh et al., 2023] and DiffEnc [Nielsen et al., 2023]. In NDMs, the noise schedule is fixed, but the mean of each marginal $q(\mathbf{w}_t|\mathbf{x})$ is learned, while DiffEnc adds a correction term to q . Diffusion normalizing flows (DNFs) represent an earlier effort where q is a normalizing flow trained by backpropagating through sampling. In Table 3.3, we compare against NDMs, DiffEnc, and DNFs on the CIFAR-10 dataset, using the authors’ published results; note that their published ImageNet numbers are either not available or are reported on a different dataset version that is not comparable. Our approach to learned diffusion outperforms previous and concurrent work.

Table 3.2: Likelihood in bits per dimension (BPD) on the test set of CIFAR-10 and ImageNet. Results with “/” means they are not reported in the original papers. Model types are autoregressive (AR), normalizing flows (Flow), diffusion models (Diff). We only compare with results achieved without data augmentation.

Model	Type	CIFAR-10 ($\downarrow$)	ImageNet ($\downarrow$)
PixelCNN [Van den Oord et al., 2016]	AR	3.03	3.83
Image Transformer [Parmar et al., 2018]	AR	2.90	3.77
DDPM [Ho et al., 2020]	Diff	≤ 3.69	/
ScoreFlow [Song et al., 2021b]	Diff	2.83	3.76
VDM [Kingma et al., 2021]	Diff	≤ 2.65	≤ 3.72
Flow Matching [Lipman et al., 2022]	Flow	2.99	/
Reflected Diffusion Models [Lou and Ermon, 2023]	Diff	2.68	3.74
MuLAN (Ours)	Diff	2.55 ± 10^{-3}	3.67 ± 10^{-3}

Table 3.3: Likelihood in bits per dimension (bpd) on CIFAR-10 for learned diffusion methods.

Model	NLL ($\downarrow$)
DNF [Zhang and Chen, 2021]	3.04
NDM [Bartosh et al., 2023]	≤ 2.70
DiffEnc [Nielsen et al., 2023]	≤ 2.62
MuLAN	2.55

3.3.4 Ablation Analysis And Additional Experiments

Due to the expensive cost of training, we only performed ablation studies on CIFAR-10 with a reduced batch size of 64 and trained the model for 2.5M training steps. In Fig. 3.2 we ablate each component of MuLAN: when we remove the conditioning on an auxiliary latent space from MuLAN so that we have a multivariate noise schedule that is solely conditioned on time t , our performance becomes comparable to that of VDM, on which our model is based. Changing to a scalar noise schedule based on latent variable $\mathbf{z}$ initially underperforms compared to VDM. This drop aligns with our likelihood formula (Eq. 3.5) which includes $D_{\text{KL}}(q_{\phi}(\mathbf{z}|\mathbf{x})|p_{\theta}(\mathbf{z}))$, an extra term not in VDM. The input-conditioned scalar schedule doesn’t offer any advantage over the scalar schedule used in VDM. This is due to the reasons

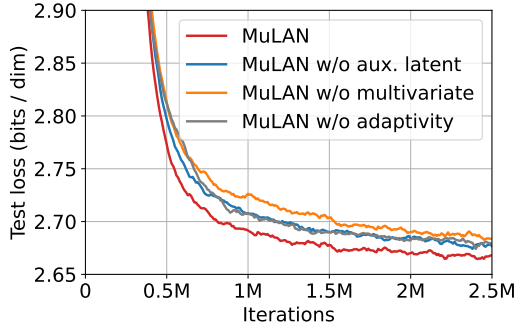


Figure 3.2: In MuLan w/o aux. latent, the noise isn’t conditioned on a latent. MuLan w/o multivariate uses a scalar noise schedule. MuLan w/o adaptivity has a linear schedule and no auxiliary latents.

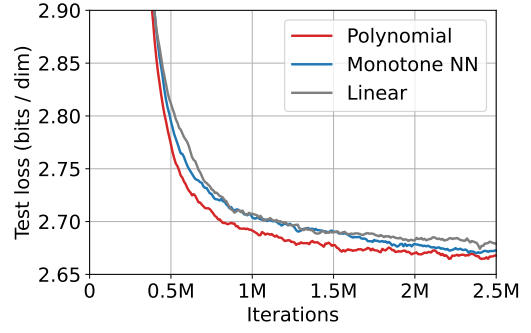


Figure 3.3: MuLan with different noise schedule parameterizations: polynomial, monotonic neural network, and linear. Our proposed polynomial parameterization performs the best.

Figure 3.4: Ablating components of MuLan on CIFAR-10 over 2.5M steps with batch size of 64.

outlined in Sec. 3.2.5.

Perceptual Quality. While perceptual quality is not the focus of this work, we report FID numbers for the VDM model and MuLan (Table 3.1). We use RK45 ODE solver to generate samples by solving the reverse time Flow ODE (Eq. A.63). We observe that MuLan does not degrade FIDs, while improving log-likelihood estimation. Note that MuLan does not incorporate many tricks that improve FID such as exponential moving averages, truncations, specialized learning schedules, etc.; our FID numbers can be improved in future work using these techniques.

Loss curves for different noise schedules. We investigate different parameterizations of the noise schedule in Fig. 3.3. Among polynomial, linear, and monotonic neural network, we find that the polynomial parameterization yields the best performance. The polynomial noise schedule is a novel component introduced

in our work. The reason why a polynomial function works better than a linear or a monotonic neural network as proposed by VDM is rooted in Occam’s razor. In Suppl. A.5.2, we show that a degree 5 polynomial is the simplest polynomial that satisfies several desirable properties, including monotonicity and having a derivative that equals zero exactly twice. More expressive models (e.g., monotonic 3-layer MLPs) are more difficult to optimize.

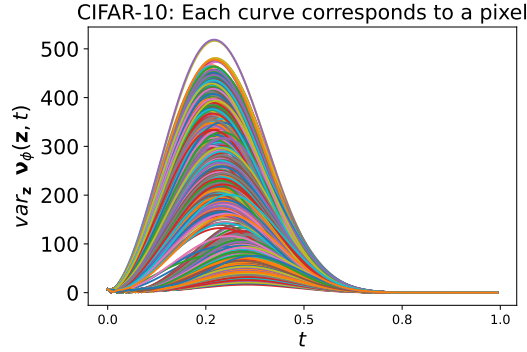


Figure 3.5: Noise schedule visualizations for MuLAN on CIFAR-10. In this figure, we plot the variance of $\nu_\phi(\mathbf{z}, t)$ across different $\mathbf{z} \sim p_\theta(\mathbf{z})$ where each curve represents the SNR corresponding to an input dimension.

Examining the noise schedule. Since the noise schedule, $\gamma_\phi(\mathbf{z}, t)$ is multivariate, we expect to learn different noise schedules for different input dimensions and different inputs $\mathbf{z} \sim p_\theta(\mathbf{z})$. In Fig. 3.5, we take our best trained model on CIFAR-10 and visualize the variance of the noise schedule at each point in time for different pixels, where the variance is taken on 128 samples $\mathbf{z} \sim p_\theta(\mathbf{z})$.

We note an increased variation in the early portions of the noise schedule. However, on an absolute scale, the variance of this noise is smaller than we expected. We also tried to visualize noise schedules across different dataset images and across different areas of the same image; refer to Fig. A.20. We also generated synthetic datasets in which each datapoint contained only high frequencies or only low frequencies, and with random masking applied to parts of the data

points; see Suppl. A.8. Surprisingly, none of these experiments revealed human-interpretable patterns in the learned schedule, although we did observe clear differences in likelihood estimation. We hypothesize that other architectures and other forms of conditioning may reveal interpretable patterns of variation; however, we leave this exploration to future work.

Replacing the noise schedules in a trained denoising model. We also confirm experimentally our claim that the learning objective is not invariant to the multivariate noise schedule. We replace the noise schedule in the trained denoising model with two alternatives: MULAN with scalar noise schedule, and a linear noise schedule: $\gamma_\phi(\mathbf{z}, t) = \gamma_{\min} + t(\gamma_{\max} - \gamma_{\min})\mathbf{1}_d$; see Kingma et al. [2021]. For both the noise schedules the likelihood reduces to the same value as that of the VDM: 2.65.

3.4 Related Work

Diffusion models have emerged in recent years as powerful tools for modeling complex distributions [Sohl-Dickstein et al., 2015, Ho et al., 2020], extending flow-based methods [Song et al., 2020, Kingma and Dhariwal, 2018, Si et al., 2023]. The noise schedule, which determines the amount and type of noise added at each step, plays a critical role in diffusion models. Chen [2023] empirically demonstrate that different noise schedules can significantly impact the generated image quality using various handcrafted noise schedules. Kingma et al. [2021] showed that the likelihood of a diffusion model remains invariant to the noise schedule with a scalar noise schedule. In this work we show that the ELBO is no longer invariant to multivariate noise schedules.

Recent works explored multivariate noise schedules (including blurring, masking, etc.) [Hooeboom and Salimans, 2022, Rissanen et al., 2022, Pearl et al., 2023, Du et al., 2023], yet none have delved into learning the noise schedule conditioned on the input data itself. Likewise, conditional noise processes are typically not learned [Lee et al., 2021, Popov et al., 2021, Yang et al., 2024] and their conditioner (e.g., a prompt) is always available. Auxiliary variable models [Yang and Mandt, 2023, Wang et al., 2023] add semantic latents in p , but not in q , and they don’t condition or learn q . In contrast, we learn multivariate noise conditioned on latent context.

Diffusion normalizing flows (DNFs) [Zhang and Chen, 2021] learn a q parameterized by a normalizing flow; however, such q do not admit tractable marginals and require sampling full data-to-noise trajectories from q , which is expensive. Concurrent work on neural diffusion models (NDMs) and DiffEnc admits tractable marginals q with learned means and univariate schedules; this yields more expressive q than ours but requires computing losses in a modified space that precludes using a noise parameterization and certain sampling strategies. Empirically, MuLAN performs better with fewer parameters (Suppl. A.1).

Optimal transport techniques seek to learn a noise process that minimizes the transport cost from data to noise, which in practice produces smoother diffusion trajectories that facilitate sampling. Schrödinger bridges [Shi et al., 2024b, De Bortoli et al., 2021, Wang et al., 2021, Peluchetti, 2023] learn expressive q do not admit analytical marginals, require computing full data-to-noise trajectories and involve iterative optimization (e.g., sinkhorn), which can be slow. Rectification Lee et al. [2023] seeks diffusion paths that are close to linear; this improves sampling, while our method chooses paths that improve log-likelihood. See Suppl. A.1 for more detailed comparisons.

3.5 Conclusion

We introduced MULAN, a context-adaptive noise process that applies Gaussian noise at varying rates across input data. Our theory challenges the prevailing assumption that the likelihood of diffusion models is independent of the noise schedule; we show that this independence holds only for univariate schedules.

Our evaluation spans multiple image datasets, where MULAN consistently outperforms state-of-the-art generative diffusion models. These results highlight the importance of designing richer, data-dependent noise schedules for improving both likelihood estimation and image generation quality [Parmar et al., 2018, Song et al., 2020]. A stronger fit to the underlying data distribution also benefits downstream applications of generative modeling, such as compression and decision-making [Nguyen and Grover, 2022, Deshpande et al., 2022, Deshpande and Kuleshov, 2023, Rastogi et al., 2023]. **Recent work in language generation further supports this direction: Shi et al. [2024a] show that learning input-conditioned noise schedules can tighten likelihood bounds for language models**, underscoring the broader value of adaptive noise design across modalities.

CHAPTER 4

SIMPLE DIFFUSION LANGUAGE MODELS

4.1 Introduction

Diffusion models excel at producing realistic, high-quality images and have received significant attention as potential tools for generating discrete data, such as text [Austin et al., 2021, Li et al., 2021, Lou et al., 2024], biological sequences [Avdeyev et al., 2023, Rastogi and Schiff, 2023], and graphs [Sun and Yang, 2023, Vignac et al., 2022]. However, discrete diffusion methods exhibit a performance gap relative to AR models [Austin et al., 2021, Gulrajani and Hashimoto, 2024, He et al., 2022, Lou et al., 2024], especially in language modeling. The standard measure of language modeling performance is log-likelihood: when controlling for parameter count, prior work reports a sizable log-likelihood gap between AR and diffusion models.

In this chapter, we introduce a simplified discrete diffusion framework tailored to the most commonly used noising processes, such as masking and uniform noise described in [Chapter 2](#). we show that simple masked diffusion language modeling (MDLM) combined with effective training recipes is more performant than previously thought [Austin et al., 2021, He et al., 2022, Zheng et al., 2023b]. We develop a well-engineered MDLM implementation that significantly improves discrete diffusion log-likelihood; we further improve likelihood using a simple substitution-based parameterization of the reverse diffusion process that enables deriving a Rao-Blackwellized continuous-time variational lower bound (ELBO) with improved tightness [Sahoo et al. \[2023\]](#). Interestingly, our objective has a simple form: it is a weighted average of masked language modeling (MLM) losses [Devlin et al., 2018], and can be used to endow BERT-style, encoder-only models with

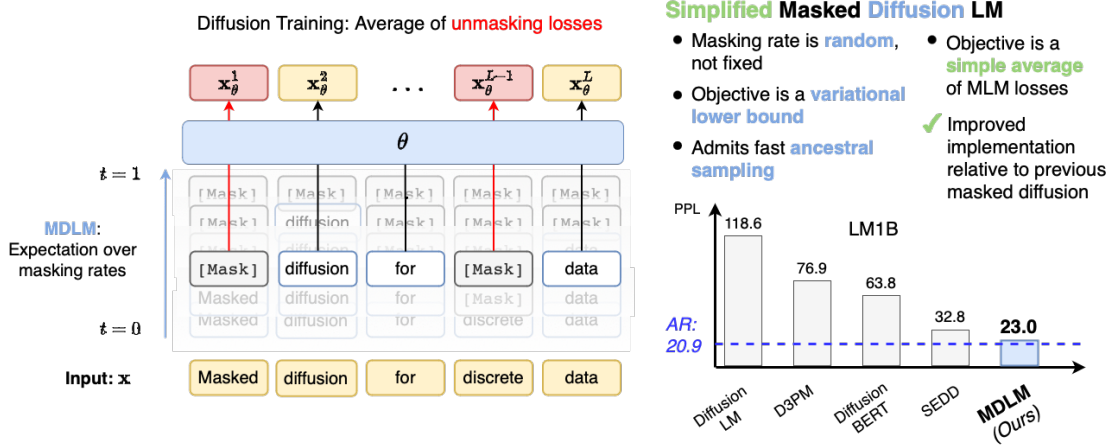


Figure 4.1: (Left) Our proposed masked diffusion language model (MDLM) is trained using a weighted average of masked cross entropy losses. (Top Right) In comparison to masked language models (MLM), MDLM’s objective correspond to a principled variational lower bound, and supports generation via ancestral sampling. (Bottom Right) Perplexity (PPL) on One Billion Words (LM1B) benchmark.

principled generation capabilities. We complement this framework with efficient samplers—including ones that can generate semi-autoregressively like a typical language model.

Our masked diffusion models achieve a new state-of-the-art among diffusion models on language modeling benchmarks and approach the perplexity of AR models within 15-25%. Surprisingly, simple engineering choices significantly improve performance in both our models and simple baselines that were previously thought to perform poorly. Our framework also extends to non-language domains, including biological sequence modeling. We pre-train DNA sequence models and observe similar or higher downstream performance compared to classical BERT-style training, while also introducing generative capabilities that classical masked DNA language models lack. Taken together, these developments establish a cohesive MDLM framework that advances the empirical performance of discrete diffusion models and clarifies their theoretical foundations. The well-engineered implemen-

tation we provide consistently outperforms existing diffusion models across major benchmarks (LM1B[Chelba et al., 2014], OWT[Gokaslan et al., 2019], DNA [Consortium et al., 2009]) and substantially improves the strength of earlier baselines [Austin et al., 2021, He et al., 2022]. Central to this framework is a substitution-based parameterization of the reverse unmasking process, which enables a simple continuous-time, Rao-Blackwellized objective with tighter and lower-variance ELBO estimates, resulting in further gains. We also introduce fast sampling methods that support semi-autoregressive generation and surpass the performance of prior SAR approaches.

4.2 Interpolating Discrete Diffusion Processes

While previous work on discrete diffusion supports general forward processes (e.g., general Q_t in D3PM), absorbing state (i.e., masking) diffusion consistently achieves the best performance [Austin et al., 2021, Lou et al., 2024]. In this work, instead of supporting general noise processes, we focus on masking and derive tight Rao-Blackwellized objectives that outperform general approaches and do not require CTMC theory. In this section, we first define the diffusion process for a categorical random variable. Later in Sec. 4.2.1, we extend this process to sequences containing multiple such categorical variables. We denote our overall approach as Masked Diffusion Language Models (MDLM).

Notation. We denote scalar discrete random variables with K categories as ‘one-hot’ column vectors and define $\mathcal{V} \in \{\mathbf{x} \in \{0, 1\}^K : \sum_{i=1}^K \mathbf{x}_i = 1\}$ as the set of all such vectors. Define $\text{Cat}(\cdot; \boldsymbol{\pi})$ as the categorical distribution over K classes with probabilities given by $\boldsymbol{\pi} \in \Delta^K$, where Δ^K denotes the K -simplex. We also assume

that the K -th category corresponds to a special [MASK] token and let $\mathbf{m} \in \mathcal{V}$ be the one-hot vector for this mask, i.e., $\mathbf{m}_K = 1$. Additionally, let $\mathbf{1} = \{1\}^K$ and $\langle \mathbf{a}, \mathbf{b} \rangle$ and $\mathbf{a} \odot \mathbf{b}$ respectively denote the dot and Hadamard products between two vectors $\mathbf{a}$ and $\mathbf{b}$.

We restrict our attention to forward processes q that interpolate between clean data $\mathbf{x} \in \mathcal{V}$ and a target distribution $\text{Cat}(\cdot; \boldsymbol{\pi})$, forming a direct extension of Gaussian diffusion. Let q define a sequence of increasingly noisy latent variables $\mathbf{z}_t \in \mathcal{V}$, where the time step t runs from $t = 0$ (least noisy) to $t = 1$ (most noisy). The marginal of $\mathbf{z}_t$ conditioned on $\mathbf{x}$ at time t is

$$q(\mathbf{z}_t | \mathbf{x}) = \text{Cat}(\mathbf{z}_t; \alpha_t \mathbf{x} + (1 - \alpha_t) \boldsymbol{\pi}), \quad (4.1)$$

where $\alpha_t \in [0, 1]$ is a strictly decreasing function in t , with $\alpha_0 \approx 1$ and $\alpha_1 \approx 0$; see Suppl. B.5.1 for details. This implies transition probabilities $q(\mathbf{z}_t | \mathbf{z}_s) = \text{Cat}(\mathbf{z}_t; \alpha_{t|s} \mathbf{z}_s + (1 - \alpha_{t|s}) \boldsymbol{\pi})$, where $\alpha_{t|s} = \alpha_t / \alpha_s$. This indicates that during each diffusion step from $s \rightarrow t$, a fraction $(1 - \alpha_{t|s})$ of the probability mass is transferred to the prior distribution $\boldsymbol{\pi}$. The reverse posterior is given as (see Suppl. B.4 for details):

$$q(\mathbf{z}_s | \mathbf{z}_t, \mathbf{x}) = \text{Cat} \left(\mathbf{z}_s; \frac{[\alpha_{t|s} \mathbf{z}_t + (1 - \alpha_{t|s}) \mathbf{1} \boldsymbol{\pi}^\top \mathbf{z}_t] \odot [\alpha_s \mathbf{x} + (1 - \alpha_s) \boldsymbol{\pi}]}{\alpha_t \mathbf{z}_t^\top \mathbf{x} + (1 - \alpha_t) \mathbf{z}_t^\top \boldsymbol{\pi}} \right). \quad (4.2)$$

While Eq. 4.1 and Eq. 4.2 represent a special case of the more general diffusion processes proposed in D3PM [Austin et al., 2021], we show below that they yield a simplified variational lower bound objective and admit straightforward continuous time extensions.

4.2.1 Masked Diffusion

Next, we focus on masking processes and derive a simple Rao-Blackwellized objective for this choice of q . This objective incurs lower variance during training and improves tightness.

Forward Masking Process

In masked (i.e., absorbing state) diffusion, we set $\boldsymbol{\pi} = \mathbf{m}$. At each noising step, t , the input $\mathbf{x}$ transitions to a ‘masked’ state $\mathbf{m}$ with some probability. If an input transitions to $\mathbf{m}$ at any time t' , it will remain in this state for all $t > t' : q(\mathbf{z}_t \mid \mathbf{z}_{t'} = \mathbf{m}) = \text{Cat}(\mathbf{z}_t; \mathbf{m})$. At time T , all inputs are masked with probability 1.

The marginal of the forward process (4.1) is given by

$$q(\mathbf{z}_t \mid \mathbf{x}) = \text{Cat}(\mathbf{z}_t; \alpha_t \mathbf{x} + (1 - \alpha_t) \mathbf{m}).$$

Using properties of the masking process, the posterior $q(\mathbf{z}_s \mid \mathbf{z}_t, \mathbf{x})$ simplifies (4.2); see Suppl. B.1.2:

$$q(\mathbf{z}_s \mid \mathbf{z}_t, \mathbf{x}) = \begin{cases} \text{Cat}(\mathbf{z}_s; \mathbf{z}_t) & \mathbf{z}_t \neq \mathbf{m}, \\ \text{Cat}\left(\mathbf{z}_s; \frac{(1 - \alpha_s) \mathbf{m} + (\alpha_s - \alpha_t) \mathbf{x}}{1 - \alpha_t}\right) & \mathbf{z}_t = \mathbf{m}. \end{cases} \quad (4.3)$$

Reverse Unmasking Process

The reverse process inverts the noise process defined by q . We consider both a finite number of steps T , as well as a continuous time model corresponding to $T \rightarrow \infty$.

We begin with the discrete-time case for which the generative model is expressed as

$$p_\theta(\mathbf{x}) = \int_{\mathbf{z}} p_\theta(\mathbf{z}_1) p_\theta(\mathbf{x}|\mathbf{z}_0) \prod_{i=1}^T p_\theta(\mathbf{z}_s|\mathbf{z}_t) d\mathbf{z}_{0:1}.$$

The optimal form for $p_\theta(\mathbf{z}_s|\mathbf{z}_t)$ matches the true posterior in (4.3): this follows immediately from the definition of the diffusion objective in (2.10), which is a sum of terms of the form $D_{\text{KL}}(q(\mathbf{z}_s|\mathbf{z}_t, \mathbf{x})||p_\theta(\mathbf{z}_s|\mathbf{z}_t))$. However, (4.3) is conditioned on $\mathbf{x}$, which we do not know. Therefore, we introduce a model $\mathbf{x}_\theta(\mathbf{z}_t, t) : \mathcal{V} \times [0, 1] \rightarrow \Delta^K$ that approximates $\mathbf{x}$ with a neural network. We can also omit explicit dependence of $\mathbf{x}_\theta$ on time t , which simplifies sampling, yielding a 2x inference speed-up (see Suppl. B.5.2).

SUBS Parameterization

The specific parameterization for $p_\theta(\mathbf{z}_s|\mathbf{z}_t)$ that we use is

$$p_\theta(\mathbf{z}_s|\mathbf{z}_t) = q(\mathbf{z}_s|\mathbf{z}_t, \mathbf{x} = \mathbf{x}_\theta(\mathbf{z}_t, t)) = \begin{cases} \text{Cat}(\mathbf{z}_s; \mathbf{z}_t), & \mathbf{z}_t \neq \mathbf{m}, \\ \text{Cat}\left(\mathbf{z}_s; \frac{(1-\alpha_s)\mathbf{m} + (\alpha_s - \alpha_t)\mathbf{x}_\theta(\mathbf{z}_t, t)}{1 - \alpha_t}\right), & \mathbf{z}_t = \mathbf{m}, \end{cases} \quad (4.4)$$

Furthermore, we induce 2 key properties of the absorbing state diffusion process into our denoising model, $\mathbf{x}_\theta(\mathbf{z}_t, t)$: an unmasked token remains unchanged during reverse diffusion, and the clean input is never masked. We implement these as substitutions to the output of $\mathbf{x}_\theta(\mathbf{z}_t, t)$, hence we call our parameterization SUBS.

Zero Masking Probabilities First, notice that by definition, $\langle \mathbf{x}, \mathbf{m} \rangle = 0$. For this reason, we design the denoising network such that $\langle \mathbf{x}_\theta(\mathbf{z}_t, t), \mathbf{m} \rangle = 0$, i.e., we substitute the logit index corresponding to the [MASK] token with $-\infty$.

Carry-Over Unmasking Second, if $\mathbf{z}_t$ is unmasked, then we desire $\mathbf{x}_\theta(\mathbf{z}_t, t) = \mathbf{z}_t$, i.e., unmasked latents are ‘carried over’. We accomplish this by substituting the output of our network to simply copy unmasked inputs.

In Suppl. B.2.1, we show that “Zero Masking Probabilities” property simplifies the D3PM’s NELBO Eq. B.27 to Eq. B.29, and “Carry-Over Unmasking” further simplifies Eq. B.29 to Eq. B.31 whose continuous time equivalent is the simplified NELBO Eq. 4.7. Table 4.8 shows that each simplification leads to an improved likelihood.

Rao-Blackwellized Likelihood Bounds

Recall from Eq. 2.10 that the diffusion training objective has the form $\mathcal{L}_{\text{recons}} + \mathcal{L}_{\text{diffusion}} + \mathcal{L}_{\text{prior}}$. For the simplified reverse process in Eq. 4.4, the discrete-time diffusion loss for finite T simplifies to (Suppl. B.2.1):

$$\begin{aligned}\mathcal{L}_{\text{diffusion}} &= \sum_{i=1}^T \mathbb{E}_q[\text{D}_{\text{KL}}(q(\mathbf{z}_{s(i)}|\mathbf{z}_{t(i)}, \mathbf{x})||p_\theta(\mathbf{z}_{s(i)}|\mathbf{z}_{t(i)}))] \\ &= \sum_{i=1}^T \mathbb{E}_q \left[\frac{\alpha_{t(i)} - \alpha_{s(i)}}{1 - \alpha_{t(i)}} \log \langle \mathbf{x}_\theta(\mathbf{z}_{t(i)}), \mathbf{x} \rangle \right]\end{aligned}\quad (4.5)$$

Note that this objective is simpler and more well-behaved than the expression one would obtain for $\text{D}_{\text{KL}}(q(\mathbf{z}_s|\mathbf{z}_t, \mathbf{x})||p_\theta(\mathbf{z}_s|\mathbf{z}_t))$ under the parameterization induced by using $p_\theta(\mathbf{z}_s|\mathbf{z}_t) = q(\mathbf{z}_s|\mathbf{z}_t, \mathbf{x} = \mathbf{x}_\theta(\mathbf{z}_t, t))$ from Eq. 4.2, which is similar to what is used by D3PM [Austin et al., 2021] (see Suppl. B.1.2):

$$\begin{aligned}&\left[\frac{\alpha_s - \alpha_t}{1 - \alpha_t} \log \frac{\alpha_t \langle \mathbf{x}_\theta(\mathbf{z}_t, t), \mathbf{m} \rangle + (1 - \alpha_t)}{(1 - \alpha_t) \langle \mathbf{x}_\theta(\mathbf{z}_t, t), \mathbf{x} \rangle} \right. \\ &\quad \left. + \frac{1 - \alpha_s}{1 - \alpha_t} \log \frac{(1 - \alpha_s)(\alpha_t \langle \mathbf{x}_\theta(\mathbf{z}_t, t), \mathbf{m} \rangle + (1 - \alpha_t))}{(1 - \alpha_t)(\alpha_s \langle \mathbf{x}_\theta(\mathbf{z}_t, t), \mathbf{m} \rangle + (1 - \alpha_s))} \right] \langle \mathbf{z}_t, \mathbf{m} \rangle\end{aligned}\quad (4.6)$$

We refer to the process of obtaining Eq. 4.5 in lieu of Eq. 4.6 as a form of Rao-Blackwellization. Specifically, we analytically compute expectations such as

$\langle \mathbf{x}_\theta(\mathbf{z}_t, t), \mathbf{m} \rangle = 0$ in order to simplify objective Eq. 4.6 to obtain Eq. 4.5. Without analytical simplifications, a model must learn θ such that $\langle \mathbf{x}_\theta(\mathbf{z}_t, t), \mathbf{m} \rangle = 0$ holds. Unlike in regular Rao-Blackwellization, simplifications are possible because of modeling choices for $\mathbf{x}_\theta(\mathbf{z}_t, t)$ (zero masking probabilities and carry-over unmasking). In that sense, our approach has similarities to graphical modeling, where incorporating conditional independencies into p_θ sets certain log-likelihood terms to zero. However, our approach also empirically helps reduce variance, hence we refer to it as Rao-Blackwellization, somewhat abusing the usual terminology.

Continuous-Time Likelihood Bounds

Previous works have shown empirically and mathematically that increasing the number of steps T yields a tighter approximation to the ELBO [Kingma et al., 2021]. Following a similar argument, we form a continuous extension of Eq. 4.5 by taking $T \rightarrow \infty$ (see Suppl. B.2.2), which yields the following NELBO, $\mathcal{L}_{\text{NELBO}}^\infty$:

$$\mathcal{L}_{\text{NELBO}}^\infty = \mathbb{E}_q \int_{t=0}^{t=1} \frac{\alpha'_t}{1 - \alpha_t} \log \langle \mathbf{x}_\theta(\mathbf{z}_t, t), \mathbf{x} \rangle dt \quad (4.7)$$

Invariance to the noise schedule The function α_t is invertible due to the monotonicity assumption in Sec. 4.2, and so we can perform the following change of variables in Eq. 4.7: $\gamma \equiv \log(1 - \alpha_t)$. Thus, the diffusion loss can be equivalently expressed as $\mathcal{L}_{\text{NELBO}}^\infty = -\mathbb{E}_q \int_{\gamma=-\infty}^{\gamma=0} \log \langle \mathbf{x}_\theta(\mathbf{z}_\gamma, \gamma), \mathbf{x} \rangle d\gamma$; see Suppl. B.5.1 for details. This new formulation demonstrates that the diffusion loss is invariant to the functional form of α_t , which we verify empirically in Suppl. B.5.1.

Masked Diffusion Language Models (MDLM)

Next, we apply masked diffusion to language modeling over sequences $\mathbf{x}^{1:L}$ of L tokens, with $\mathbf{x}^\ell$ denoting the ℓ -th token. We make the assumption that the forward noising process is applied independently across a sequence and that, conditioned on a sequence of latents $\mathbf{z}_t^{1:L}$, the denoising process factorizes independently across tokens, i.e., $p_\theta(\mathbf{z}_s^{1:L} \mid \mathbf{z}_t^{1:L}) = \prod_{\ell=1}^L p_\theta(\mathbf{z}_s^\ell \mid \mathbf{z}_t^{1:L})$. Thus, we use a single model to compute $\mathbf{x}_\theta^\ell(\mathbf{z}_t^{1:L}, t)$ for each ℓ from a masked sequence $\mathbf{z}_t$, optimizing:

$$\mathcal{L}_{\text{NELBO}}^\infty = \mathbb{E}_q \int_{t=0}^{t=1} \frac{\alpha'_t}{1 - \alpha_t} \sum_{\ell} \log \langle \mathbf{x}_\theta^\ell(\mathbf{z}_t^{1:L}, t), \mathbf{x}^\ell \rangle dt \quad (4.8)$$

Interestingly, our objective has a simple form: it is the weighted average of masked language modeling (MLM) losses [Devlin et al., 2018]. Thus our work establishes a connection between generative diffusion models and encoder-only BERT models. Our objective enables principled selection of a (randomized) masking rate, and also endows BERT-style models with principled generation capabilities; see Sec. 4.5. The full training algorithm is provided in Suppl. B.2.3.

Note: Although Eq. 4.8 imposes a loss on all tokens, unmasked tokens don’t contribute to the loss, as they are copied over by the denoising network due to “carry-over unmasking” (Sec. 4.2.1), effectively reducing $\log \langle \mathbf{x}_\theta^\ell(\mathbf{z}_t^{1:L}, t), \mathbf{x}^\ell \rangle$ to zero.

Training Considerations for Masked Diffusion

One of the key contributions of our work is a well-engineered implementation of masked diffusion models. Our experiments demonstrate that these improvements greatly boost performance even for methods previously thought to perform poorly, e.g., Austin et al. [2021]. Below we briefly summarize these implementation details. First, we find that tokenization is critical to performance. Small vocabularies, such

as the 8k vocabulary in [Austin et al. \[2021\]](#), result in longer-range dependencies that decrease the performance of both diffusion and AR models. Additionally, by focusing on masked diffusion, we are able to provide a numerically stable implementation of the objective function. Namely, since previous formulations of discrete diffusion were constructed to accommodate a wide range of limiting distributions [[Austin et al., 2021](#)], the objective was implemented by materializing the full transition matrices $\bar{Q}_t$ and posterior probabilities. In contrast, we evaluate $D_{\text{KL}}[q(\mathbf{z}_s | \mathbf{z}_t, \mathbf{x}) || p_\theta(\mathbf{z}_s | \mathbf{z}_t)]$ by examining only the masked token indices rather than comparing the full true and approximate posterior distributions.

Furthermore, we modernize the architecture for the denoising network relative to D3PM [[Austin et al., 2021](#)]. In lieu of the T5 architecture used in D3PM, we use the diffusion transformer (DiT) introduced in [Peebles and Xie \[2023\]](#), which integrates time step conditioning into a standard encoder-only transformer [[Vaswani et al., 2017](#)] and uses rotary positional embeddings [[Su et al., 2021](#)]. In addition, we implement a low-discrepancy sampler that reduces the variance of the ELBO, similar to [Kingma et al. \[2021\]](#) and draws correlated samples t_i rather than performing i.i.d. sampling.

4.2.2 Uniform-state Diffusion

In Uniform-state diffusion, we set $\boldsymbol{\pi} = \mathbf{1}/K$. The forward posterior is:

$$q_t(\mathbf{z}_t | \mathbf{x}) = \text{Cat} \left(\mathbf{z}_t; \alpha_t \mathbf{x} + (1 - \alpha_t) \frac{\mathbf{1}}{K} \right). \quad (4.9)$$

For this forward process, the true reverse posterior Eq. 4.2 for a timestep $s < t$ simplifies to:

$$q_{s|t}(\cdot | \mathbf{z}_t, \mathbf{x}) = \text{Cat} \left(\cdot; \frac{K\alpha_t \mathbf{z}_t \odot \mathbf{x} + (\alpha_{t|s} - \alpha_t) \mathbf{z}_t}{K\alpha_t \langle \mathbf{z}_t, \mathbf{x} \rangle + 1 - \alpha_t} + \frac{(\alpha_s - \alpha_t) \mathbf{x} + (1 - \alpha_{t|s})(1 - \alpha_s) \mathbf{1}/K}{K\alpha_t \langle \mathbf{z}_t, \mathbf{x} \rangle + 1 - \alpha_t} \right) \quad (4.10)$$

where $\alpha_{t|s} = \alpha_t / \alpha_s$. Since $\mathbf{x}$ is unavailable during inference, we approximate it with a neural network $\mathbf{x}_\theta : \mathcal{V} \times [0, 1] \rightarrow \Delta^K$ with parameters θ . The resulting approximate reverse posterior is:

$$\begin{aligned} p_{s|t}^\theta(\cdot | \mathbf{z}_t) &= q_{s|t}(\cdot | \mathbf{z}_t, \mathbf{x} = \mathbf{x}_\theta(\mathbf{z}_t, t)) \\ &= \text{Cat} \left(\cdot; \frac{K\alpha_t \mathbf{z}_t \odot \mathbf{x}_\theta(\mathbf{z}_t, t) + (\alpha_{t|s} - \alpha_t) \mathbf{z}_t}{K\alpha_t \langle \mathbf{z}_t, \mathbf{x}_\theta(\mathbf{z}_t, t) \rangle + 1 - \alpha_t} + \frac{(\alpha_s - \alpha_t) \mathbf{x}_\theta(\mathbf{z}_t, t) + (1 - \alpha_{t|s})(1 - \alpha_s) \mathbf{1}/K}{K\alpha_t \langle \mathbf{z}_t, \mathbf{x}_\theta(\mathbf{z}_t, t) \rangle + 1 - \alpha_t} \right) \end{aligned} \quad (4.11)$$

The goal is to learn an approximate reverse process p_θ which minimizes the Negative Evidence Lower Bound (NELBO). In Suppl. C.1.4, we show that for the simplified reverse process in Eq. 4.11, the diffusion training objective Eq. 2.10 simplifies as $T \rightarrow \infty$ simplifies to:

$$\text{NELBO}^{\text{Uniform}}(q, p_\theta; \mathbf{x}) = \mathbb{E}_{t \sim \mathcal{U}[0,1], q_t(\mathbf{z}_t | \mathbf{x}; \alpha_t)} f_{\text{Duo}}(\mathbf{z}_t, \mathbf{x}_\theta(\mathbf{z}_t, t), \alpha_t; \mathbf{x}), \quad (4.12)$$

where f_{Duo} ¹ is:

$$\begin{aligned} f_{\text{Duo}} = -\frac{\alpha_t'}{K\alpha_t} &\left[\frac{K}{\bar{\mathbf{x}}_i} - \frac{K}{(\bar{\mathbf{x}}_\theta)_i} - (\kappa_t \mathbb{1}_{\mathbf{z}_t=\mathbf{x}} + \mathbb{1}_{\mathbf{z}_t \neq \mathbf{x}}) \sum_j \log \frac{(\bar{\mathbf{x}}_\theta)_i}{(\bar{\mathbf{x}}_\theta)_j} - K \frac{\alpha_t}{1 - \alpha_t} \log \frac{(\bar{\mathbf{x}}_\theta)_i}{(\bar{\mathbf{x}}_\theta)_m} \mathbb{1}_{\mathbf{z}_t \neq \mathbf{x}} \right. \\ &\left. - \left((K-1)\kappa_t \mathbb{1}_{\mathbf{z}_t=\mathbf{x}} - \frac{1}{\kappa_t} \mathbb{1}_{\mathbf{z}_t \neq \mathbf{x}} \right) \log \kappa_t \right] \end{aligned} \quad (4.13)$$

This objective is simpler and more stable than the expressions proposed in D3PM [Austin et al., 2021], SEDD [Lou et al., 2024], and UDLM [Schiff et al., 2024]. In the remainder of this chapter, we focus on masked diffusion and defer the discussion of Uniform-state diffusion to Chapter 5.

¹The subscript ‘‘Duo’’ refers to the Duo algorithm, which we introduce and analyze in detail in Chapter 5.

4.3 Inference and Sampling in Masked Diffusion Language Models

4.3.1 Efficient Ancestral Sampling

To generate a sequence of length L , the reverse diffusion process starts with the sequence $\mathbf{z}_{t=1}^{1:L}$ where $\mathbf{z}_{t=1}^\ell = \mathbf{m}$, for all $\ell \in \{1, \dots, L\}$. Then the subsequent latents, $\mathbf{z}_t^{1:L}$ are generated by discretizing the reverse diffusion process with some finite T . Given $\mathbf{z}_t^{1:L}$, we construct $\mathbf{z}_s^{1:L}$ by sampling each token $\mathbf{z}_s^\ell$ independently from the distribution $p_\theta(\mathbf{z}_s^\ell | \mathbf{z}_t^{1:L})$ given in Eq. 4.4.

Note that in the reverse process, unmasked tokens remain unchanged. Thus, if no new tokens in $\mathbf{z}_s^{1:L}$ become unmasked (which can occur often in early denoising stages for large T), then $\mathbf{z}_s^{1:L} = \mathbf{z}_t^{1:L}$. Additionally if the denoising model, $\mathbf{x}_\theta(\mathbf{z}_t^{1:L})$ is not conditioned on time, then we can simply draw a new sample from $p_\theta(\mathbf{z}_{s-1/T}^{1:L} | \mathbf{z}_s^{1:L})$ using the previously computed and cached value $\mathbf{x}_\theta(\mathbf{z}_t^{1:L})$. This means we have effectively “skipped” over the time step s , saving a function call to the denoising network. Note that SEDD [Lou et al. \[2024\]](#) does not support this caching because the denoising network models time-dependent rates, which requires conditioning on time.

4.3.2 Semi-Autoregressive Masked Diffusion Language Models

Our method also admits an effective semi-autoregressive (SAR) decoding method that allows the model to generate sequences of arbitrary length Han et al. [2022], Si et al. [2023]. Let $\tilde{\mathbf{x}}^{1:L}$ represent the output from sampling a sequence of L tokens using the reverse diffusion process described above. To generate additional $L' < L$ tokens, we propose a generation algorithm in which the latter $L - L'$ tokens $\tilde{\mathbf{x}}^{L':L}$ are used as a prefix for an additional round of generation. Given the carry-over unmasking described in Sec. 4.2.1, these prefix tokens will simply be copied over at each decoding step. The remaining tokens are generated as above with $\mathbf{z}_s^\ell \sim p_\theta(\mathbf{z}_s^\ell \mid \mathbf{z}_t^{L:L+L'})$ for all $\ell \in \{L + 1, \dots, L + L'\}$, with $\mathbf{z}_{t=1}^{L-L':L}$ initialized to $\tilde{\mathbf{x}}^{L-L':L}$ as opposed to being initialized as masked tokens $\mathbf{m}$. At the end of this process, we have produced $L + L'$ tokens $\text{concat}[\tilde{\mathbf{x}}^{1:L}, \tilde{\mathbf{x}}^{L+1:L+L'}]$, where $\text{concat}[\cdot]$ denotes concatenation along the sequence length dimension. This process can repeat indefinitely, with the prefix shifted for every new round of generation.

4.4 Experiments

4.4.1 Masked Diffusion Language Models

Experimental Setup We evaluate MDLM as a generative model of language and as a representation model via fine-tuning on downstream tasks.

For language modeling likelihood evaluation, we conduct experiments on two datasets: The One Billion Words Dataset (LM1B; [Chelba et al., 2014]) and

OpenWebText (OWT; [Gokaslan et al., 2019]). We use the `bert-base-uncased` tokenizer for LM1B, and report perplexities on the test split. Models have a context size of 128. For OWT, which does not have a pre-defined split, we reserve the last 100K documents as a held-out validation set and report perplexities on this set. We use the GPT2 tokenizer [Radford et al., 2019] for OWT. Models have a context size of 1,024. We utilize the transformer architecture from Lou et al. [2024], which augments the diffusion transformer [Peebles and Xie, 2023] with rotary embeddings [Su et al., 2021]. MDLM was trained for 1M or 10M steps (corresponding to 33B, 327B tokens, respectively) on LM1B and 1M steps on OWT (which corresponds to 262B tokens). The corresponding AR baseline was trained for half the number of steps to ensure similar number of tokens seen (details in Suppl. B.4.2). Full hyperparameters are given in Suppl. B.4.4. On OWT, we train with and without time step conditioning.

For representation learning, we pre-train models on the C4 dataset [Raffel et al., 2020b], then fine-tune and evaluate models on the GLUE benchmark [Wang et al., 2019]. Models have a context size of 128. We use the `bert-base-uncased` tokenizer for the representation learning experiments. We utilize the MosaicBERT architecture from Portes et al. [2024], an extension of the original BERT architecture [Devlin et al., 2018]. We pre-train a bidirectional MosaicBERT using an MLM objective for 37B tokens of C4, as well as a causal variant on the same data. We further fine-tune MosaicBERT model using the MDLM for 327M tokens, less than 1% of the pre-training data. We provide the full hyperparameters in Suppl. B.4.6.

Likelihood Evaluation On LM1B, MDLM outperforms all previous diffusion methods (Table 4.1). Compared to the SEDD baseline reported by Lou et al. [2024], trained for 33B tokens, MDLM, which we train for the same amount, achieves a

Table 4.1: Test perplexities (PPL; $\downarrow$) on LM1B. [†]Reported in He et al. [2022]. Best diffusion value is bolded.

		Parameters	PPL ($\downarrow$)
<i>Autoregressive</i>	Transformer-X Base [Dai et al., 2019]	0.46B	23.5
	OmniNet _T [Tay et al., 2021]	100M	21.5
<i>Diffusion</i>	BERT-Mouth [Wang and Cho, 2019] [†]	110M	≤ 142.89
	D3PM (absorb) [Austin et al., 2021]	70M	≤ 76.90
	Diffusion-LM [Li et al., 2022] [†]	80M	≤ 118.62
	DiffusionBert [He et al., 2022]	110M	≤ 63.78
	SEDD [Lou et al., 2024] (33B tokens)	110M	≤ 32.79
<i>Autoregressive (Retrained)</i>	Transformer (33B tokens)	110M	22.32
	Transformer (327B tokens)		20.86
<i>Diffusion (Ours)</i>	MDLM (33B tokens)	110M	≤ 27.04
	MDLM (327B tokens)		$\leq \mathbf{23.00}$

Table 4.2: Test perplexities (PPL; $\downarrow$) on OWT for models trained for 262B tokens. [†] denotes retrained models.

	PPL ($\downarrow$)
AR [†]	17.54
SEDD [†]	≤ 24.10
MDLM (Ours)	$\leq \mathbf{23.21}$

17% improvement on the perplexity bound. Finally, MDLM gets within 14% of an AR baseline and continues to improve with more training. We see the same trend for models trained on OWT, a larger dataset, shown in Table 4.2 – MDLM outperforms prior diffusion methods, closing the gap towards AR models. In Table B.4 we find that models trained with and without time conditioning attain similar perplexities on OWT. Additionally, Figure B.2 demonstrates the reduced variance we achieve from our objective, when compared to previous masked diffusion models such as SEDD [Lou et al., 2024].

Table 4.3: Zero-shot perplexities ($\downarrow$) of models trained for 524B tokens on OWT. All perplexities for diffusion models are upper bounds.

	PTB	Wikitext	LM1B	Lambada	AG News	Pubmed	Arxiv
AR (Retrained)	82.05	25.75	51.25	51.28	52.09	49.01	41.73
SEDD (Retrained)	100.09	34.28	68.20	49.86	62.09	44.53	38.48
MDLM (Ours)	95.26	32.83	67.01	47.52	61.15	41.89	37.37

Table 4.4: GLUE evaluation results. Evaluation measures ($\uparrow$) are F1 score for QQP and MRPC, Spearman correlations for STS-B, and accuracy for the rest. For MNLI, we report match/mismatch accuracies.

	MNLI (m/mm)	QQP	QNLI	SST-2	COLA	STS-B	MRPC	RTE	Avg
AR	80.94/80.78	86.98	86.16	90.14	33.43	84.32	83.88	47.29	74.88
BERT	84.43/85.35	88.41	90.46	92.20	54.81	88.41	89.16	61.37	81.62
+MDLM-FT	84.76/85.07	88.49	90.30	92.20	57.69	87.48	90.53	62.09	82.06

Zero-Shot Likelihood Evaluation We also explore models’ ability to generalize by taking models trained on OWT and evaluating how well they model unseen datasets. We compare the perplexities of our MDLM with SEDD [Austin et al., 2021] and an AR Transformer language model. Our zero-shot datasets include the validation splits of Penn Tree Bank (PTB; [Marcus et al., 1993]), Wikitext [Merity et al., 2016], LM1B, Lambada [Paperno et al., 2016], AG News [Zhang et al., 2015], and Scientific Papers (Pubmed and Arxiv subsets; [Cohan et al., 2018]). Full experimental details are available in Suppl. B.4.4.

MDLM consistently outperforms the SEDD diffusion parameterization. In some cases, e.g., for Lambada and Scientific Papers, MDLM attains better perplexity than AR. We hypothesize that these datasets are farther from OWT, and that diffusion models may be more robust to out-of-domain evaluation due to the unmasking-based objective.

Table 4.5: Semi-AR generative perplexity (Gen. PPL; $\downarrow$) for sequences of 2048 tokens.

	Gen. PPL ($\downarrow$)	Sec/Seq ($\downarrow$)
SSD-LM	35.43	2473.9
MDLM (Ours)	27.18	89.3

Downstream Task Evaluation We find that BERT fine-tuned with MDLM to be a generative model results in strong perplexities while preserving performance on downstream tasks. On the C4 validation set, the AR model attains perplexity (PPL) of 22, the pre-trained BERT attains a PPL upper bound of 78 (evaluated using the MDLM variational bound), and BERT + MDLM-FT attains a PPL upper bound of 35. In Table 4.4, we further find that BERT + MDLM fine-tuning has no degradation in downstream GLUE performance compared to the BERT initialization. While the perplexity of our method is higher than the AR baseline, the downstream task performance is significantly better.

Semi-Autoregressive Modeling To test the SAR decoding algorithm presented in Sec. 4.3.2, we compare to SSD-LM [Han et al., 2022] a diffusion model that was designed to generate blocks of text autoregressively. We generate 200 sequences of length 2048 tokens on a single 3090 GPU and evaluate generative perplexity under a pre-trained GPT-2 [Radford et al., 2019] model. The SSD-LM sequences are generated using blocks of 25 tokens (as implemented in their pre-trained model) and the MDLM sequences are generated using $L' = 512$. In Table 4.5, we find that in addition to achieving better generative perplexity, MDLM enables ~ 25 - 30 x faster SAR decoding relative to SSD-LM.

4.4.2 Masked Diffusion DNA Models

We also explore applications to the generative modeling of biological sequences [Deshpande et al., 2022, Rastogi and Schiff, 2023] using a state space model (SSM) backbone [Gu et al., 2021]. Namely, we build on the recently-proposed Caduceus DNA language model [Schiff et al., 2024], which uses as a backbone the data-dependent SSM Mamba block [Gu and Dao, 2023].

Experimental Setup We pre-train the encoder-only Caduceus [Schiff et al., 2024], which is an MLM, on the HG38 human reference genome [Consortium, 2009] and perform fine-tuning using our diffusion parameterization. We use a context length of 1024 tokens and follow Schiff et al. [2024] for the experimental setup, other than learning rate which was reduced to 1e-3. See Suppl. B.4.7 for full experimental details. We assess both generative performance using perplexity and downstream performance on Genomics Benchmarks [Grešová et al., 2023] across language diffusion paradigms and AR models.

Generative Performance We fine-tune the Caduceus MLM across diffusion parameterizations and compare perplexities against AR models. We report perplexity values in Table 4.6. MDLM outperforms all other diffusion language modeling schemes.

Downstream Task Fine-tuning We perform downstream evaluation with the Genomics Benchmarks [Grešová et al., 2023], a recently proposed benchmark with eight regulatory element classification tasks. As shown in Table 4.7, our generative fine-tuning paradigm preserves or improves upon downstream performance from

Table 4.6: Test perplexities (PPL; $\downarrow$) of generative fine-tuning of the Caduceus MLM [Schiff et al., 2024] on the HG38 reference genome. Best diffusion model values are bolded. Error bars indicate the difference between the maximum and minimum values across 5 random seeds used for fine-tuning.

		Params	PPL ($\downarrow$)
<i>Autoregressive (Retrained)</i>	Mamba	465K	$3.067 \pm .010$
	HyenaDNA	433K	$3.153 \pm .001$
<i>Diffusion (Retrained)</i>	Plaid	507K	$\leq 3.240 \pm .005$
	SEDD	467K	$\leq 3.216 \pm .003$
<i>Diffusion (Ours)</i>	MDLM	467K	$\leq \mathbf{3.199} \pm .010$

Table 4.7: Genomic Benchmarks. Top-1 accuracy ($\uparrow$) across 5-fold cross-validation (CV) for a pre-trained AR Mamba, and a pre-trained Caduceus model fine-tuned with different diffusion parameterizations. The best values per task are bolded and the second best are italicized. Error bars indicate the difference between the maximum and minimum values across 5 random seeds used for CV.

Model	Mamba	Caduceus	Caduceus	Caduceus	Caduceus
Fine-Tuning Objective	AR	MLM	Plaid	SEDD	MDLM (ours)
(Parameter Count)	(465K)	(467K)	(507k)	(467k)	(467k)
Mouse Enhancers	0.76 ± 0.00	0.81 ± 0.02	0.75 ± 0.08	0.78 ± 0.06	<i>0.80 ± 0.03</i>
Coding vs. Intergenic	0.90 ± 0.00	0.91 ± 0.00	<i>0.91 ± 0.00</i>	0.91 ± 0.00	0.91 ± 0.00
Human vs. Worm	0.97 ± 0.00	<i>0.97 ± 0.00</i>	0.97 ± 0.00	<i>0.97 ± 0.00</i>	<i>0.97 ± 0.00</i>
Human Enhancers Cohn	0.73 ± 0.03	0.74 ± 0.00	<i>0.74 ± 0.01</i>	0.75 ± 0.02	<i>0.74 ± 0.02</i>
Human Enhancer Ensembl	0.86 ± 0.00	0.91 ± 0.00	0.89 ± 0.00	<i>0.91 ± 0.00</i>	0.90 ± 0.00
Human Regulatory	0.86 ± 0.008	0.87 ± 0.00	<i>0.87 ± 0.01</i>	0.83 ± 0.04	<i>0.84 ± 0.00</i>
Human OCR Ensembl	0.81 ± 0.01	<i>0.82 ± 0.00</i>	0.82 ± 0.00	0.82 ± 0.01	0.82 ± 0.01
Human NonTATA Promoters	0.93 ± 0.01	<i>0.94 ± 0.01</i>	<i>0.94 ± 0.01</i>	<i>0.94 ± 0.01</i>	0.94 ± 0.01

MLM pre-training. Absorbing-state diffusion methods outperform Plaid across tasks except for the simplest task Human vs. Worm, where all methods have roughly the same performance. For tasks where the input is a biased subsample of the full genome, we observe that the correlation between perplexity and downstream performance is weaker; see Suppl. B.4.7.

Table 4.8: Test perplexities (PPL; $\downarrow$) for MDLM ablations on LM1B. For the discrete-time models, we use $T = 1000$. Standard deviation is measured over 5 seeds during evaluation.

	PPL ($\leq$)
MDLM Eq. B.35	27.04 $\pm$.01
w/o continuous time Eq. B.31	27.19 $\pm$.07
& w/o carry-over Eq. B.29	28.56 $\pm$.15
& w/o zero masking Eq. B.27	28.51 $\pm$.15

4.4.3 Ablation Analysis

In Table 4.8, we can see the effect of our streamlined masked diffusion implementation. The improvements described in Sec. 4.2.1 allow us to greatly reduce perplexity of previously discounted models, such as D3PM (see the bottom row of this table, which is mathematically equivalent to the D3PM formulation). While most works assumed that D3PM achieves mediocre log-likelihoods, we show that is incorrect: our re-implementation almost matches state-of-the-art score-based methods. This introduces a new strong baseline that opens new research opportunities. Additionally, in Table 4.8, we ablate different components of MDLM. We observe that the perplexity for MDLM trained with a discrete $T = 1000$ marginally worsens by 0.1 compared to MDLM trained in continuous time. Additionally, removing the “carry over” operation from the SUBS parameterization increases the perplexity by 1.5 points. However, further removing the “zero masking” operation does not lead to any meaningful change in perplexity. We provide further ablations for the continuous time formulation in the Appendix, showing in Table B.3 that for a pre-trained model, at inference, increasing T yields better likelihoods.

4.5 Related Work

Comparison to D3PM Masked diffusion is a strict subset of D3PM [Austin et al., 2021]; setting $Q_{t|s} = \alpha_{t|s}\mathbf{I} + (1 - \alpha_{t|s})\mathbf{1m}^\top$ in their framework yields our forward diffusion. We improve over D3PM in three ways: (1) we adopt the SUBS parameterization; (2) this allows us to derive a simplified objective that analytically simplifies certain expectations to zero; (3) we adopt well-engineered training recipes that improve performance. Both (1) and (2) are possible because we focus on masking instead of developing a general discrete diffusion framework. Our framework also supports continuous time, unlike D3PM [Austin et al., 2021].

Comparison to CTMC Most implementations of diffusion work best in continuous time. However, extending D3PM in this way requires computing the limit of the product of an infinite number of matrices $Q_T \cdot Q_{T-1} \cdots Q_t$ as $T \rightarrow \infty$, which requires advanced CTMC theory [Campbell et al., 2022]. Our work describes simple continuous-time formulations for the most common noise processes (e.g., masking and uniform $\boldsymbol{\pi}$), thus helping make an important part of the literature more accessible.

In Suppl. B.3, we show that our results are compatible with CTMC, using the rate forward matrix $R_t = \frac{\alpha'_t}{\alpha_t}(\mathbf{I} - \mathbf{1m}^\top)$. For $\mathbf{y}, \mathbf{y}' \in \mathcal{V}$, the entry $R_t(\mathbf{y}', \mathbf{y})$ represents the transition rate for $\mathbf{y} \rightarrow \mathbf{y}'$. Similarly, the reverse rate $\tilde{R}_t(\mathbf{y}', \mathbf{y})$ for the transition $\mathbf{y} \rightarrow \mathbf{y}'$ is defined as:

$$\tilde{R}_t(\mathbf{y}', \mathbf{y}) = -\frac{\alpha'_t}{1 - \alpha_t}[\mathbf{y}']^\top [\mathbf{x}_\theta(\mathbf{y}, t) - \mathbf{m}]\langle \mathbf{y}, \mathbf{m} \rangle \quad (4.14)$$

Comparison to Score Estimation Score-based approaches to diffusion [Song and Ermon, 2019] extend to discrete states, although they typically further build upon advanced CTMC theory. In particular, SEDD [Lou et al., 2024] optimizes an ELBO² that is a function of the score model, obtaining state-of-the-art log-likelihoods among diffusion models. Our approach, however, is much simpler and does not require advanced theory. Furthermore, we can extract the score for MDLM Eq. B.64, as demonstrated in Suppl. B.3.3, making it compatible with various techniques designed for score-based algorithms, such as samplers [Campbell et al., 2022], score parameterization [Lou et al., 2024], efficient designs of the denoising network [Sun et al., 2022], guidance techniques, and more.

Comparison to BERT Our work provides a principled way of making BERT generative when trained with randomized masking rates. Previous work on generating from BERT used Gibbs sampling or ad-hoc methods [Ghazvininejad et al., 2019, Liao et al., 2020, Wang and Cho, 2019]. The connection between BERT and diffusion was first made by Austin et al. [2021]: their objective effectively involves unmasking. He et al. [2022] additionally starts training from a pretrained BERT. However, both works use an objective that is similar to (4.6), which is less numerically stable than our objective (see Section 4.2.1). Austin et al. [2021] mention in their appendix that their ELBO simplifies to a weighted masking (MLM) loss similar to (4.5), but it uses a more complex formula for the weights and is limited to the discrete time setting unlike our work. Furthermore, they do not train with that objective. Our work derives a simpler expression for the average of MLM losses, implements it, and obtains better likelihoods.

²Lou et al. [2024] mention that their ELBO can be derived from Benton et al. [2022], Hanson [2007] but do not provide an explicit derivation. To address this, we present a rigorous derivation in Suppl. B.3.2 using CTMC theory.

Comparison to Alternative Diffusion LMs In contrast to this work, which defines diffusion over discrete structures, Plaid [Gulrajani and Hashimoto, 2024] and Diffusion LM [Li et al., 2022] define a Gaussian diffusion process over word embeddings. Zhang et al. [2024] and Hu et al. [2024] extend this approach to flow matching over word embeddings, enabling the design of faster samplers. Discrete Flow Matching (DFM) [Campbell et al., 2024] applies flow matching to discrete structures, using a cross-entropy loss as their training objective: $-\mathbb{E}_{q,t} \log p_{\theta}(\mathbf{x}^{1:L} | \mathbf{z}_t^{1:L})$. Similar to Chang et al. [2022], DFM’s objective, while effective, is not weighted to serve as a proper ELBO. In MDLM, however, we derive a tight, principled lower bound on the log-likelihood.

Concurrent Work Concurrent to our work, Shi et al. [2024a] and Ou et al. [2024] derive a similar simplified objective for masked diffusion processes. While Ou et al. [2024] start from a score matching perspective, we tackle this problem from a variational lens similar to Shi et al. [2024a]. Similar to Ou et al. [2024], we formulate efficient samplers in Section 4.3.1 by leveraging a time-independent denoising network.

Another difference between our method and concurrent work Shi et al. [2024a], Ou et al. [2024] is the semi-autoregressive decoding method we present in Section 4.3.2. While [Shi et al., 2024a, Ou et al., 2024] are restricted to sample sequences of a fixed length, we propose samplers to generate arbitrary lengths of text like a traditional language model. Furthermore, we establish the connection between our simplified objective and the masked language modeling (MLM) objective. As a result, we endow BERT-style models with principled generation capabilities while maintaining representation learning capabilities. Whereas [Shi et al., 2024a, Ou et al., 2024] only evaluate on NLP datasets, we show that masked diffusion is also

effective in modeling biological sequences.

4.6 Conclusion

In this chapter, we introduced an interpolating discrete diffusion framework with an emphasis on masked diffusion. With a well-engineered implementation and a simple variational objective, our method MDLM achieves state-of-the-art diffusion perplexities on language benchmarks and shows how to efficiently convert BERT-style encoders into generative models. **The approach has since been scaled to industry settings by ByteDance’s Diffusion LLM³ [Song et al., 2025] (the fastest diffusion-based LLM at the time of writing) and by Nvidia’s GenMol⁴ for drug discovery [Lee et al., 2025].**

³https://studio.seed.ai/exp/seed_diffusion/

⁴<https://build.nvidia.com/nvidia/genmol-generate>

THE DISCRETE-CONTINUOUS DIFFUSION DUALITY

5.1 Introduction

An eternal theme in mathematics is that discreteness emerges from underlying continuity. From quantum mechanics, where the quantized energy states of electrons arise as solutions to continuous wave equations, to the binary logic of digital circuits, fundamentally driven by smooth analog currents, discreteness has repeatedly and naturally emerged from an underlying continuum. Our work continues this tradition by demonstrating that a discrete diffusion process is, in fact, an emergent phenomenon of an underlying continuous Gaussian diffusion process. This perspective enables the design of faster training and sampling algorithms for discrete diffusion models.

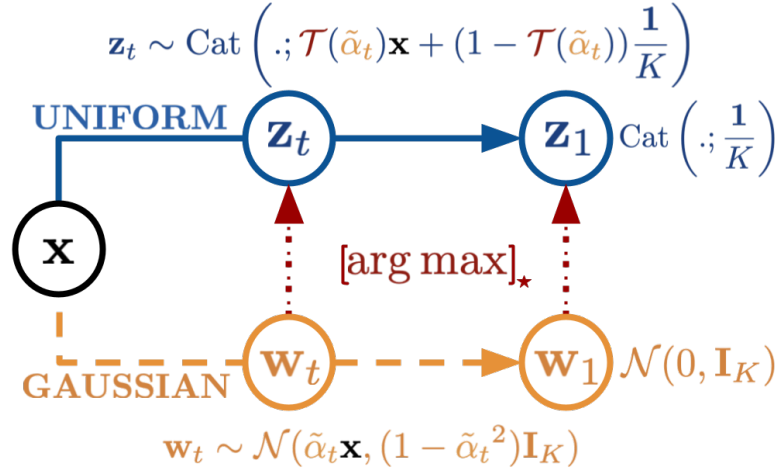


Figure 5.1: An illustration of **Uniform-state discrete diffusion** (top) and the underlying **Gaussian diffusion** (bottom). While both are separate Markov processes, applying **arg max** on Gaussian latents $\mathbf{w}_t \in \mathbb{R}^n$ converts them to discrete latents $\mathbf{z}_t \in \mathcal{V}$, transforming their marginals from $\tilde{q}_t(.|\mathbf{x}; \tilde{\alpha}_t)$ Eq. 2.11 to $q_t(.|\mathbf{x}; \mathcal{T}(\tilde{\alpha}_t))$ Eq. 4.1 and adjusting diffusion parameters from $\tilde{\alpha}_t$ to $\alpha_t = \mathcal{T}(\tilde{\alpha}_t)$ Eq. 5.3. Notably, the ELBO for Uniform-state diffusion induces a tighter bound on the likelihood than Gaussian diffusion, as established in Theorem 5.3.2.

While Gaussian diffusion is well-studied, it underperforms discrete diffusion models on tasks involving discrete data—such as text [Sahoo et al., 2024b], graphs [Liu et al., 2023a], and molecules [Lee et al., 2025]. However, from the perspective of Gaussian diffusion, the design space for discrete diffusion models remains primitive: mean parameterization for the denoising model [Sahoo et al., 2024a, Schiff et al., 2025] and slow ancestral sampling [Austin et al., 2021] are still the dominant approaches. Recent work on distilling Masked Discrete Diffusion Models (MDMs) improves sampling speed [Deschenaux and Gulcehre, 2024], but performance degrades severely in the few-step regime. Unlike Gaussian diffusion models with Probability Flow ODEs [Song et al., 2020], MDMs lack an “implicit” property: a deterministic mapping from noise to data. This property is vital for few-step distillation methods [Song et al., 2023, Frans et al., 2024], but MDMs forgo it due to their deterministic prior, which requires stochasticity during sampling.

Our objective is (1) to design a framework for discrete diffusion that enables the transfer of advanced training and inference techniques from Gaussian diffusion to discrete diffusion models. And (2) create language models that support few-step generation. To this end, we focus on Uniform-state Diffusion Models (USDMs) [Austin et al., 2021]. In this paper, we discover a remarkable property of USDMs—these emerge from Gaussian diffusion processes as illustrated in Fig. 5.1. We call this phenomenon—the Diffusion Duality which expands the design space of USDMs, making it possible to incorporate techniques developed for Gaussian diffusion. Notably, USDMs models allow token updates during reverse sampling unlike MDMs, naturally correcting earlier mistakes without requiring costly predictor-corrector [Zhao et al., 2024, Wang et al., 2025] steps—saving function evaluations (NFEs). However, these models have historically underperformed compared to MDMs [Austin et al., 2021, Lou et al., 2024], raising the key question: Can USDMs

be made competitive with MDMs? And more importantly, can the implicit property of the underlying Gaussian diffusion be leveraged for fast, few-step generation?

We answer both questions with **Duo**, a rich framework of theoretical connections between USDMs and Gaussian diffusion. Duo enriches the design space of USDMs by incorporating Gaussian diffusion, which allows us to develop efficient training strategies that accelerate the training of USDMs, significantly reducing the performance gap between MDMs and AR models on standard language generation benchmarks. Notably, we **surpass AR models on 3 out of 7 zero-shot datasets** (Table 5.2). Furthermore, this duality allows us to adapt consistency distillation [Song et al., 2023] from Gaussian to discrete diffusion, reducing NFEs from 1024 to 8 with minimal effect on sample quality (Sec. 5.5.2). Importantly, in the low-NFE regime, Duo outperforms MDMs. Our main contributions are threefold: First, we establish a theoretical connection between continuous and discrete diffusion, demonstrating that discrete diffusion arises from an underlying continuous Gaussian diffusion. This insight enables the transfer of techniques from the continuous domain to the discrete setting, opening up new possibilities. Second, our framework **doubles the training speed** of USDMs by introducing a low-variance curriculum, and third, **accelerates sampling by two orders of magnitude** by adapting efficient distillation methods from continuous diffusion models.

5.2 Background

Consistency Distillation Consistency models [Song et al., 2023, Song and Dhariwal, 2023] are a class of generative models that define a bijective mapping

between the samples from the noise distribution $\mathcal{N}(0, \mathbf{I}_K)$ and the data distribution q_{data} . They build on deterministic samplers for Gaussian diffusion [Song et al., 2020, 2021a], specifically using the Probability-Flow ODE (PF-ODE). Given a pre-trained Gaussian diffusion model $\mathbf{x}_\theta$, which requires hundreds or thousands of sampling steps, Consistency Distillation is a popular technique to distil them down to fewer steps generation that enables much faster generation. The distillation begins with a teacher model $\mathbf{x}_{\theta^-}$, often the Exponentially Moving Average (EMA) of the student model $\mathbf{x}_\theta$ obtained during the course of training. A noisy sample $\mathbf{w}_t$ is drawn from the forward process $\tilde{q}_t(\cdot|\mathbf{x})$ Eq. 2.11, and a less noisy sample $\mathbf{w}_s$ is obtained by numerically solving one PF-ODE step using $\mathbf{x}_{\theta^-}$. The student model is then trained to match the teacher’s estimate of the clean sample minimizing the following loss:

$$\mathcal{L}(\theta, \theta^-) = \lambda(t) d(\mathbf{x}_\theta(\mathbf{w}_t, t), \mathbf{x}_{\theta^-}(\mathbf{w}_s, s)), \quad (5.1)$$

where $d : \mathbb{R}^K \times \mathbb{R}^K \rightarrow \mathbb{R}^+$ denotes the error between the teacher model’s reconstruction $\mathbf{x}_{\theta^-}(\mathbf{w}_s, t)$ and the student model’s reconstruction $\mathbf{x}_\theta(\mathbf{w}_t, t)$ of the original sample and $\lambda : [0, 1] \rightarrow \mathbb{R}^+$ is a weighting function that scales the loss based on the diffusion time-step t .

5.3 The Diffusion Duality

Unlike discrete diffusion, Gaussian diffusion is replete with well-established empirical techniques, which have driven significant advances in both training [Ho et al., 2020, Salimans and Ho, 2022b, Zheng et al., 2023a] and sampling [Karras et al., 2022, Song et al., 2023, Song and Dhariwal, 2023, Yin et al., 2024]. Our goal in this section is to establish a theoretical bridge between discrete-state diffusion and

continuous-state diffusion, which will enable us to leverage tools from the latter to improve the former.

We propose a simple method to map a Gaussian latent to the discrete space: the **arg max** operator. But does this transformation of latents also transform a Gaussian diffusion process into a discrete one? A necessary and sufficient condition for this is that the marginal distribution of the discretized vector satisfies the characteristic ODE of a discrete diffusion process Eq. 2.8. We first derive a closed-form expression for this marginal and show that **arg max** maps the marginals of a Gaussian diffusion to those of a Uniform-state discrete diffusion, including a transformation of the diffusion parameters Eq. 5.2. Finally, we verify that this marginal evolves according to Eq. 5.4, establishing that **arg max** transforms a Gaussian diffusion process into a Uniform-state discrete diffusion process.

5.3.1 Gaussian Diffusion under the **arg max** Pushforward

We begin by defining a Gaussian diffusion process on $\mathbf{x} \in \mathcal{V}$ as per Eq. 2.11, with $\tilde{q}_{t=0} \approx q_{\text{data}}$ and $\tilde{q}_{t=1} = \mathcal{N}(0, \mathbf{I}_K)$. Let $\mathbf{w}_t \sim \tilde{q}_t(\cdot | \mathbf{x}; \tilde{\alpha}_t)$ be an intermediate latent at time t . Next, define the operation **arg max** : $\mathbb{R}^K \rightarrow \mathcal{V}$ to map a continuous vector $\mathbf{w} \in \mathbb{R}^K$ to the one-hot vector corresponding to the index of its largest entry in $\mathbf{w}$, i.e., $\text{arg max}(\mathbf{w}) = \text{arg max}_{\mathbf{z} \in \mathcal{V}} \mathbf{z}^\top \mathbf{w}$.

Discrete Marginals Let $\mathbf{z}_t = \text{arg max}(\mathbf{w}_t)$ and $P_t(\cdot | \mathbf{x})$ denote its conditional pmf marginalized over $\mathbf{w}_t \sim \tilde{q}_t(\cdot | \mathbf{x}; \tilde{\alpha}_t)$. In Suppl. C.1.1, we show:

$$\mathbf{z}_t \sim P_t(\cdot | \mathbf{x}; \mathcal{T}(\tilde{\alpha}_t)) = \text{Cat}\left(\cdot; \mathcal{T}(\tilde{\alpha}_t)\mathbf{x} + (1 - \mathcal{T}(\tilde{\alpha}_t))\frac{\mathbf{1}}{K}\right), \quad (5.2)$$

where the function $\mathcal{T} : [0, 1] \rightarrow [0, 1]$ is the *Diffusion Transformation operator*, defined as:

$$\mathcal{T}(\tilde{\alpha}_t) = \frac{K}{K-1} \left[\int_{-\infty}^{\infty} \phi \left(z - \frac{\tilde{\alpha}_t}{\sqrt{1 - \tilde{\alpha}_t^2}} \right) \Phi^{K-1}(z) dz - \frac{1}{K} \right], \quad (5.3)$$

where $\phi(z) = \exp(-z^2)/\sqrt{2\pi}$ is the standard Normal distribution and $\Phi(z) = \int_{-\infty}^z \phi(t)dt$ is its cumulative distribution function.

Time Evolution of Marginals Next, we examine how the discrete marginal P_t evolves with time as the continuous vector $\mathbf{w}_t$ undergoes Gaussian diffusion. In Suppl. C.1.2 we show that P_t evolves as per the following linear ODE:

$$\frac{d}{dt} P_t = - \underbrace{\frac{\mathcal{T}'(\tilde{\alpha}_t)}{K\mathcal{T}(\tilde{\alpha}_t)} [\mathbf{1}\mathbf{1}^\top - K\mathbf{I}_K]}_{Q_t} P_t \quad (5.4)$$

where $\mathcal{T}'$ represents the time derivative of $\mathcal{T}$. From Eq. 2.8, we infer that Eq. 5.4 characterizes a Uniform-state discrete diffusion process with diffusion parameter $\mathcal{T}(\tilde{\alpha}_t)$. It is important to note that while the marginals of the discretized latents evolve according to a Markovian Uniform-state discrete diffusion process, a Gaussian diffusion trajectory after discretization, might not map to a discrete diffusion trajectory. We discuss this in detail in Suppl. C.1.3.

Duality The implications of Eq. 5.2 and Eq. 5.4 are quite profound. These reveal a fundamental connection between Uniform-state discrete diffusion and Gaussian diffusion bridged by the **arg max** operator: **The arg max operation transforms Gaussian diffusion into Uniform-state diffusion**, with the diffusion parameters related by Eq. 5.3. More formally, this can be expressed as:

$$q_t(\mathbf{z}_t|\mathbf{x}; \mathcal{T}(\tilde{\alpha}_t)) = [\text{arg max}]_{\star} \tilde{q}_t(\mathbf{w}_t|\mathbf{x}; \tilde{\alpha}_t) \quad (5.5)$$

where the $\star$ operator denotes the *pushforward* of the K -dimensional Gaussian density $\tilde{q}_t$ under the **arg max** map, yielding a categorical distribution q_t with K classes.

Thus, as $\mathbf{x}$ diffuses in the discrete space via a Uniform-state process, there exists an underlying continuous-space representation in which $\mathbf{x}$ follows Gaussian diffusion, as illustrated in Fig. 5.1.

5.3.2 Discrete–Gaussian Samplers and Likelihoods

We now show that, given an approximate reverse process for a Gaussian diffusion, $\bar{p}_t^\theta$, there exists a reverse process in the discrete domain, p_t^θ , such that

$$p_t^\theta = [\mathbf{arg\,max}]_\star \bar{p}_t^\theta \quad \forall t \in [0, 1]. \quad (5.6)$$

The processes p_t^θ and $\bar{p}_t^\theta$ share the same denoising transformer with learnable parameters θ defined in the Gaussian space. Our goal is to express both the transition kernel of the discrete reverse process and the associated denoising model in terms of the Gaussian reverse process.

Theorem 5.3.1. *The reverse transition kernel $p_{s|t}^\theta$ in the discrete space that ensures $(p_t^\theta = [\mathbf{arg\,max}]_\star \bar{p}_t^\theta)_{t \in [0, 1]}$ is given by*

$$p_{s|t}^\theta(\cdot | \mathbf{z}_t) = [\mathbf{arg\,max}]_\star \int \bar{p}_{s|t}^\theta(\mathbf{w}_s | \mathbf{w}_t) \frac{\bar{p}_t^\theta(\mathbf{w}_t)}{p_t^\theta(\mathbf{z}_t)} \mathbb{1}_{\mathbf{arg\,max}(\mathbf{w}_t) = \mathbf{z}_t} d\mathbf{w}_t \quad (5.7)$$

Proof. We prove the claim by mathematical induction on t .

Base case. For $t = 1$ we have the discrete prior $p_{t=1}^\theta(\cdot) = \text{Cat}(\cdot; \mathbf{1}/K)$ and the Gaussian prior $\bar{p}_{t=1}^\theta(\cdot) = \mathcal{N}(0, I_K)$. By symmetry, for $\mathbf{w} \sim \mathcal{N}(0, I_K)$ each index is

equally likely to be the value of $\arg \max(\mathbf{w})$, so

$$p_{t=1}^\theta = [\arg \max]_\star \bar{p}_{t=1}^\theta. \quad (5.8)$$

Inductive step. Assume that for some t the relation

$$p_t^\theta = [\arg \max]_\star \bar{p}_t^\theta$$

holds. Let $s = t - \delta$ with $0 < \delta < t$. We show that

$$p_s^\theta = [\arg \max]_\star \bar{p}_s^\theta,$$

and that Eq. 5.7 yields a reverse process with this property.

We start from the marginal at time s and expand it via the joint with $\mathbf{z}_t$:

$$\begin{aligned} p_s^\theta(\mathbf{z}_s) &= \sum_{\mathbf{z}_t} p_s^\theta(\mathbf{z}_s, \mathbf{z}_t) \\ &= \sum_{\mathbf{z}_t} \left[p_{s|t}^\theta(\mathbf{z}_s | \mathbf{z}_t) p_t^\theta(\mathbf{z}_t) \right] \end{aligned}$$

Substituting $p_{s|t}^\theta$ from Eq. 5.7, we get:

$$= \sum_{\mathbf{z}_t} \left[[\arg \max]_\star \int \bar{p}_{s|t}^\theta(\mathbf{w}_s | \mathbf{w}_t) \frac{\bar{p}_t^\theta(\mathbf{w}_t)}{\bar{p}_t^\theta(\mathbf{z}_t)} \mathbb{1}_{\arg \max(\mathbf{w}_t) = \mathbf{z}_t} d\mathbf{w}_t \cancel{p_t^\theta(\mathbf{z}_t)} \right]$$

$\because$ pushforward is linear [Kolmogorov, 1950, Stroock, 1999], we get:

$$\begin{aligned} &= [\arg \max]_\star \int \bar{p}_{s|t}^\theta(\mathbf{w}_s | \mathbf{w}_t) \bar{p}_t^\theta(\mathbf{w}_t) \underbrace{\sum_{\mathbf{z}_t} \left[\mathbb{1}_{\arg \max(\mathbf{w}_t) = \mathbf{z}_t} \right]}_{=1} d\mathbf{w}_t \\ &= [\arg \max]_\star \int \bar{p}_{s|t}^\theta(\mathbf{w}_s | \mathbf{w}_t) \bar{p}_t^\theta(\mathbf{w}_t) d\mathbf{w}_t \\ &= [\arg \max]_\star \bar{p}_s^\theta(\mathbf{w}_s). \end{aligned} \quad (5.9)$$

Thus, the discrete marginal at time s is the pushforward of the Gaussian marginal under $\arg \max$. Since $s = t - \delta$ was arbitrary, this completes the induction and

hence the proof that $\left(p_t^\theta = [\arg \max]_\star \bar{p}_t^\theta \right)_{t \in [0,1]}$.

Next, recall that the transition kernel of the Uniform-state discrete diffusion process is given by Eq. 4.11. We now show that there exists a denoising model $\mathbf{x}_\theta$ for which Eq. 4.11 coincides with Eq. 5.7. To this end, we equate these two kernels and solve for $\mathbf{x}_\theta(\mathbf{z}_t, t)$, which yields

$$[\mathbf{x}_\theta(\mathbf{z}_t, t)]_i = \begin{cases} \beta, & \text{if } [\mathbf{z}_t]_i = 1 \\ \frac{[\gamma]_j(K\alpha_t\beta + 1 - \alpha_t) - (1 - \alpha_{t|s})(1 - \alpha_s)/K}{\alpha_s - \alpha_t}, & \text{otherwise.} \end{cases}$$

where

$$\beta = \frac{[\gamma]_i(1 - \alpha_t) - (1 - \alpha_{t|s})(1 - \alpha_s)/K - (\alpha_{t|s} - \alpha_t)}{K\alpha_t + (\alpha_s - \alpha_t) - K\alpha_t[\gamma]_i}, \quad (5.10)$$

$$\gamma = [\arg \max]_\star \int \bar{p}_{s|t}^\theta(\mathbf{w}_s | \mathbf{w}_t) \frac{\bar{p}_t^\theta(\mathbf{w}_t)}{p_t^\theta(\mathbf{z}_t)} \mathbb{1}_{\arg \max(\mathbf{w}_t) = \mathbf{z}_t} d\mathbf{w}_t, \quad (5.11)$$

and $[\cdot]_k$ denotes the k^{th} entry in a vector.

With the denoising model defined in Eq. 5.10, the standard ancestral sampler for USDMs (based on Eq. 4.11) coincides with the marginal-preserving reverse process in Eq. 5.7. Consequently, the resulting denoising model guarantees that the marginals of the Uniform-state diffusion process, p_t^θ , match the pushforward of the underlying Gaussian diffusion marginals, $\bar{p}_t^\theta$, under $\arg \max$ for all $t \in [0, 1]$.

Likelihood Note that these two processes are separate Markov chains with no transitions between them, and they induce distinct marginal distributions over the data, leading to different log-likelihoods.

Theorem 5.3.2. *Let p_{data}^θ and $\bar{p}_{data}^\theta$ denote the approximations to the training data distribution induced by the Uniform-state diffusion and its underlying Gaussian diffusion, respectively. Then the marginal likelihood of p_{data}^θ under the true data distribution q_{data} is at least as high as that of $\bar{p}_{data}^\theta$:*

$$\underbrace{\mathbb{E}_{\mathbf{x} \sim q_{data}} \log p_{data}^\theta(\mathbf{x})}_{\text{Discrete Likelihood}} \geq \underbrace{\mathbb{E}_{\mathbf{x} \sim q_{data}} \log \bar{p}_{data}^\theta(\mathbf{x})}_{\text{Gaussian Likelihood}} \quad (5.12)$$

Proof. Before proceeding, we recall a standard result from [Csiszár \[1964\]](#), [Cover and Thomas \[2012\]](#): for a broad class of statistical divergences D —such as the Kullback–Leibler (KL) divergence, Total Variation Distance (TVD), or Rényi divergence—and for any Markov kernel $k : X \rightarrow Y$ mapping distributions on a measurable space X to distributions on another measurable space Y , the following inequality holds:

$$D([k]_{\star}q, [k]_{\star}p) \leq D(q, p), \quad (5.13)$$

where q and p denote probability distributions on X , and $[k]_{\star}$ denotes the pushforward operation induced by the kernel k .

Let p_{data}^{θ} and $\bar{p}_{\text{data}}^{\theta}$ denote the approximations to the training data distribution induced by Uniform-state diffusion and its underlying Gaussian diffusion process by the marginal preserving samplers defined in Eq. 5.7, respectively. The likelihoods of the Uniform-state diffusion process ($\log p_{\text{data}}^{\theta}$) and its underlying Gaussian diffusion process ($\log \bar{p}_{\text{data}}^{\theta}$) are related as follows:

$$D_{\text{KL}}([\text{arg max}]_{\star} q_{\text{data}}, [\text{arg max}]_{\star} \bar{p}_{\text{data}}^{\theta}(\mathbf{x})) \leq D_{\text{KL}}(q_{\text{data}}, \bar{p}_{\text{data}}^{\theta}) \quad (\text{From Eq. 5.13})$$

$\because q_{\text{data}}$ defines a distribution over categorical random variables, we get:

$$\begin{aligned} \implies D_{\text{KL}}(q_{\text{data}}, [\text{arg max}]_{\star} \bar{p}_{\text{data}}^{\theta}) &\leq D_{\text{KL}}(q_{\text{data}}, \bar{p}_{\text{data}}^{\theta}) \\ \implies \sum_{\mathbf{x}} q_{\text{data}}(\mathbf{x}) \log \frac{q_{\text{data}}(\mathbf{x})}{[\text{arg max}]_{\star} \bar{p}_{\text{data}}^{\theta}(\mathbf{x})} &\leq \sum_{\mathbf{x}} q_{\text{data}}(\mathbf{x}) \log \frac{q_{\text{data}}(\mathbf{x})}{\bar{p}_{\text{data}}^{\theta}(\mathbf{x})} \\ \implies \sum_{\mathbf{x}} \left[\cancel{q_{\text{data}}(\mathbf{x}) \log q_{\text{data}}(\mathbf{x})} - q_{\text{data}}(\mathbf{x}) \log [\text{arg max}]_{\star} \bar{p}_{\text{data}}^{\theta}(\mathbf{x}) \right] \\ &\leq \sum_{\mathbf{x}} \left[\cancel{q_{\text{data}}(\mathbf{x}) \log q_{\text{data}}(\mathbf{x})} - q_{\text{data}}(\mathbf{x}) \log \bar{p}_{\text{data}}^{\theta}(\mathbf{x}) \right] \\ \implies \sum_{\mathbf{x}} \left[-q_{\text{data}}(\mathbf{x}) \log [\text{arg max}]_{\star} \bar{p}_{\text{data}}^{\theta}(\mathbf{x}) \right] &\leq \sum_{\mathbf{x}} \left[-q_{\text{data}}(\mathbf{x}) \log \bar{p}_{\text{data}}^{\theta}(\mathbf{x}) \right] \\ \implies \mathbb{E}_{\mathbf{x} \sim q_{\text{data}}} \log \underbrace{[\text{arg max}]_{\star} \bar{p}_{\text{data}}^{\theta}(\mathbf{x})}_{\equiv p_{\text{data}}^{\theta}(\mathbf{x})} &\geq \mathbb{E}_{\mathbf{x} \sim q_{\text{data}}} \log \bar{p}_{\text{data}}^{\theta}(\mathbf{x}) \\ \implies \underbrace{\mathbb{E}_{\mathbf{x} \sim q_{\text{data}}} \log p_{\text{data}}^{\theta}(\mathbf{x})}_{\text{Discrete Likelihood}} &\geq \underbrace{\mathbb{E}_{\mathbf{x} \sim q_{\text{data}}} \log \bar{p}_{\text{data}}^{\theta}(\mathbf{x})}_{\text{Gaussian Likelihood}} \end{aligned} \quad (5.14)$$

The key insight from Eq. 5.14 is that, **for a given Gaussian diffusion process, there exists an equivalent discrete diffusion process that induces a higher marginal likelihood on the training data.** Since USDMs provide an improved likelihood estimate, it is advantageous to design the denoising model to operate on discrete latents. Consequently, we adopt Eq. 4.12 as our training and evaluation objective.

5.3.3 Duo: Sampling and Improved Training Objective

In this subsection, we describe the discrete-space samplers for Duo and introduce an improved low-variance training objective.

Sampler To sample from Duo, we use ancestral sampling for USDMs Eq. 4.11. Furthermore, to improve text quality, we propose *Greedy-Tail Sampler*, that improves the sample quality by decreasing the sample entropy in a manner similar to nucleus sampling in AR models [Holtzman et al., 2020]. Specifically, during the final denoising step, instead of sampling the clean sequence via $\tilde{\mathbf{x}} \sim p_{0|\delta}^\theta(\cdot)$, we perform greedy decoding: $\tilde{\mathbf{x}} = \text{arg max}(p_{0|\delta}^\theta(\cdot))$ where δ denotes the time discretization.

Rao-Blackwellized NELBO As described in Sec. 4.2.2 of Chapter 4, we derive an equivalent Rao-Blackwellized objective, given in Eq. C.20. This objective avoids materializing one-hot vectors, unlike prior work [Lou et al., 2024, Schiff et al., 2025], and reduces the variance of the training objective, which leads to faster optimization and lower perplexity (Sec. 5.5.1). The resulting reformulation provides an efficient, low-variance expression for computing the NELBO for USDMs, while incurring only minimal memory overhead. We provide the full derivation in Suppl. C.1.4 and

ablate this design choice in Table 5.3.

Sequence-Level Discrete Diffusion We extend our discrete diffusion framework to sequences $\mathbf{x}^{1:L} \sim q_{\text{data}}$ of length L . The forward process and the reverse process factorize independently over tokens as:

$$q_t(\mathbf{z}_t^{1:L} | \mathbf{x}^{1:L}; \alpha_t) = \prod_{\ell \in [L]} q_t(\mathbf{z}_t^\ell | \mathbf{x}^\ell; \alpha_t)$$

based on Eq. 4.9, and

$$p_{s|t}^\theta(\mathbf{z}_s^{1:L} | \mathbf{z}_t^{1:L}) = \prod_{\ell \in [L]} q_{s|t}(\mathbf{z}_s^\ell | \mathbf{z}_t^{1:L}, \mathbf{x}_\theta^\ell(\mathbf{z}_t^{1:L}, t))$$

based on Eq. 4.10, respectively. Here, $\mathbf{x}_\theta : \mathcal{V}^L \times [0, 1] \rightarrow \Delta^L$ denotes the denoising model. Consequently, the sequence-level NELBO decomposes into a sum of token-level losses:

$$\begin{aligned} \text{NELBO}(\mathbf{q}, p_\theta; \mathbf{x}^{1:L}) \\ = \mathbb{E}_{t \sim \mathcal{U}[0,1], \mathbf{q}_t} \sum_{\ell \in [L]} f_{\text{Duo}}(\mathbf{z}_t^\ell, \mathbf{x}_\theta^\ell(\mathbf{z}_t^{1:L}, t), \alpha_t; \mathbf{x}^\ell). \end{aligned} \quad (5.15)$$

We exploit this duality to incorporate Gaussian diffusion into the design space of USDMs. This allows us to design a low-variance training algorithm that leads to faster training (Sec. 5.4.1) and a distillation scheme that unlocks few-step generation in diffusion language models (Sec. 5.4.2).

5.4 Applications

We now present two applications where discrete diffusion models benefit from leveraging the underlying Gaussian diffusion. In Sec. 5.4.1, we introduce a curriculum

learning strategy that reduces training variance and leads to faster training. Then, in Sec. 5.4.2, we propose a distillation algorithm that cuts the number of sampling steps by two orders of magnitude with minimal impact on sample quality.

5.4.1 Faster Training using Curriculum Learning

Curriculum learning [Bengio et al., 2009] gradually exposes models to increasingly complex data, starting with simpler, easier-to-denoise noise patterns and progressing to more challenging ones. Here, we design a curriculum for USDMs by exploiting the underlying Gaussian diffusion.

Similar to relaxation methods in discrete gradient estimation [Jang et al., 2017, Maddison et al., 2017], our curriculum is centered around annealing the temperature parameter of a smooth approximation of $\arg \max$. We reformulate the NELBO for discrete diffusion in terms of $\arg \max$ over Gaussian latents (Sec. 5.4.1). The denoising model is then trained to operate on the $\arg \max$ of these Gaussian variables. We then relax the $\arg \max$ using a tempered softmax (Sec. 5.4.1), which yields a lower-variance but biased estimator of the ELBO. Initially, the model operates on fully relaxed, continuous Gaussian latents. As training progresses, the temperature is gradually decreased, transitioning the model’s inputs from soft (continuous) to hard (discrete). By the end of the curriculum, the model effectively operates on discrete latents, closing the gap between training and inference-time behavior.

Discrete NELBO with Gaussian Latents

Consider the discrete diffusion NELBO Eq. 5.15, which marginalizes f_{Duo} over the discrete latents $\mathbf{z}_t^{1:L} \sim q_t(\cdot | \mathbf{x}^{1:L}; \alpha_t)$. Our goal is to re-express this objective in terms of Gaussian latents $\mathbf{w}_t^{1:L} \sim \tilde{q}_t(\cdot | \mathbf{x}^{1:L}; \tilde{\alpha}_t)$ such that marginalizing over $\mathbf{w}_t^{1:L}$ yields the same numerical value for the NELBO. In Suppl. C.2.1, we show:

$$\begin{aligned}
& \text{NELBO}(q, p_\theta; \mathbf{x}^{1:L}) \\
&= \mathbb{E}_{t, q_t(\mathbf{z}_t^{1:L} | \mathbf{x}^{1:L}; \alpha_t)} \sum_{\ell \in [L]} f_{\text{Duo}}(\mathbf{z}_t^\ell, \mathbf{x}_\theta^\ell(\mathbf{z}_t^{1:L}, t), \alpha_t; \mathbf{x}^\ell) \\
&= \mathbb{E}_{t, \tilde{q}_t(\mathbf{w}_t^{1:L} | \mathbf{x}; \tilde{\alpha}_t)} \sum_{\ell \in [L]} f_{\text{Duo}}\left(\mathbf{z}_t^\ell := \text{arg max}(\mathbf{w}_t^\ell), \mathbf{x}_\theta\left([\text{arg max}(\mathbf{w}_t^{\ell'})]_{\ell'=1}^L, t\right), \alpha_t := \mathcal{T}(\tilde{\alpha}_t); \mathbf{x}^\ell\right),
\end{aligned} \tag{5.16}$$

where $\alpha_t = \mathcal{T}(\tilde{\alpha}_t)$ is obtained via Eq. 5.3 from the Gaussian diffusion coefficient $\tilde{\alpha}_t$ and also verify this empirically. As discussed in Sec. 5.3, these are distinct Markov chains whose marginal distributions are related only through Eq. 5.5. This reparameterization underpins our curriculum learning strategy which we present in the next section.

Low-Variance Training Loss

To reduce training variance, we replace $\text{arg max}(\mathbf{w}_t^\ell)$ in the denoising model input Eq. 5.16 with a tempered softmax. We argue that this substitution eases recovery of the clean sequence from its noisy counterpart, and that the difficulty of this recovery is regulated by the temperature parameter.

As shown in prior work [Jang et al., 2017, Maddison et al., 2017], arg max is a limiting case of softmax :

$$\text{arg max}(\mathbf{w}_t^\ell) = \lim_{\tau \rightarrow 0^+} \text{softmax}(\mathbf{w}_t^\ell / \tau). \tag{5.17}$$

We relax this operation by setting the temperature parameter $\tau > 0$. While computing the NELBO in Eq. 5.16, note that the discrete diffusion parameter $\mathcal{T}(\tilde{\alpha}_t)$ spans the interval $[0, 1]$, as does its Gaussian counterpart $\tilde{\alpha}_t$. The diffusion transformation operator $\mathcal{T}$ Eq. 5.3 has a crucial property: as the vocabulary size K increases, a small sub-interval $[a, 1]_{0 \leq a < 1}$ within the domain of $\mathcal{T}$ is sufficient to map onto the full range $[0, 1]$. For instance, in Suppl. C.3.7, we observe that for $K = 30K$, when $\tilde{\alpha}_t \in [0.85, 1]$, the corresponding $\alpha_t = \mathcal{T}(\tilde{\alpha}_t)$ nearly spans the entire interval $[0, 1]$. This observation is counter-intuitive: since the Gaussian latents mostly resemble $\mathbf{x}$, one might expect the discrete NELBO to approach zero when evaluated with $\tilde{\alpha}_t$ restricted to such a narrow range. However, in practice, the NELBO remains largely unchanged. Why? The key reason lies in the discretization step. Even small amounts of Gaussian noise in $\mathbf{w}_\ell^t$ can cause the output of the $\arg \max$ operation to change drastically, as it is highly sensitive to perturbations. As a result, much of the extra signal is lost due to discretization. To mitigate this, we allow the denoising model $\mathbf{x}_\theta$ to access the continuous latent $\mathbf{w}_t^\ell$ through a tempered softmax in Eq. 5.17. This relaxation helps preserve more of the signal, making the reconstruction task easier. In this way, the temperature parameter τ effectively controls the difficulty of the learning problem.

Hence, unlike prior discrete diffusion methods, we design the denoising model $\mathbf{x}_\theta : \Delta^L \cup \mathcal{V}^L \times [0, 1] \rightarrow \Delta^L$ to handle both continuous latents and discrete latents; see Suppl. C.3.2 for details. During training, we sample $t \sim \mathcal{U}[\beta, \gamma]_{0 \leq \beta < \gamma \leq 1}$ from a sub-interval so that $\tilde{\alpha}_t \in [a, b]_{0 \leq a < b \leq 1}$. Following Arriola et al. [2025], we set $b < 1$ as $\tilde{\alpha}_t = 1$ doesn't provide much training signal. Thus, we propose the following

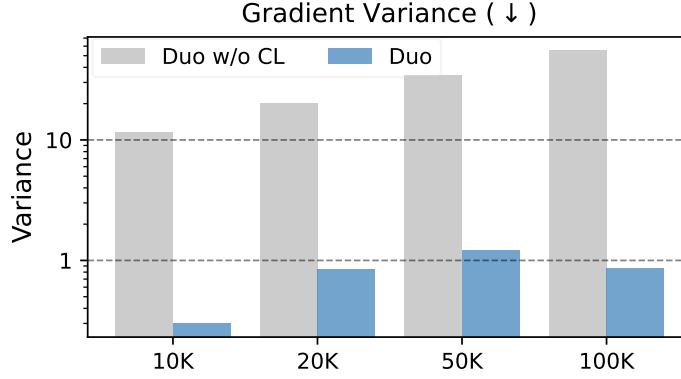


Figure 5.2: Curriculum learning drastically lowers the gradient variance in Duo trained with a fixed $\tau = 0.001$. The figure shows the summed gradient variance of the 100 weights with the highest variance, comparing Duo with CL (blue) and without CL (grey).

training loss:

$$\mathcal{L}_{\text{train}} = \mathbb{E}_{\mathbf{x}, t \sim \mathcal{U}[\beta, \gamma], \tilde{q}_t} \sum_{\ell \in [L]} f_{\text{Duo}} \left(\mathbf{z}_t^\ell := \text{arg max}(\mathbf{w}_t^\ell), \right. \\ \left. \mathbf{x}_\theta([\text{softmax}(\mathbf{w}_t^{\ell'} / \tau)]_{\ell'=1}^L, t), \alpha_t := \mathcal{T}(\tilde{\alpha}_t); \mathbf{x}^\ell \right). \quad (5.18)$$

This loss doesn’t correspond to a valid NELBO because the denoising model operates on a continuous-time random variable (r.v.), while the loss is defined for a discrete diffusion process. **It only becomes a valid NELBO in the limiting case $\lim_{\tau \rightarrow 0+}$ with $\beta = 0$ and $\gamma = 1$.** During evaluation, we evaluate the model as a discrete diffusion model using Eq. 5.15. As shown in Figure 5.2 and Table C.1, this approach results in lower training variance compared to previous MDMs and USDMs, ultimately improving model likelihood (Sec. 5.5.1).

5.4.2 Discrete Consistency Distillation

In this section, we present a new method to exploit the duality property of USDMs. This duality enables USDMs to adopt Consistency Distillation—a technique devel-

oped for Gaussian diffusion models to distil them into few-step generation models. However, standard discrete diffusion models cannot use this approach due to the absence of such PF-ODEs. To address this, we introduce Discrete Consistency Distillation (DCD), which sidesteps this limitation by utilizing the PF-ODE of the underlying Gaussian diffusion model to construct deterministic trajectories.

Deterministic Discrete Trajectories (DDT) Consistency Distillation relies on a PF-ODE parameterized by the denoising model. In our setting, the trained discrete denoiser $\mathbf{x}_\theta$ cannot be used to parameterize the ODE in the Gaussian space, as it operates only on discrete samples—the temperature τ in Eq. 5.18 is reduced to zero by the end of the training. To circumvent this, we construct a deterministic trajectory in Gaussian space by reversing the PF-ODE using an *optimal denoiser*, and then project this trajectory to the discrete domain. Let $\mathcal{P}_{\text{ODE}}$ denote such a trajectory. Given a clean data point $\mathbf{x}^{1:L} \sim q_{\text{data}}$ and Gaussian noise $\boldsymbol{\epsilon}^{1:L} = \{\boldsymbol{\epsilon}^\ell \sim \mathcal{N}(0, \mathbf{I}_K) \forall \ell \in [L]\}$, $\mathcal{P}_{\text{ODE}}(\mathbf{x}^{1:L}, \boldsymbol{\epsilon}^{1:L}) = \left\{ [\tilde{\alpha}_t \mathbf{x}^\ell + \sqrt{1 - \tilde{\alpha}_t^2} \boldsymbol{\epsilon}^\ell]_{\ell=1}^L \right\}_{t \in [0,1]}$; see Suppl. C.2.2 for a detailed discussion. Next, we project this trajectory to the discrete space via the **arg max** operator:

$$\mathcal{P}_{\text{DDT}}(\mathbf{x}^{1:L}, \boldsymbol{\epsilon}^{1:L}) = \left\{ [\text{arg max}(\tilde{\alpha}_t \mathbf{x}^\ell + \sqrt{1 - \tilde{\alpha}_t^2} \boldsymbol{\epsilon}^\ell)]_{\ell=1}^L \right\}_{t \in [0,1]}. \quad (5.19)$$

$\mathcal{P}_{\text{DDT}}$ serves as a proxy for the absence of a proper PF-ODE defined in the discrete space; see Fig. C.1 for an illustrative example.

Distillation Given a teacher model $\mathbf{x}_{\theta^-}$, our goal is to distill it into a student model $\mathbf{x}_\theta$ that generates samples of similar quality but in fewer steps. To perform distillation, we sample an adjacent pair of latents $(\mathbf{z}_s^{1:L}, \mathbf{z}_t^{1:L}) \sim \{(\mathbf{z}_{j-\delta}^{1:L}, \mathbf{z}_j^{1:L}) | \mathbf{z}^{1:L}_{\{\cdot\}} \in \mathcal{P}_{\text{DDT}}(\mathbf{x}^{1:L}, \boldsymbol{\epsilon}^{1:L}), j \in [\delta, 1]\}$ for a given step size $\delta \in [0, 1]$. Here, $\mathbf{z}_t^{1:L}$ is noisier than $\mathbf{z}_s^{1:L}$ and serves as the input to the student model. Let $\mathbf{x}_{\theta^-}(\mathbf{z}_s^{1:L}, s)$ and $\mathbf{x}_\theta(\mathbf{z}_t^{1:L}, t)$

Algorithm 1 Discrete Consistency Distillation (DCD)

Input: Dataset $\mathcal{D}$, learning rate η , number of distillation rounds N , number of training iterations per round M , ema μ , weights of the denoising model θ , weights of the EMA model θ_{ema} , discretization step δ .

```
for  $i = 1$  to  $N$  do
   $\theta^- \leftarrow \text{stopgrad}(\theta)$ 
  for  $j = 1$  to  $M$  do
    Sample  $\mathbf{x}^{1:L} \sim \mathcal{D}$ ,  $t \sim \mathcal{U}[0, 1]$ , and  $\epsilon^\ell \sim \mathcal{N}(0, I_K)$ .
     $s \leftarrow \max(t - \delta, 0)$ 
     $\mathbf{z}_s^{1:L} \leftarrow [\arg \max(\tilde{\alpha}_s \mathbf{x}^\ell + \sqrt{1 - \tilde{\alpha}_s^2} \epsilon^\ell)]_{\ell=1}^L$ 
     $\mathbf{z}_t^{1:L} \leftarrow [\arg \max(\tilde{\alpha}_t \mathbf{x}^\ell + \sqrt{1 - \tilde{\alpha}_t^2} \epsilon^\ell)]_{\ell=1}^L$ 
     $\mathcal{L}_{\text{DCD}}(\theta; \theta^-) \leftarrow \text{D}_{\text{KL}}(\mathbf{x}_\theta(\mathbf{z}_t^{1:L}, t), \mathbf{x}_{\theta^-}(\mathbf{z}_s^{1:L}, s))$ 
     $\theta \leftarrow \theta - \eta \nabla_\theta \mathcal{L}_{\text{DCD}}(\theta; \theta^-)$ 
     $\theta_{\text{ema}} \leftarrow \text{stopgrad}(\mu \theta_{\text{ema}} + (1 - \mu) \theta)$ 
  end for
   $\delta \leftarrow 2 \cdot \delta$ 
end for
return  $\theta_{\text{ema}}$ 
```

denote the output distributions over clean samples produced by the teacher and the student models, respectively. Following [Deschenaux and Gulcehre \[2024\]](#), we train the student by minimizing the KL divergence between these distributions:

$$\mathcal{L}_{\text{DCD}}(\theta; \theta^-) = \sum_{\ell \in [L]} \text{D}_{\text{KL}}(\mathbf{x}_\theta^\ell(\mathbf{z}_t^{1:L}, t), \mathbf{x}_{\theta^-}^\ell(\mathbf{z}_s^{1:L}, s)). \quad (5.20)$$

The distillation process proceeds in N rounds, each consisting of M training steps. At the end of each round, the teacher weights are updated with the current student weights. The full procedure is outlined in [Algo. 1](#).

5.5 Experiments

We evaluate Duo on standard language modeling benchmarks, training on LM1B [[Chelba et al., 2014](#)] and OpenWebText (OWT) [[Gokaslan et al., 2019](#)] with sequence packing [[Raffel et al., 2020a](#)]. We train our models for 1M steps

with a batch size of 512 on both datasets. For LM1B, we use a context length of 128 with the `bert-base-uncased` tokenizer [Devlin et al., 2018] with sequence packing [Arriola et al., 2025, Austin et al., 2021] and without it [Sahoo et al., 2024a, Lou et al., 2024, He et al., 2022]. For OWT, we use a context length of 1024 with the GPT-2 tokenizer [Radford et al., 2019]. Following Sahoo et al. [2024a], we reserve the last 100K documents for validation. Our model is a 170M-parameter modified diffusion transformer (DiT) [Peebles and Xie, 2023] with rotary positional encoding [Su et al., 2023] and adaptive layer norm for conditioning on diffusion time, consistent with prior work [Lou et al., 2024, Sahoo et al., 2024a]. Training is conducted on 8×H100s with `bf16` precision. We train Duo using Eq. 5.18, which requires computing the integral in Eq. 5.3. To reduce computation overhead, we pre-compute and cache 100K $(\tilde{\alpha}_t, \mathcal{T}(\tilde{\alpha}_t))$. The Gaussian diffusion parameter $\tilde{\alpha}_t$ is parameterized using a linear schedule i.e., $(\tilde{\alpha}_t = 1 - t)_{t \in [0,1]}$.

5.5.1 Improved Training

Our experiments show that (1) the proposed curriculum learning strategy (Sec. 5.4.1) **accelerates training by 2× and achieves a new state-of-the-art among USDMs** (Table 5.1), and (2) Duo performs competitively with Absorbing State diffusion across major language modeling benchmarks, even **surpassing AR models on 3/7 zero-shot PPL benchmarks** (Table 5.2).

Experimental Setup The primary baselines for Duo are the leading USDMs (SEDD Uniform [Lou et al., 2024] and UDLM [Schiff et al., 2025]) and Gaussian diffusion method, PLAID [Gulrajani and Hashimoto, 2024]. Additionally, we compare Duo with an AR model and MDMs such as MDLM [Sahoo et al., 2024a]

(SOTA), SEDD Absorb [Lou et al., 2024], and D3PM Absorb [Austin et al., 2021]. While training Duo, we set τ as a function of the training iteration n for Duo: $\tau = 0.001$ for the first 500K steps ($n < 500\text{K}$) with $\beta = 0.03$ and $\gamma = 0.15$ in Eq. 5.18, and $\tau = 0$ for the remaining steps up to 1M ($n \geq 500\text{K}$). To compute PPL for Duo, we use Eq. 5.15 with $\alpha_t = 1 - t$.

Bias-variance Tradeoff First, we study the effect of τ on the training dynamics in Fig. C.4 by training Duo on the LM1B dataset over 150K steps with a fixed $\tau \in \{0, 0.001, 0.01, 0.1\}$. Here, $\tau = 0$ (blue) corresponds to Eq. 5.15, i.e., no curriculum. Recall that a larger τ introduces more bias but reduces training variance. Ideally, τ should strike a balance—minimizing both the bias (measured by deviation from the blue curve) and the variance in the loss curve. For $\tau = 0.1$ (red), the loss drops sharply, indicating excessive bias, making it suboptimal. As τ decreases to 0.01 (orange) and 0.001 (purple), the loss curves become more stable. Among them, $\tau = 0.001$ is the most desirable, as it closely follows the blue curve (low bias) while exhibiting significantly lower variance.

Faster Training Notably, after just 10K steps of fine-tuning as a discrete diffusion model, i.e., at 510K steps, Duo achieves a PPL of 35.2—almost 1.5 points better than UDLM trained for 1M steps—indicating that curriculum learning accelerates convergence by at least $2\times$. In Fig. 5.2, we compare the summed gradient variance of the top 100 weights with the highest variance for Duo with (blue) and without (grey) curriculum. For these weights, we notice that curriculum learning reduces the gradient variance by an order of magnitude, which also manifests as lower loss variance in Fig. C.3 and Table C.1.

Likelihood Evaluation On LM1B and OWT (Table 5.1), Duo outperforms previous USDMs and Gaussian diffusion models, notably SEDD Uniform and UDLM and shrinks the gap with absorbing diffusion below 2 PPL points. On LM1B, We retrained Plaid which attained a PPL of 89.9 in 100K steps while Duo achieves a PPL of 43.0 in the same number of steps. This result is excluded from the table due to incomplete training; see Suppl. C.3.1 for details.

Zero-Shot Likelihood Evaluation We measure the zero-shot generalization of the models trained on OWT by evaluating their PPL on 7 other datasets. Following Sahoo et al. [2024a], our zero-shot datasets include the validation splits of Penn Tree Bank (PTB; Marcus et al. [1993]), WikiText [Merity et al., 2016], LM1B, Lambada [Paperno et al., 2016], AG News [Zhang et al., 2015], and Scientific papers from ArXiv and Pubmed [Cohan et al., 2018]. We observe that Duo outperforms SEDD Uniform and Plaid across all benchmarks. It achieves a better PPL than SEDD Absorbing on 4/7 datasets, MDLM on 1/7, most notably, outperforming an autoregressive transformer on 3/7 datasets.

Ablation Duo introduces two key improvements over UDLM: (i) a Rao-Blackwellized ELBO Eq. C.20 and (ii) a low-variance training curriculum. As shown in Table 5.3, the overall 3-point improvement in PPL comes roughly equally from both components, with Eq. C.20 accounting for about 1.7 points and the remainder from the curriculum.

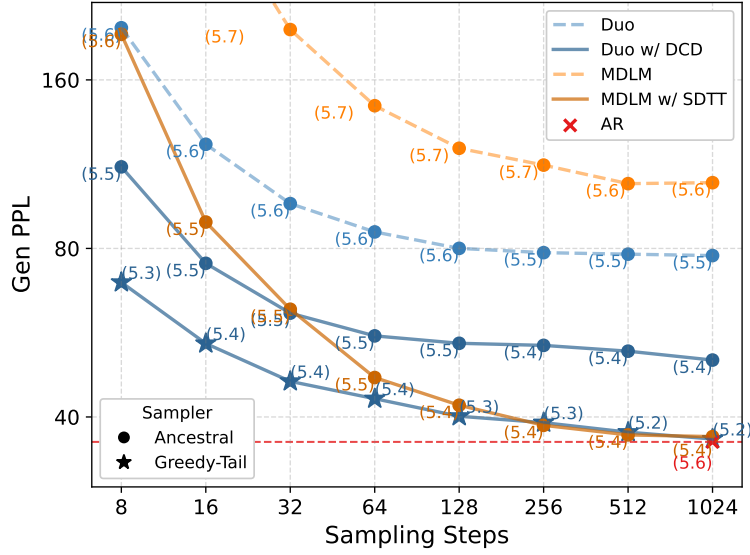


Figure 5.3: Sample quality comparison of Duo vs. MDLM. Duo outperforms MDLM in Gen PPL ($\downarrow$) for base models and in low-NFE regime after 5 distillation rounds.

5.5.2 Improved Sampling

Our sampling experiments show that for undistilled models, (1) **Duo generates higher-quality samples than all previous diffusion models** (Fig. C.5); (2) combining DCD with the Greedy-Tail sampler **reduces the number of sampling steps by two orders of magnitude** (Fig. 5.4); and (3) the distilled Duo model outperforms a distilled MDLM model, especially in the low NFE regime.

Experimental Setup We distill Duo on OWT using DCD, following the same setup as our main baseline—MDLM distilled with SDTT [Deschenaux and Gulcehre, 2024], a distillation method for MDMs. We run $N = 5$ distillation rounds, starting with discretization step $\delta = 1/512$ in Algo. 1 and doubling it every $M = 10K$ steps. To assess sample quality, we report GPT-2 Large generative perplexity (Gen PPL) and average sequence entropy for diversity. As noted by Zheng et al. [2024], masked diffusion models can suffer from low diversity and misleading Gen PPL

Table 5.1: Test perplexities (PPL; $\downarrow$) on LM1B. *Reported in He et al. [2022]. Best uniform/Gaussian diffusion value is bolded. ¶ Denotes the dataset didn’t incorporate sentence packing. †Reported in Arriola et al. [2025]. For diffusion models, we report the bound on the likelihood. Best diffusion value is underlined. ‡Denotes retrained models.

	LM1B [¶]	LM1B	OWT
<i>Autoregressive</i>			
Transformer [‡]	22.3	22.8 [†]	17.5
<i>Diffusion (absorbing state)</i>			
BERT-Mouth* [Wang and Cho, 2019]	-	142.9	-
D3PM Absorb [Austin et al., 2021]	-	76.9	-
DiffusionBert [He et al., 2022]	-	63.8	-
SEDD Absorb [‡] [Lou et al., 2024]	32.7	-	24.1
MDLM [Sahoo et al., 2024a]	<u>27.0</u>	<u>31.8[†]</u>	<u>23.2</u>
<i>Diffusion (Uniform-state / Gaussian)</i>			
D3PM Uniform [Austin et al., 2021]	-	137.9	-
Diffusion-LM* [Li et al., 2022]	-	118.6	-
SEDD Uniform [Lou et al., 2024]	40.3	-	29.7
UDLM [‡] [Schiff et al., 2025]	31.3	36.7	27.4
Duo (Ours)	29.9	33.7	25.2

Table 5.2: Zero-shot perplexities ($\downarrow$) of models trained for 1M steps on OWT. All perplexities for diffusion models are upper bounds. † Taken from Sahoo et al. [2024a]. ¶ Taken from [Lou et al., 2024] models were trained for 1.3Msteps as opposed to the baselines that were trained for 1Msteps. All perplexities for diffusion models are upper bounds. Best uniform / Gaussian diffusion values are **bolded** and diffusion values better than AR are underlined. ‡ denotes retrained model.

	PTB	Wikitext	LM1B	Lambada	AG News	Pubmed	Arxiv
<i>Autoregressive</i>							
Transformer [†]	82.05	25.75	51.25	51.28	52.09	49.01	41.73
<i>Diffusion (absorbing state)</i>							
SEDD Absorb [†]	100.09	34.28	68.20	<u>49.86</u>	62.09	<u>44.53</u>	<u>38.48</u>
D3PM Absorb [¶]	200.82	50.86	138.92	93.47	-	-	-
MDLM [†]	95.26	32.83	67.01	<u>47.52</u>	61.15	<u>41.89</u>	<u>37.37</u>
<i>Diffusion (Uniform-state / Gaussian)</i>							
SEDD Uniform [‡]	105.51	41.10	82.62	57.29	82.64	55.89	50.86
Plaid [¶]	142.60	50.86	91.12	57.28	-	-	-
UDLM [‡]	112.82	39.42	77.59	53.57	80.96	50.98	44.08
Duo (Ours)	89.35	33.57	73.86	<u>49.78</u>	67.81	<u>44.48</u>	<u>40.39</u>

Table 5.3: Ablation of two key components of Duo: (i) a low-variance training curriculum (Alg. 1), and (ii) an improved training loss Eq. C.20 in the form of a Rao-Blackwellized version of Eq. C.19.

Method	PPL ($\downarrow$)
Duo	33.7
& w/o CL (Alg. 1)	35.0
& w/o improved training loss (Eq. C.20)	36.7

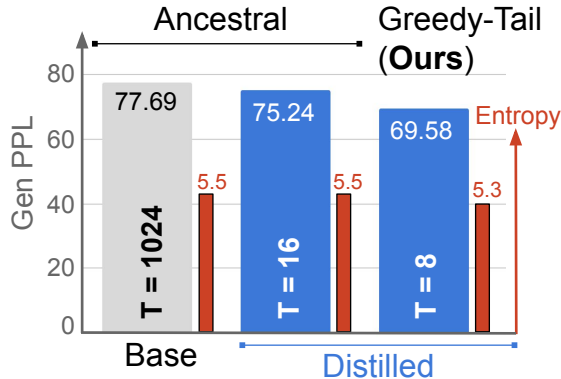


Figure 5.4: Sample quality comparison between the base Duo model and Duo distilled for 5 rounds using our DCD algorithm. The distilled model matches the base model’s sample quality in just 16 steps (vs. 1024) with ancestral sampling. With our Greedy-Tail sampler, sampling steps can be further reduced to 8, achieving slightly better Gen PPL and lower entropy.

under low-precision sampling. To address this, we use `float64` precision in all sampling experiments.

Sample Quality In Fig. C.5, Duo consistently outperforms all MDMs and USDMs in Gen PPL across sampling steps $T \in \{8, \dots, 1024\}$, with particularly strong performance at low NFEs. In Fig. 5.3, we compare Duo distilled with DCD to MDLM distilled with SDTT after 5 rounds (entropy values in parentheses). Under ancestral sampling, Duo performs significantly better than MDLM for $T \leq 32$. This is because MDMs generate many tokens independently and cannot revise them, leading to incoherence at low NFEs. In contrast, USDMs are self-correcting: they

can fix earlier errors in later denoising steps. At higher NFEs, MDLM outperforms the distilled Duo. While MDLM (with SDTT) matches the Gen PPL of the AR model, its lower entropy (5.4 vs. 5.6) indicates reduced diversity.

Greedy-Tail vs Ancestral Sampler In Fig. 5.3, the Greedy-Tail sampler improves Gen PPL by reducing sample entropy. In Fig. 5.4, we observe that the ancestral sampler enables a $64\times$ speedup (reducing NFEs from 1024 to 16) while maintaining Gen PPL. Greedy-Tail offers an even greater $128\times$ speedup, with a slight drop in entropy. Interestingly, as shown in Fig. C.7 and Fig. C.8, each distillation round improves both Gen PPL and diversity when using the Greedy-Tail sampler—unlike ancestral sampling—suggesting that Greedy-Tail is particularly effective for distilled models.

Ablation In Algo. 1, we use the denoising model weights as the teacher, deviating from the common practice of using EMA weights in consistency models. To test this choice, we modify Algo. 1 to use EMA weights (Algo. 3) instead. As shown in Fig. C.6, using the denoising model as a teacher leads to a better distilled model.

5.6 Related Work

Diffusion Language Models Prior work on discrete diffusion language models operates strictly in discrete space [Austin et al., 2021, Lou et al., 2024, Sahoo et al., 2024a, Schiff et al., 2025, Arriola et al., 2025]. In contrast, we expand their design space by incorporating continuous-state Gaussian diffusion, showing that training USDMs in this setting not only accelerates convergence but also improves model

likelihood. Li et al. [2022], Han et al. [2022], Dieleman et al. [2022], Gulrajani and Hashimoto [2024] use Gaussian diffusion for language modeling by injecting noise into the continuous embeddings of discrete tokens. Duo takes a different approach: it defines the diffusion process directly over one-hot token representations, rather than their embeddings. This hybrid treatment achieves better performance than methods confined to either domain alone.

Distillation for Faster Sampling Distillation techniques in Gaussian diffusion models [Luhman and Luhman, 2021, Salimans and Ho, 2022b, Song et al., 2023] rely on deterministic PF-ODE trajectories, which are unavailable for discrete diffusion. SDTT [Deschenaux and Gulcehre, 2024] addresses this by distilling along stochastic trajectories. Our method, Discrete Consistency Distillation 5.4.2, instead constructs deterministic PF-ODE trajectories in Gaussian space and maps them to the discrete domain via **arg max**, yielding distilled models that surpass prior methods in few-step generation.

Argmax Differentiation The softmax annealing trick was introduced by Jang et al. [2017], Maddison et al. [2017] to enable backpropagation through the **arg max** operation. Here, we repurpose it in a different context: instead of feeding either a clean or noisy token (the **arg max** of a Gaussian latent) into the denoising model, we approximate **arg max** with a tempered softmax, yielding a superposition of the two. This provides a low-variance training curriculum for USDMs.

5.7 Conclusion

In this work, we established a theoretical connection between continuous-space Gaussian diffusion models and discrete-space Uniform-state diffusion models. We leveraged this connection to achieve a $2\times$ speed-up in training (Sec. 5.5.1) convergence and a two-orders-of-magnitude improvement in sampling speed (Sec. 5.5.2). While USDMs trail MDMs in terms of perplexity, we showed that they outperform in few-step generation. We hope that our theoretical foundation opens up new avenues of future research which would leverage this connection to improve USDMs by borrowing techniques from Gaussian diffusion—a connection non-existent for MDMs.

CHAPTER 6

DISCUSSION AND CONCLUDING REMARK

6.1 Conclusion

This dissertation set out to address a central question: *Can diffusion language models be made competitive with traditional autoregressive approaches?* Through three complementary lines of work, we have shown that discrete diffusion models can substantially close the likelihood gap to autoregressive models, achieve competitive or superior performance on several benchmarks, and support few-step sampling while preserving much of the flexibility that makes diffusion attractive.

First, we demonstrated that the choice of noise process is not a benign design detail. By introducing context-adaptive noise schedules, we established that the long-held invariance of the diffusion likelihood to the noise schedule only holds under restrictive, univariate assumptions. Allowing the noise schedule to depend on the input leads to strictly tighter likelihoods and improved sample quality. Although this work was developed primarily in the continuous domain, the underlying message is broad: **richer, data-dependent forward processes provide a powerful lever for improving both training objectives and generative performance across modalities, including language.**

Second, we proposed a simplified framework for discrete diffusion that unifies Masked and Uniform-state processes and yields tighter, lower-variance likelihood bounds through Rao–Blackwellization. Within this framework, MDLM achieves state-of-the-art perplexities among diffusion language models and approaches, or even surpasses, autoregressive baselines on several natural language benchmarks.

At a systems level, this framework shows how to efficiently convert BERT-style encoders into high-quality generative models, and it has already informed large-scale deployments in industry. These results provide **the first concrete evidence that diffusion language models can move from a purely conceptual alternative to a practically competitive modeling paradigm.**

Third, we established a theoretical duality between discrete Uniform-state diffusion and continuous Gaussian diffusion. This connection allows classical Gaussian techniques, such as efficient denoising parameterizations, improved training objectives, and distillation-based fast samplers, to transfer into the discrete setting. Building on this duality, we obtained a twofold speedup in training convergence and orders-of-magnitude faster sampling for Uniform-state diffusion, and **we introduced the first few-step discrete diffusion samplers via discrete consistency distillation.** While Uniform-state models do not yet match the best masked diffusion models in terms of perplexity, they excel in the few-step regime, highlighting a promising direction for latency-sensitive applications.

Taken together, these contributions advance a unified view of diffusion language modeling. They show that (i) the forward noising process is a learnable object with real impact on likelihood and quality, (ii) discrete diffusion can be trained with principled, low-variance objectives that scale to realistic language tasks, and (iii) the gap between discrete and continuous diffusion is narrower than it may appear, which enables systematic transfer of ideas across domains. This foundation opens the door to diffusion language models that are not only competitive with autoregressive models on standard metrics, but also offer unique advantages in parallel generation, iterative refinement, and controllable editing.

6.2 Limitations and Future Directions

Despite these advances, several limitations remain, which also suggest fruitful directions for future work.

Training Variance, Scaling, and Robustness. Although MDLM and related models approach or surpass autoregressive perplexities on benchmarks for maths and code, diffusion language models still trail the strongest autoregressive systems on complex downstream tasks such as multi-turn conversations [Nie et al., 2025]. Training objectives for discrete diffusion exhibit higher gradient variance than autoregressive counterparts which contributes to the likelihood gap wrt AR models. Extending the proposed bounds and parameterizations to larger models, longer contexts, and more diverse datasets, while maintaining stability, is an open challenge. Moreover, most evaluations focus on likelihood and perplexity; the behavior of diffusion language models under distribution shift, adversarial prompts, and instruction-following settings is less well understood.

Efficiency and inference constraints. Parallel generation does not automatically translate into practical speedups. In MDLM, each denoising step is more expensive than a single autoregressive decoding step, and techniques that improve sample quality, such as remasking, increase the number of function evaluations. Current architectures also limit the ability to revise already generated tokens without restarting the diffusion trajectory, which constrains interactive editing and fine-grained control. Uniform-state diffusion with discrete consistency distillation alleviates some of these issues by enabling few-step sampling, but it still exhibits a quality gap relative to the autoregressive baselines. Designing architectures and

objectives that combine high-quality few-step generation with flexible token-level editing remains an important direction.

Expressiveness of corruption processes. This thesis focuses on two particular classes of discrete diffusion processes: masked diffusion and Uniform-state diffusion. While these processes provide clean theoretical structure and practical tractability, they are still highly restrictive and may not capture the inductive biases required in many domains. In settings such as molecular design, program synthesis, or structured document generation, the underlying objects have rich combinatorial and semantic structure that is not well modeled by simple masking or uniform corruption. More expressive, domain-specific corruption processes that respect chemical validity, syntactic constraints, or other structural invariants could yield stronger likelihoods, better sample quality, and more reliable downstream performance. Designing and analyzing such structured diffusion processes remains an important direction for future work.

Broader applicability and alignment. Finally, while the proposed methods already show impact in applications such as large-scale language modeling and molecular design, their behavior in safety-critical and high-stakes domains has not been systematically studied. Questions of controllability, alignment with human preferences, and robustness to harmful instructions remain largely open for diffusion-based language models. Integrating diffusion objectives with reinforcement learning or preference-based fine-tuning, and understanding how the diffusion dynamics interact with alignment signals, is an important line of future work.

In summary, this dissertation demonstrates that diffusion language models can be substantially improved in both quality and efficiency by rethinking the role

of the noise process, the training objective, and the relationship between discrete and continuous diffusion. At the same time, it highlights the open problems in scaling, stability, efficiency, and alignment. Addressing these challenges will be crucial for realizing the full potential of diffusion-based language models as a viable and complementary alternative to autoregressive approaches.

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APPENDIX A

CHAPTER 1 OF APPENDIX

A.1 Comparing to Previous Work

MuLAN is the first method to introduce a learned adaptive noise process. A widely held assumption is that the ELBO objective of a diffusion model is invariant to the noise process [Kingma et al., 2021]. We dispel this assumption: we show that when input-conditioned noise is combined with (a) multivariate noise, (b) a novel polynomial parameterization, and (c) auxiliary variables, a learned noise process yields an improved variational posterior and a tighter ELBO. This approach sets a new state-of-the-art in density estimation. While (a), (c) were proposed in other contexts, we leverage them as subcomponents of a novel algorithm. We elaborate further on this below.

A.1.1 Diffusion Models with Custom Noise

The noise schedule, which determines the amount and type of noise added at each step, plays a critical role in diffusion models. Chen [2023] empirically demonstrate that different noise schedules can significantly impact the generated image quality using various handcrafted noise schedules. Kingma et al. [2021] showed that the likelihood of a diffusion model remains invariant to the noise schedule with a scalar noise schedule. In this work we show that the ELBO is no longer invariant to multivariate noise schedules.

Recent works, including Hoogeboom and Salimans [2022], Rissanen et al. [2022],

Pearl et al. [2023], have explored per-pixel noise schedules (including blurring and other types of noising), yet none have delved into learning or conditioning the noise schedule on the input data itself. The shared components among these models are summarized and compared in Table A.1.

A.1.2 Advanced Diffusion Models

Yang et al. [2024] proposes noise processes that are conditioned on an external context (e.g., text). We also propose context-conditioned noise processes; however, their setting is that of conditional generation, where the context is always available at training and inference time, and the context represents external data. Our paper instead looks at unconditional generation, and we condition the noising process on the image itself that we want to generate (via latent variable) and learn how to apply noise across an image as a function of the image.

Lee et al. [2021], Popov et al. [2021] proposed using a data-dependent prior: however, they do not learn q and their noise process is not adaptive to the input $\mathbf{x}$. Thus they propose a fairly different set of methods from what we introduce.

Yang and Mandt [2023], Wang et al. [2023] have explored diffusion models with an auxiliary latent space, where the denoising network is conditioned on a latent distribution. Our paper also incorporate auxiliary latents, but unlike previous works, we add them to both p and q and we also focus on learning the process q (as opposed to doing representation learning using the auxiliary learned space). Lastly, our algorithm relies on many other components including a custom noise schedule, multivariate noise, etc. The shared components among these models are summarized and compared in Table A.1.

Table A.1: MuLAN is a noise schedule that can be integrated into any diffusion model such as VDM [Kingma et al., 2021] or InfoDiffusion [Wang et al., 2023]. The shared components between MuLAN and these models are summarized and compared in this table.

Method	learned noise	multivariate noise	input conditioned noise	auxiliary latents	noise schedule
VDM [Kingma et al., 2021]	Yes	No	No	No	Monotonic neural network
Blurring Diffusion Model [Hoogeboom and Salimans, 2022]	No	Yes	No	No	Frequency scaling
Inverse Heat Dissipation [Risänen et al., 2022]	No	Yes	No	No	Exponential
SVNR [Pearl et al., 2023]	No	Yes	No	No	Linear
InfoDiffusion [Wang et al., 2023]	No	No	No	In denoising process	Cosine
Lossy Compression [Yang and Mandt, 2023]	No	No	No	In denoising process	Linear / Cosine
MuLAN (Ours)	Yes	Yes	Yes	In noising and denoising process	Polynomial

A.1.3 Learned Diffusion

Diffusion Normalizing Flow (DNF) uses the following forward process:

$$d\mathbf{w}_t = \mathbf{f}_\theta(\mathbf{w}_t, t)dt + g(t)d\mathbf{w}, \quad (\text{A.1})$$

where the drift term $\mathbf{f}_\theta : \mathbb{R}^d \times \mathbb{R} \rightarrow \mathbb{R}^d$ is parameterized by a neural network with parameters θ and the diffusion term $g(t) \in \mathbb{R}^+$ is a scalar constant and $\mathbf{w}$ is the

standard Brownian motion. However, in MuLAN, the forward process is given by

$$d\mathbf{w}_t = \mathbf{f}_\theta(\mathbf{z}, t) \odot \mathbf{w}_t dt + \mathbf{g}_\theta(\mathbf{z}, t) \odot d\mathbf{w}; \mathbf{z} \sim q_\phi(\mathbf{z}|\mathbf{x}), \quad (\text{A.2})$$

where $\mathbf{z} \in \{0, 1\}^m$ is the auxiliary latent vector, $\mathbf{f}_\theta : \mathbb{R}^m \times \mathbb{R} \rightarrow \mathbb{R}^d$ and $\mathbf{g}_\theta : \mathbb{R}^m \times \mathbb{R} \rightarrow \mathbb{R}^d$ are parameterized by a neural network. Notice that the drift term in DNF, $\mathbf{f}_\theta(\mathbf{x}, t)$, is a non-linear function in $\mathbf{x}$, and the same holds for MuLAN since in the drift term, $\mathbf{f}_\theta(\mathbf{z}, t) \odot \mathbf{x}$, $\mathbf{z}$ and $\mathbf{x}$ depend on $\mathbf{x}$. Additionally, the diffusion coefficient, $\mathbf{g}_\theta(\mathbf{z}, t)$, is multivariate and conditioned on $\mathbf{x}$ via $\mathbf{z}$. The two parameterizations are different: on one hand, DNF admits more general classes of neural networks because it does not require marginals to be tractable. On the other hand MuLAN admits a more flexible noise model $\mathbf{g}_\theta(\mathbf{z}, t)$ and admits more efficient training (see the summarized Table A.2 below).

MuLAN has the advantage that it is simulation free; which means that given a data $\mathbf{x}$, the noisy sample $\mathbf{w}_t$ can be computed in closed form; however, in Diffusion Normalizing Flow, to compute $\mathbf{w}_t$, one needs to simulate the forward SDE which is resource intensive and limits its scalability to larger denoising models. While MuLAN optimizes the ELBO, DNF optimizes an approximation for the ELBO. In particular, the DNF training objective does not involve a term that accounts for the entropy of the encoder. Thus, the objective is closer to that of a normal auto-encoder in that regard.

Table A.2: The key differences between MuLAN and DNF is listed below.

Aspect	Property	Diffusion Normalizing Flow	MuLAN
Drift Term	Multivariate	Yes	Yes
	Adaptive	Yes	Yes
	Learnable	Yes	Yes
Diffusion Term	Multivariate	No	Yes
	Adaptive	No	Yes
	Learnable	No	Yes
Simulation Free Training		No	Yes
Exact ELBO Optimization		No	Yes
NLL ($\downarrow$)	CIFAR-10	3.04	2.55

Other concurrent work seeks to improve log-likelihood estimation by learning the forward diffusion process in a simulation-free setting. In neural diffusion models (NDMs), the noise schedule is fixed, but the mean of each marginal $q(\mathbf{w}_t|\mathbf{x})$ is learned. This requires denoising $\mathbf{x}$ in a transformed space, which prevents using noise parameterization, a design choice that is important for performance. Their denoising family also induces a parameterization that limits the kinds of samplers that can be used. Lastly, NDMs use a model that is 2x larger than a regular diffusion model, while ours only adds 10% more parameters.

The DiffEnc framework adds an extra learned correction term to q to adjust the mean of each marginal $q(\mathbf{w}_t|\mathbf{x})$. This noise choice also requires using certain parameterizations for $\mathbf{x}$ that are not compatible with noise parameterization; while their approach supports v-parameterization, it also requires training a mean parameterization network. Similarly to NDMs, the noise schedule remains fixed, while the mean of each marginal is adjusted by the network. Our approach towards learning the noise schedule yields better empirical performance and is, in our opinion, simpler; it can also be combined with this prior work on learning the marginals’ means.

A.1.4 Optimal Transport

In techniques based on optimal transport, the goal is to learn a noise process that minimizes the transport cost from data to noise, which in practice produces smoother diffusion trajectories that facilitate sampling.

Minimizing Trajectory Curvature (MTC) of ODE-based generative models [Lee et al. \[2023\]](#), the primary goal is to design the forward diffusion process that is

optimal for fast sampling; however, MuLAN strives to learn a forward process that optimizes for log-likelihood. In the former, the marginals $\mathbf{w}_t$ in the forward process are given as

$$\mathbf{w}_t = (1 - t)\mathbf{x} + t\mathbf{z}; \mathbf{z} \sim q_\phi(\mathbf{z}|\mathbf{x}) \quad (\text{A.3})$$

where $\mathbf{w}_t, \mathbf{x}, F \in \mathbb{R}^d$. However for MuLAN the marginals are $\mathbf{w}_t = \boldsymbol{\alpha}_\phi(\mathbf{z}, t) \odot \mathbf{x} + \sqrt{1 - \boldsymbol{\alpha}_\phi^2(\mathbf{z}, t)} \odot \epsilon$; $\epsilon \sim \mathcal{N}(0, \mathbf{I}_d)$; $\mathbf{z} \sim q_\phi(\mathbf{z}|\mathbf{x})$ where $\boldsymbol{\alpha}_\phi(\mathbf{z}, t) : \mathbb{R}^d \times \mathbb{R} \rightarrow \mathbb{R}_{\geq 0}^d$, $\mathbf{z} \in \{0, 1\}^m$, $\epsilon \in \mathbb{R}^d$. Notice that in the MTC formula, the coefficient of $\mathbf{x}$, the time integral of the drift term, is a scalar and linear function of, and is independent of the input $\mathbf{x}$. In MuLAN, that term is a multivariate non-linear function in t , and conditioned on $\mathbf{x}$ via the auxiliary latent variable $\mathbf{z}$. This implies that the forward diffusion process in MuLAN is more expressive than MTC. The simplistic forward process in MTC enables faster sampling whereas the richer / more expressive forward process in MuLAN leads to improved likelihood estimates. Table A.3 summarizes the key differences.

Table A.3: Comparison between Minimizing Trajectory Curvature and MuLAN methods.

Aspect	Property	Minimizing Trajectory Curvature	MuLAN
Goal		Design faster sampler	Improve log-likelihood
Drift Term	Learnable	No	Yes
	Linearity	Linear in time t , linear in $\mathbf{x}$	Non-linear in time t , Non-linear in $\mathbf{z}$ (and hence $\mathbf{x}$)
	Dimensionality	Scalar	Multivariate
	Adaptive	No	Yes
Diffusion Term	Linearity	Linear in time t	Non-linear in time t
	Dimensionality	Multivariate	Multivariate
	Learnable	Yes	Yes
	Adaptive	Yes	Yes

An alternative approach to learning a forward process that performs optimal transport is via the theory of Schrodinger bridges [Shi et al., 2024b, De Bortoli et al., 2021, Wang et al., 2021, Peluchetti, 2023]. Similarly to the DNF framework,

these methods do not admit analytical marginals and therefore involve computing full trajectories from noisy and clean data. Additionally, they are typically trained using an iterative procedure that generalizes the sinkhorn algorithm and involves iteratively training q and p . As such, these types of methods are typically more expensive to train and competitive results on standard benchmarks (e.g., CIFAR10, ImageNet) are not yet available to our knowledge.

A.2 Standard Diffusion Models

We have a Gaussian diffusion process that begins with the data $\mathbf{x}$, and defines a sequence of increasingly noisy versions of $\mathbf{x}$ which we call the latent variables $\mathbf{w}_t$, where t runs from $t = 0$ (least noisy) to $t = 1$ (most noisy). Given, T , we discretize time uniformly into T timesteps each with a width $1/T$. We define $t(i) = i/T$ and $s(i) = (i - 1)/T$.

A.2.1 Forward Process

$$q(\mathbf{w}_t | \mathbf{w}_s) = \mathcal{N}(\alpha_{t|s} \mathbf{w}_s, \sigma_{t|s}^2 \mathbf{I}_n) \quad (\text{A.4})$$

where

$$\alpha_{t|s} = \frac{\alpha_t}{\alpha_s} \quad (\text{A.5})$$

$$\sigma_{t|s}^2 = \sigma_t^2 - \alpha_{t|s}^2 \sigma_s^2 \quad (\text{A.6})$$

A.2.2 Reverse Process

Kingma et al. [2021] show that the distribution $q(\mathbf{w}_s|\mathbf{w}_t, \mathbf{x})$ is also gaussian,

$$q(\mathbf{w}_s|\mathbf{w}_t, \mathbf{x}) = \mathcal{N}\left(\boldsymbol{\mu}_q = \frac{\alpha_{t|s}\sigma_s^2}{\sigma_t^2}\mathbf{w}_t + \frac{\sigma_{t|s}^2\alpha_s}{\sigma_t^2}\mathbf{x}, \boldsymbol{\Sigma}_q = \frac{\sigma_s^2\sigma_{t|s}^2}{\sigma_t^2}\mathbf{I}_n\right) \quad (\text{A.7})$$

Since during the reverse process, we don't have access to $\mathbf{x}$, we approximate it using a neural network $\mathbf{x}_\theta(\mathbf{w}_t, t)$ with parameters θ . Thus,

$$p_\theta(\mathbf{w}_s|\mathbf{w}_t) = \mathcal{N}\left(\boldsymbol{\mu}_p = \frac{\alpha_{t|s}\sigma_s^2}{\sigma_t^2}\mathbf{w}_t + \frac{\sigma_{t|s}^2\alpha_s}{\sigma_t^2}\mathbf{x}_\theta(\mathbf{w}_t, t), \boldsymbol{\Sigma}_p = \frac{\sigma_s^2\sigma_{t|s}^2}{\sigma_t^2}\mathbf{I}_n\right) \quad (\text{A.8})$$

A.2.3 Variational Lower Bound

This corruption process q is the following markov-chain as $q(\mathbf{w}_{0:1}) = q(\mathbf{x}) \left(\prod_{i=1}^T q(\mathbf{w}_{t(i)}|\mathbf{w}_{s(i)}) \right)$. In the Reverse Process, or the Denoising Process, p_θ , a neural network (with parameters θ) is used to denoise the noising process q . The Reverse Process factorizes as:

$$p_\theta(\mathbf{w}_{0:1}) = p_\theta(\mathbf{w}_1) \prod_{i=1}^T p_\theta(\mathbf{w}_{s(i)}|\mathbf{w}_{t(i)}).$$

Let $\mathbf{x}_\theta(\mathbf{w}_t, t)$ be the reconstructed input by a neural network from $\mathbf{w}_t$. Similar to Sohl-Dickstein et al. [2015], Kingma et al. [2021] we decompose the negative lower

bound (VLB) as:

$$\begin{aligned}
-\log p_\theta(\mathbf{x}) &\leq \underbrace{\mathbb{E}_{q_\phi} \left[-\log \frac{p_\theta(\mathbf{w}_{t(0):t(T)})}{q(\mathbf{w}_{t(1):t(T)}|\mathbf{x})} \right]}_{\text{ELBO}(p_\theta(\mathbf{w}_{0:1}), q(\mathbf{w}_{0:1})) \text{ defined in Eq. 2.9}} \\
&= \mathbb{E}_{\mathbf{w}_{t(1)} \sim q(\mathbf{w}_{t(1)}|\mathbf{x})} [-\log p_\theta(\mathbf{w}_0|\mathbf{w}_{t(1)})] \\
&\quad + \sum_{i=2}^T \mathbb{E}_{\mathbf{w}_{t(i)} \sim q(\mathbf{w}_{t(i)}|\mathbf{x})} \text{D}_{\text{KL}}[p_\theta(\mathbf{w}_{s(i)}|\mathbf{w}_{t(i)}) \| q(\mathbf{w}_{s(i)}|\mathbf{w}_{t(i)}, \mathbf{x})] \\
&\quad + \text{D}_{\text{KL}}[p_\theta(\mathbf{w}_1) \| q_\phi(\mathbf{w}_1|\mathbf{x})] \\
&= \underbrace{\mathbb{E}_{\mathbf{w}_{t(1)} \sim q(\mathbf{w}_{t(1)}|\mathbf{x})} [-\log p_\theta(\mathbf{w}_0|\mathbf{w}_{t(1)})]}_{\mathcal{L}_{\text{recons}}} \\
&\quad + \underbrace{\frac{T}{2} \mathbb{E}_{\epsilon \sim \mathcal{N}(0, \mathbf{I}_n), i \sim U\{2, T\}} \text{D}_{\text{KL}}[p_\theta(\mathbf{w}_{s(i)}|\mathbf{w}_{t(i)}) \| q(\mathbf{w}_{s(i)}|\mathbf{w}_{t(i)}, \mathbf{x})]}_{\mathcal{L}_{\text{diffusion}}} \\
&\quad + \underbrace{\text{D}_{\text{KL}}[p_\theta(\mathbf{w}_1) \| q(\mathbf{w}_1|\mathbf{x})]}_{\mathcal{L}_{\text{prior}}} \tag{A.9}
\end{aligned}$$

The prior loss, $\mathcal{L}_{\text{prior}}$, and reconstruction loss, $\mathcal{L}_{\text{recons}}$, can be (stochastically and differentiably) estimated using standard techniques; see [Kingma and Welling \[2013\]](#). The diffusion loss, $\mathcal{L}_{\text{diffusion}}$, varies with the formulation of the noise schedule. We provide an exact formulation for it in the subsequent sections.

A.2.4 Diffusion Loss

For brevity, we use the notation s for $s(i)$ and t for $t(i)$. From Eq. A.19 and Eq. A.20 we get the following expression for $q(\mathbf{w}_s|\mathbf{w}_t, \mathbf{x})$:

$$\begin{aligned} & D_{\text{KL}}(q(\mathbf{w}_s|\mathbf{w}_t, \mathbf{x})||p_\theta(\mathbf{w}_s|\mathbf{w}_t)) \\ &= \frac{1}{2} \left((\boldsymbol{\mu}_q - \boldsymbol{\mu}_p)^\top \boldsymbol{\Sigma}_\theta^{-1} (\boldsymbol{\mu}_q - \boldsymbol{\mu}_p) + \text{tr} \left(\boldsymbol{\Sigma}_q \boldsymbol{\Sigma}_p^{-1} - \mathbf{I}_n \right) - \log \left| \frac{\boldsymbol{\Sigma}_q}{\boldsymbol{\Sigma}_p} \right| \right) \\ &= \frac{1}{2} (\boldsymbol{\mu}_q - \boldsymbol{\mu}_p)^\top \boldsymbol{\Sigma}_\theta^{-1} (\boldsymbol{\mu}_q - \boldsymbol{\mu}_p) \end{aligned}$$

Substituting $\boldsymbol{\mu}_q, \boldsymbol{\Sigma}_q, \boldsymbol{\mu}_p, \boldsymbol{\Sigma}_p$ from equation A.8 and equation A.7, we get:

$$= \frac{1}{2} (\nu(s) - \nu(t)) \|\mathbf{x} - \mathbf{x}_\theta(\mathbf{w}_t, t)\|_2^2 \quad (\text{A.10})$$

Thus $\mathcal{L}_{\text{diffusion}}$ is given by

$$\begin{aligned} & \mathcal{L}_{\text{diffusion}} \\ &= \lim_{T \rightarrow \infty} \frac{T}{2} \mathbb{E}_{\epsilon \sim \mathcal{N}(0, \mathbf{I}_n), i \sim U\{2, T\}} D_{\text{KL}}[p_\theta(\mathbf{w}_{s(i)}|\mathbf{w}_{t(i)})||q_\phi(\mathbf{w}_{s(i)}|\mathbf{w}_{t(i)}, \mathbf{x})] \\ &= \lim_{T \rightarrow \infty} \frac{1}{2} \sum_{i=2}^T \mathbb{E}_{\epsilon \sim \mathcal{N}(0, \mathbf{I}_n)} (\nu(s) - \nu(t)) \|\mathbf{x} - \mathbf{x}_\theta(\mathbf{w}_t, t)\|_2^2 \\ &= \frac{1}{2} \mathbb{E}_{\epsilon \sim \mathcal{N}(0, \mathbf{I}_n)} \left[\lim_{T \rightarrow \infty} \sum_{i=2}^T (\nu(s) - \nu(t)) \|\mathbf{x} - \mathbf{x}_\theta(\mathbf{w}_t, t)\|_2^2 \right] \\ &= \frac{1}{2} \mathbb{E}_{\epsilon \sim \mathcal{N}(0, \mathbf{I}_n)} \left[\lim_{T \rightarrow \infty} \sum_{i=2}^T T (\nu(s) - \nu(t)) \|\mathbf{x} - \mathbf{x}_\theta(\mathbf{w}_t, t)\|_2^2 \frac{1}{T} \right] \end{aligned}$$

Substituting $\lim_{T \rightarrow \infty} T(\nu(s) - \nu(t)) = \frac{d}{dt} \nu(t) \equiv \nu'(t)$; see Kingma et al. [2021]

$$= \frac{1}{2} \mathbb{E}_{\epsilon \sim \mathcal{N}(0, \mathbf{I}_n)} \left[\int_0^1 \nu'(t) \|\mathbf{x} - \mathbf{x}_\theta(\mathbf{w}_t, t)\|_2^2 dt \right] \quad (\text{A.11})$$

In practice instead of computing the integral is computed by MC sampling.

$$= -\frac{1}{2} \mathbb{E}_{\epsilon \sim \mathcal{N}(0, \mathbf{I}_n), t \sim U[0, 1]} \left[\nu'(t) \|\mathbf{x} - \mathbf{x}_\theta(\mathbf{w}_t, t)\|_2^2 \right] \quad (\text{A.12})$$

A.3 Multivariate Noise Schedule

For a multivariate noise schedule we have $\boldsymbol{\alpha}_t, \boldsymbol{\sigma}_t \in \mathbb{R}_+^d$ where $t \in [0, 1]$. $\boldsymbol{\alpha}_t, \boldsymbol{\sigma}_t$ are vectors. The timesteps s, t satisfy $0 \leq s < t \leq 1$. Furthermore, we use the following notations where arithmetic division represents element wise division between 2 vectors:

$$\boldsymbol{\alpha}_{t|s} = \frac{\boldsymbol{\alpha}_t}{\boldsymbol{\alpha}_s} \quad (\text{A.13})$$

$$\boldsymbol{\sigma}_{t|s}^2 = \boldsymbol{\sigma}_t^2 - \boldsymbol{\alpha}_{t|s}^2 \boldsymbol{\sigma}_s^2 \quad (\text{A.14})$$

A.3.1 Forward Process

$$q(\mathbf{w}_t | \mathbf{w}_s) = \mathcal{N}(\boldsymbol{\alpha}_{t|s} \mathbf{w}_s, \text{diag}(\boldsymbol{\sigma}_{t|s}^2)) \quad (\text{A.15})$$

Change of variables. We can write $\mathbf{w}_t$ explicitly in terms of the signal-to-noise ratio, $\boldsymbol{\nu}(t)$, and input $\mathbf{x}$ in the following manner:

$$\boldsymbol{\nu}_t = \frac{\boldsymbol{\alpha}_t^2}{\boldsymbol{\sigma}_t^2}$$

We know $\alpha_t^2 = 1 - \sigma_t^2$ for Variance Preserving process; see Sec. 2.

$$\begin{aligned} \implies \frac{1 - \sigma_t^2}{\sigma_t^2} &= \boldsymbol{\nu}_t \\ \implies \sigma_t^2 &= \frac{1}{1 + \boldsymbol{\nu}_t} \quad \text{and} \quad \boldsymbol{\alpha}_t^2 = \frac{\boldsymbol{\nu}_t}{1 + \boldsymbol{\nu}_t} \end{aligned} \quad (\text{A.16})$$

$$\nu_t = \frac{\alpha_t^2}{\sigma_t^2}$$

We know $\alpha_t^2 = 1 - \sigma_t^2$ for Variance Preserving process; see Sec. 2.

$$\begin{aligned} \implies \frac{1 - \sigma_t^2}{\sigma_t^2} &= \nu_t \\ \implies \sigma_t^2 &= \frac{1}{1 + \nu_t} \quad \text{and} \quad \alpha_t^2 = \frac{\nu_t}{1 + \nu_t} \end{aligned} \quad (\text{A.17})$$

Thus, we write $\mathbf{w}_t$ in terms of the signal-to-noise ratio in the following manner:

$$\begin{aligned}\mathbf{w}_{\nu(t)} &= \boldsymbol{\alpha}_t \mathbf{x} + \boldsymbol{\sigma}_t \boldsymbol{\epsilon}_t; \quad \boldsymbol{\epsilon}_t \sim \mathcal{N}(0, \mathbf{I}_n) \\ &= \frac{\sqrt{\nu(t)}}{\sqrt{1 + \nu(t)}} \mathbf{x} + \frac{1}{\sqrt{1 + \nu(t)}} \boldsymbol{\epsilon}_t\end{aligned}\quad \text{Using Eq. A.16} \quad (\text{A.18})$$

A.3.2 Reverse Process

The distribution of $\mathbf{w}_t$ given $\mathbf{w}_s$ is given by:

$$q(\mathbf{w}_s | \mathbf{w}_t, \mathbf{x}) = \mathcal{N}\left(\boldsymbol{\mu}_q = \frac{\boldsymbol{\alpha}_{t|s} \boldsymbol{\sigma}_s^2}{\boldsymbol{\sigma}_t^2} \mathbf{w}_t + \frac{\boldsymbol{\sigma}_{t|s}^2 \boldsymbol{\alpha}_s}{\boldsymbol{\sigma}_t^2} \mathbf{x}, \quad \boldsymbol{\Sigma}_q = \text{diag}\left(\frac{\boldsymbol{\sigma}_s^2 \boldsymbol{\sigma}_{t|s}^2}{\boldsymbol{\sigma}_t^2}\right)\right) \quad (\text{A.19})$$

Let $\mathbf{x}_\theta(\mathbf{w}_t, t)$ be the neural network approximation for $\mathbf{x}$. Then we get the following reverse process:

$$p_\theta(\mathbf{w}_s | \mathbf{w}_t) = \mathcal{N}\left(\boldsymbol{\mu}_p = \frac{\boldsymbol{\alpha}_{t|s} \boldsymbol{\sigma}_s^2}{\boldsymbol{\sigma}_t^2} \mathbf{w}_t + \frac{\boldsymbol{\sigma}_{t|s}^2 \boldsymbol{\alpha}_s}{\boldsymbol{\sigma}_t^2} \mathbf{x}_\theta(\mathbf{w}_t, t), \quad \boldsymbol{\Sigma}_p = \text{diag}\left(\frac{\boldsymbol{\sigma}_s^2 \boldsymbol{\sigma}_{t|s}^2}{\boldsymbol{\sigma}_t^2}\right)\right) \quad (\text{A.20})$$

A.3.3 Diffusion Loss

For brevity we use the notation s for $s(i)$ and t for $t(i)$. From Eq. A.19 and Eq. A.20 we get the following expression for $q(\mathbf{w}_s|\mathbf{w}_t, \mathbf{x})$:

$$\begin{aligned} & D_{\text{KL}}(q(\mathbf{w}_s|\mathbf{w}_t, \mathbf{x})||p_\theta(\mathbf{w}_s|\mathbf{w}_t)) \\ &= \frac{1}{2} \left((\boldsymbol{\mu}_q - \boldsymbol{\mu}_p)^\top \boldsymbol{\Sigma}_\theta^{-1} (\boldsymbol{\mu}_q - \boldsymbol{\mu}_p) + \text{tr}(\boldsymbol{\Sigma}_q \boldsymbol{\Sigma}_p^{-1} - \mathbf{I}_n) - \log \frac{|\boldsymbol{\Sigma}_q|}{|\boldsymbol{\Sigma}_p|} \right) \\ &= \frac{1}{2} (\boldsymbol{\mu}_q - \boldsymbol{\mu}_p)^\top \boldsymbol{\Sigma}_\theta^{-1} (\boldsymbol{\mu}_q - \boldsymbol{\mu}_p) \end{aligned}$$

Substituting $\boldsymbol{\mu}_q, \boldsymbol{\mu}_p, \boldsymbol{\Sigma}_p$ from equation A.20 and equation A.19.

$$\begin{aligned} &= \frac{1}{2} \left(\frac{\boldsymbol{\sigma}_{t|s}^2 \boldsymbol{\alpha}_s}{\boldsymbol{\sigma}_t^2} \mathbf{x} - \frac{\boldsymbol{\sigma}_{t|s}^2 \boldsymbol{\alpha}_s}{\boldsymbol{\sigma}_t^2} \mathbf{x}_\theta(\mathbf{w}_t, t) \right)^\top \text{diag} \left(\frac{\boldsymbol{\sigma}_s^2 \boldsymbol{\sigma}_{t|s}^2}{\boldsymbol{\sigma}_t^2} \right)^{-1} \left(\frac{\boldsymbol{\sigma}_{t|s}^2 \boldsymbol{\alpha}_s}{\boldsymbol{\sigma}_t^2} \mathbf{x} - \frac{\boldsymbol{\sigma}_{t|s}^2 \boldsymbol{\alpha}_s}{\boldsymbol{\sigma}_t^2} \mathbf{x}_\theta(\mathbf{w}_t, t) \right) \\ &= \frac{1}{2} (\mathbf{x} - \mathbf{x}_\theta(\mathbf{w}_t, t))^\top \text{diag} \left(\frac{\boldsymbol{\sigma}_{t|s}^2 \boldsymbol{\alpha}_s}{\boldsymbol{\sigma}_t^2} \right)^\top \text{diag} \left(\frac{\boldsymbol{\sigma}_s^2 \boldsymbol{\sigma}_{t|s}^2}{\boldsymbol{\sigma}_t^2} \right)^{-1} \text{diag} \left(\frac{\boldsymbol{\sigma}_{t|s}^2 \boldsymbol{\alpha}_s}{\boldsymbol{\sigma}_t^2} \right) (\mathbf{x} - \mathbf{x}_\theta(\mathbf{w}_t, t)) \\ &= \frac{1}{2} (\mathbf{x} - \mathbf{x}_\theta(\mathbf{w}_t, t))^\top \text{diag} \left(\frac{\boldsymbol{\sigma}_{t|s}^2 \boldsymbol{\alpha}_s}{\boldsymbol{\sigma}_t^2} \odot \frac{\boldsymbol{\sigma}_t^2}{\boldsymbol{\sigma}_s^2 \boldsymbol{\sigma}_{t|s}^2} \odot \frac{\boldsymbol{\sigma}_{t|s}^2 \boldsymbol{\alpha}_s}{\boldsymbol{\sigma}_t^2} \right) (\mathbf{x} - \mathbf{x}_\theta(\mathbf{w}_t, t)) \\ &= \frac{1}{2} (\mathbf{x} - \mathbf{x}_\theta(\mathbf{w}_t, t))^\top \text{diag} \left(\frac{\boldsymbol{\sigma}_{t|s}^2 \boldsymbol{\alpha}_s^2}{\boldsymbol{\sigma}_t^2 \boldsymbol{\sigma}_s^2} \right) (\mathbf{x} - \mathbf{x}_\theta(\mathbf{w}_t, t)) \end{aligned}$$

Simplifying the expression using Eq. A.13 and Eq. A.14 we get,

$$= \frac{1}{2} (\mathbf{x} - \mathbf{x}_\theta(\mathbf{w}_t, t))^\top \text{diag} \left(\frac{\boldsymbol{\alpha}_s^2}{\boldsymbol{\sigma}_s^2} - \frac{\boldsymbol{\alpha}_t^2}{\boldsymbol{\sigma}_t^2} \right) (\mathbf{x} - \mathbf{x}_\theta(\mathbf{w}_t, t))$$

Using the relation $\boldsymbol{\nu}(t) = \boldsymbol{\alpha}_t^2 / \boldsymbol{\sigma}_t^2$ we get,

$$= \frac{1}{2} (\mathbf{x} - \mathbf{x}_\theta(\mathbf{w}_t, t))^\top \text{diag} (\boldsymbol{\nu}(s) - \boldsymbol{\nu}(t)) (\mathbf{x} - \mathbf{x}_\theta(\mathbf{w}_t, t)) \quad (\text{A.21})$$

Like Kingma et al. [2021] we train the model in the continuous domain with

$T \rightarrow \infty$.

$\mathcal{L}_{\text{diffusion}}$

$$\begin{aligned}
&= \lim_{T \rightarrow \infty} \frac{1}{2} \sum_{i=2}^T \mathbb{E}_{\epsilon \sim \mathcal{N}(0, \mathbf{I}_n)} D_{\text{KL}}(q(\mathbf{w}_{s(i)} | \mathbf{w}_{t(i)}, \mathbf{x}) \| p_{\theta}(\mathbf{w}_{s(i)} | \mathbf{w}_{t(i)})) \\
&= \lim_{T \rightarrow \infty} \frac{1}{2} \sum_{i=2}^T \mathbb{E}_{\epsilon \sim \mathcal{N}(0, \mathbf{I}_n)} (\mathbf{x} - \mathbf{x}_{\theta}(\mathbf{w}_{t(i)}, t(i)))^{\top} \text{diag}(\boldsymbol{\nu}_{s(i)} - \boldsymbol{\nu}_{t(i)}) (\mathbf{x} - \mathbf{x}_{\theta}(\mathbf{w}_{t(i)}, t)) \\
&= \frac{1}{2} \mathbb{E}_{\epsilon \sim \mathcal{N}(0, \mathbf{I}_n)} \left[\lim_{T \rightarrow \infty} \sum_{i=2}^T (\mathbf{x} - \mathbf{x}_{\theta}(\mathbf{w}_{t(i)}, t(i)))^{\top} \text{diag}(\boldsymbol{\nu}_{s(i)} - \boldsymbol{\nu}_{t(i)}) (\mathbf{x} - \mathbf{x}_{\theta}(\mathbf{w}_{t(i)}, t)) \right] \\
&= \frac{1}{2} \mathbb{E}_{\epsilon \sim \mathcal{N}(0, \mathbf{I}_n)} \left[\lim_{T \rightarrow \infty} \sum_{i=2}^T T (\mathbf{x} - \mathbf{x}_{\theta}(\mathbf{w}_{t(i)}, t(i)))^{\top} \text{diag}(\boldsymbol{\nu}_{s(i)} - \boldsymbol{\nu}_{t(i)}) (\mathbf{x} - \mathbf{x}_{\theta}(\mathbf{w}_{t(i)}, t)) \frac{1}{T} \right] \\
&\text{Let } \lim_{T \rightarrow \infty} T(\boldsymbol{\nu}_{s(i)} - \boldsymbol{\nu}_{t(i)}) = \frac{d}{dt} \boldsymbol{\nu}(t) \text{ denote the scalar derivative of the vector } \boldsymbol{\nu}(t) \text{ w.r.t } t \\
&= \frac{1}{2} \mathbb{E}_{\epsilon \sim \mathcal{N}(0, \mathbf{I}_n)} \left[\int_0^1 (\mathbf{x} - \mathbf{x}_{\theta}(\mathbf{w}_t, t))^{\top} \text{diag} \left(\frac{d}{dt} \boldsymbol{\nu}(t) \right) (\mathbf{x} - \mathbf{x}_{\theta}(\mathbf{w}_t, t)) dt \right] \tag{A.22}
\end{aligned}$$

In practice instead of computing the integral is computed by MC sampling.

$$= -\frac{1}{2} \mathbb{E}_{\epsilon \sim \mathcal{N}(0, \mathbf{I}_n), t \sim U[0, 1]} \left[(\mathbf{x} - \mathbf{x}_{\theta}(\mathbf{w}_t, t))^{\top} \text{diag} \left(\frac{d}{dt} \boldsymbol{\nu}(t) \right) (\mathbf{x} - \mathbf{x}_{\theta}(\mathbf{w}_t, t)) \right] \tag{A.23}$$

A.3.4 Vectorized Representation of the diffusion loss

Let $\boldsymbol{\nu}(t)$ be the vectorized representation of the diagonal entries of the matrix $\boldsymbol{\nu}(t)$. We can rewrite the integral in [A.22](#) in the following vectorized form where $\odot$

denotes element wise multiplication and $\langle, \rangle$ denotes dot product between 2 vectors.

$$\begin{aligned}
\mathcal{L}_{\text{diffusion}} &= -\frac{1}{2} \int_0^1 (\mathbf{x} - \mathbf{x}_\theta(\mathbf{w}_t, t))^\top \text{diag} \left(\frac{d}{dt} \boldsymbol{\nu}(t) \right) (\mathbf{x} - \mathbf{x}_\theta(\mathbf{w}_t, t)) dt \\
&= -\frac{1}{2} \int_0^1 \left\langle (\mathbf{x} - \mathbf{x}_\theta(\mathbf{w}_t, t)) \odot (\mathbf{x} - \mathbf{x}_\theta(\mathbf{w}_t, t)), \frac{d}{dt} \boldsymbol{\nu}(t) \right\rangle dt
\end{aligned}$$

Using change of variables as mentioned in Sec. 3.2.2 we have

$$= -\frac{1}{2} \int_0^1 \left\langle (\mathbf{x} - \tilde{\mathbf{x}}_\theta(\mathbf{w}_{\boldsymbol{\nu}(t)}, \boldsymbol{\nu}(t))) \odot (\mathbf{x} - \tilde{\mathbf{x}}_\theta(\mathbf{w}_{\boldsymbol{\nu}(t)}, \boldsymbol{\nu}(t))), \frac{d}{dt} \boldsymbol{\nu}(t) \right\rangle dt$$

Let $\mathbf{f}_\theta(\mathbf{x}_0, \boldsymbol{\nu}(t)) = (\mathbf{x} - \tilde{\mathbf{x}}_\theta(\mathbf{w}_{\boldsymbol{\nu}(t)}, \boldsymbol{\nu}(t))) \odot (\mathbf{x} - \tilde{\mathbf{x}}_\theta(\mathbf{w}_{\boldsymbol{\nu}(t)}, \boldsymbol{\nu}(t)))$

$$= \int_0^1 \left\langle \mathbf{f}_\theta(\mathbf{x}_0, \boldsymbol{\nu}(t)), \frac{d}{dt} \boldsymbol{\nu}(t) \right\rangle dt \quad (\text{A.24})$$

Thus $\mathcal{L}_{\text{diffusion}}$ can be interpreted as the amount of work done along the trajectory $\boldsymbol{\nu}(0) \rightarrow \boldsymbol{\nu}(1)$ in the presence of a vector field $\mathbf{f}_\theta(\mathbf{x}, \boldsymbol{\nu}(\mathbf{z}, t))$. From the perspective of thermodynamics, this is precisely equal to the amount of heat lost into the environment during the process of transition between 2 equilibria via the noise schedule specified by $\boldsymbol{\nu}(t)$.

A.3.5 Log likelihood and Noise Schedules: A Thermodynamics perspective

A diffusion model characterizes a quasi-static process that occurs between two equilibrium distributions: $q(\mathbf{x}) \rightarrow q(\mathbf{w}_1)$, via a stochastic trajectory [Sohl-Dickstein et al., 2015]. According to Spinney and Ford [2012], it is demonstrated that the diffusion schedule or the noising process plays a pivotal role in determining the "measure of irreversibility" for this stochastic trajectory which is expressed as $\log \frac{P_F(\mathbf{w}_{0:1})}{P_B(\mathbf{w}_{1:0})}$. $P_F(\mathbf{w}_{0:1})$ represents the probability of observing the forward path $\mathbf{w}_{0:1}$

and $P_B(\mathbf{w}_{1:0})$ represents the probability of observing the reverse path $\mathbf{w}_{1:0}$. It's worth noting that $\log \frac{P_F(\mathbf{w}_{0:1})}{P_B(\mathbf{w}_{1:0})}$ corresponds precisely to the ELBO Eq. 2.9 that we optimize when training a diffusion model. Consequently, thermodynamics asserts that the noise schedule indeed has an impact on the log-likelihood of the diffusion model which contradicts Kingma et al. [2021].

A.4 Multivariate Noise Schedule Conditioned on Context

Let's say we have a context variable $\mathbf{c} \in \mathbb{R}^m$ that captures high level information about $\mathbf{x}$. $\boldsymbol{\alpha}_t(\mathbf{c}), \boldsymbol{\sigma}_t(\mathbf{c}) \in \mathbb{R}_+^d$ are vectors. The timesteps s, t satisfy $0 \leq s < t \leq 1$. Furthermore, we use the following notations:

$$\boldsymbol{\alpha}_{t|s}(\mathbf{c}) = \frac{\boldsymbol{\alpha}_t(\mathbf{c})}{\boldsymbol{\alpha}_s(\mathbf{c})} \quad (\text{A.25})$$

$$\boldsymbol{\sigma}_{t|s}^2(\mathbf{c}) = \boldsymbol{\sigma}_t^2(\mathbf{c}) - \frac{\boldsymbol{\alpha}_{t|s}^2(\mathbf{c})}{\boldsymbol{\sigma}_s^2(\mathbf{c})} \quad (\text{A.26})$$

The forward process for such a method is given as:

$$q_\phi(\mathbf{w}_t | \mathbf{w}_s, \mathbf{c}) = \mathcal{N}(\boldsymbol{\alpha}_{t|s}(\mathbf{c})\mathbf{w}_s, \boldsymbol{\sigma}_{t|s}^2(\mathbf{c})) \quad (\text{A.27})$$

The distribution of $\mathbf{w}_t$ given $\mathbf{w}_s$ is given by (the derivation is similar to Hooeboom and Salimans [2022]):

$$\begin{aligned} q_\phi(\mathbf{w}_s | \mathbf{w}_t, \mathbf{x}, \mathbf{c}) \\ = \mathcal{N}\left(\boldsymbol{\mu}_q = \frac{\boldsymbol{\alpha}_{t|s}(\mathbf{c})\boldsymbol{\sigma}_s^2(\mathbf{c})}{\boldsymbol{\sigma}_t^2(\mathbf{c})}\mathbf{w}_t + \frac{\boldsymbol{\sigma}_{t|s}^2(\mathbf{c})\boldsymbol{\alpha}_s(\mathbf{c})}{\boldsymbol{\sigma}_t^2(\mathbf{c})}\mathbf{x}, \boldsymbol{\Sigma}_q = \text{diag}\left(\frac{\boldsymbol{\sigma}_s^2(\mathbf{c})\boldsymbol{\sigma}_{t|s}^2(\mathbf{c})}{\boldsymbol{\sigma}_t^2(\mathbf{c})}\right)\right) \end{aligned} \quad (\text{A.28})$$

A.4.1 Context Available During Inference

Even though $\mathbf{c}$ represents the input $\mathbf{x}$, it could be available during inference. For example $\mathbf{c}$ could be class labels [Dhariwal and Nichol, 2021] or preexisting embeddings from an auto-encoder [Preechakul et al., 2022].

Reverse Process: Approximate

Let $\mathbf{x}_\theta(\mathbf{w}_t, \mathbf{c}, t)$ be an approximation for $\mathbf{x}$. Then we get the following reverse process (for brevity we write $\mathbf{x}_\theta(\mathbf{w}_t, \mathbf{c}, t)$ as $\mathbf{x}_\theta$):

$$p_\theta(\mathbf{w}_s | \mathbf{w}_t, \mathbf{c}) = \mathcal{N} \left(\boldsymbol{\mu}_p = \frac{\boldsymbol{\alpha}_{t|s}(\mathbf{c}) \boldsymbol{\sigma}_s^2(\mathbf{c})}{\boldsymbol{\sigma}_t^2(\mathbf{c})} \mathbf{w}_t + \frac{\boldsymbol{\sigma}_{t|s}^2(\mathbf{c}) \boldsymbol{\alpha}_s(\mathbf{c})}{\boldsymbol{\sigma}_t^2(\mathbf{c})} \mathbf{x}_\theta, \boldsymbol{\Sigma}_p = \text{diag} \left(\frac{\boldsymbol{\sigma}_s^2(\mathbf{c}) \boldsymbol{\sigma}_{t|s}^2(\mathbf{c})}{\boldsymbol{\sigma}_t^2(\mathbf{c})} \right) \right) \quad (\text{A.29})$$

Diffusion Loss

Similar to the derivation of multi-variate $\mathcal{L}_{\text{diffusion}}$ in Eq. A.21 we can derive $\mathcal{L}_{\text{diffusion}}$ for this case too:

$$\mathcal{L}_{\text{diffusion}} = -\frac{1}{2} \mathbb{E}_{\epsilon \sim \mathcal{N}(0, \mathbf{I}_n), t \sim U[0,1]} \left[(\mathbf{x} - \mathbf{x}_\theta(\mathbf{w}_t, \mathbf{c}, t))^\top \text{diag} \left(\frac{d}{dt} \boldsymbol{\nu}(t) \right) (\mathbf{x} - \mathbf{x}_\theta(\mathbf{w}_t, \mathbf{c}, t)) \right] \quad (\text{A.30})$$

Limitations of this method. This approach is very limited where the diffusion process is only conditioned on class labels. Using pre-existing embeddings like Diff-AE [Preechakul et al., 2022] is also not possible in general and is only limited to tasks such as attribute manipulation in datasets.

A.4.2 Context Unavailable During Inference

If the context $\mathbf{c}$, is an explicit function of the input $\mathbf{x}$ things become challenging because $\mathbf{x}$ isn't available during the inference stage. For this reason, Eq. A.28 can't be used to parameterize $\boldsymbol{\mu}_p, \boldsymbol{\Sigma}_p$ in $p_\theta(\mathbf{w}_s|\mathbf{w}_t)$. Let $p_\theta(\mathbf{w}_s|\mathbf{w}_t) = \mathcal{N}(\boldsymbol{\mu}_p(\mathbf{w}_t, t), \boldsymbol{\Sigma}_p(\mathbf{w}_t, t))$ where $\boldsymbol{\mu}_p, \boldsymbol{\Sigma}_p$ are parameterized directly by a neural network. Using Eq. 3.3 we get the following diffusion loss:

$$\begin{aligned} \mathcal{L}_{\text{diffusion}} &= T \mathbb{E}_{i \sim U[0, T]} D_{\text{KL}} \left(q(\mathbf{w}_{s(i)}|\mathbf{w}_{t(i)}, \mathbf{x}) || p_\theta(\mathbf{w}_{s(i)}|\mathbf{w}_{t(i)}) \right) \\ &= \mathbb{E}_{q_\phi} \left(\underbrace{\frac{T}{2} (\boldsymbol{\mu}_q - \boldsymbol{\mu}_p)^\top \boldsymbol{\Sigma}_\theta^{-1} (\boldsymbol{\mu}_q - \boldsymbol{\mu}_p)}_{\text{term 1}} + \underbrace{\frac{T}{2} \left(\text{tr}(\boldsymbol{\Sigma}_q \boldsymbol{\Sigma}_p^{-1} - \mathbf{I}_n) - \log \frac{|\boldsymbol{\Sigma}_q|}{|\boldsymbol{\Sigma}_p|} \right)}_{\text{term 2}} \right) \end{aligned} \quad (\text{A.31})$$

Reverse Process: Approximate

Due to the challenges associated with parameterizing $\boldsymbol{\mu}_p, \boldsymbol{\Sigma}_p$ directly using a neural network we parameterize $\mathbf{c}$ using a neural network that approximates $\mathbf{c}$ in the reverse process. Let $\mathbf{x}_\theta(\mathbf{w}_t, t)$ be an approximation for $\mathbf{x}$. Then we get the following reverse Rrocess (for brevity we write $\mathbf{x}_\theta(\mathbf{w}_t, t)$ as $\mathbf{x}_\theta$, and $\mathbf{c}_\theta$ denotes an approximation to $\mathbf{c}$ in the reverse process.):

$$\begin{aligned} p_\theta(\mathbf{w}_s|\mathbf{w}_t) &= \mathcal{N} \left(\boldsymbol{\mu}_p = \frac{\boldsymbol{\alpha}_{t|s}(\mathbf{c}_\theta) \boldsymbol{\sigma}_s^2(\mathbf{c}_\theta)}{\boldsymbol{\sigma}_t^2(\mathbf{c}_\theta)} \mathbf{w}_t + \frac{\boldsymbol{\sigma}_{t|s}^2(\mathbf{c}_\theta) \boldsymbol{\alpha}_s(\mathbf{c}_\theta)}{\boldsymbol{\sigma}_t^2(\mathbf{c}_\theta)} \mathbf{x}_\theta, \boldsymbol{\Sigma}_p = \text{diag} \left(\frac{\boldsymbol{\sigma}_s^2(\mathbf{c}_\theta) \boldsymbol{\sigma}_{t|s}^2(\mathbf{c}_\theta)}{\boldsymbol{\sigma}_t^2(\mathbf{c}_\theta)} \right) \right) \end{aligned} \quad (\text{A.32})$$

Diffusion Loss

Consider the limiting case where $T \rightarrow \infty$. Let's analyze the 2 terms in Eq. A.31 separately.

Term 1. Using Eq. 3.3 and Eq. 3.5, term 1 in Eq. A.31 simplifies in the following manner:

$$\begin{aligned} & \lim_{T \rightarrow \infty} \frac{T}{2} (\boldsymbol{\mu}_q - \boldsymbol{\mu}_p)^\top \boldsymbol{\Sigma}_\theta^{-1} (\boldsymbol{\mu}_q - \boldsymbol{\mu}_p) \\ &= \lim_{T \rightarrow \infty} \frac{T}{2} \sum_{i=1}^d \frac{((\boldsymbol{\mu}_q)_i - (\boldsymbol{\mu}_p)_i)^2}{(\boldsymbol{\Sigma}_\theta)_i} \end{aligned} \quad (\text{A.33})$$

Substituting $1 / T$ as δ

$$\begin{aligned} &= \lim_{\delta \rightarrow 0^+} \sum_{i=1}^d \frac{1}{\delta \sigma_i^2(\mathbf{x}_\theta, t - \delta) \left(1 - \frac{\nu_i(\mathbf{x}_\theta, t)}{\nu_i(\mathbf{x}_\theta, t - \delta)}\right)} \times \\ & \quad \left[\frac{\alpha_i(\mathbf{x}, t - \delta)}{\alpha_i(\mathbf{x}, t)} \frac{\nu_i(\mathbf{x}, t)}{\nu_i(\mathbf{x}, t - \delta)} \mathbf{z}_t + \alpha_i(\mathbf{x}, t - \delta) \left(1 - \frac{\nu_i(\mathbf{x}, t)}{\nu_i(\mathbf{x}, t - \delta)}\right) \mathbf{w}_i \right. \\ & \quad \left. - \frac{\alpha_i(\mathbf{x}_\theta, t - \delta)}{\alpha_i(\mathbf{x}_\theta, t)} \frac{\nu_i(\mathbf{x}_\theta, t)}{\nu_i(\mathbf{x}_\theta, t - \delta)} \mathbf{z}_t + \alpha_i(\mathbf{x}_\theta, t - \delta) \left(1 - \frac{\nu_i(\mathbf{x}_\theta, t)}{\nu_i(\mathbf{x}_\theta, t - \delta)}\right) (\mathbf{x}_\theta)_i \right]^2 \end{aligned} \quad (\text{A.34})$$

Let's consider the scalar case for simplicity. Substituting $\delta = 1/T$, we get:

$$\begin{aligned} & \lim_{\delta \rightarrow 0} \frac{1}{2\delta \sigma^2(\mathbf{x}_\theta, t - \delta) \left(1 - \frac{\nu(\mathbf{x}_\theta, t)}{\nu(\mathbf{x}_\theta, t - \delta)}\right)} \times \\ & \quad \left[\frac{\alpha(\mathbf{x}, t - \delta)}{\alpha(\mathbf{x}, t)} \frac{\nu(\mathbf{x}, t)}{\nu(\mathbf{x}, t - \delta)} \mathbf{z}_t + \alpha(\mathbf{x}, t - \delta) \left(1 - \frac{\nu(\mathbf{x}, t)}{\nu(\mathbf{x}, t - \delta)}\right) \mathbf{x} \right. \\ & \quad \left. - \frac{\alpha(\mathbf{x}_\theta, t - \delta)}{\alpha(\mathbf{x}_\theta, t)} \frac{\nu(\mathbf{x}_\theta, t)}{\nu(\mathbf{x}_\theta, t - \delta)} \mathbf{z}_t + \alpha(\mathbf{x}_\theta, t - \delta) \left(1 - \frac{\nu(\mathbf{x}_\theta, t)}{\nu(\mathbf{x}_\theta, t - \delta)}\right) \mathbf{x}_\theta \right]^2 \end{aligned} \quad (\text{A.35})$$

Notice that this equation is indeterminate for when we substitute $\delta = 0$. For simplicity, we rewrite Eq. A.35 as a product of 2 expressions below and apply

L'Hospital rule on each separately:

$$\begin{aligned} \text{expression 1: } & \left[\lim_{\delta \rightarrow 0} \frac{1}{\delta} \times \left[\frac{\alpha(\mathbf{x}, t - \delta)}{\alpha(\mathbf{x}, t)} \frac{\nu(\mathbf{x}, t)}{\nu(\mathbf{x}, t - \delta)} \mathbf{z}_t + \alpha(\mathbf{x}, t - \delta) \left(1 - \frac{\nu(\mathbf{x}, t)}{\nu(\mathbf{x}, t - \delta)} \right) \mathbf{x} \right. \right. \\ & \quad \left. \left. - \frac{\alpha(\mathbf{x}_\theta, t - \delta)}{\alpha(\mathbf{x}_\theta, t)} \frac{\nu(\mathbf{x}_\theta, t)}{\nu(\mathbf{x}_\theta, t - \delta)} \mathbf{z}_t + \alpha(\mathbf{x}_\theta, t - \delta) \left(1 - \frac{\nu(\mathbf{x}_\theta, t)}{\nu(\mathbf{x}_\theta, t - \delta)} \right) \mathbf{x}_\theta \right] \right]^2 \end{aligned} \quad (\text{A.36})$$

$$\text{expression 2: } \lim_{\delta \rightarrow 0} \frac{\delta}{2\sigma^2(\mathbf{x}_\theta, t - \delta) \left(1 - \frac{\nu(\mathbf{x}_\theta, t)}{\nu(\mathbf{x}_\theta, t - \delta)} \right)} = -\frac{\nu(\mathbf{x}_\theta, t)}{2\sigma^2(\mathbf{x}_\theta, t) \nu'(\mathbf{x}_\theta, t)} \quad (\text{A.37})$$

Applying L'Hospital rule in expression 1 we get,

$$\begin{aligned} \frac{d}{d\delta} \left(\frac{\alpha(\mathbf{x}, t - \delta)}{\alpha(\mathbf{x}, t)} \frac{\nu(\mathbf{x}, t)}{\nu(\mathbf{x}, t - \delta)} \right) \Big|_{\delta=0} &= \frac{\nu(\mathbf{x}, t) - \nu(\mathbf{x}, t) \alpha'(\mathbf{x}, t) + \alpha(\mathbf{x}, t) \nu'(\mathbf{x}, t)}{\alpha(\mathbf{x}, t) \nu^2(\mathbf{x}, t)} \\ &= \frac{-\alpha'(\mathbf{x}, t)}{\alpha(\mathbf{x}, t)} + \frac{\nu'(\mathbf{x}, t)}{\nu(\mathbf{x}, t)} \end{aligned} \quad (\text{A.38})$$

$$(\text{A.39})$$

And,

$$\frac{d}{d\delta} \alpha(\mathbf{x}, t - \delta) \left(1 - \frac{\nu(\mathbf{x}, t)}{\nu(\mathbf{x}, t - \delta)} \right) \Big|_{\delta=0} = -\alpha(\mathbf{x}, t) \frac{\nu'(\mathbf{x}, t)}{\nu(\mathbf{x}, t)} \quad (\text{A.40})$$

Thus, term 1 simplifies to:

$$\begin{aligned} & \sum_{i=1}^d \left[\left(\frac{-\alpha_i'(\mathbf{x}, t)}{\alpha_i(\mathbf{x}, t)} + \frac{\nu_i'(\mathbf{x}, t)}{\nu_i(\mathbf{x}, t)} + \frac{\alpha_i'(\mathbf{x}_\theta, t)}{\alpha_i(\mathbf{x}_\theta, t)} - \frac{\nu_i'(\mathbf{x}_\theta, t)}{\nu_i(\mathbf{x}_\theta, t)} \right) \mathbf{z}_t \right. \\ & \quad \left. - \alpha_i(\mathbf{x}, t) \frac{\nu_i'(\mathbf{x}, t)}{\nu_i(\mathbf{x}, t)} \mathbf{x} + \alpha_i(\mathbf{x}_\theta, t) \frac{\nu_i'(\mathbf{x}_\theta, t)}{\nu_i(\mathbf{x}_\theta, t)} \mathbf{x}_\theta \right]^2 \times \frac{\nu_i(\mathbf{x}, t)}{2\sigma_i^2(\mathbf{x}_\theta, t) \nu_i'(\mathbf{x}, t)} \\ &= \Lambda^\top \text{diag} \left(\frac{1}{2\sigma^2(\mathbf{x}_\theta, t)} \odot \frac{\nu(\mathbf{x}, t)}{\nu'(\mathbf{x}, t)} \right) \Lambda \\ & \text{where } \Lambda = \left[\left(\frac{-\alpha'(\mathbf{x}, t)}{\alpha(\mathbf{x}, t)} + \frac{\nu'(\mathbf{x}, t)}{\nu(\mathbf{x}, t)} + \frac{\alpha'(\mathbf{x}_\theta, t)}{\alpha(\mathbf{x}_\theta, t)} - \frac{\nu'(\mathbf{x}_\theta, t)}{\nu(\mathbf{x}_\theta, t)} \right) \mathbf{z}_t - \alpha(\mathbf{x}, t) \frac{\nu'(\mathbf{x}, t)}{\nu(\mathbf{x}, t)} \mathbf{x} + \alpha(\mathbf{x}_\theta, t) \frac{\nu'(\mathbf{x}_\theta, t)}{\nu(\mathbf{x}_\theta, t)} \mathbf{x}_\theta \right] \end{aligned} \quad (\text{A.41})$$

Term 2. For the second term we have the following:

$$\begin{aligned}
& \lim_{T \rightarrow \infty} \frac{T}{2} \left(\text{tr} (\mathbf{\Sigma}_q \mathbf{\Sigma}_p^{-1} - \mathbf{I}_n) - \log \frac{|\mathbf{\Sigma}_q|}{|\mathbf{\Sigma}_p|} \right) \\
&= \lim_{T \rightarrow \infty} \frac{T}{2} \left[\text{tr} \left(\text{diag} \left(\boldsymbol{\sigma}^2(\mathbf{c}, s) \left(1 - \frac{\nu(\mathbf{c}, t)}{\nu(\mathbf{c}, s)} \right) \right) / \text{diag} \left(\boldsymbol{\sigma}^2(\mathbf{c}_\theta, s) \left(1 - \frac{\nu(\mathbf{c}_\theta, t)}{\nu(\mathbf{c}_\theta, s)} \right) \right) - \mathbf{I}_n \right) \right. \\
&\quad \left. - \log \frac{\left| \text{diag} \left(\boldsymbol{\sigma}^2(\mathbf{c}, s) \left(1 - \frac{\nu(\mathbf{c}, t)}{\nu(\mathbf{c}, s)} \right) \right) \right|}{\left| \text{diag} \left(\boldsymbol{\sigma}^2(\mathbf{c}_\theta, s) \left(1 - \frac{\nu(\mathbf{c}_\theta, t)}{\nu(\mathbf{c}_\theta, s)} \right) \right) \right|} \right] \\
&= \lim_{T \rightarrow \infty} \frac{T}{2} \sum_{i=1}^d \left(\frac{\boldsymbol{\sigma}_i^2(\mathbf{c}, s) \left(1 - \frac{\nu_i(\mathbf{c}, t)}{\nu_i(\mathbf{c}, s)} \right)}{\boldsymbol{\sigma}_i^2(\mathbf{c}_\theta, s) \left(1 - \frac{\nu_i(\mathbf{c}_\theta, t)}{\nu_i(\mathbf{c}_\theta, s)} \right)} - 1 - \log \frac{\boldsymbol{\sigma}_i^2(\mathbf{c}, s) \left(1 - \frac{\nu_i(\mathbf{c}, t)}{\nu_i(\mathbf{c}, s)} \right)}{\boldsymbol{\sigma}_i^2(\mathbf{c}_\theta, s) \left(1 - \frac{\nu_i(\mathbf{c}_\theta, t)}{\nu_i(\mathbf{c}_\theta, s)} \right)} \right) \tag{A.42}
\end{aligned}$$

(A.43)

Let $p_i = \frac{\boldsymbol{\sigma}_i^2(\mathbf{c}, s) \left(1 - \frac{\nu_i(\mathbf{c}, t)}{\nu_i(\mathbf{c}, s)} \right)}{\boldsymbol{\sigma}_i^2(\mathbf{c}_\theta, s) \left(1 - \frac{\nu_i(\mathbf{c}_\theta, t)}{\nu_i(\mathbf{c}_\theta, s)} \right)}$. The sequence $\lim_{T \rightarrow \infty} \frac{T}{2} \sum_{i=1}^d (p_i - 1 - \log p_i)$ converges iff $\lim_{T \rightarrow \infty} \sum_{i=1}^d (p_i - 1 - \log p_i) = 0$. Notice that the function $f(x) = x - 1 - \log x \geq 0 \quad \forall x \in \mathbb{R}$ and the equality holds for $x = 1$. Thus, the condition $\lim_{T \rightarrow \infty} \frac{T}{2} \sum_{i=1}^d (p_i - 1 - \log p_i)$ holds iff $\lim_{T \rightarrow \infty} p_i = 0 \quad \forall i \in \{1, \dots, d\}$. Thus,

$$\begin{aligned} \lim_{T \rightarrow \infty} p_i &= 1 \\ \implies \lim_{T \rightarrow \infty} \left(\frac{\sigma_i^2(\mathbf{c}, s) \left(1 - \frac{\nu_i(\mathbf{c}, t)}{\nu_i(\mathbf{c}, s)}\right)}{\sigma_i^2(\mathbf{c}_\theta, s) \left(1 - \frac{\nu_i(\mathbf{c}_\theta, t)}{\nu_i(\mathbf{c}_\theta, s)}\right)} \right) &= 1 \end{aligned}$$

Substituting $1/T$ as δ ,

$$\begin{aligned} \implies \lim_{\delta \rightarrow 0^+} \left(\frac{\sigma_i^2(\mathbf{c}, t - \delta) \left(1 - \frac{\nu_i(\mathbf{c}, t)}{\nu_i(\mathbf{c}, t - \delta)}\right)}{\sigma_i^2(\mathbf{c}_\theta, t - \delta) \left(1 - \frac{\nu_i(\mathbf{c}_\theta, t)}{\nu_i(\mathbf{c}_\theta, t - \delta)}\right)} \right) &= 1 \\ \implies \frac{\sigma_i^2(\mathbf{c}, t)}{\sigma_i^2(\mathbf{c}_\theta, t)} \lim_{\delta \rightarrow 0^+} \left(\frac{1 - \frac{\nu_i(\mathbf{c}, t)}{\nu_i(\mathbf{c}, t - \delta)}}{1 - \frac{\nu_i(\mathbf{c}_\theta, t)}{\nu_i(\mathbf{c}_\theta, t - \delta)}} \right) &= 1 \end{aligned}$$

Applying L'Hospital rule,

$$\begin{aligned} \implies \frac{\sigma_i^2(\mathbf{c}, t)}{\sigma_i^2(\mathbf{c}_\theta, t)} \left(\frac{\frac{-\nu_i'(\mathbf{c}, t)}{\nu_i(\mathbf{c}, t)}}{\frac{-\nu_i'(\mathbf{c}_\theta, t)}{\nu_i(\mathbf{c}_\theta, t)}} \right) &= 1 \\ \implies \frac{\sigma_i^2(\mathbf{c}, t)}{\sigma_i^2(\mathbf{c}_\theta, t)} \left(\frac{\nu_i'(\mathbf{c}, t) \nu_i(\mathbf{c}_\theta, t)}{\nu_i(\mathbf{c}, t) \nu_i'(\mathbf{c}_\theta, t)} \right) &= 1 \end{aligned} \tag{A.44}$$

In the vector form the above equation can be written as,

$$\frac{\sigma_t^2(\mathbf{c}) \boldsymbol{\nu}_t(\mathbf{c}_\theta) \nabla_t \boldsymbol{\nu}(\mathbf{c}, t)}{\sigma_t^2(\mathbf{c}_\theta) \boldsymbol{\nu}_t(\mathbf{c}) \nabla_t \boldsymbol{\nu}(\mathbf{c}_\theta, t)} = \mathbf{1}_d \tag{A.45}$$

Eq. A.45 holds if:

- $\mathbf{x}_\theta = \mathbf{x}$ i.e. the unet can perfectly map $\mathbf{w}_t$ to $\mathbf{x} \forall t \in [0, 1]$ which is unrealistic.
- Clever parameterizations for σ, α, ν that ensure Eq. A.45 holds.

Because of aforementioned challenges we evaluate this method with finite $T = 1000$.

We demonstrate the performance of the model empirically in Fig. A.1.

Recovering VDM

If we substitute $\boldsymbol{\nu}_t(\mathbf{c}), \boldsymbol{\nu}_t(\mathbf{c}_\theta)$ with $\boldsymbol{\nu}(t)$ (since the SNR isn't conditioned on the context $\mathbf{c}$), $\boldsymbol{\sigma}_t(\mathbf{c}_\theta), \boldsymbol{\sigma}_t(\mathbf{c})$ with σ_t and $\boldsymbol{\alpha}_t(\mathbf{c}_\theta), \boldsymbol{\alpha}_t(\mathbf{c})$ with $\boldsymbol{\alpha}_t$, Eq. A.33 reduces to the intermediate loss in VDM i.e. $\frac{1}{2}(\mathbf{x}_\theta - \mathbf{x})^\top \nabla_t \boldsymbol{\nu}(t) (\mathbf{x}_\theta - \mathbf{x})$ and Eq. A.42 reduces to 0.

A.4.3 Experimental results

In Fig. A.1 we demonstrate that the multivariate diffusion processes where $\mathbf{c}$ = “class labels” or $\mathbf{c} = \mathbf{x}$ perform worse than VDM. Since a continuous time formulation i.e. $T \rightarrow \infty$ for the case when $\mathbf{c} = \mathbf{x}$ isn't possible (unlike MuLAN or VDM) we evaluate these models in the discrete time setting where we use $T = 1000$. Furthermore we also ablate $T = 10k, 100k$ for $\mathbf{c} = \mathbf{x}$ to show that the VLB degrades with increasing T whereas for VDM and MuLAN it improves for increasing T; see Kingma et al. [2021]. This empirical observation is consistent with our mathematical insights earlier. As these models consistently exhibit inferior performance w.r.t VDM, in line with our initial conjectures, we refrain from training them beyond 300k iterations due to the substantial computational cost involved.

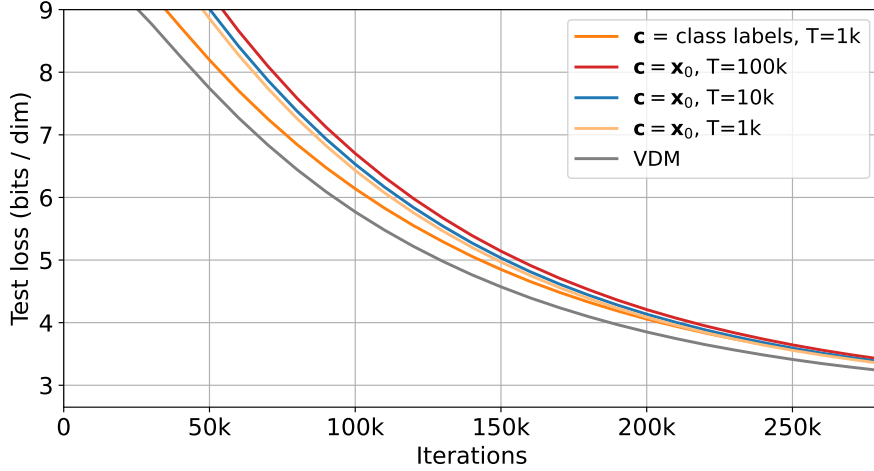


Figure A.1: For $\mathbf{c}$ = “class labels” or $\mathbf{c} = \mathbf{x}$ the likelihood estimates are worse than VDM. For $\mathbf{c} = \mathbf{x}$, we see that the VLB degrades with increasing T , but for VDM and MuLAN, it improves with increasing T . This empirical observation is consistent with our mathematical insights earlier. As these models consistently exhibit inferior performance w.r.t VDM, in line with our initial conjectures, we refrain from training them beyond 300k iterations due to the substantial computational cost involved.

A.5 MuLAN: MULTivariate Latent Auxiliary variable Noise Schedule

A.5.1 Parameterization in the reverse process

Noise parameterization

Since the forward pass is given by $\mathbf{w}_t = \alpha_t(\mathbf{z})\mathbf{x} + \sigma_t(\mathbf{z})\epsilon_t$, we can write the noise ϵ_t in terms of $\mathbf{x}, \mathbf{w}_t$ in the following manner:

$$\epsilon_t = \frac{\mathbf{w}_t - \alpha_t(\mathbf{z})\mathbf{x}}{\sigma_t(\mathbf{z})} \quad (\text{A.46})$$

Following Dhariwal and Nichol [2021], Kingma et al. [2021], instead of parameterizing $\mathbf{x}_\theta(\mathbf{w}_t, \mathbf{z}, t)$ using a neural network, we use Eq. A.46 to parameterize the

Table A.4: Likelihood in bits per dimension (BPD) (mean and 95% confidence interval), on the test set of CIFAR-10 computed using VLB estimate.

parameterization	Num training steps	CIFAR-10 ($\downarrow$)
Noise parameterization	10M	2.60 ± 10^{-3}
v-parameterization	8M	2.59 ± 10^{-3}

denoising model in terms of a noise prediction model $\epsilon_\theta(\mathbf{w}_t, \mathbf{z}, t)$,

$$\epsilon_\theta(\mathbf{w}_t, \mathbf{z}, t) = \frac{\mathbf{w}_t - \alpha_t(\mathbf{z})\mathbf{x}_\theta(\mathbf{w}_t, \mathbf{z}, t)}{\sigma_t(\mathbf{z})} \quad (\text{A.47})$$

Velocity parameterization

Following [Salimans and Ho \[2022a\]](#), [Zheng et al. \[2023a\]](#), we explore another parameterization of the denoising network which is given by

$$\mathbf{v}_\theta(\mathbf{w}_t, \mathbf{z}, t) = \frac{\alpha_t(\mathbf{z})\mathbf{w}_t - \mathbf{x}_\theta(\mathbf{w}_t, \mathbf{z}, t)}{\sigma_t(\mathbf{z})} \quad (\text{A.48})$$

In practice, v-parameterization leads to a better performance than noise parameterization; as illustrated in [Table A.4](#).

A.5.2 Polynomial Noise Schedule

Let $f(x; \psi)$ be a scalar-valued polynomial of degree n with coefficients $\psi \in \mathbb{R}^{n+1}$ expressed as:

$$f(x; \psi) = \psi_n x^n + \psi_{n-1} x^{n-1} + \cdots + \psi_1 x + \psi_0,$$

and denote its derivative with respect to x as $\frac{d}{dx}f(x; \psi)$, represented by $f'(x; \psi)$.

Now we'd like to find least n such that $f(x; \psi)$ satisfies the following properties:

1. $f(x; \psi)$ is monotonically increasing, i.e. $f'(x; \psi) \geq 0 \forall x \in \mathbb{R}, \psi \in \mathbb{R}^{n+1}$.

$$2. f'(x_1; \psi) = 0, f'(x_2; \psi) = 0 \exists x_1, x_2 \in \mathbb{C}, x_1 \neq x_2, \forall \psi \in \mathbb{R}^{n+1}.$$

For the **first** condition to hold, we can design $f'(x; \psi)$ such that it's a perfect square with real / imaginary roots. That way $f'(x; \psi) \geq 0 \forall x \in \mathbb{R}, \psi \in \mathbb{R}^{n+1}$. This means that $f'(x; \psi)$ is an even degree polynomial, i.e. the degree of $f'(x; \psi)$ can take the following values: $2, 4, \dots$. Also, note that at least half of the roots of $f'(x; \psi)$ are repeated since $f'(x; \psi)$ can be expressed as a perfect square, i.e., if $f'(x; \psi)$ has a degree 2 then it has exactly 1 unique root (real / imaginary), if $f'(x; \psi)$ has a degree 4 then it has at most 2 unique roots (real / imaginary), and so on.

For the **second** condition to hold, $f'(x; \psi)$ needs to have at least 2 unique roots $\exists \psi \in \mathbb{R}^{n+1}$. For this reason $f'(x; \psi)$ is a polynomial of degree 4. Thus, $f'(x; \psi)$ can be written as $f'(x; \psi) = (\psi_3 x^2 + \psi_2 x + \psi_1)^2$. This ensures that $\exists \psi \in \mathbb{R}^5$ s.t. $f'(x; \psi) = 0$ twice in $x \in \mathbb{R}$, and $f'(x; \psi) \geq 0 \forall \psi \in \mathbb{R}^5$.

Thus, $f(x; \psi)$ takes the following functional form:

$$\begin{aligned} f(x; \psi) &= \int (\psi_3 x^2 + \psi_2 x + \psi_1)^2 dx \\ &= \frac{\psi_3^2}{5} x^5 + \frac{\psi_3 \psi_2}{2} x^4 + \frac{\psi_2^2 + 2\psi_3 \psi_1}{3} x^3 + \psi_2 \psi_1 x^2 + \psi_1^2 x + \text{constant}. \end{aligned} \quad (\text{A.49})$$

For the above-mentioned reasons we express $\gamma_\phi(\mathbf{c}, t) : \mathbb{R}^m \times [0, 1] \rightarrow \mathbb{R}^d$ as a degree 5 polynomial in t . We define neural networks $\mathbf{a}_\phi(\mathbf{c}) : \mathbb{R}^m \rightarrow \mathbb{R}^d$, $\mathbf{b}_\phi(\mathbf{c}) : \mathbb{R}^m \rightarrow \mathbb{R}^d$, and $\mathbf{d}_\phi(\mathbf{c}) : \mathbb{R}^m \rightarrow \mathbb{R}^d$ with parameters ϕ . Let $f_\phi : \mathbb{R}^m \times [0, 1] \rightarrow \mathbb{R}^d$ be defined as:

$$f_\phi(\mathbf{c}, t) = \frac{\mathbf{a}_\phi^2(\mathbf{c})}{5} t^5 + \frac{\mathbf{a}_\phi(\mathbf{c}) \mathbf{b}_\phi(\mathbf{c})}{2} t^4 + \frac{\mathbf{b}_\phi^2(\mathbf{c}) + 2\mathbf{a}_\phi(\mathbf{c}) \mathbf{d}_\phi(\mathbf{c})}{3} t^3 + \mathbf{b}_\phi(\mathbf{c}) \mathbf{d}_\phi(\mathbf{c}) t^2 + \mathbf{d}_\phi^2(\mathbf{c}) t$$

where the multiplication and division operations are elementwise. The the noise schedule, $\gamma(\mathbf{c}, t)$, is given as follows:

$$\gamma_\phi(\mathbf{c}, t) = \gamma_{\min} + (\gamma_{\max} - \gamma_{\min}) \frac{f_\phi(\mathbf{c}, t)}{f_\phi(\mathbf{c}, t=1)} \quad (\text{A.50})$$

Notice that $\gamma_\phi(\mathbf{c}, t)$ has these interesting properties:

- Is increasing in $t \in [0, 1]$ which is crucial as mentioned in Sec. 3.2.5.
- $\gamma_\phi(\mathbf{c}, t)$ has end points at $t = 0$ and $t = 1$ which the user can specify via $\gamma_{\min}$ and $\gamma_{\max}$. Specifically, $\gamma_\phi(\mathbf{c}, t = 0) = \gamma_{\min} \mathbf{1}_d$ and $\gamma_\phi(\mathbf{c}, t = 1) = \gamma_{\max} \mathbf{1}_d$.
- Its time-derivative i.e. $\nabla_t \gamma_\phi(\mathbf{c}, t)$ **can** be zero twice in $t \in [0, 1]$. This isn't a necessary condition but it's nice to have a flexible noise schedule whose time-derivative can be 0 at the beginning and the end of the diffusion process.

A.5.3 Variational Lower Bound

In this section we derive the VLB. For ease of reading we use the notation $\mathbf{w}_t$ to denote $\mathbf{w}_{t(i)}$ and $\mathbf{w}_{t-1}$ to denote $\mathbf{w}_{t(i-1)} \equiv \mathbf{w}_{s(i)}$ in the following derivation.

$$\begin{aligned}
& -\log p_\theta(\mathbf{x}) \\
& \leq \mathbb{E}_{q_\phi} \left[-\log \frac{p_\theta(\mathbf{z}, \mathbf{w}_{0:T})}{q_\phi(\mathbf{z}, \mathbf{w}_{1:T}|\mathbf{x})} \right] \\
& = \mathbb{E}_{q_\phi} \left[-\log \frac{p_\theta(\mathbf{w}_{0:T-1}|\mathbf{z}, \mathbf{w}_T)}{q_\phi(\mathbf{z}, \mathbf{w}_{1:T}|\mathbf{x})} - \log p_\theta(\mathbf{w}_T) - \log p_\theta(\mathbf{z}) \right] \\
& = \mathbb{E}_{q_\phi} \left[-\log \frac{p_\theta(\mathbf{w}_{0:T-1}|\mathbf{z}, \mathbf{w}_T)}{q_\phi(\mathbf{w}_{1:T}|\mathbf{z}, \mathbf{x})} - \log \frac{1}{q_\phi(\mathbf{z}|\mathbf{x})} - \log p_\theta(\mathbf{w}_T) - \log p_\theta(\mathbf{z}) \right] \\
& = \mathbb{E}_{q_\phi} \left[-\log \frac{p_\theta(\mathbf{w}_{0:T-1}|\mathbf{z}, \mathbf{w}_T)}{q_\phi(\mathbf{w}_{1:T}|\mathbf{z}, \mathbf{x})} - \log p_\theta(\mathbf{w}_T) - \log \frac{p_\theta(\mathbf{z})}{q_\phi(\mathbf{z}|\mathbf{x})} \right] \\
& = \mathbb{E}_{q_\phi} \left[-\sum_{t=1}^T \log \frac{p_\theta(\mathbf{w}_{t-1}|\mathbf{z}, \mathbf{w}_t)}{q_\phi(\mathbf{w}_t|\mathbf{w}_{t-1}, \mathbf{z}, \mathbf{x})} - \log p_\theta(\mathbf{w}_T) - \log \frac{p_\theta(\mathbf{z})}{q_\phi(\mathbf{z}|\mathbf{x})} \right] \\
& = \mathbb{E}_{q_\phi} \left[-\log \frac{p_\theta(\mathbf{w}_0|\mathbf{z}, \mathbf{w}_1)}{q_\phi(\mathbf{w}_1|\mathbf{w}_0, \mathbf{z})} - \sum_{t=2}^T \log \frac{p_\theta(\mathbf{w}_{t-1}|\mathbf{z}, \mathbf{w}_t)}{q_\phi(\mathbf{w}_t|\mathbf{w}_{t-1}, \mathbf{z}, \mathbf{x})} - \log p_\theta(\mathbf{w}_T) - \log \frac{p_\theta(\mathbf{z})}{q_\phi(\mathbf{z}|\mathbf{x})} \right] \\
& = \mathbb{E}_{q_\phi} \left[-\log \frac{p_\theta(\mathbf{w}_0|\mathbf{z}, \mathbf{w}_1)}{q_\phi(\mathbf{w}_1|\mathbf{w}_0, \mathbf{z})} - \sum_{t=2}^T \log \frac{p_\theta(\mathbf{w}_{t-1}|\mathbf{z}, \mathbf{w}_t) q_\phi(\mathbf{w}_{t-1}|\mathbf{z}, \mathbf{x})}{q_\phi(\mathbf{w}_{t-1}|\mathbf{w}_t, \mathbf{z}, \mathbf{x}) q_\phi(\mathbf{w}_t|\mathbf{z}, \mathbf{x})} - \log p_\theta(\mathbf{w}_T) - \log \frac{p_\theta(\mathbf{z})}{q_\phi(\mathbf{z}|\mathbf{x})} \right] \\
& = \mathbb{E}_{q_\phi} \left[-\log \frac{p_\theta(\mathbf{w}_0|\mathbf{z}, \mathbf{w}_1)}{q_\phi(\mathbf{w}_1|\mathbf{w}_0, \mathbf{z})} - \sum_{t=2}^T \log \frac{p_\theta(\mathbf{w}_{t-1}|\mathbf{z}, \mathbf{w}_t)}{q_\phi(\mathbf{w}_{t-1}|\mathbf{w}_t, \mathbf{z}, \mathbf{x})} - \sum_{t=2}^T \log \frac{q_\phi(\mathbf{w}_{t-1}|\mathbf{z}, \mathbf{x})}{q_\phi(\mathbf{w}_t|\mathbf{z}, \mathbf{x})} - \log p_\theta(\mathbf{w}_T) - \log \frac{p_\theta(\mathbf{z})}{q_\phi(\mathbf{z}|\mathbf{x})} \right] \\
& = \mathbb{E}_{q_\phi} \left[-\log \frac{p_\theta(\mathbf{w}_0|\mathbf{z}, \mathbf{w}_1)}{q_\phi(\mathbf{w}_1|\mathbf{w}_0, \mathbf{z})} - \sum_{t=2}^T \log \frac{p_\theta(\mathbf{w}_{t-1}|\mathbf{z}, \mathbf{w}_t)}{q_\phi(\mathbf{w}_{t-1}|\mathbf{w}_t, \mathbf{z}, \mathbf{x})} - \log \frac{q(\mathbf{w}_1|\mathbf{z}, \mathbf{x})}{q_\phi(\mathbf{w}_T|\mathbf{z}, \mathbf{x})} - \log p_\theta(\mathbf{w}_T) - \log \frac{p_\theta(\mathbf{z})}{q_\phi(\mathbf{z}|\mathbf{x})} \right] \\
& = \mathbb{E}_{q_\phi} \left[-\log p_\theta(\mathbf{w}_0|\mathbf{z}, \mathbf{w}_1) - \sum_{t=2}^T \log \frac{p_\theta(\mathbf{w}_{t-1}|\mathbf{z}, \mathbf{w}_t)}{q_\phi(\mathbf{w}_{t-1}|\mathbf{w}_t, \mathbf{z}, \mathbf{x})} - \log \frac{1}{q_\phi(\mathbf{w}_T|\mathbf{z}, \mathbf{x})} - \log p_\theta(\mathbf{w}_T) - \log \frac{p_\theta(\mathbf{z})}{q_\phi(\mathbf{z}|\mathbf{x})} \right] \\
& = \mathbb{E}_{q_\phi} \left[-\log p_\theta(\mathbf{w}_0|\mathbf{z}, \mathbf{w}_1) - \sum_{t=2}^T \log \frac{p_\theta(\mathbf{w}_{t-1}|\mathbf{z}, \mathbf{w}_t)}{q_\phi(\mathbf{w}_{t-1}|\mathbf{w}_t, \mathbf{z}, \mathbf{x})} - \log \frac{p_\theta(\mathbf{w}_T)}{q_\phi(\mathbf{w}_T|\mathbf{z}, \mathbf{x})} - \log \frac{p_\theta(\mathbf{z})}{q_\phi(\mathbf{z}|\mathbf{x})} \right] \\
& = \mathbb{E}_{q_\phi} \left[\underbrace{-\log p_\theta(\mathbf{w}_0|\mathbf{z}, \mathbf{w}_1)}_{\mathcal{L}_{\text{recons}}} + \underbrace{\sum_{t=2}^T \text{D}_{\text{KL}}[p_\theta(\mathbf{w}_{t-1}|\mathbf{z}, \mathbf{w}_t) \| q_\phi(\mathbf{w}_{t-1}|\mathbf{w}_t, \mathbf{z}, \mathbf{x})]}_{\mathcal{L}_{\text{diffusion}}} \right] \\
& \quad + \mathbb{E}_{q_\phi} \left[\underbrace{\text{D}_{\text{KL}}[p_\theta(\mathbf{w}_T) \| q_\phi(\mathbf{w}_T|\mathbf{z}, \mathbf{x})]}_{\mathcal{L}_{\text{prior}}} + \underbrace{\text{D}_{\text{KL}}[p_\theta(\mathbf{z}) \| q(\mathbf{z}|\mathbf{x})]}_{\mathcal{L}_{\text{latent}}} \right] \tag{A.51}
\end{aligned}$$

Switching back to the notation used throughout the paper, the VLB is given as:

$$\begin{aligned}
& -\log p_\theta(\mathbf{x}) \\
& \leq \mathbb{E}_{q_\phi} \left[\underbrace{-\log p_\theta(\mathbf{w}_0|\mathbf{z}, \mathbf{w}_1)}_{\mathcal{L}_{\text{recons}}} + \underbrace{\sum_{i=2}^T D_{\text{KL}}[p_\theta(\mathbf{w}_{s(i)}|\mathbf{z}, \mathbf{w}_{t(i)})||q_\phi(\mathbf{w}_{s(i)}|\mathbf{w}_{t(i)}, \mathbf{z}, \mathbf{x})]}_{\mathcal{L}_{\text{diffusion}}} \right] \\
& + \mathbb{E}_{q_\phi} \left[\underbrace{D_{\text{KL}}[p_\theta(\mathbf{w}_1)||q_\phi(\mathbf{w}_1|\mathbf{z}, \mathbf{x})]}_{\mathcal{L}_{\text{prior}}} + \underbrace{D_{\text{KL}}[p_\theta(\mathbf{z})||q_\phi(\mathbf{z}|\mathbf{x})]}_{\mathcal{L}_{\text{latent}}} \right] \tag{A.52}
\end{aligned}$$

A.5.4 Diffusion Loss

To derive the diffusion loss, $\mathcal{L}_{\text{diffusion}}$ in Eq. 3.8, we first derive an expression for $D_{\text{KL}}(q_\phi(\mathbf{w}_s|\mathbf{z}, \mathbf{w}_t, \mathbf{x})||p_\theta(\mathbf{w}_s|\mathbf{z}, \mathbf{w}_t))$ using Eq. 3.3 and Eq. 3.5 in the following manner (details in Suppl. A.5):

$$\begin{aligned}
& D_{\text{KL}}(q_\phi(\mathbf{w}_s|\mathbf{z}, \mathbf{w}_t, \mathbf{x})||p_\theta(\mathbf{w}_s|\mathbf{z}, \mathbf{w}_t)) \\
& = \frac{1}{2} \left((\boldsymbol{\mu}_{q_\phi} - \boldsymbol{\mu}_p)^\top \boldsymbol{\Sigma}_\theta^{-1} (\boldsymbol{\mu}_{q_\phi} - \boldsymbol{\mu}_p) + \text{tr} \left(\boldsymbol{\Sigma}_{q_\phi} \boldsymbol{\Sigma}_p^{-1} - \mathbf{I}_n \right) - \log \frac{|\boldsymbol{\Sigma}_{q_\phi}|}{|\boldsymbol{\Sigma}_p|} \right) \\
& = \frac{1}{2} \left((\mathbf{x} - \mathbf{x}_\theta)^\top \text{diag}(\boldsymbol{\nu}(\mathbf{z}, s) - \boldsymbol{\nu}(\mathbf{z}, t)) (\mathbf{x} - \mathbf{x}_\theta) \right) \tag{A.53}
\end{aligned}$$

Let $\lim_{T \rightarrow \infty} T(\boldsymbol{\nu}_s(\mathbf{z}) - \boldsymbol{\nu}_t(\mathbf{z})) = -\nabla_t \boldsymbol{\nu}(\mathbf{z}, t)$ be the partial derivative of the vector $\boldsymbol{\nu}(\mathbf{z}, t)$ w.r.t scalar t . Then we derive the diffusion loss, $\mathcal{L}_{\text{diffusion}}$, for the continuous case in the following manner (for brevity we use the notation s for $s(i) = (i-1)/T$

and t for $t(i) = i/T$:

$\mathcal{L}_{\text{diffusion}}$

$$= \lim_{T \rightarrow \infty} \frac{1}{2} \sum_{i=2}^T \mathbb{E}_{\epsilon \sim \mathcal{N}(0, \mathbf{I}_n)} D_{\text{KL}}(q(\mathbf{w}_s | \mathbf{w}_t, \mathbf{x}, \mathbf{z}) \| p_{\theta}(\mathbf{w}_s | \mathbf{w}_t, \mathbf{z}))$$

Using Eq. A.53 we get,

$$\begin{aligned} &= \lim_{T \rightarrow \infty} \frac{1}{2} \sum_{i=2}^T \mathbb{E}_{\epsilon \sim \mathcal{N}(0, \mathbf{I}_n)} (\mathbf{x} - \mathbf{x}_{\theta}(\mathbf{w}_t, t(i)))^{\top} \text{diag}(\boldsymbol{\nu}(s(i), \mathbf{z}) - \boldsymbol{\nu}(t(i), \mathbf{z})) (\mathbf{x} - \mathbf{x}_{\theta}(\mathbf{w}_t, t(i))) \\ &= \frac{1}{2} \mathbb{E}_{\epsilon \sim \mathcal{N}(0, \mathbf{I}_n)} \left[\lim_{T \rightarrow \infty} \sum_{i=2}^T T (\mathbf{x} - \mathbf{x}_{\theta}(\mathbf{w}_t, t(i)))^{\top} \text{diag}(\boldsymbol{\nu}(s(i), \mathbf{z}) - \boldsymbol{\nu}(t(i), \mathbf{z})) (\mathbf{x} - \mathbf{x}_{\theta}(\mathbf{w}_t, t(i))) \frac{1}{T} \right] \end{aligned}$$

Using the fact that $\lim_{T \rightarrow \infty} T(\boldsymbol{\nu}(s, \mathbf{z}) - \boldsymbol{\nu}(t, \mathbf{z})) = -\nabla_t \boldsymbol{\nu}(t, \mathbf{z})$ we get,

$$= -\frac{1}{2} \mathbb{E}_{t \sim \{0, \dots, 1\}} [(\mathbf{x} - \mathbf{x}_{\theta}(\mathbf{w}_t, t))^{\top} (\nabla_t \boldsymbol{\nu}_t(\mathbf{z})) (\mathbf{x} - \mathbf{x}_{\theta}(\mathbf{w}_t, t))]$$

Substituting $\mathbf{x} = \boldsymbol{\alpha}_t^{-1}(\mathbf{z})(\mathbf{w}_t - \boldsymbol{\sigma}_t(\mathbf{z})\boldsymbol{\epsilon}_t)$ from Eq. A.46 and

Substituting $\mathbf{x}_{\theta}(\mathbf{w}_t, \mathbf{z}, t) = \boldsymbol{\alpha}_t^{-1}(\mathbf{z})(\mathbf{w}_t - \boldsymbol{\sigma}_t(\mathbf{z})\boldsymbol{\epsilon}_{\theta}(\mathbf{w}_t, t))$ from Eq. A.47 we get,

$$= -\frac{1}{2} \mathbb{E}_{t \sim [0, 1]} \left[(\boldsymbol{\epsilon}_t - \boldsymbol{\epsilon}_{\theta}(\mathbf{w}_t, t))^{\top} \left(\frac{\boldsymbol{\sigma}_t^2(\mathbf{z})}{\boldsymbol{\alpha}_t^2(\mathbf{z})} \times \nabla_t \boldsymbol{\nu}_t(\mathbf{z}) \right) (\boldsymbol{\epsilon}_t - \boldsymbol{\epsilon}_{\theta}(\mathbf{w}_t, t)) \right]$$

Let $\boldsymbol{\nu}^{-1}(\mathbf{z}, t)$ denote the reciprocal of the values in the vector $\boldsymbol{\nu}(\mathbf{z}, t)$.

$$= -\frac{1}{2} \mathbb{E}_{t \sim [0, 1]} [(\boldsymbol{\epsilon}_t - \boldsymbol{\epsilon}_{\theta}(\mathbf{w}_t, t))^{\top} \text{diag}(\boldsymbol{\nu}^{-1}(t)(\mathbf{z}) \nabla_t \boldsymbol{\nu}_t(\mathbf{z})) (\boldsymbol{\epsilon}_t - \boldsymbol{\epsilon}_{\theta}(\mathbf{w}_t, t))]$$

Substituting $\boldsymbol{\nu}(\mathbf{z}, t) = \exp(-\boldsymbol{\gamma}(\mathbf{z}, t))$ from Sec. A.5.1

$$\begin{aligned} &= -\frac{1}{2} \mathbb{E}_{t \sim [0, 1]} [(\boldsymbol{\epsilon}_t - \boldsymbol{\epsilon}_{\theta}(\mathbf{w}_t, t))^{\top} \text{diag}(\exp(\boldsymbol{\gamma}(\mathbf{z}, t)) \nabla_t \exp(-\boldsymbol{\gamma}(\mathbf{z}, t))) (\boldsymbol{\epsilon}_t - \boldsymbol{\epsilon}_{\theta}(\mathbf{w}_t, t))] \\ &= \frac{1}{2} \mathbb{E}_{t \sim [0, 1]} [(\boldsymbol{\epsilon}_t - \boldsymbol{\epsilon}_{\theta}(\mathbf{w}_t, t))^{\top} \text{diag}(\exp(\boldsymbol{\gamma}(\mathbf{z}, t)) \exp(-\boldsymbol{\gamma}(\mathbf{z}, t)) \nabla_t \boldsymbol{\gamma}(\mathbf{z}, t)) (\boldsymbol{\epsilon}_t - \boldsymbol{\epsilon}_{\theta}(\mathbf{w}_t, t))] \\ &= \frac{1}{2} \mathbb{E}_{t \sim [0, 1]} [(\boldsymbol{\epsilon}_t - \boldsymbol{\epsilon}_{\theta}(\mathbf{w}_t, t))^{\top} \text{diag}(\nabla_t \boldsymbol{\gamma}(\mathbf{z}, t)) (\boldsymbol{\epsilon}_t - \boldsymbol{\epsilon}_{\theta}(\mathbf{w}_t, t))] \end{aligned} \tag{A.54}$$

A.5.5 Recovering VDM from the Vectorized Representation of the diffusion loss

Notice that we recover the loss function in VDM when $\boldsymbol{\nu}(\mathbf{z}, t) = \nu(t) \mathbf{1}_d$ where $\nu_t \in \mathbb{R}^+$ and $\mathbf{1}_d$ represents a vector of 1s of size d and the noising schedule isn't conditioned on $\mathbf{z}$.

$$\begin{aligned}
\int_0^1 \langle \mathbf{f}_\theta(\mathbf{x}_0, \boldsymbol{\nu}(\mathbf{z}, t)), \frac{d}{dt} \boldsymbol{\nu}(t) \rangle dt &= \int_0^1 \langle \mathbf{f}_\theta(\mathbf{x}_0, \boldsymbol{\nu}(t)), \frac{d}{dt} (\nu(t) \mathbf{1}_n) \rangle dt \\
&= \int_0^1 \langle \mathbf{f}_\theta(\mathbf{x}_0, \boldsymbol{\nu}(t)), \mathbf{1}_d \rangle \nu'(t) dt \\
&= \int_0^1 \nu'(t) \|\mathbf{f}_\theta(\mathbf{x}_0, \boldsymbol{\nu}(t))\|_1 dt \\
&= \int_0^1 \nu'(t) \|\mathbf{x} - \tilde{\mathbf{x}}_\theta(\mathbf{w}_{\boldsymbol{\nu}(t)}, \boldsymbol{\nu}(t))\|_2^2 dt \quad (\text{A.55})
\end{aligned}$$

$\int_0^1 \frac{d}{dt} \nu(t) \|\mathbf{x} - \tilde{\mathbf{x}}_\theta(\mathbf{w}_{\boldsymbol{\nu}(t)}, \boldsymbol{\nu}(t))\|_2^2 dt$ denotes the diffusion loss, $\mathcal{L}_{\text{diffusion}}$, as used in VDM; see [Kingma et al. \[2021\]](#).

A.6 Subset Sampling

Sampling a subset of k items from a collection of collection of n items, $x_1, x_2, \dots, x_n$ belongs to a category of algorithms called reservoir algorithms. In weighted reservoir sampling, every x_i is associated with a weight $w_i \geq 0$. The probability associated with choosing the sequence $S_{\text{wrs}} = [i_1, i_2, \dots, i_k]$ be a tuple of indices. Then the probability associated with sampling this sequence is

$$p(S_{\text{wrs}} | \mathbf{w}) = \frac{w_{i_1}}{Z} \frac{w_{i_2}}{Z - w_{i_1}} \cdots \frac{w_{i_k}}{Z - \sum_{j=1}^{k-1} w_{i_j}} \quad (\text{A.56})$$

[Efrimidis and Spirakis \[2006\]](#) give an algorithm for weighted reservoir sampling where each item is assigned a random key $r_i = u_i^{\frac{1}{w_i}}$ where u_i is drawn from a uniform distribution $[0, 1]$ and w_i is the weight of item x_i . Let $\text{TopK}(\mathbf{r}, k)$ which takes keys $\mathbf{r} = [r_1, r_2, \dots, r_n]$ and returns a sequence $[i_1, i_2, \dots, i_k]$. [Efrimidis and Spirakis \[2006\]](#) proved that $\text{TopK}(\mathbf{r}, k)$ is distributed according to $p(S_{\text{wrs}} | \mathbf{w})$.

Let's represent a subset $S \in \{0, 1\}^n$ with exactly k non-zero elements that are

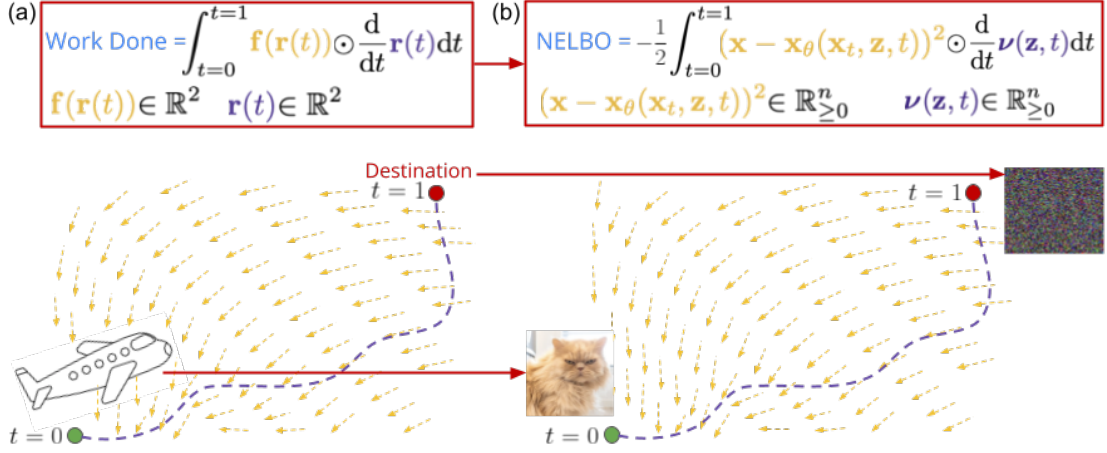


Figure A.2: **(a)** Imagine piloting a plane across a region with cyclones and strong winds, as shown in Fig. A.2. Plotting a direct, straight-line course through these adverse weather conditions requires more fuel and effort due to increased resistance. By navigating around the cyclones and winds, however, the plane reaches its destination with less energy, even if the route is longer. This intuition translates into mathematical and physical terms. The plane’s trajectory is denoted by $\mathbf{r}(t) \in \mathbb{R}_+^n$, while the forces acting on it are represented by $\mathbf{f}(\mathbf{r}(t)) \in \mathbb{R}^n$. The work required to navigate is given by $\int_0^1 \mathbf{f}(\mathbf{r}(t)) \cdot \frac{d}{dt} \mathbf{r}(t) dt$. Here, the work depends on the trajectory because $\mathbf{f}(\mathbf{r}(t))$ is not a conservative field.

(b) This concept also applies to the diffusion NELBO. From Eq. 3.13, it’s clear that the trajectory $\mathbf{r}(t)$ is parameterized by the noise schedule $\boldsymbol{\nu}(\mathbf{z}, t)$, which is influenced by complex forces, $\mathbf{f}$ (analogous to weather patterns), represented by the dimension-wise reconstruction error of the denoising model, $(\mathbf{x} - \mathbf{x}_\theta(\mathbf{w}_t, \mathbf{z}, t))^2$. Thus, the diffusion loss, $\mathcal{L}_{\text{diffusion}}$, can be interpreted as the work done along the trajectory $\boldsymbol{\nu}(\mathbf{z}, t)$ in the presence of these vector field forces $\mathbf{f}$. By learning the noise schedule, we can avoid “high-resistance” paths (those where the loss accumulates rapidly), thereby minimizing the overall “energy” expended, as measured by the NELBO.

equal to 1. Then the probability associated with sampling S is given as,

$$p(S|\mathbf{w}) = \sum_{S_{\text{wrs}} \in \Pi(S)} p(S_{\text{wrs}}|\mathbf{w}) \quad (\text{A.57})$$

where $\Pi(S)$ denotes all possible permutations of the sequence S . By ignoring the ordering of the elements in S_{wrs} we can sample using the same algorithm. Xie and Ermon [2019] show that this sampling algorithm is equivalent to $\text{TopK}(\hat{\mathbf{r}}, k)$ where $\hat{\mathbf{r}} = [\hat{r}_1, \hat{r}_2, \dots, \hat{r}_n]$ where $\hat{r}_i = -\log(-\log(r_i)) = \log w_i + \text{Gumbel}(0, 1)$. This holds

true because the monotonic transformation $-\log(-\log(x))$ preserves the ordering of the keys and thus $\text{TopK}(\mathbf{r}, k) \equiv \text{TopK}(\hat{\mathbf{r}}, k)$.

Sum of Gamma Distribution. Niepert et al. [2021] show that adding SOG noise instead of Gumbel noise leads to better performance.

Niepert et al. [2021] show that $\mathbf{z} \sim p_{\theta}(\mathbf{z}; \theta)$ is equivalent to $\mathbf{z} = \arg \max_{y \in Y} \langle \theta + \epsilon_g, y \rangle$ where ϵ_g is a sample from Sum-of-Gamma distribution given by

$$\text{SOG}(k, \tau, s) = \frac{\tau}{k} \left(\sum_{i=1}^s \text{Gamma}\left(\frac{1}{k}, \frac{k}{i}\right) - \log s \right), \quad (\text{A.58})$$

where s is a positive integer and $\text{Gamma}(\alpha, \beta)$ is the Gamma distribution with (α, β) as the shape and scale parameters.

And hence, given logits $\log \mathbf{w}$, we sample a k -hot vector using $\text{TopK}(\log \mathbf{w} + \epsilon)$. We choose a categorical prior with uniform distribution across n classes. Thus the KL loss term is given by:

$$-\sum_{i=1}^n \frac{w_i}{Z} \log \left(n \frac{w_i}{Z} \right) \quad (\text{A.59})$$

A.7 Experiment Details

A.7.1 Model Architecture

Denoising network. Our model architecture is extremely similar to VDM. The UNet of our pixel-space diffusion has an unchanged architecture from Kingma et al. [2021]. This structure is specifically designed for optimal performance in maximum likelihood training. We employ features from VDM such as the elimination of

internal downsampling/upsampling processes and the integration of Fourier features to enhance fine-scale prediction accuracy. In alignment with the configurations suggested by Kingma et al. (2021), our approach varies depending on the dataset: For CIFAR-10, we employ a U-Net with a depth of 32 and 128 channels; for ImageNet-32, the U-Net also has a depth of 32, but the channel count is increased to 256. Additionally, all these models incorporate a dropout rate of 0.1 in their intermediate layers.

Encoder network. $q_\phi(\mathbf{z}|\mathbf{x})$ is modeled using a sequence of 4 Resnet blocks with a channel count of 128 for CIFAR-10 and 256 for ImageNet-32 with a drop out of 0.1 in their intermediate layers.

Noise schedule. For polynomial noise schedule, we use an MLP that maps the latent vector $\mathbf{z}$ to $\mathbf{a}_\phi(\mathbf{z}), \mathbf{b}_\phi(\mathbf{z}), \mathbf{c}(\mathbf{z})$; see Eq. A.5.2 for details. The MLP has 2 hidden layers of size 3072 with `swish` activation function. The final layer is a linear mapping to 3×3072 values corresponding to $\mathbf{a}_\phi(\mathbf{z}), \mathbf{b}_\phi(\mathbf{z}), \mathbf{c}(\mathbf{z})$. Note that $\mathbf{a}_\phi(\mathbf{z}), \mathbf{b}_\phi(\mathbf{z}), \mathbf{c}(\mathbf{z})$ have the same dimensionality of 3072.

A.7.2 Hardware.

For the ImageNet experiments, we used a single GPU node with 8-A100s. For the cifar-10 experiments, the models were trained on 4 GPUs spanning several GPUs types like V100, A5000s, A6000s, and 3090s with `float32` precision.

A.7.3 Hyperparameters

We follow the same default training settings as [Kingma et al. \[2021\]](#). For all our experiments, we use the Adam [\[Kingma and Ba, 2014\]](#) optimizer with learning rate 2×10^{-4} , exponential decay rates of $\beta_1 = 0.9$, $\beta_2 = 0.99$ and decoupled weight decay [\[Loshchilov and Hutter, 2017\]](#) coefficient of 0.01. We also maintain an exponential moving average (EMA) of model parameters with an EMA rate of 0.9999 for evaluation. For other hyperparameters, we use fixed start and end times which satisfy $\gamma_{\min} = -13.3$, $\gamma_{\max} = 5.0$, which is used in [Kingma et al. \[2021\]](#), [Zheng et al. \[2023a\]](#).

A.8 Datasets and Visualizations

In this section we provide a brief description of the datasets used in the paper and visualize the generated samples and the noise schedules.

A.8.1 CIFAR-10

The CIFAR-10 dataset [\[Krizhevsky et al., 2009\]](#) is a collection of images consisting of 60,000 32×32 color images in 10 different classes, with each class representing a distinct object or scene. The dataset is divided into 50,000 training images and 10,000 test images, with each class having an equal representation in both sets. The classes in CIFAR-10 include: Airplane, Automobile, Bird, Cat, Deer, Dog, Frog, Horse, Ship, Truck.

Randomly generated samples for the CIFAR-10 dataset are provided in [Fig. A.3](#)

for MuLAN and Fig. A.4 for VDM. We visualize the noise schedule in Fig. A.20.



Figure A.3: MuLAN with velocity reparameterization after 8M training iterations.



Figure A.4: VDM after 10M training iterations.

Figure A.5: CIFAR-10 samples generated by different methods.

A.8.2 ImageNet-32

ImageNet-32 is a dataset derived from ImageNet [Deng et al., 2009], where the original images have been resized to a resolution of 32×32 . This dataset comprises 1,281,167 training samples and 50,000 test samples, distributed across 1,000 labels.

Randomly generated samples for the ImageNet dataset are provided in Fig. A.6 for MuLAN and Fig. A.7 for VDM. We visualize the noise schedule in Fig. A.20.



Figure A.6: MuLAN with noise parameterization after 2M training iterations.



Figure A.7: VDM after 2M training iterations.

A.8.3 Frequency

To see if MuLAN learns different noise schedules for parts of the images with different frequencies, we modify the images in the CIFAR-10 dataset where we modify an image where we randomly remove the low frequency component an image or remove the high frequency with equal probability. Fig. A.8 shows the training

samples. MuLAN was trained for 500K steps. The samples generated by MuLAN is shown in Fig. A.9. The corresponding noise schedules is shown in Fig. A.20. As compared to CIFAR-10, we notice that the spatial variation in the noise schedule increases (SNRs for all the pixels form a wider band) while the variance of the noise schedule across instances decreases slightly.

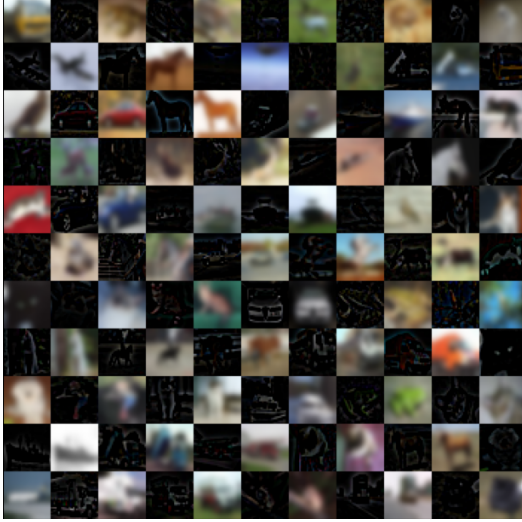


Figure A.8: Training samples.



Figure A.9: Samples generated by MuLAN with noise parameterization after 500K training iterations.

Figure A.10: Frequency Split CIFAR-10 dataset.

A.8.4 Frequency-2

To see if MuLAN learns different noise schedules for images with different frequencies, we modify the images in the CIFAR-10 dataset where we modify an image where we randomly remove the low frequency component an image or remove the high frequency with equal probability. Fig. A.8 shows the training samples. MuLAN was trained for 500K steps. The samples generated by MuLAN is shown in Fig. A.9. The corresponding noise schedules is shown in Fig. A.20. As compared to CIFAR-10,

we notice that the spatial variation in the noise schedule increases (SNRs for all the pixels form a wider band) and the variance of the noise schedule across instances increases as well.

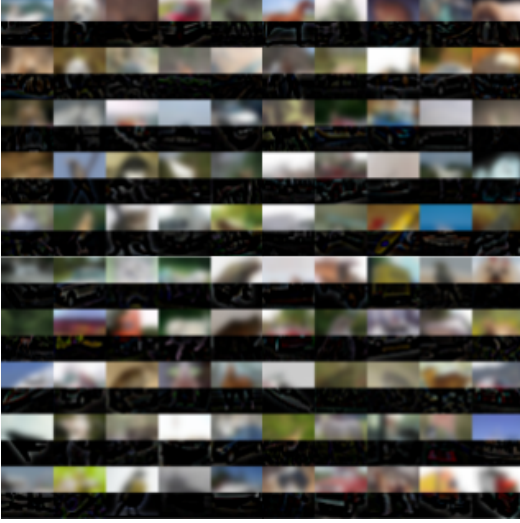


Figure A.11: Training samples.



Figure A.12: Samples generated by MuLAN with noise parameterization after 500K training iterations.

Figure A.13: Frequency Split-2 CIFAR-10 dataset.

A.8.5 CIFAR-10: Intensity

To see if MuLAN learns different noise schedules for images with different intensities, we modify the images in the CIFAR-10 dataset where we randomly convert an image into a low intensity or a high intensity image with equal probability. Originally, the CIFAR10 images are in the range $[0, 255]$. To convert an image into a low intensity image we multiply all pixel values by 0.5. To convert an image into a high intensity image we multiply all the pixel values by 0.5 and add 127.5 to them. Fig. A.14 shows the training samples. MuLAN was trained for 500K steps. The samples generated by MuLAN is shown in Fig. A.15. The corresponding noise schedules is

shown in Fig. A.20. As compared to CIFAR-10, we notice that the spatial variation in the noise schedule slightly increases (SNRs for all the pixels form a wider band) while the variance of the noise schedule across instances slightly decreases.



Figure A.14: Training samples.



Figure A.15: Samples generated by MuLAN with noise parameterization after 500K training iterations.

Figure A.16: Intensity CIFAR-10 dataset.

A.8.6 Mask

We modify the CIFAR-10 dataset where we randomly mask (i.e. replace with **0s**) the top of an image or the bottom half of an image with equal probability. Fig. A.17 shows the training samples. MuLAN was trained for 500K steps. The samples generated by MuLAN is shown in Fig. A.18. The corresponding noise schedules is shown in Fig. A.20. As compared to CIFAR-10, we notice that the spatial variation in the noise schedule slightly increases (SNRs for all the pixels form a wider band) while the variance of the noise schedule across instances decreases.



Figure A.17: Training samples.



Figure A.18: Samples generated by MuLAN with noise parameterization after 500K training iterations.

Figure A.19: Intensity CIFAR-10 dataset.

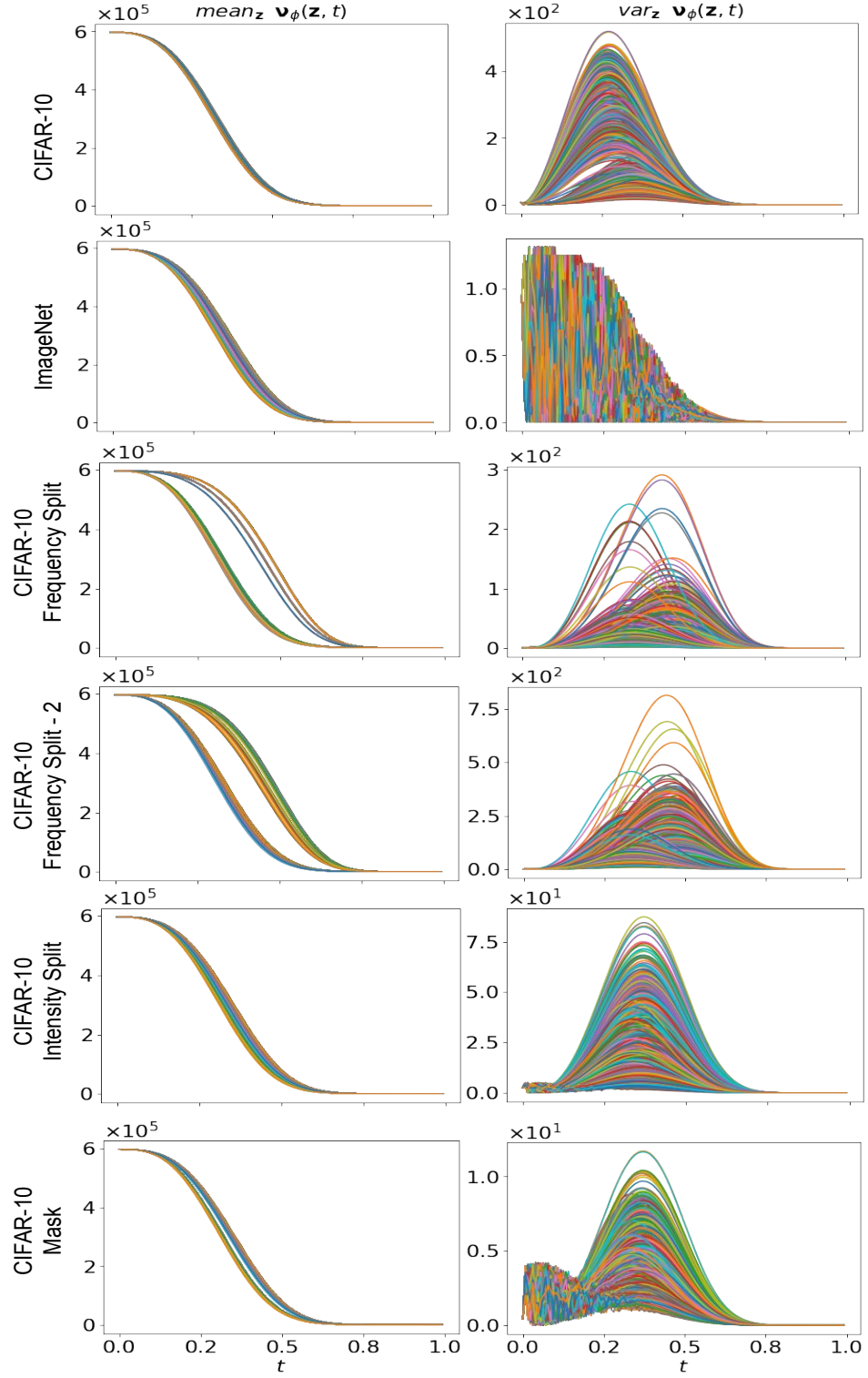


Figure A.20: signal-to-noise ratio for different datasets.

A.9 Likelihood Estimation

We used both Variance Lower Bound (VLB) and ODE-based methods to compute BPD.

A.9.1 VLB Estimate

In the VLB-based approach, we employ Eq. 3.8. To compute $\mathcal{L}_{\text{diffusion}}$, we use $T = 128$ in Eq. 3.9, discretizing the timesteps, $t \in [0, 1]$ into 128 bins.

A.9.2 Exact likelihood computation using Probability Flow ODE

A diffusion process whose marginal is given by (the same as in Eq. 3.1),

$$q(\mathbf{w}_t|\mathbf{x}) = \mathcal{N}(\mathbf{w}_t; \boldsymbol{\alpha}_t \mathbf{x}, \text{diag}(\boldsymbol{\sigma}_t^2)), \quad \mathbf{x} \sim q_0(\mathbf{x}), \quad (\text{A.60})$$

can be modeled as the solution to an Itô Stochastic Differential Equation (SDE):

$$d\mathbf{w}_t = \mathbf{f}(t)\mathbf{w}_t dt + \mathbf{g}(t)d\mathbf{w}_t, \quad \mathbf{x} \sim q_0(\mathbf{x}), \quad (\text{A.61})$$

where $\mathbf{f}(t) \in \mathbb{R}^d, \mathbf{g}(t) \in \mathbb{R}^d$ take the following expressions [Song et al., 2020]:

$$\begin{aligned} \mathbf{f}(t) &= \frac{d}{dt} \log \boldsymbol{\alpha}_t, \\ \mathbf{g}^2(t) &= \frac{d}{dt} \boldsymbol{\sigma}_t^2 - 2\boldsymbol{\sigma}_t^2 \frac{d}{dt} \log \boldsymbol{\alpha}_t \end{aligned}$$

The corresponding reverse process, Eq. 3.3, can also be modelled by an equivalent reverse-time SDE:

$$d\mathbf{w}_t = [\mathbf{f}(t) - \mathbf{g}(t)^2 \nabla_{\mathbf{x}_t} \log q(\mathbf{w}_t|\mathbf{x})]dt + \mathbf{g}(t)d\bar{\mathbf{w}}_t, \quad \mathbf{w}_1 \sim p_\theta(\mathbf{w}_1), \quad (\text{A.62})$$

where $\bar{\mathbf{w}}$ is a standard Wiener process when time flows backwards from $1 \rightarrow 0$, and dt is an infinitesimal negative timestep. Song et al. [2020] show that the marginals of Eq. A.62 can be described by the following Ordinary Differential Equation (ODE) in the reverse process:

$$d\mathbf{w}_t = \left[\mathbf{f}(t)\mathbf{w}_t - \frac{1}{2}\mathbf{g}^2(t)\nabla_{\mathbf{x}_t} \log q(\mathbf{w}_t) \right] dt. \quad (\text{A.63})$$

This ODE, also called the probability flow ODE, allows us to compute the exact likelihood on any input data via the instantaneous change of variables formula as proposed in Chen et al. [2018]. Note that during the reverse process, the term $q(\mathbf{w}_t)$ is unknown and is approximated by parameterized by $p_\theta(\mathbf{w}_t)$. For the probability flow defined in Eq. A.63, Chen et al. [2018] show that the log-likelihood of $p_\theta(\mathbf{x})$ can be computed using the following equation:

$$\begin{aligned} \log p_\theta(\mathbf{x}) &= \log p_\theta(\mathbf{w}_1) - \int_{t=0}^{t=1} \text{tr}(\nabla_{\mathbf{x}_t} \mathbf{h}_\theta(\mathbf{w}_t, t)) dt, \\ \text{where } \mathbf{h}_\theta(\mathbf{w}_t, t) &\equiv \mathbf{f}(t)\mathbf{w}_t - \frac{1}{2}\mathbf{g}^2(t)\nabla_{\mathbf{x}_t} \log p_\theta(\mathbf{w}_t) \end{aligned} \quad (\text{A.64})$$

Probability Flow ODE for MuLAN.

Similarly for the forward process conditioned on the auxiliary latent variable, $\mathbf{z}$,

$$q_\phi(\mathbf{w}_t|\mathbf{x}, \mathbf{z}) = \mathcal{N}(\mathbf{w}_t; \boldsymbol{\alpha}_t(\mathbf{z})\mathbf{x}, \text{diag}(\boldsymbol{\sigma}_t^2(\mathbf{z}))), \quad \mathbf{x} \sim q_0(\mathbf{x}), \quad \mathbf{z} \sim q_\phi(\mathbf{z}|\mathbf{x}), \quad (\text{A.65})$$

we can extend Eq. A.61 in the following manner,

$$d\mathbf{w}_t = \mathbf{f}(\mathbf{z}, t)\mathbf{w}_t dt + \mathbf{g}(\mathbf{z}, t)d\mathbf{w}_t, \quad \mathbf{x} \sim q_0(\mathbf{x}), \quad \mathbf{z} \sim q_\phi(\mathbf{z}|\mathbf{x}), \quad (\text{A.66})$$

to obtain the corresponding SDE formulation. Notice that the random variable $\mathbf{z}$ in the above equation doesn't have a subscript t , and hence, $\mathbf{z}$ is drawn from $q_\phi(\mathbf{z}|\mathbf{x})$ once and the same $\mathbf{z}$ is used as $\mathbf{x}$ diffuses to $\mathbf{w}_1$. The expressions for

$\mathbf{f}(\mathbf{z}, t) : \mathbb{R}^m \times [0, 1] \rightarrow \mathbb{R}^d$, $\mathbf{g}(\mathbf{z}, t) : \mathbb{R}^m \times [0, 1] \rightarrow \mathbb{R}^d$ is given as follows:

$$\begin{aligned}\mathbf{f}(\mathbf{z}, t) &= \frac{d}{dt} \log \boldsymbol{\alpha}_t(\mathbf{z}), \\ \mathbf{g}^2(\mathbf{z}, t) &= \frac{d}{dt} \boldsymbol{\sigma}_t^2(\mathbf{z}) - 2\boldsymbol{\sigma}_t^2(\mathbf{z}) \frac{d}{dt} \log \boldsymbol{\alpha}_t(\mathbf{z})\end{aligned}$$

Recall that $\boldsymbol{\alpha}_t^2(\mathbf{z}) = \text{sigmoid}(-\gamma_\phi(\mathbf{z}, t))$, $\boldsymbol{\sigma}_t^2(\mathbf{z}) = \text{sigmoid}(\gamma_\phi(\mathbf{z}, t))$. Substituting these in the above equations, the expressions for $\mathbf{f}(\mathbf{z}, t)$ and $\mathbf{g}^2(\mathbf{z}, t)$ simplify to the following:

$$\begin{aligned}\mathbf{f}(\mathbf{z}, t) &= -\frac{1}{2} \text{sigmoid}(\gamma_\phi(\mathbf{z}, t)) \frac{d}{dt} \gamma_\phi(\mathbf{z}, t), \\ \mathbf{g}^2(\mathbf{z}, t) &= \text{sigmoid}(\gamma_\phi(\mathbf{z}, t)) \frac{d}{dt} \gamma_\phi(\mathbf{z}, t)\end{aligned}$$

The corresponding reverse-time SDE is given as:

$$d\mathbf{w}_t = [\mathbf{f}(t) - \mathbf{g}(t)^2 \nabla_{\mathbf{x}_t} \log q_\phi(\mathbf{w}_t | \mathbf{x}, \mathbf{z})] dt + \mathbf{g}(t) d\bar{\mathbf{w}}_t, \quad \mathbf{w}_1 \sim p_\theta(\mathbf{w}_1), \quad \mathbf{z} \sim p_\theta(\mathbf{z}), \quad (\text{A.67})$$

where $\bar{\mathbf{w}}$ is a standard Wiener process when time flows backwards from $1 \rightarrow 0$, and dt is an infinitesimal negative timestep. Given, $\mathbf{s}_\theta(\mathbf{w}_t, \mathbf{z})$, an approximation to the true score function, $\nabla_{\mathbf{x}_t} \log q_\phi(\mathbf{w}_t | \mathbf{x}, \mathbf{z})$, [Song et al. \[2020\]](#) show that the marginals of Eq. [A.67](#) can be described by the following Ordinary Differential Equation (ODE):

$$d\mathbf{w}_t = \left[\mathbf{f}(\mathbf{z}, t) - \frac{1}{2} \mathbf{g}^2(\mathbf{z}, t) \mathbf{s}_\theta(\mathbf{w}_t, \mathbf{z}) \right] dt, \quad (\text{A.68})$$

[Zheng et al. \[2023a\]](#) show that the score function, $\mathbf{s}_\theta(\mathbf{w}_t, \mathbf{z})$, for the noise and the v-parameterization is given as follows:

$$\mathbf{s}_\theta(\mathbf{w}_t, \mathbf{z}) = \begin{cases} -\frac{\boldsymbol{\epsilon}_\theta(\mathbf{w}_t, t)}{\boldsymbol{\sigma}_t(\mathbf{z})} & \text{Noise parameterization (Sec. [A.5.1](#)) (A.69a)} \\ -\mathbf{w}_t - \exp\left(-\frac{1}{2} \gamma_\phi(\mathbf{z}, t)\right) \mathbf{v}_\theta(\mathbf{w}_t, \mathbf{z}, t) & \text{v-parameterization (Sec. [A.5.1](#)) (A.69b)} \end{cases}$$

Applying the change of variables formula [Chen et al., 2018] on Eq. A.68, $\log p_\theta(\mathbf{x}|\mathbf{z})$ can be computed in the following manner:

$$\begin{aligned} \log p_\theta(\mathbf{x}|\mathbf{z}) &= \log p_\theta(\mathbf{w}_1) - \int_{t=0}^{t=1} \text{tr}(\nabla_{\mathbf{x}_t} \mathbf{h}_\theta(\mathbf{w}_t, \mathbf{z}, t)) dt, \\ \text{where } \mathbf{h}_\theta(\mathbf{w}_t, \mathbf{z}, t) &\equiv \mathbf{f}(\mathbf{z}, t) - \frac{1}{2} \mathbf{g}^2(\mathbf{z}, t) \mathbf{s}_\theta(\mathbf{w}_t, \mathbf{z}) \end{aligned} \quad (\text{A.70})$$

The expression for log-likelihood (Eq. 3.7) is as follows,

$$\begin{aligned} &\log p_\theta(\mathbf{x}) \\ &\geq \mathbb{E}_{q_\phi(\mathbf{z}|\mathbf{x})} [\log p_\theta(\mathbf{x}|\mathbf{z})] - \text{D}_{\text{KL}}(q_\phi(\mathbf{z}|\mathbf{x}) \| p_\theta(\mathbf{z})) \\ \text{Using Eq. A.70,} \\ &= \mathbb{E}_{q_\phi(\mathbf{z}|\mathbf{x})} \left[\log p_\theta(\mathbf{w}_1) - \int_{t=0}^{t=1} \text{tr}(\nabla_{\mathbf{x}_t} \mathbf{h}_\theta(\mathbf{w}_t, t, \mathbf{z})) dt \right] - \text{D}_{\text{KL}}(q_\phi(\mathbf{z}|\mathbf{x}) \| p_\theta(\mathbf{z})) \end{aligned} \quad (\text{A.71})$$

Computing $\text{tr}(\nabla_{\mathbf{x}_t} \mathbf{h}_\theta(\mathbf{w}_t, t, \mathbf{z}))$ is expensive and we follow Chen et al. [2018], Zheng et al. [2023a], Grathwohl et al. [2018] to estimate it with Skilling-Hutchinson trace estimator [Skilling, 1989, Hutchinson, 1989]. In particular, we have

$$\text{tr}(\nabla_{\mathbf{x}_t} \mathbf{h}_\theta(\mathbf{w}_t, t, \mathbf{z})) = \mathbb{E}_{p(\epsilon)} [\epsilon^\top \nabla_{\mathbf{x}_t} \mathbf{h}_\theta(\mathbf{w}_t, t, \mathbf{z}) \epsilon], \quad (\text{A.72})$$

where the random variable ϵ satisfies $\mathbb{E}_{p(\epsilon)}[\epsilon] = \mathbf{0}$ and $\text{Cov}_{p(\epsilon)}[\epsilon] = \mathbf{I}$. Common choices for $p(\epsilon)$ include Rademacher or Gaussian distributions. Notably, the term $\nabla_{\mathbf{x}_t} \mathbf{h}_\theta(\mathbf{w}_t, t, \mathbf{z}) \epsilon$ can be computed efficiently using “Jacobian-vector-product” computation in JAX. In our experiments, we follow the exact evaluation procedure for computing likelihood as outlined in Song et al. [2020], Grathwohl et al. [2018]. Specifically, for the computation of Eq. A.72, we employ a Rademacher distribution for $p(\epsilon)$. To calculate the integral in Eq. A.71, we utilize the RK45 ODE solver [Dormand and Prince, 1980] provided by `scipy.integrate.solve_ivp` with `atol=1e-5` and `rtol=1e-5`.

Dequantization.

Real-world datasets for images or texts often consist of discrete data. Attempting to learn a continuous density model directly on these discrete data points can lead to degenerate outcomes [Uria et al., 2013] and fail to provide meaningful density estimations. Dequantization [Salimans et al., 2017, Ho et al., 2020, Zheng et al., 2023a] is a common solution in such cases. To elaborate, let x_0 represent 8-bit discrete data scaled to $[-1, 1]$. Dequantization methods assume that we have trained a continuous model distribution p_θ for x_0 , and define the discrete model distribution by

$$P_\theta(\mathbf{x}) = \int_{[-\frac{1}{256}, \frac{1}{256}]^d} p_\theta(\mathbf{x} + u) du.$$

To train $P_\theta(\mathbf{x})$ by maximum likelihood estimation, variational dequantization [Ho et al., 2020, Zheng et al., 2023a] introduces a dequantization distribution $q(u|\mathbf{x})$ and jointly train p_{model} and $q(u|\mathbf{x})$ by a variational lower bound:

$$\log P_\theta(\mathbf{x}) \geq \mathbb{E}_{q(u|\mathbf{x})}[p_\theta(\mathbf{x} + u) - \log q(u|\mathbf{x})]. \quad (\text{A.73})$$

Truncated Normal Dequantization. Zheng et al. [2023a] show that truncated Normal distribution,

$$q(u|\mathbf{x}) = \mathcal{TN}\left(\mathbf{0}, \mathbf{I}, -\frac{1}{256}, \frac{1}{256}\right)$$

with mean $\mathbf{0}$, covariance $\mathbf{I}$, and bounds $[-\frac{1}{256}, \frac{1}{256}]$ along each dimension, leads to a better likelihood estimate. Thus, Eq. A.73 simplifies to the following (for details

please refer to section A. in [Zheng et al. \[2023a\]](#)):

$$\log P_\theta(\mathbf{x}) \geq \mathbb{E}_{\hat{\epsilon} \sim \mathcal{TN}(\mathbf{0}, \mathbf{I}, -\tau, \tau)} \left[\log p_\theta \left(\mathbf{x} + \frac{\sigma_\epsilon}{\alpha_\epsilon} \hat{\epsilon} \right) \right] + \frac{d}{2} (1 + \log(2\pi\sigma_\epsilon^2)) - 0.01522 \times d \quad (\text{A.74})$$

$$\text{with } \frac{\sigma_\epsilon}{\alpha_\epsilon} = \exp\left(-\frac{1}{2} \times 13.3\right),$$

$$\sigma_\epsilon = \text{sqrt}(\text{sigmoid}(-13.3)), \text{ and } \tau = 3.$$

$\log p_\theta \left(\mathbf{x} + \frac{\sigma_\epsilon}{\alpha_\epsilon} \hat{\epsilon} \right)$ is evaluated using Eq. [A.71](#).

Importance Weighted Estimator. Eq. [A.74](#) can also be extended to obtain an importance weighted likelihood estimator to get a tighter bound on the likelihood. The variational bound is given by (for details please refer to section A. in [Zheng et al. \[2023a\]](#)):

$$\begin{aligned} \log P_\theta(\mathbf{x}) &\geq \mathbb{E}_{\hat{\epsilon}^{(1)}, \dots, \hat{\epsilon}^{(K)} \sim \mathcal{TN}(\mathbf{0}, \mathbf{I}, -\tau, \tau)} \left[\log \left(\frac{1}{K} \sum_{i=1}^K \frac{p_\theta \left(\mathbf{x} + \frac{\sigma_\epsilon}{\alpha_\epsilon} \hat{\epsilon}^{(i)} \right)}{q(\hat{\epsilon}^{(i)})} \right) \right] + d \log \sigma_\epsilon \quad (\text{A.75}) \\ &\text{with } \frac{\sigma_\epsilon}{\alpha_\epsilon} = \exp\left(-\frac{1}{2} \times 13.3\right), \log \sigma_\epsilon = \frac{1}{2}(-13.3 + \text{softplus}(-13.3)), \\ &q(\hat{\epsilon}) = \frac{1}{(2\pi Z)^2} \exp\left(-\frac{1}{2} \|\hat{\epsilon}\|_2^2\right), Z = 0.9974613, \text{ and } \tau = 3. \end{aligned}$$

Note that for $K = 1$, Eq. [A.75](#) is equivalent to Eq. [A.74](#); see [Zheng et al. \[2023a\]](#). $\log p_\theta \left(\mathbf{x} + \frac{\sigma_\epsilon}{\alpha_\epsilon} \hat{\epsilon} \right)$ is evaluated using Eq. [A.71](#). In Table [A.5](#), we report BPD values for MuLAN on CIFAR10 (8M training steps, v-parameterization) and ImageNet (2M training steps, noise parameterization) using both the VLB-based approach, and the ODE-based approach with $K = 1$ and $K = 20$ importance samples.

Table A.5: NLL (mean and 95% Confidence Interval for MuLAN) on CIFAR10 (8M training steps, v-parameterization) and ImageNet (2M training steps, noise parameterization) using both the VLB-based approach, and the ODE-based approach. $K = 1$ means that we do not use importance weighted estimator since Eq. A.75 is equivalent to Eq. A.74 for this case; see Zheng et al. [2023a].

Likelihood Estimation type	CIFAR-10 ($\downarrow$)	Imagenet ($\downarrow$)
VLB-based	2.59 ± 10^{-3}	3.71 ± 10^{-3}
ODE-based ($K = 1$; Eq. A.74)	$2.59 \pm 3 \times 10^{-4}$	3.71 ± 10^{-3}
ODE-based ($K = 20$; Eq. A.75)	$2.55 \pm 3 \times 10^{-4}$	3.67 ± 10^{-3}

APPENDIX B

SIMPLE DIFFUSION LANGUAGE MODELS

B.1 Discrete time ELBO

This section is organized as follows: First, we derive the expressions for the true posterior and the approximate posterior as outlined in Suppl. B.1.1. We then simplify these expressions specifically for the case of absorbing state diffusion in Suppl. B.1.2. Finally, we derive the expression for the ELBO for absorbing state diffusion in Suppl. B.1.2.

B.1.1 Generic case

Given the state transition matrix Q_t , prior $\boldsymbol{\pi}$, and the latent variables $\mathbf{z}_s$ and $\mathbf{z}_t$, where $s < t$, let

$$Q_{t|s} = \alpha_{t|s} \mathbf{I} + (1 - \alpha_{t|s}) \mathbf{1} \boldsymbol{\pi}^\top. \quad (\text{B.1})$$

$$q(\mathbf{z}_t | \mathbf{z}_s)$$

Thus, the marginals in Eq. 2.7 correspond to the following forward process:

$$\begin{aligned} q(\mathbf{z}_t | \mathbf{z}_s) &= \text{Cat}(\mathbf{z}_t; Q_{t|s}^\top \mathbf{z}_s) \\ &= \text{Cat}(\mathbf{z}_t; [\alpha_{t|s} \mathbf{I} + (1 - \alpha_{t|s}) \mathbf{1} \boldsymbol{\pi}^\top]^\top \mathbf{z}_s) \\ &= \text{Cat}(\mathbf{z}_t; \alpha_{t|s} \mathbf{z}_s + (1 - \alpha_{t|s}) \boldsymbol{\pi} \mathbf{1}^\top \mathbf{z}_s) && \because \mathbf{1}^\top \mathbf{z}_s = 1 \\ &= \text{Cat}(\mathbf{z}_t; \alpha_{t|s} \mathbf{z}_s + (1 - \alpha_{t|s}) \boldsymbol{\pi}). \end{aligned} \quad (\text{B.2})$$

The above equation indicates that during each diffusion step from $s \rightarrow t$, a fraction $(1 - \alpha_{t|s})$ of the probability mass is transferred to the prior distribution $\boldsymbol{\pi}$.

$$q(\mathbf{z}_s | \mathbf{z}_t, \mathbf{x})$$

Austin et al. [2021] show that the posterior corresponding to Eq. B.2 is given as follows:

$$q(\mathbf{z}_s | \mathbf{z}_t, \mathbf{x}) = \text{Cat} \left(\mathbf{z}_s; \frac{Q_{t|s} \mathbf{z}_t \odot Q_s^\top \mathbf{x}}{\mathbf{z}_t^\top Q_t^\top \mathbf{x}} \right), \quad (\text{B.3})$$

which we simplify to the following:

$$\begin{aligned} q(\mathbf{z}_s | \mathbf{z}_t, \mathbf{x}) &= \text{Cat} \left(\mathbf{z}_s; \frac{[\alpha_{t|s} \mathbf{I} + (1 - \alpha_{t|s}) \mathbf{1} \boldsymbol{\pi}^\top] \mathbf{z}_t \odot [\alpha_s \mathbf{I} + (1 - \alpha_s) \mathbf{1} \boldsymbol{\pi}^\top]^\top \mathbf{x}}{\mathbf{z}_t^\top [\alpha_t \mathbf{I} + (1 - \alpha_t) \mathbf{1} \boldsymbol{\pi}^\top]^\top \mathbf{x}} \right) \\ &= \text{Cat} \left(\mathbf{z}_s; \frac{[\alpha_{t|s} \mathbf{z}_t + (1 - \alpha_{t|s}) \mathbf{1} \boldsymbol{\pi}^\top \mathbf{z}_t] \odot [\alpha_s \mathbf{x} + (1 - \alpha_s) \boldsymbol{\pi}]}{\mathbf{z}_t^\top [\alpha_t \mathbf{x} + (1 - \alpha_t) \boldsymbol{\pi} \mathbf{1}^\top \mathbf{x}]} \right) \\ &= \text{Cat} \left(\mathbf{z}_s; \frac{[\alpha_{t|s} \mathbf{z}_t + (1 - \alpha_{t|s}) \mathbf{1} \boldsymbol{\pi}^\top \mathbf{z}_t] \odot [\alpha_s \mathbf{x} + (1 - \alpha_s) \boldsymbol{\pi}]}{\alpha_t \mathbf{z}_t^\top \mathbf{x} + (1 - \alpha_t) \mathbf{z}_t^\top \boldsymbol{\pi}} \right). \quad \because \mathbf{1}^\top \mathbf{x} = 1 \quad (\text{B.4}) \end{aligned}$$

$$p_\theta(\mathbf{z}_s | \mathbf{z}_t)$$

Austin et al. [2021] approximate the reverse process in the following manner:

$$p_\theta(\mathbf{z}_s | \mathbf{z}_t) = q(\mathbf{z}_s | \mathbf{z}_t, \mathbf{x} = \mathbf{x}_\theta(\mathbf{z}_t, t)) = \text{Cat} \left(\mathbf{z}_s; \frac{Q_{t|s} \mathbf{z}_t \odot Q_s^\top \mathbf{x}_\theta(\mathbf{z}_t, t)}{\mathbf{z}_t^\top Q_t^\top \mathbf{x}_\theta(\mathbf{z}_t, t)} \right). \quad (\text{B.5})$$

where $\mathbf{x}_\theta(\mathbf{z}_t, t) : \mathcal{V} \times [0, 1] \rightarrow \Delta^K$ is an approximation for $\mathbf{x}$.

B.1.2 Absorbing state

For the absorbing state diffusion process we have $\boldsymbol{\pi} = \mathbf{m}$.

$$q(\mathbf{z}_s | \mathbf{z}_t, \mathbf{x})$$

Since, $\mathbf{z}_t \in \{\mathbf{x}, \mathbf{m}\}$, takes only 2 values we consider the separate cases: $\mathbf{z}_t = \mathbf{x}$ and $\mathbf{z}_t = \mathbf{m}$.

Case 1. Consider the case $\mathbf{z}_t = \mathbf{x}$ i.e. $\mathbf{z}_t$ is unmasked. From Eq. B.4, we have the following:

$$\begin{aligned}
& q(\mathbf{z}_s | \mathbf{z}_t = \mathbf{x}, \mathbf{x}) \\
&= \text{Cat} \left(\mathbf{z}_s; \frac{[\alpha_{t|s}\mathbf{x} + (1 - \alpha_{t|s})\mathbf{1}\mathbf{m}^\top\mathbf{x}] \odot [\alpha_s\mathbf{x} + (1 - \alpha_s)\mathbf{m}]}{\alpha_t\mathbf{x}^\top\mathbf{x} + (1 - \alpha_t)\mathbf{x}^\top\mathbf{m}} \right) \\
&\because \mathbf{x}^\top\mathbf{m} = 0, \text{ we get:} \\
&= \text{Cat} \left(\mathbf{z}_s; \frac{[\alpha_{t|s}\mathbf{x}] \odot [\alpha_s\mathbf{x} + (1 - \alpha_s)\mathbf{m}]}{\alpha_t} \right) \\
&\because \mathbf{x} \odot \mathbf{m} = \mathbf{0} \text{ and } \alpha_t = \alpha_{t|s}\alpha_s, \text{ we get:} \\
&= \text{Cat} \left(\mathbf{z}_s; \frac{\alpha_t\mathbf{x}}{\alpha_t} \right) \\
&\because \alpha_t = \alpha_{t|s}\alpha_s, \text{ we get:} \qquad \qquad \qquad = \text{Cat}(\mathbf{z}_s; \mathbf{x}) \quad (\text{B.6})
\end{aligned}$$

Thus, we have the following:

$$q(\mathbf{z}_s | \mathbf{z}_t = \mathbf{x}, \mathbf{x}) = \text{Cat}(\mathbf{z}_s; \mathbf{x}). \quad (\text{B.7})$$

Case 2. Consider the case $\mathbf{z}_t = \mathbf{m}$. By substituting $\mathbf{z}_t = \mathbf{m}$ and $\boldsymbol{\pi} = \mathbf{m}$ in Eq. B.4, $q(\mathbf{z}_s | \mathbf{z}_t, \mathbf{x})$ simplifies to the following:

$$\begin{aligned}
q(\mathbf{z}_s | \mathbf{z}_t = \mathbf{m}, \mathbf{x}) &= \text{Cat} \left(\frac{(\alpha_{t|s}\mathbf{m} + (1 - \alpha_{t|s})\mathbf{1}) \odot (\alpha_s\mathbf{x} + (1 - \alpha_s)\mathbf{m})}{(1 - \alpha_t)} \right) \\
&= \text{Cat} \left(\frac{(\alpha_{t|s}(1 - \alpha_s)\mathbf{m} + (1 - \alpha_{t|s})(1 - \alpha_s)\mathbf{m} + (\alpha_s - \alpha_t)\mathbf{x})}{(1 - \alpha_t)} \right) \\
&= \text{Cat} \left(\mathbf{z}_s; \frac{(1 - \alpha_s)\mathbf{m} + (\alpha_s - \alpha_t)\mathbf{x}}{1 - \alpha_t} \right) \quad (\text{B.8})
\end{aligned}$$

Note that the above categorical distribution is non-zero for $\mathbf{z}_s \in \{\mathbf{x}, \mathbf{m}\}$ and zero for every other value. The non-zero values are specified as follows:

$$q(\mathbf{z}_s = \mathbf{x} | \mathbf{z}_t = \mathbf{m}, \mathbf{x}) = \frac{\alpha_s - \alpha_t}{1 - \alpha_t} \quad (\text{B.9})$$

$$q(\mathbf{z}_s = \mathbf{m} | \mathbf{z}_t = \mathbf{m}, \mathbf{x}) = \frac{1 - \alpha_s}{1 - \alpha_t} \quad (\text{B.10})$$

Combining Cases 1 and 2, we get:

$$q(\mathbf{z}_s | \mathbf{z}_t, \mathbf{x}) = \begin{cases} \text{Cat}(\mathbf{z}_s; \mathbf{z}_t) & \mathbf{z}_t \neq \mathbf{m}, \\ \text{Cat}\left(\mathbf{z}_s; \frac{(1-\alpha_s)\mathbf{m} + (\alpha_s - \alpha_t)\mathbf{x}}{1 - \alpha_t}\right) & \mathbf{z}_t = \mathbf{m}. \end{cases} \quad (\text{B.11})$$

$$p_\theta(\mathbf{z}_s | \mathbf{z}_t)$$

For the absorbing state diffusion process with $\boldsymbol{\pi} = \mathbf{m}$, we want to simplify the Eq. [B.5](#). For this reason, we consider 2 cases: first, when $\mathbf{z}_t \neq \mathbf{m}$ (**case 1**), second, when $\mathbf{z}_t = \mathbf{m}$ (**case 2**).

Case 1. Consider the case when $\mathbf{z}_t \neq \mathbf{m}$. Eq. B.5 simplifies to the following:

$$\begin{aligned}
p_\theta(\mathbf{z}_s | \mathbf{z}_t \neq \mathbf{m}) &= \text{Cat} \left(\mathbf{z}_s; \frac{Q_{t|s} \mathbf{z}_t \odot Q_s^\top \mathbf{x}_\theta(\mathbf{z}_t, t)}{\mathbf{z}_t^\top Q_t^\top \mathbf{x}_\theta(\mathbf{z}_t, t)} \right) \\
&= \text{Cat} \left(\mathbf{z}_s; \frac{Q_{t|s} \mathbf{z}_t \odot Q_s^\top \mathbf{x}_\theta(\mathbf{z}_t, t)}{[Q_t \mathbf{z}_t]^\top \mathbf{x}_\theta(\mathbf{z}_t, t)} \right) \\
&= \text{Cat} \left(\mathbf{z}_s; \frac{[\alpha_{t|s} \mathbf{z}_t] \odot [\alpha_s \mathbf{I} + (1 - \alpha_s) \mathbf{m} \mathbf{1}^\top] \mathbf{x}_\theta(\mathbf{z}_t, t)}{[\alpha_t \mathbf{z}_t]^\top \mathbf{x}_\theta(\mathbf{z}_t, t)} \right) \\
&= \text{Cat} \left(\mathbf{z}_s; \frac{[\alpha_{t|s} \mathbf{z}_t] \odot [\alpha_s \mathbf{x}_\theta(\mathbf{z}_t, t) + (1 - \alpha_s) \mathbf{m} \langle \mathbf{1}, \mathbf{x}_\theta(\mathbf{z}_t, t) \rangle]}{\alpha_t \langle \mathbf{z}_t, \mathbf{x}_\theta(\mathbf{z}_t, t) \rangle} \right) \\
&\text{since } \langle \mathbf{1}, \mathbf{x}_\theta(\mathbf{z}_t, t) \rangle = 1, \text{ we have the following:} \\
&= \text{Cat} \left(\mathbf{z}_s; \frac{[\alpha_{t|s} \mathbf{z}_t] \odot [\alpha_s \mathbf{x}_\theta(\mathbf{z}_t, t) + (1 - \alpha_s) \mathbf{m}]}{\alpha_t \langle \mathbf{z}_t, \mathbf{x}_\theta(\mathbf{z}_t, t) \rangle} \right) \\
&\text{since } \mathbf{z}_t \odot \mathbf{m} = \mathbf{0}, \text{ we have the following:} \\
&= \text{Cat} \left(\mathbf{z}_s; \frac{\alpha_t \mathbf{z}_t \odot \mathbf{x}_\theta(\mathbf{z}_t, t)}{\alpha_t \langle \mathbf{z}_t, \mathbf{x}_\theta(\mathbf{z}_t, t) \rangle} \right) \\
&= \text{Cat}(\mathbf{z}_s; \mathbf{z}_t) \tag{B.12}
\end{aligned}$$

Case 2. Consider the case when $\mathbf{z}_t = \mathbf{m}$. Eq. B.5 simplifies to the following:

$$\begin{aligned}
p_\theta(\mathbf{z}_s | \mathbf{z}_t = \mathbf{m}) &= \text{Cat} \left(\mathbf{z}_s; \frac{Q_{t|s} \mathbf{m} \odot Q_s^\top \mathbf{x}_\theta(\mathbf{z}_t, t)}{\mathbf{m}^\top Q_t^\top \mathbf{x}_\theta(\mathbf{z}_t, t)} \right) \\
&= \text{Cat} \left(\mathbf{z}_s; \frac{Q_{t|s} \mathbf{m} \odot Q_s^\top \mathbf{x}_\theta(\mathbf{z}_t, t)}{[Q_t \mathbf{m}]^\top \mathbf{x}_\theta(\mathbf{z}_t, t)} \right) \\
&= \text{Cat} \left(\mathbf{z}_s; \frac{[\alpha_{t|s} \mathbf{m} + (1 - \alpha_{t|s}) \mathbf{1}] \odot [\alpha_s \mathbf{I} + (1 - \alpha_s) \mathbf{m} \mathbf{1}^\top] \mathbf{x}_\theta(\mathbf{z}_t, t)}{[\alpha_t \mathbf{m} + (1 - \alpha_t) \mathbf{1}]^\top \mathbf{x}_\theta(\mathbf{z}_t, t)} \right) \\
&= \text{Cat} \left(\mathbf{z}_s; \frac{[\alpha_{t|s} \mathbf{m} + (1 - \alpha_{t|s}) \mathbf{1}] \odot [\alpha_s \mathbf{x}_\theta(\mathbf{z}_t, t) + (1 - \alpha_s) \mathbf{m} \langle \mathbf{1}, \mathbf{x}_\theta(\mathbf{z}_t, t) \rangle]}{\alpha_t \langle \mathbf{m}, \mathbf{x}_\theta(\mathbf{z}_t, t) \rangle + (1 - \alpha_t) \langle \mathbf{1}, \mathbf{x}_\theta(\mathbf{z}_t, t) \rangle} \right) \\
&= \text{Cat} \left(\mathbf{z}_s; \frac{[\alpha_{t|s} \mathbf{m} + (1 - \alpha_{t|s}) \mathbf{1}] \odot [\alpha_s \mathbf{x}_\theta(\mathbf{z}_t, t) + (1 - \alpha_s) \mathbf{m}]}{\alpha_t \langle \mathbf{x}_\theta(\mathbf{z}_t, t), \mathbf{m} \rangle + (1 - \alpha_t)} \right) \\
&= \text{Cat} \left(\mathbf{z}_s; \frac{\alpha_t \mathbf{m} \odot \mathbf{x}_\theta(\mathbf{z}_t, t) + (\alpha_s - \alpha_t) \mathbf{x}_\theta(\mathbf{z}_t, t) + (1 - \alpha_s) \mathbf{m}}{\alpha_t \langle \mathbf{x}_\theta(\mathbf{z}_t, t), \mathbf{m} \rangle + (1 - \alpha_t)} \right) \tag{B.13}
\end{aligned}$$

Note that the above categorical distribution, we can obtain the values for $p_\theta(\mathbf{z}_s = \mathbf{x} | \mathbf{z}_t = \mathbf{m})$ and $p_\theta(\mathbf{z}_s = \mathbf{m} | \mathbf{z}_t = \mathbf{m})$ which are as follows:

$$p_\theta(\mathbf{z}_s = \mathbf{x} | \mathbf{z}_t = \mathbf{m}) = \frac{(\alpha_s - \alpha_t) \langle \mathbf{x}_\theta(\mathbf{z}_t, t), \mathbf{x} \rangle}{\alpha_t \langle \mathbf{x}_\theta(\mathbf{z}_t, t), \mathbf{m} \rangle + (1 - \alpha_t)} \quad (\text{B.14})$$

$$p_\theta(\mathbf{z}_s = \mathbf{m} | \mathbf{z}_t = \mathbf{m}) = \frac{\alpha_s \langle \mathbf{x}_\theta(\mathbf{z}_t, t), \mathbf{m} \rangle + (1 - \alpha_s)}{\alpha_t \langle \mathbf{x}_\theta(\mathbf{z}_t, t), \mathbf{m} \rangle + (1 - \alpha_t)} \quad (\text{B.15})$$

As a sanity check, we can verify that Eq. B.14 reduces to Eq. B.9, and Eq. B.15 reduces to Eq. B.10 if our denoising network can reconstruct $\mathbf{x}$ perfectly, i.e., $\mathbf{x}_\theta(\mathbf{z}_t, t) = \mathbf{x}$.

Combining Eq. B.12 and Eq. B.13, we get the following expression for the reverse process parameterization:

$$p_\theta(\mathbf{z}_s | \mathbf{z}_t) = \begin{cases} \text{Cat}(\mathbf{z}_s; \mathbf{z}_t) & \mathbf{z}_t \neq \mathbf{m}, \\ \text{Cat}\left(\mathbf{z}_s; \frac{\alpha_t \mathbf{m} \odot \mathbf{x}_\theta(\mathbf{z}_t, t) + (\alpha_s - \alpha_t) \mathbf{x}_\theta(\mathbf{z}_t, t) + (1 - \alpha_s) \mathbf{m}}{\alpha_t \langle \mathbf{x}_\theta(\mathbf{z}_t, t), \mathbf{m} \rangle + (1 - \alpha_t)}\right) & \mathbf{z}_t = \mathbf{m}. \end{cases} \quad (\text{B.16})$$

Diffusion Loss

For a given T , Let $\mathcal{L}_T = \mathbb{E}_{t \in \{\frac{1}{T}, \frac{2}{T}, \dots, 1\}} \mathbb{E}_{q(\mathbf{z}_t | \mathbf{x})} T \text{D}_{\text{KL}}(q(\mathbf{z}_s | \mathbf{z}_t, \mathbf{x}) \| p_\theta(\mathbf{z}_s | \mathbf{z}_t))$ denote the diffusion loss. We break down the computation of $\text{D}_{\text{KL}}(q(\mathbf{z}_s | \mathbf{z}_t, \mathbf{x}) \| p_\theta(\mathbf{z}_s | \mathbf{z}_t))$ into 2 cases: $\mathbf{z}_t = \mathbf{x}$ (**case 1**) and $\mathbf{z}_t = \mathbf{m}$ (**case 2**).

Case 1: consider the case $\mathbf{z}_t = \mathbf{x}$. Let's simplify $\text{D}_{\text{KL}}(q(\mathbf{z}_s | \mathbf{z}_t = \mathbf{x}, \mathbf{x}) \| p_\theta(\mathbf{z}_s | \mathbf{z}_t = \mathbf{x}))$.

$$\begin{aligned} & \text{D}_{\text{KL}}(q(\mathbf{z}_s | \mathbf{z}_t = \mathbf{x}, \mathbf{x}) \| p_\theta(\mathbf{z}_s | \mathbf{z}_t = \mathbf{x})) \\ &= \text{D}_{\text{KL}}(\mathbf{z}_t \| \mathbf{z}_t) && \text{From Eq. B.11 and Eq. B.12} \\ &= 0 \end{aligned} \quad (\text{B.17})$$

Case 2: Consider the case $\mathbf{z}_t = \mathbf{m}$. Let's simplify $D_{\text{KL}}(q(\mathbf{z}_s|\mathbf{z}_t = \mathbf{m}, \mathbf{x})||p_\theta(\mathbf{z}_s|\mathbf{z}_t = \mathbf{m}))$.

$$\begin{aligned}
& D_{\text{KL}}(q(\mathbf{z}_s|\mathbf{z}_t = \mathbf{m}, \mathbf{x})||p_\theta(\mathbf{z}_s|\mathbf{z}_t = \mathbf{m})) \\
&= \sum_{\mathbf{z}_s} q(\mathbf{z}_s|\mathbf{z}_t = \mathbf{m}, \mathbf{x}) \log \frac{q(\mathbf{z}_s|\mathbf{z}_t = \mathbf{m}, \mathbf{x})}{p_\theta(\mathbf{z}_s|\mathbf{z}_t = \mathbf{m})} \\
&= \sum_{\mathbf{z}_s \in \{\mathbf{x}, \mathbf{m}\}} q(\mathbf{z}_s|\mathbf{z}_t = \mathbf{m}, \mathbf{x}) \log \frac{q(\mathbf{z}_s|\mathbf{z}_t = \mathbf{m}, \mathbf{x})}{p_\theta(\mathbf{z}_s|\mathbf{z}_t = \mathbf{m})} \\
&= \underbrace{q(\mathbf{z}_s = \mathbf{x}|\mathbf{z}_t = \mathbf{m}, \mathbf{x}) \log \frac{q(\mathbf{z}_s = \mathbf{x}|\mathbf{z}_t = \mathbf{m}, \mathbf{x})}{p_\theta(\mathbf{z}_s = \mathbf{x}|\mathbf{z}_t = \mathbf{m})}}_{\text{Simplify using Eq. B.9 and Eq. B.14}} \\
&\quad + \underbrace{q(\mathbf{z}_s = \mathbf{m}|\mathbf{z}_t = \mathbf{m}, \mathbf{x}) \log \frac{q(\mathbf{z}_s = \mathbf{m}|\mathbf{z}_t = \mathbf{m}, \mathbf{x})}{p_\theta(\mathbf{z}_s = \mathbf{m}|\mathbf{z}_t = \mathbf{m})}}_{\text{Simplify using Eq. B.10 and Eq. B.15}} \\
&= \frac{\alpha_s - \alpha_t}{1 - \alpha_t} \log \frac{\alpha_t \langle \mathbf{x}_\theta(\mathbf{z}_t, t), \mathbf{m} \rangle + (1 - \alpha_t)}{(1 - \alpha_t) \langle \mathbf{x}_\theta(\mathbf{z}_t, t), \mathbf{x} \rangle} \\
&\quad + \frac{1 - \alpha_s}{1 - \alpha_t} \log \frac{(1 - \alpha_s)(\alpha_t \langle \mathbf{x}_\theta(\mathbf{z}_t, t), \mathbf{m} \rangle + (1 - \alpha_t))}{(1 - \alpha_t)(\alpha_s \langle \mathbf{x}_\theta(\mathbf{z}_t, t), \mathbf{m} \rangle + (1 - \alpha_s))} \tag{B.18}
\end{aligned}$$

Thus, $D_{\text{KL}}(q(\mathbf{z}_s|\mathbf{z}_t, \mathbf{x})||p_\theta(\mathbf{z}_s|\mathbf{z}_t))$ can be written in the following manner where $\langle \mathbf{z}_t, \mathbf{x} \rangle$ evaluates to 1 if $\mathbf{z}_t = \mathbf{x}$ and $\langle \mathbf{z}_t, \mathbf{m} \rangle$ evaluates to 1 if $\mathbf{z}_t = \mathbf{m}$:

$$\begin{aligned}
& D_{\text{KL}}(q(\mathbf{z}_s|\mathbf{z}_t, \mathbf{x})||p_\theta(\mathbf{z}_s|\mathbf{z}_t)) \\
&= \underbrace{D_{\text{KL}}(q(\mathbf{z}_s|\mathbf{z}_t = \mathbf{x}, \mathbf{x})||p_\theta(\mathbf{z}_s|\mathbf{z}_t = \mathbf{x}))}_{=0, \text{ from Eq. B.17}} \langle \mathbf{z}_t, \mathbf{x} \rangle \\
&\quad + \underbrace{D_{\text{KL}}(q(\mathbf{z}_s|\mathbf{z}_t = \mathbf{m}, \mathbf{x})||p_\theta(\mathbf{z}_s|\mathbf{z}_t = \mathbf{m}))}_{\text{Given by Eq. B.18}} \langle \mathbf{z}_t, \mathbf{m} \rangle \tag{B.19}
\end{aligned}$$

Thus, we derive the diffusion loss, $\mathcal{L}_T$, in the following manner:

$$\begin{aligned}
\mathcal{L}_T &= \mathbb{E}_{t \in \{\frac{1}{T}, \frac{2}{T}, \dots, 1\}} \mathbb{E}_{q(\mathbf{z}_t | \mathbf{x})} T \text{D}_{\text{KL}}(q(\mathbf{z}_s | \mathbf{z}_t, \mathbf{x}) \| p_\theta(\mathbf{z}_s | \mathbf{z}_t)) \\
&= \mathbb{E}_{t \in \{\frac{1}{T}, \frac{2}{T}, \dots, 1\}} \mathbb{E}_{q(\mathbf{z}_t | \mathbf{x})} T \left[\frac{\alpha_s - \alpha_t}{1 - \alpha_t} \log \frac{\alpha_t \langle \mathbf{x}_\theta(\mathbf{z}_t, t), \mathbf{m} \rangle + (1 - \alpha_t)}{(1 - \alpha_t) \langle \mathbf{x}_\theta(\mathbf{z}_t, t), \mathbf{x} \rangle} \right. \\
&\quad \left. + \frac{1 - \alpha_s}{1 - \alpha_t} \log \frac{(1 - \alpha_s)(\alpha_t \langle \mathbf{x}_\theta(\mathbf{z}_t, t), \mathbf{m} \rangle + (1 - \alpha_t))}{(1 - \alpha_t)(\alpha_s \langle \mathbf{x}_\theta(\mathbf{z}_t, t), \mathbf{m} \rangle + (1 - \alpha_s))} \right] \langle \mathbf{z}_t, \mathbf{m} \rangle
\end{aligned} \tag{B.20}$$

Note that $\mathcal{L}_T$ is 0 if $\mathbf{z}_t$ is an unmasked token i.e. $\mathbf{z}_t = \mathbf{x}$.

NELBO

Austin et al. [2021], Sohl-Dickstein et al. [2015] model α_i as $(\alpha_i)_{i \in \{1, \dots, T\}} = 1 - \frac{i}{T}$ given latents $\mathbf{z}_{1, \dots, T}$. However, in this paper, we denote the latents as $\mathbf{z}_{t(0), \dots, t(T)}$; and hence, the $\alpha_{t(i)}$ are given as follows:

$$\begin{aligned}
(\alpha_i)_{i \in \{1, \dots, T\}} &= 1 - \frac{i}{T} \\
\implies (\alpha_i)_{i \in \{1, \dots, T+1\}} &= 1 - \frac{i}{T+1} && \text{For } T+1 \text{ latents} \\
\implies (\alpha_i)_{i \in \{0, \dots, T\}} &= 1 - \frac{i+1}{T+1} && \text{Offsetting the indices by 1.} \\
\implies (\alpha_{t(i)})_{i \in \{0, \dots, T\}} &= 1 - \frac{i+1}{T+1} && \text{Switching the notations from } \alpha_i \text{ to } \alpha_{t(i)}. \tag{B.21}
\end{aligned}$$

Consequently, from Equation B.21, we derive that

$$\alpha_{t(0)} = \frac{T}{T+1}, \tag{B.22}$$

$$\alpha_{t(T)} = 0. \tag{B.23}$$

Thus we have the following:

$$\mathbf{z}_{t(0)} \sim \text{Cat}(\cdot; \alpha_{t=0} \mathbf{x} + (1 - \alpha_{t=0}) \mathbf{m}) = \text{Cat}\left(\cdot; \frac{T}{T+1} \mathbf{x} + \frac{1}{T+1} \mathbf{m}\right), \tag{B.24}$$

$$q(\mathbf{z}_{t(T)} | \mathbf{x}) = \text{Cat}(\cdot; \alpha_{t=1} \mathbf{x} + (1 - \alpha_{t=1}) \mathbf{m}) = \text{Cat}(\cdot; \mathbf{m}), \tag{B.25}$$

$$p_\theta(\mathbf{z}_{t(T)}) = \text{Cat}(\cdot; \mathbf{m}) \tag{B.26}$$

The NELBO Eq. 2.10 simplifies to the following:

$$\begin{aligned}
& \mathbb{E}_q \left[-\log p_\theta(\mathbf{x}|\mathbf{z}_{t(0)}) + \underbrace{\mathcal{L}_T}_{\text{Compute using Eq. B.20}} \right] + \underbrace{D_{\text{KL}}[q(\mathbf{z}_{t(T)}|\mathbf{x})||p_\theta(\mathbf{z}_{t(T)})]}_{=0 \text{ using Eq. B.25 and Eq. B.26}} \\
&= \mathbb{E}_{q,t} \left[-\log p_\theta(\mathbf{x}|\mathbf{z}_{t(0)}) + T \left[\frac{\alpha_s - \alpha_t}{1 - \alpha_t} \log \frac{\alpha_t \langle \mathbf{x}_\theta(\mathbf{z}_t, t), \mathbf{m} \rangle + (1 - \alpha_t)}{(1 - \alpha_t) \langle \mathbf{x}_\theta(\mathbf{z}_t, t), \mathbf{x} \rangle} \right. \right. \\
&\quad \left. \left. + \frac{1 - \alpha_s}{1 - \alpha_t} \log \frac{(1 - \alpha_s)(\alpha_t \langle \mathbf{x}_\theta(\mathbf{z}_t, t), \mathbf{m} \rangle + (1 - \alpha_t))}{(1 - \alpha_t)(\alpha_s \langle \mathbf{x}_\theta(\mathbf{z}_t, t), \mathbf{m} \rangle + (1 - \alpha_s))} \right] \langle \mathbf{z}_t, \mathbf{m} \rangle \right] \quad (\text{B.27})
\end{aligned}$$

B.2 MDLM

In this section, we show how SUBS parameterization can simplify the functional form of the NELBO as defined in Eq. B.27.

B.2.1 Rao-Blackwellization

We employ the RB techniques as described in Sec. 4.2.1 to simplify the NELBO Eq. B.27 to Eq. B.29 using RB2, and further to Eq. B.31 using RB1.

Zero Masking Probabilities

Using “Zero Masking Probabilities” (RB2) from Sec. 4.2.1, we set $\langle \mathbf{x}_\theta(\mathbf{z}_t, t), \mathbf{m} \rangle = 0$ in Eq. B.20 to obtain the following simplified diffusion loss:

$$\begin{aligned}
\mathcal{L}_T^{\text{RB2}} &= \mathbb{E}_{t \in \{\frac{1}{T}, \frac{2}{T}, \dots, 1\}} \mathbb{E}_{q(\mathbf{z}_t|\mathbf{x})} T \left[\frac{\alpha_s - \alpha_t}{1 - \alpha_t} \log \frac{1}{\langle \mathbf{x}_\theta(\mathbf{z}_t, t), \mathbf{x} \rangle} \right] \langle \mathbf{z}_t, \mathbf{m} \rangle \\
&= \mathbb{E}_{t \in \{\frac{1}{T}, \frac{2}{T}, \dots, 1\}} \mathbb{E}_{q(\mathbf{z}_t|\mathbf{x})} T \left[\frac{\alpha_t - \alpha_s}{1 - \alpha_t} \log \langle \mathbf{x}_\theta(\mathbf{z}_t, t), \mathbf{x} \rangle \right] \langle \mathbf{z}_t, \mathbf{m} \rangle. \quad (\text{B.28})
\end{aligned}$$

The corresponding Rao-Blackwellized NELBO is given as:

$$\begin{aligned} & \mathbb{E}_q \left[-\log p_\theta(\mathbf{x}|\mathbf{z}_{t(0)}) + \underbrace{\mathcal{L}_T^{\text{RB2}}}_{\text{Compute using Eq. B.28}} \right] + \underbrace{D_{\text{KL}}[q(\mathbf{z}_{t(T)}|\mathbf{x})||p_\theta(\mathbf{z}_{t(T)})]}_{=0 \text{ using Eq. B.25 and Eq. B.26}} \\ &= \mathbb{E}_{q,t} \left[-\log p_\theta(\mathbf{x}|\mathbf{z}_{t(0)}) + T \left[\frac{\alpha_t - \alpha_s}{1 - \alpha_t} \log \langle \mathbf{x}_\theta(\mathbf{z}_t, t), \mathbf{x} \rangle \right] \langle \mathbf{z}_t, \mathbf{m} \rangle \right] \end{aligned} \quad (\text{B.29})$$

Carry Over Unmasking

Notice that the term $\langle \mathbf{z}_t, \mathbf{m} \rangle$ in Eq. B.28 is intended to reduce the diffusion loss to zero when $\mathbf{z}_t = \mathbf{x}$. Now, we will demonstrate that, by applying ‘‘Carry Over Unmasking’’ (RB1) from Sec. 4.2.1, $\langle \mathbf{z}_t, \mathbf{m} \rangle$ can be removed from Eq. B.28.

Recall that RB1 guarantees $\mathbf{x}_\theta(\mathbf{z}_t, t) = \mathbf{x}$ when $\mathbf{z}_t = \mathbf{x}$. Thus, with the RB1 parameterization, the diffusion loss in Eq. B.28 becomes zero for $\mathbf{z}_t = \mathbf{x}$, as $\log \langle \mathbf{x}_\theta(\mathbf{z}_t, t), \mathbf{x} \rangle = 0$. Consequently, $\langle \mathbf{z}_t, \mathbf{m} \rangle$ can be safely omitted from Eq. B.29, yielding the following diffusion loss:

$$\mathcal{L}_T^{\text{RB2} + \text{RB1}} = \mathbb{E}_{t \in \{\frac{1}{T}, \frac{2}{T}, \dots, 1\}} \mathbb{E}_{q(\mathbf{z}_t|\mathbf{x})} T \left[\frac{\alpha_t - \alpha_s}{1 - \alpha_t} \log \langle \mathbf{x}_\theta(\mathbf{z}_t, t), \mathbf{x} \rangle \right] \quad (\text{B.30})$$

NELBO

Thus, we have the following NELBO:

$$\begin{aligned} & \mathbb{E}_q \left[-\log p_\theta(\mathbf{x}|\mathbf{z}_{t(0)}) + \underbrace{\mathcal{L}_T^{\text{RB2} + \text{RB1}}}_{\text{Compute using Eq. B.30}} \right] + \underbrace{D_{\text{KL}}[q(\mathbf{z}_{t(T)}|\mathbf{x})||p_\theta(\mathbf{z}_{t(T)})]}_{=0 \text{ using Eq. B.25 and Eq. B.26}} \\ &= \mathbb{E}_{q,t} \left[-\log p_\theta(\mathbf{x}|\mathbf{z}_{t(0)}) + T \left[\frac{\alpha_t - \alpha_s}{1 - \alpha_t} \log \langle \mathbf{x}_\theta(\mathbf{z}_t, t), \mathbf{x} \rangle \right] \right] \end{aligned} \quad (\text{B.31})$$

Comparing Eq. B.31 and Eq. B.29. Note that due to RB1, $\log p_\theta(\mathbf{x}|\mathbf{z}_{t(0)})$ in Eq. B.31 reduces to 0 every time $\mathbf{z}_{t(0)} = \mathbf{x}$ as explained in Eq. B.33. However,

this is not the case in Eq. B.29, even though it has a functionally similar expression to Eq. B.31. Because of this reason Eq. B.31 should lead to a better likelihood estimate and we empirically verify this in Table 4.8.

B.2.2 Continuous Time

Diffusion Loss

To derive the continuous-time diffusion loss, $\mathcal{L}_{\text{diffusion}}^\infty$, we consider the limiting case $\lim_{T \rightarrow \infty} \mathcal{L}_T^{\text{RB2} + \text{RB1}}$ Eq. B.30:

$$\begin{aligned}
& \mathcal{L}_{\text{diffusion}}^\infty \\
&= \lim_{T \rightarrow \infty} \mathcal{L}_T^{\text{RB2} + \text{RB1}} \\
&= \mathbb{E}_{t \in \{\frac{1}{T}, \frac{2}{T}, \dots, 1\}, q(\mathbf{z}_t | \mathbf{x})} \left[\lim_{T \rightarrow \infty} T \frac{\alpha_t - \alpha_s}{1 - \alpha_t} \log \langle \mathbf{x}_\theta(\mathbf{z}_t, t), \mathbf{x} \rangle \right] \\
&= \mathbb{E}_{t \sim \mathcal{U}[0,1], q(\mathbf{z}_t | \mathbf{x})} \left[\frac{\alpha'_t}{1 - \alpha_t} \log \langle \mathbf{x}_\theta(\mathbf{z}_t, t), \mathbf{x} \rangle \right] \quad \text{Using } \lim_{T \rightarrow \infty} T(\alpha_t - \alpha_s) = \alpha'_t
\end{aligned} \tag{B.32}$$

Reconstruction Loss

For the continuous time case, from Eq. B.24, we have

$$\begin{aligned}
\mathbf{z}_{t(0)} &\sim \lim_{T \rightarrow \infty} \text{Cat} \left(\cdot; \frac{T}{T+1} \mathbf{x} + \frac{1}{T+1} \mathbf{m} \right) \\
&\implies \mathbf{z}_{t(0)} \sim \text{Cat}(\cdot; \mathbf{x}) \\
&\implies \mathbf{z}_{t(0)} = \mathbf{x}
\end{aligned} \tag{B.33}$$

Thus, the reconstruction loss reduces to 0 in the following manner:

$$\begin{aligned}
\mathcal{L}_{\text{recons}} &= -\log p_{\theta}(\mathbf{x}|\mathbf{z}_{t(0)}) \\
&= -\log p_{\theta}(\mathbf{x}|\mathbf{z}_{t(0)} = \mathbf{x}) && \text{From Eq. B.33} \\
&= -\log \langle \mathbf{x}_{\theta}(\mathbf{x}, t(0)), \mathbf{x} \rangle \\
&= -\log \langle \mathbf{x}, \mathbf{x} \rangle && \text{Due to "carry-over unmasking" } \mathbf{x}_{\theta}(\mathbf{x}, t(0)) = \mathbf{x} \\
&= 0. && \text{(B.34)}
\end{aligned}$$

NELBO

Thus, we have the following NELBO:

$$\begin{aligned}
&\mathbb{E}_q \left[\underbrace{-\log p_{\theta}(\mathbf{x}|\mathbf{z}_{t(0)})}_{= 0 \text{ from Eq. B.34}} + \underbrace{\mathcal{L}_{\text{diffusion}}^{\infty}}_{\text{Compute using Eq. B.30}} \right] + \underbrace{D_{\text{KL}}[q(\mathbf{z}_{t(T)}|\mathbf{x})||p_{\theta}(\mathbf{z}_{t(T)}, t)]}_{= 0 \text{ using Eq. B.25 and Eq. B.26}} \\
&= \mathbb{E}_{q,t} \left[\frac{\alpha'_t}{1 - \alpha_t} \log \langle \mathbf{x}_{\theta}(\mathbf{z}_t, t), \mathbf{x} \rangle \right] && \text{(B.35)}
\end{aligned}$$

B.2.3 Final Algorithm

In Algorithm 2, we present the training algorithm for MDLM.

B.3 Concrete Score Matching

In the previous section, we defined the discrete diffusion process as a Discrete-Time Markov Chain (DTMC) with a finite set of T states, $\mathbf{z}_{\{0, \frac{1}{T}, \dots, 1\}}$, and a state transition matrix Q_t . To derive the continuous-time ELBO, we simply take the limit as $T \rightarrow \infty$.

Algorithm 2 Training MDLM

repeat

$\mathbf{x}^{1:L} \sim q(\mathbf{x})$ ▷ Sample a sentence.

$t \sim \mathcal{U}[0, 1]$ ▷ Sample a time step.

$\mathbf{z}_t^\ell \sim \text{Cat}(\mathbf{z}_t^\ell; \alpha_t \mathbf{x}^\ell + (1 - \alpha_t) \mathbf{m}) \forall 1 \leq \ell \leq L$

 ▷ Mask Each token $\mathbf{x}^\ell$ independently to obtain the latent $\mathbf{z}_t^{1:L}$.

Take gradient descent step on

$$\nabla_{\theta} \frac{\alpha'_t}{1 - \alpha_t} \sum_{\ell} \log \langle \mathbf{x}_{\theta}^{\ell}(\mathbf{z}_t^{1:L}, t), \mathbf{x}^{\ell} \rangle$$

until converged

In contrast, [Campbell et al. \[2022\]](#) and [Lou et al. \[2024\]](#) defined the discrete diffusion process as a Continuous-Time Markov Chain (CTMC), where the forward corruption process is specified by the rate change matrix $R_t \in \mathbb{R}^{|\mathcal{V}| \times |\mathcal{V}|}$. The transition rate matrix entry $R_t(\mathbf{y}', \mathbf{y})$ can be thought of as the instantaneous rate at which one state $\mathbf{y}$ transitions to another state $\mathbf{y}'$. With this formulation, the forward posterior $q(\mathbf{z}_t | \mathbf{z}_s)$ and the true reverse posterior $q(\mathbf{z}_s | \mathbf{z}_t, \mathbf{x})$ can be expressed in terms of the rate change matrix as follows:

$$q(\mathbf{z}_t = \mathbf{y}' | \mathbf{z}_s = \mathbf{y}) = \delta_{\mathbf{y}', \mathbf{y}} + R_t(\mathbf{y}', \mathbf{y}) \frac{1}{T} + \mathcal{O}\left(\frac{1}{T^2}\right) \quad (\text{B.36})$$

$$q(\mathbf{z}_s = \mathbf{y}' | \mathbf{z}_t = \mathbf{y}, \mathbf{x}) = \delta_{\mathbf{y}', \mathbf{y}} + \tilde{R}_t(\mathbf{y}', \mathbf{y}) \frac{1}{T} + \mathcal{O}\left(\frac{1}{T^2}\right) \quad (\text{B.37})$$

where δ is the Kroenecker delta and $\mathcal{O}(\frac{1}{T^2})$ represents higher order terms of $\frac{1}{T^2}$. In Sec. [B.3.1](#), we show how to express the rate matrix R_t in terms of the state transition matrix Q_t . [Lou et al. \[2024\]](#) propose a continuous time ELBO for this process. They mention that this expression can be derived from [Benton et al. \[2022\]](#), though they do not provide an explicit derivation. For this reason, we present a rigorous derivation in Sec. [B.3.2](#) and further demonstrate that, under the SUBS parameterization in Sec. [4.2.1](#), this formula reduces to our proposed continuous-time ELBO, given by Eq. [4.7](#).

For the remainder of this section, we switch to the notation $q_{t|s}(\mathbf{y}'|\mathbf{y})$ to denote $q(\mathbf{z}_t = \mathbf{y}'|\mathbf{z}_s = \mathbf{y})$, $q_{s|t}(\mathbf{y}'|\mathbf{y})$ for $q(\mathbf{z}_s = \mathbf{y}'|\mathbf{z}_t = \mathbf{y})$, and $q(\mathbf{z}_t = \mathbf{y}|\mathbf{x})$ for $q_t(\mathbf{y}|\mathbf{x})$, aligning with the notation typically used in the CTMC literature.

B.3.1 Extracting the Rate Matrix

Here, we aim to express the rate change matrix R_t in terms of the state transition matrix Q_t . To do this, we first represent the forward transition $q_{t|s}$ in terms of Q_t and R_t separately, allowing us to illustrate their relationship.

Using Eq. B.1, we can write $q_{t|s}$ as follows:

$$\begin{aligned} q_{t|s}(\mathbf{y}'|\mathbf{y}) &= [\mathbf{y}']^\top [\alpha_{t|s} \mathbf{I} + (1 - \alpha_{t|s}) \mathbf{1} \mathbf{m}^\top]^\top \mathbf{y} \\ &= [\mathbf{y}']^\top [\alpha_{t|s} \mathbf{y} + (1 - \alpha_{t|s}) \mathbf{m} \mathbf{1}^\top \mathbf{y}] \\ &= [\mathbf{y}']^\top [\alpha_{t|s} \mathbf{y} + (1 - \alpha_{t|s}) \mathbf{m}] \\ &= \alpha_{t|s} [\mathbf{y}']^\top \mathbf{y} + (1 - \alpha_{t|s}) [\mathbf{y}']^\top \mathbf{m} \end{aligned} \tag{B.38}$$

Now let's analyze all possible combinations for the tuple $(\mathbf{y}', \mathbf{y})$:

1. Case $(\mathbf{y}' = \mathbf{x}, \mathbf{y} = \mathbf{x})$: Using Eq. B.38, we find that $q_{t|s}(\mathbf{x}|\mathbf{x}) = \alpha_{t|s}$ for the DTMC. By Eq. B.36, we have $q_{t|s}(\mathbf{x}|\mathbf{x}) = 1 + R_t(\mathbf{x}, \mathbf{x}) \frac{1}{T}$ as $T \rightarrow \infty$, since the higher-order terms $\mathcal{O}\left(\frac{1}{T^2}\right)$ vanish in the limit. Thus, we get:

$$\begin{aligned} \lim_{T \rightarrow \infty} \left[1 + R_t(\mathbf{x}, \mathbf{x}) \frac{1}{T} \right] &= \lim_{T \rightarrow \infty} \alpha_{t|s} \\ \implies R_t(\mathbf{x}, \mathbf{x}) &= \lim_{T \rightarrow \infty} T(\alpha_{t|s} - 1) = \frac{\alpha'_t}{\alpha_t} \end{aligned} \tag{B.39}$$

2. Case ($\mathbf{y}' = \mathbf{m}, \mathbf{y} \in \mathcal{V} - \{\mathbf{m}\}$): Similarly, using Eq. B.38 and Eq. B.36, we have $q_{t|s}(\mathbf{m}|\mathbf{y} \neq \mathbf{m}) = 1 - \alpha_{t|s}$ and $q_{t|s}(\mathbf{x}|\mathbf{y} \neq \mathbf{m}) = R_t(\mathbf{m}, \mathbf{y} \neq \mathbf{m})\frac{1}{T}$. Thus,

$$\begin{aligned} \lim_{T \rightarrow \infty} \left[R_t(\mathbf{m}, \mathbf{y} \neq \mathbf{m}) \frac{1}{T} \right] &= \lim_{T \rightarrow \infty} [1 - \alpha_{t|s}] \\ \implies R_t(\mathbf{m}, \mathbf{y} \neq \mathbf{m}) &= \lim_{T \rightarrow \infty} T(1 - \alpha_{t|s}) = -\frac{\alpha'_t}{\alpha_t} \end{aligned} \quad (\text{B.40})$$

3. Case ($\mathbf{y}' = \mathbf{m}, \mathbf{y} = \mathbf{m}$): Using Eq. B.38 and Eq. B.36, we find $q_{t|s}(\mathbf{m}|\mathbf{m}) = 1$ and $q_{t|s}(\mathbf{m}|\mathbf{m}) = 1 + R_t(\mathbf{m}, \mathbf{m})\frac{1}{T} + \mathcal{O}\left(\frac{1}{T^2}\right)$. Since these two expressions must be equal for any T , it follows that

$$R_t(\mathbf{m}, \mathbf{m}) = 0. \quad (\text{B.41})$$

Note that when $R_t(\mathbf{m}, \mathbf{m})$ is constant, the term $\mathcal{O}\left(\frac{1}{T^2}\right)$ reduces to zero, as it includes higher-order time derivatives of R_t .

4. Case ($\mathbf{y}' = \mathbf{x}, \mathbf{y} \in \mathcal{V} - \{\mathbf{x}\}$): In the context of absorbing state diffusion, these states are never observed. Thus,

$$R_t(\mathbf{y}' = \mathbf{x}, \mathbf{y} \in \mathcal{V} - \{\mathbf{m}, \mathbf{x}\}) = 0 \quad (\text{B.42})$$

5. Case ($\mathbf{y}' \in \mathcal{V} - \{\mathbf{m}, \mathbf{x}\}, \mathbf{y} \in \{\mathbf{m}, \mathbf{x}\}$): In the context of absorbing state diffusion, these states are never observed. Thus,

$$R_t(\mathbf{y}' \in \mathcal{V} - \{\mathbf{m}, \mathbf{x}\}, \mathbf{y} \in \{\mathbf{m}, \mathbf{x}\}) = 0 \quad (\text{B.43})$$

Finally, we can express the forward rate matrix as:

$$\boxed{R_t = \frac{\alpha'_t}{\alpha_t} (\mathbf{I} - \mathbf{m}\mathbf{1}^\top)} \quad (\text{B.44})$$

It can be seen that the columns of this matrix sum to zero, i.e.,

$$\sum_{\mathbf{y}' \in \mathcal{V}} R_t(\mathbf{y}', \mathbf{y}) = 0 \implies R_t(\mathbf{y}, \mathbf{y}) = \sum_{\mathbf{y}' \neq \mathbf{y}} R_t(\mathbf{y}', \mathbf{y}), \quad (\text{B.45})$$

which ensures that the probability mass is preserved in the forward diffusion process.

Similarly, the reverse rate matrix $\tilde{R}_t$ can be written in terms of the forward rate matrix R_t as follows [Lou et al., 2024]:

$$\tilde{R}_t(\mathbf{y}', \mathbf{y}) = \begin{cases} \frac{q_t(\mathbf{y}')}{q_t(\mathbf{y})} R_t(\mathbf{y}, \mathbf{y}') & \mathbf{y}' \neq \mathbf{y} \\ -\sum_{\tilde{\mathbf{y}} \neq \mathbf{y}} \frac{q_t(\tilde{\mathbf{y}})}{q_t(\mathbf{y})} R_t(\mathbf{y}, \tilde{\mathbf{y}}) & \mathbf{y}' = \mathbf{y}. \end{cases} \quad (\text{B.46})$$

where $q_t(\cdot)$ denotes the marginal distribution of the data distribution p_{data} at diffusion timestep t .

B.3.2 NELBO

Meng et al. [2022] introduced the term “concrete score” for the term $q_t(\mathbf{y}')/q_t(\mathbf{y})$ that appears in $\tilde{R}_t$. Since this quantity is not directly accessible in the reverse diffusion process, we approximate it using a neural network, $\mathbf{s}_\theta : \mathcal{V} \rightarrow \mathcal{V}$, with parameters θ . The approximate reverse posterior $p_{s|t}^\theta$ can then be expressed in terms of the approximate reverse rate matrix $\tilde{R}_t$ in the following manner:

$$p_{s|t}^\theta(\mathbf{y}'|\mathbf{y}) = \delta_{\mathbf{y}', \mathbf{y}} + \tilde{R}_t^\theta(\mathbf{y}', \mathbf{y}) \frac{1}{T} + \mathcal{O}\left(\frac{1}{T^2}\right) \quad (\text{B.47})$$

$$\tilde{R}_t^\theta(\mathbf{y}', \mathbf{y}) = \begin{cases} \mathbf{s}_\theta(\mathbf{y})_{\mathbf{y}'} R_t(\mathbf{y}, \mathbf{y}') & \mathbf{y}' \neq \mathbf{y} \\ -\sum_{\tilde{\mathbf{y}} \neq \mathbf{y}} \mathbf{s}_\theta(\mathbf{y})_{\tilde{\mathbf{y}}} R_t(\mathbf{y}, \tilde{\mathbf{y}}) & \mathbf{y}' = \mathbf{y} \end{cases} \quad (\text{B.48})$$

where $\mathbf{s}_\theta(\mathbf{y})_{\mathbf{y}'}$ denotes the approximate concrete score $q_t(\mathbf{y}')/q_t(\mathbf{y})$. Lou et al. [2024]

propose the following NELBO to train such a model:

$$\mathbb{E}_{t \in [0,1], \mathbf{y} \sim q_t(\cdot|\mathbf{x})} \left[\sum_{\mathbf{y}' \neq \mathbf{y}} R_t(\mathbf{y}, \mathbf{y}') \left(\mathbf{s}_\theta(\mathbf{y})_{\mathbf{y}'} - \frac{q_t(\mathbf{y}'|\mathbf{x})}{q_t(\mathbf{y}|\mathbf{x})} \log \mathbf{s}_\theta(\mathbf{y})_{\mathbf{y}'} + K \left(\frac{q_t(\mathbf{y}'|\mathbf{x})}{q_t(\mathbf{y}|\mathbf{x})} \right) \right) \right] \quad (\text{B.49})$$

where $K(a) = a \log a - a$. They mention that this expression can be derived from [Benton et al. \[2022\]](#), though they do not provide an explicit derivation. For this reason, we present a rigorous derivation in the following section.

Proof. Let's focus on the diffusion loss for this process. As mentioned in the previous section, the reconstruction and prior loss terms reduce to zero. The continuous-time diffusion loss is given by:

$$\begin{aligned}
& \lim_{T \rightarrow \infty} T \mathbb{E}_{t \in \{\frac{1}{T}, \frac{2}{T}, \dots, 1\}, \mathbf{y} \sim q_t(\cdot | \mathbf{x})} [\text{D}_{\text{KL}}(q_{s|t}(\mathbf{y}' | \mathbf{y}, \mathbf{x}) \| p_{s|t}(\mathbf{y}' | \mathbf{y}))] \\
&= \lim_{T \rightarrow \infty} T \mathbb{E}_{t \in \{\frac{1}{T}, \frac{2}{T}, \dots, 1\}, \mathbf{y} \sim q_t(\cdot | \mathbf{x})} \left[\sum_{\mathbf{y}'} q_{s|t}(\mathbf{y}' | \mathbf{y}, \mathbf{x}) \log \frac{q_{s|t}(\mathbf{y}' | \mathbf{y}, \mathbf{x})}{p_{s|t}(\mathbf{y}' | \mathbf{y}, \mathbf{x})} \right] \\
&= \mathbb{E}_{t \in [0, 1], \mathbf{y} \sim q_t(\cdot | \mathbf{x})} \left[\lim_{T \rightarrow \infty} T \sum_{\mathbf{y}'} q_{s|t}(\mathbf{y}' | \mathbf{y}, \mathbf{x}) \log \frac{q_{s|t}(\mathbf{y}' | \mathbf{y}, \mathbf{x})}{p_{s|t}(\mathbf{y}' | \mathbf{y}, \mathbf{x})} \right] \\
&= \mathbb{E}_{t \in [0, 1], \mathbf{y} \sim q_t(\cdot | \mathbf{x})} \left[\underbrace{\lim_{T \rightarrow \infty} T q_{s|t}(\mathbf{y} | \mathbf{y}, \mathbf{x}) \log \frac{q_{s|t}(\mathbf{y} | \mathbf{y}, \mathbf{x})}{p_{s|t}(\mathbf{y} | \mathbf{y}, \mathbf{x})}}_{\text{Term 1}} + \underbrace{\lim_{T \rightarrow \infty} T \sum_{\mathbf{y}' \neq \mathbf{y}} q_{s|t}(\mathbf{y}' | \mathbf{y}, \mathbf{x}) \log \frac{q_{s|t}(\mathbf{y}' | \mathbf{y}, \mathbf{x})}{p_{s|t}(\mathbf{y}' | \mathbf{y}, \mathbf{x})}}_{\text{Term 2}} \right] \quad (\text{B.50})
\end{aligned}$$

Let's simplify these two terms separately. For the derivation, we'll rely on two key observations: In the limiting case as $T \rightarrow \infty$, it follows from Eq. [B.37](#) that $\lim_{T \rightarrow \infty} q_{s|t}(\mathbf{y} | \mathbf{y}, \mathbf{x}) = 1$ and from Eq. [B.47](#) that $\lim_{T \rightarrow \infty} p_{s|t}^\theta(\mathbf{y} | \mathbf{y}, \mathbf{x}) = 1$.

Term 1:

$$\lim_{T \rightarrow \infty} T q_{s|t}(\mathbf{y}|\mathbf{y}, \mathbf{x}) \log \frac{q_{s|t}(\mathbf{y}|\mathbf{y}, \mathbf{x})}{p_{s|t}(\mathbf{y}|\mathbf{y}, \mathbf{x})}$$

$\because \lim_{T \rightarrow \infty} q_{s|t}(\mathbf{y}|\mathbf{y}, \mathbf{x}) = 1$; hence,

$$= \lim_{T \rightarrow \infty} T \log \frac{q_{s|t}(\mathbf{y}|\mathbf{y}, \mathbf{x})}{p_{s|t}(\mathbf{y}|\mathbf{y}, \mathbf{x})}$$

The above term is in $\infty \times 0$ indeterminate form; therefore,

$$= \lim_{T \rightarrow \infty} T \left(\log q_{s|t}(\mathbf{y}|\mathbf{y}, \mathbf{x}) - \log p_{s|t}^\theta(\mathbf{y}|\mathbf{y}, \mathbf{x}) \right)$$

Substituting $q_{s|t}$ and $p_{s|t}^\theta$ from Eq. B.37 and Eq. B.47, we get:

$$= \lim_{T \rightarrow \infty} T \left[\log \left(1 + \tilde{R}_t(\mathbf{y}, \mathbf{y}) \frac{1}{T} + \mathcal{O} \left(\frac{1}{T^2} \right) \right) - \log \left(1 + \tilde{R}_t^\theta(\mathbf{y}, \mathbf{y}) \frac{1}{T} + \mathcal{O} \left(\frac{1}{T^2} \right) \right) \right]$$

Applying the Taylor series expansion for $\log(1+x)$, we get:

$$\begin{aligned} &= \lim_{T \rightarrow \infty} T \left[\tilde{R}_t(\mathbf{y}, \mathbf{y}) \frac{1}{T} + \mathcal{O} \left(\frac{1}{T^2} \right) - \tilde{R}_t^\theta(\mathbf{y}, \mathbf{y}) \frac{1}{T} - \mathcal{O} \left(\frac{1}{T^2} \right) \right] \\ &= \tilde{R}_t(\mathbf{y}, \mathbf{y}) + \lim_{T \rightarrow \infty} T \mathcal{O} \left(\frac{1}{T^2} \right) - \tilde{R}_t^\theta(\mathbf{y}, \mathbf{y}) - \lim_{T \rightarrow \infty} T \mathcal{O} \left(\frac{1}{T^2} \right) \end{aligned}$$

$\because \lim_{T \rightarrow \infty} T \mathcal{O} \left(\frac{1}{T^2} \right) = 0$, we get:

$$= \tilde{R}_t(\mathbf{y}, \mathbf{y}) - \tilde{R}_t^\theta(\mathbf{y}, \mathbf{y})$$

Using Eq. B.46 and Eq. B.48, we get:

$$= - \sum_{\mathbf{y}' \neq \mathbf{y}} R_t(\mathbf{y}, \mathbf{y}') \left(\frac{q_t(\mathbf{y}'|\mathbf{x})}{q_t(\mathbf{y}|\mathbf{x})} - \mathbf{s}_\theta(\mathbf{y})_{\mathbf{y}'} \right) \quad (\text{B.51})$$

Term 2:

$$\lim_{T \rightarrow \infty} T \sum_{\mathbf{y}' \neq \mathbf{y}} q_{s|t}(\mathbf{y}'|\mathbf{y}, \mathbf{x}) \log \frac{q_{s|t}(\mathbf{y}'|\mathbf{y}, \mathbf{x})}{p_{s|t}(\mathbf{y}'|\mathbf{y}, \mathbf{x})}$$

Substituting $q_{s|t}$ and $p_{s|t}^\theta$ from Eq. B.37 and Eq. B.47, we get:

$$\begin{aligned} &= \lim_{T \rightarrow \infty} T \sum_{\mathbf{y}' \neq \mathbf{y}} \left[\left[\frac{q_s(\mathbf{y}'|\mathbf{x})}{q_t(\mathbf{y}|\mathbf{x})} R_t(\mathbf{y}, \mathbf{y}') \frac{1}{T} + \mathcal{O}\left(\frac{1}{T^2}\right) \right] \log \frac{\frac{q_s(\mathbf{y}'|\mathbf{x})}{q_t(\mathbf{y}|\mathbf{x})} R_t(\mathbf{y}, \mathbf{y}') \frac{1}{T} + \mathcal{O}\left(\frac{1}{T^2}\right)}{\mathbf{s}_\theta(\mathbf{y})_{\mathbf{y}'} R_t(\mathbf{y}, \mathbf{y}') \frac{1}{T} + \mathcal{O}\left(\frac{1}{T^2}\right)} \right] \\ &= \lim_{T \rightarrow \infty} \sum_{\mathbf{y}' \neq \mathbf{y}} \left[\left[\frac{q_s(\mathbf{y}'|\mathbf{x})}{q_t(\mathbf{y}|\mathbf{x})} R_t(\mathbf{y}, \mathbf{y}') + T \mathcal{O}\left(\frac{1}{T^2}\right) \right] \log \frac{\frac{q_s(\mathbf{y}'|\mathbf{x})}{q_t(\mathbf{y}|\mathbf{x})} R_t(\mathbf{y}, \mathbf{y}') + T \mathcal{O}\left(\frac{1}{T^2}\right)}{\mathbf{s}_\theta(\mathbf{y})_{\mathbf{y}'} R_t(\mathbf{y}, \mathbf{y}') + T \mathcal{O}\left(\frac{1}{T^2}\right)} \right] \\ &\because \lim_{T \rightarrow \infty} T \mathcal{O}\left(\frac{1}{T^2}\right) = 0, \text{ we get:} \\ &= \sum_{\mathbf{y}' \neq \mathbf{y}} \frac{q_s(\mathbf{y}'|\mathbf{x})}{q_t(\mathbf{y}|\mathbf{x})} R_t(\mathbf{y}, \mathbf{y}') \log \frac{\frac{q_s(\mathbf{y}'|\mathbf{x})}{q_t(\mathbf{y}|\mathbf{x})} R_t(\mathbf{y}, \mathbf{y}')}{\mathbf{s}_\theta(\mathbf{y})_{\mathbf{y}'} R_t(\mathbf{y}, \mathbf{y}')} \\ &= \sum_{\mathbf{y}' \neq \mathbf{y}} \frac{q_t(\mathbf{y}'|\mathbf{x})}{q_t(\mathbf{y}|\mathbf{x})} R_t(\mathbf{y}, \mathbf{y}') \left(\log \frac{q_t(\mathbf{y}'|\mathbf{x})}{q_t(\mathbf{y}|\mathbf{x})} - \log \mathbf{s}_\theta(\mathbf{y})_{\mathbf{y}'} \right) \end{aligned} \quad (\text{B.52})$$

Finally, plugging Eq. B.51 and Eq. B.52 into Eq. B.50 yields us the NELBO as proposed in Lou et al. [2024]:

$$\begin{aligned} &\mathbb{E}_{t \in [0,1], \mathbf{y} \sim q_t(\cdot|\mathbf{x})} \left[- \sum_{\mathbf{y}' \neq \mathbf{y}} R_t(\mathbf{y}, \mathbf{y}') \left(\frac{q_t(\mathbf{y}'|\mathbf{x})}{q_t(\mathbf{y}|\mathbf{x})} - \mathbf{s}_\theta(\mathbf{y})_{\mathbf{y}'} \right) \right. \\ &\quad \left. + \sum_{\mathbf{y}' \neq \mathbf{y}} \frac{q_t(\mathbf{y}'|\mathbf{x})}{q_t(\mathbf{y}|\mathbf{x})} R_t(\mathbf{y}, \mathbf{y}') \left(\log \frac{q_t(\mathbf{y}'|\mathbf{x})}{q_t(\mathbf{y}|\mathbf{x})} - \log \mathbf{s}_\theta(\mathbf{y})_{\mathbf{y}'} \right) \right] \\ &= \mathbb{E}_{t \in [0,1], \mathbf{y} \sim q_t(\cdot|\mathbf{x})} \left[\sum_{\mathbf{y}' \neq \mathbf{y}} R_t(\mathbf{y}, \mathbf{y}') \left(- \frac{q_t(\mathbf{y}'|\mathbf{x})}{q_t(\mathbf{y}|\mathbf{x})} + \mathbf{s}_\theta(\mathbf{y})_{\mathbf{y}'} \right. \right. \\ &\quad \left. \left. + \frac{q_t(\mathbf{y}'|\mathbf{x})}{q_t(\mathbf{y}|\mathbf{x})} \log \frac{q_t(\mathbf{y}'|\mathbf{x})}{q_t(\mathbf{y}|\mathbf{x})} - \frac{q_t(\mathbf{y}'|\mathbf{x})}{q_t(\mathbf{y}|\mathbf{x})} \log \mathbf{s}_\theta(\mathbf{y})_{\mathbf{y}'} \right) \right] \\ &= \mathbb{E}_{t \in [0,1], \mathbf{y} \sim q_t(\cdot|\mathbf{x})} \left[\sum_{\mathbf{y}' \neq \mathbf{y}} R_t(\mathbf{y}, \mathbf{y}') \left(\mathbf{s}_\theta(\mathbf{y})_{\mathbf{y}'} - \frac{q_t(\mathbf{y}'|\mathbf{x})}{q_t(\mathbf{y}|\mathbf{x})} \log \mathbf{s}_\theta(\mathbf{y})_{\mathbf{y}'} + K \left(\frac{q_t(\mathbf{y}'|\mathbf{x})}{q_t(\mathbf{y}|\mathbf{x})} \right) \right) \right] \end{aligned} \quad (\text{B.53})$$

where $K(a) = a \log a - a$. This concludes the proof.

B.3.3 Concrete Score for MDLM

Given a latent variable $\mathbf{z}_t$ and the output of the denoising model, $\mathbf{x}_\theta(\mathbf{z}_t, t)$ parameterized using SUBS, we aim to recover the concrete score $\mathbf{s}_\theta(\mathbf{z}_t) \in (\mathbb{R}^+ + \{0\})^{|\mathcal{V}|}$. Note that $\mathbf{s}_\theta(\mathbf{z}_t)_\mathbf{y}$ is the ratio $\frac{p_t(\mathbf{y})}{p_t(\mathbf{z}_t)}$ in the reverse process. Since $\mathbf{x}_\theta$ approximates $\mathbf{x}$, we use $p_t(\mathbf{y}) = q_t(\mathbf{y}|\mathbf{x}_\theta(\mathbf{z}_t, t))$; therefore,

$$\mathbf{s}_\theta(\mathbf{z}_t)_\mathbf{y} = \frac{p_t(\mathbf{y})}{p_t(\mathbf{z}_t)} = \frac{q_t(\mathbf{y}|\mathbf{x}_\theta(\mathbf{z}_t, t))}{q_t(\mathbf{z}_t|\mathbf{x}_\theta(\mathbf{z}_t, t))}. \quad (\text{B.54})$$

To obtain the score, we first compute $q_t(\mathbf{y}|\mathbf{x}_\theta(\mathbf{z}_t, t))$ for all possible $\mathbf{y}$ and $\mathbf{z}_t$. Using Eq. 4.1, we derive the following expressions under the SUBS parameterization:

$$q_t(\mathbf{y} \neq \mathbf{m}|\mathbf{x}_\theta(\mathbf{z}_t = \mathbf{m}, t)) = \alpha_t \langle \mathbf{x}_\theta(\mathbf{z}_t, t), \mathbf{y} \rangle \quad (\text{B.55})$$

$$q_t(\mathbf{y} \notin \{\mathbf{m}, \mathbf{z}_t\}|\mathbf{x}_\theta(\mathbf{z}_t \neq \mathbf{m}, t)) = 0 \quad (\text{B.56})$$

$$q_t(\mathbf{y} = \mathbf{z}_t|\mathbf{x}_\theta(\mathbf{z}_t \neq \mathbf{m}, t)) = \alpha_t \quad (\text{B.57})$$

$$q_t(\mathbf{y} = \mathbf{m}|\mathbf{x}_\theta(\mathbf{z}_t \in \mathcal{V}, t)) = 1 - \alpha_t \quad (\text{B.58})$$

Plugging these into Eq. B.54, we get:

$$\mathbf{s}_\theta(\mathbf{z}_t = \mathbf{m})_{\mathbf{y} \neq \mathbf{m}} = \frac{\alpha_t}{1 - \alpha_t} \langle \mathbf{x}_\theta(\mathbf{z}_t, t), \mathbf{y} \rangle \quad \text{Using Eq. B.58 and Eq. B.55} \quad (\text{B.59})$$

$$\mathbf{s}_\theta(\mathbf{z}_t = \mathbf{m})_{\mathbf{y} = \mathbf{m}} = 1 \quad \text{Using Eq. B.58} \quad (\text{B.60})$$

$$\mathbf{s}_\theta(\mathbf{z}_t \neq \mathbf{m})_{\mathbf{y} = \mathbf{m}} = \frac{1 - \alpha_t}{\alpha_t} \quad \text{Using Eq. B.57 and Eq. B.58} \quad (\text{B.61})$$

$$\mathbf{s}_\theta(\mathbf{z}_t \neq \mathbf{m})_{\mathbf{y} = \mathbf{z}_t} = 1 \quad \text{Using Eq. B.57} \quad (\text{B.62})$$

$$\mathbf{s}_\theta(\mathbf{z}_t \neq \mathbf{m})_{\mathbf{y} \notin \{\mathbf{m}, \mathbf{z}_t\}} = 0 \quad \text{Using Eq. B.56 and Eq. B.57} \quad (\text{B.63})$$

These can be consolidated into the following expression:

$$\boxed{\mathbf{s}_\theta(\mathbf{z}_t)_\mathbf{y} = \mathbf{y}^\top \left[\delta_{\mathbf{z}_t, \mathbf{m}} \frac{\alpha_t}{1 - \alpha_t} \mathbf{x}_\theta(\mathbf{z}_t, t) + (1 - \delta_{\mathbf{z}_t, \mathbf{m}}) \frac{1 - \alpha_t}{\alpha_t} \mathbf{m} + \mathbf{z}_t \right]} \quad (\text{B.64})$$

B.3.4 Reverse Rate Matrix for MDLM

We can formulate the reverse rate matrix for MDLM using Eq. B.64 and Eq. B.44.

Recall that the reverse rate matrix $\tilde{R}_t(\mathbf{y}', \mathbf{y})$ is given by:

$$\tilde{R}_t(\mathbf{y}', \mathbf{y}) = \begin{cases} \mathbf{s}_\theta(\mathbf{y})_{\mathbf{y}'} R_t(\mathbf{y}, \mathbf{y}') & \mathbf{y}' \neq \mathbf{y} \\ -\sum_{\tilde{\mathbf{y}} \neq \mathbf{y}} \mathbf{s}_\theta(\mathbf{y})_{\mathbf{y}'} R_t(\mathbf{y}, \tilde{\mathbf{y}}) & \mathbf{y}' = \mathbf{y}. \end{cases}$$

Let's examine the cases where $\mathbf{y} = \mathbf{m}$ and $\mathbf{y} \neq \mathbf{m}$.

Case $\mathbf{y} = \mathbf{m}$: For $\mathbf{y}' \neq \mathbf{m}$, the reverse rate $\tilde{R}_t(\mathbf{y}', \mathbf{y} = \mathbf{m})$ is given by:

$$\begin{aligned} & \tilde{R}_t(\mathbf{y}' \neq \mathbf{m}, \mathbf{y} = \mathbf{m}) \\ &= \mathbf{s}_\theta(\mathbf{y} = \mathbf{m})_{\mathbf{y}' \neq \mathbf{m}} R_t(\mathbf{y} = \mathbf{m}, \mathbf{y}') \\ & \text{Using Eq. B.59 and Eq. B.44, we get:} \\ &= \frac{\alpha_t}{1 - \alpha_t} \langle \mathbf{x}_\theta(\mathbf{z}_t, t), \mathbf{y}' \rangle \left[-\frac{\alpha'_t}{\alpha_t} \right] \\ &= -\frac{\alpha'_t}{1 - \alpha_t} \langle \mathbf{x}_\theta(\mathbf{z}_t, t), \mathbf{y}' \rangle \end{aligned} \tag{B.65}$$

For $\mathbf{y}' = \mathbf{m}$, the reverse rate $\tilde{R}_t(\mathbf{y}', \mathbf{y} = \mathbf{m})$ is given by:

$$\begin{aligned} & \tilde{R}_t(\mathbf{y}' = \mathbf{m}, \mathbf{y} = \mathbf{m}) \\ &= -\sum_{\tilde{\mathbf{y}} \neq \mathbf{m}} \tilde{R}_t(\tilde{\mathbf{y}}, \mathbf{y} = \mathbf{m}) \end{aligned}$$

Using Eq. B.65, we get:

$$\begin{aligned} &= \sum_{\tilde{\mathbf{y}} \neq \mathbf{m}} \frac{\alpha'_t}{1 - \alpha_t} \langle \mathbf{x}_\theta(\mathbf{z}_t, t), \tilde{\mathbf{y}} \rangle \\ &= \frac{\alpha'_t}{1 - \alpha_t} \sum_{\tilde{\mathbf{y}} \neq \mathbf{m}} \langle \mathbf{x}_\theta(\mathbf{z}_t, t), \tilde{\mathbf{y}} \rangle \end{aligned}$$

“zero-masking probability” in Sec. 4.2.1 $\implies \sum_{\tilde{\mathbf{y}} \neq \mathbf{m}} \langle \mathbf{x}_\theta(\mathbf{z}_t, t), \tilde{\mathbf{y}} \rangle = 1$; hence,

$$= \frac{\alpha'_t}{1 - \alpha_t}. \tag{B.66}$$

Case $\mathbf{y} \neq \mathbf{m}$: For $\mathbf{y}' \neq \mathbf{y}$ we have:

$$\begin{aligned}
& \tilde{R}_t(\mathbf{y}' \notin \{\mathbf{y}, \mathbf{m}\}, \mathbf{y} \neq \mathbf{m}) \\
&= \mathbf{s}_\theta(\mathbf{y} \neq \mathbf{m})_{\mathbf{y}' \notin \{\mathbf{y}, \mathbf{m}\}} \underbrace{R_t(\mathbf{y} \neq \mathbf{m}, \mathbf{y}' \notin \{\mathbf{y}, \mathbf{m}\})}_{= 0 \text{ from Eq. B.44}} \\
&= 0
\end{aligned} \tag{B.67}$$

For $\mathbf{y}' = \mathbf{m}$, we have:

$$\begin{aligned}
& \tilde{R}_t(\mathbf{y}' = \mathbf{m}, \mathbf{y} \neq \mathbf{m}) \\
&= \mathbf{s}_\theta(\mathbf{y} \neq \mathbf{m})_{\mathbf{y}' = \mathbf{m}} \underbrace{R_t(\mathbf{y} \neq \mathbf{m}, \mathbf{y}' = \mathbf{m})}_{= 0 \text{ from Eq. B.44}} \\
&= 0
\end{aligned} \tag{B.68}$$

Thus, for $\mathbf{y}' = \mathbf{y}$, we have:

$$\begin{aligned}
& \tilde{R}_t(\mathbf{y}' = \mathbf{y}, \mathbf{y} \neq \mathbf{m}) \\
&= - \sum_{\tilde{\mathbf{y}} \neq \mathbf{y}} \tilde{R}_t(\tilde{\mathbf{y}}, \mathbf{y} \neq \mathbf{m}) \\
&= - \underbrace{\tilde{R}_t(\tilde{\mathbf{y}} = \mathbf{m}, \mathbf{y} \neq \mathbf{m})}_{= 0 \text{ from Eq. B.68}} - \underbrace{\sum_{\tilde{\mathbf{y}} \notin \{\mathbf{y}, \mathbf{m}\}} \tilde{R}_t(\tilde{\mathbf{y}}, \mathbf{y} \neq \mathbf{m})}_{= 0 \text{ from Eq. B.67}} \\
&= 0
\end{aligned} \tag{B.69}$$

Summarizing Eq. B.65, Eq. B.66, Eq. B.67, Eq. B.68, Eq. B.69, we have:

$$\begin{aligned}
\tilde{R}_t(\mathbf{y}', \mathbf{y}) &= \begin{cases} -\langle \mathbf{x}_\theta(\mathbf{y}, t), \mathbf{y}' \rangle \frac{\alpha'_t}{1-\alpha_t} & \mathbf{y}' \neq \mathbf{m}, \mathbf{y} = \mathbf{m} \\ \frac{\alpha'_t}{1-\alpha_t} & \mathbf{y}' = \mathbf{m}, \mathbf{y} = \mathbf{m} \\ 0 & \text{otherwise.} \end{cases} \\
&= \boxed{-\frac{\alpha'_t}{1-\alpha_t} [\mathbf{y}']^\top [\mathbf{x}_\theta(\mathbf{y}, t) - \mathbf{m}] \langle \mathbf{y}, \mathbf{m} \rangle}
\end{aligned} \tag{B.70}$$

B.3.5 Deriving MDLM's NELBO via CTMC

Now, we aim to show that substituting the expression for the rate matrix R_t in terms of state transition matrix Q_t from Eq. B.44 into Eq. B.53 and switching from score-parameterization to the SUBS parameterization (Sec. 4.2.1) yields the simplified NELBO for MDLM as given by Eq. 4.7. We present the proof below. Recall that the term $\langle \mathbf{a}, \mathbf{b} \rangle$ denotes the dot product of two vectors $\mathbf{a}$ and $\mathbf{b}$. When $\mathbf{a}$ and $\mathbf{b}$ represent two *one-hot* vectors, this quantity evaluates to 1 if $\mathbf{a} = \mathbf{b}$ and 0 otherwise.

Proof. Recall that for absorbing state diffusion, $\mathbf{y}$ takes only two possible values, i.e., $\mathbf{y} \in \{\mathbf{x}, \mathbf{m}\}$. Thus, we expand Eq. B.53 as follows:

$$\begin{aligned}
& \mathbb{E}_{t \in [0,1], \mathbf{y} \sim q_t(\cdot|\mathbf{x})} \left[\langle \mathbf{y}, \mathbf{x} \rangle \left[\sum_{\mathbf{y}' \neq \mathbf{x}} R_t(\mathbf{x}, \mathbf{y}') \left(\mathbf{s}_\theta(\mathbf{x})_{\mathbf{y}'} - \frac{q_t(\mathbf{y}'|\mathbf{x})}{q_t(\mathbf{x}|\mathbf{x})} \log \mathbf{s}_\theta(\mathbf{x})_{\mathbf{y}'} + K \left(\frac{q_t(\mathbf{y}'|\mathbf{x})}{q_t(\mathbf{x}|\mathbf{x})} \right) \right) \right] \right. \\
& \quad \left. + \langle \mathbf{y}, \mathbf{m} \rangle \left[\sum_{\mathbf{y}' \neq \mathbf{m}} R_t(\mathbf{m}, \mathbf{y}') \left(\mathbf{s}_\theta(\mathbf{m})_{\mathbf{y}'} - \frac{q_t(\mathbf{y}'|\mathbf{x})}{q_t(\mathbf{m}|\mathbf{x})} \log \mathbf{s}_\theta(\mathbf{m})_{\mathbf{y}'} + K \left(\frac{q_t(\mathbf{y}'|\mathbf{x})}{q_t(\mathbf{m}|\mathbf{x})} \right) \right) \right] \right] \\
& \because R_t(\mathbf{x}, \mathbf{y}' \neq \mathbf{x}) = 0 \text{ from Eq. B.42, we get:} \\
& = \mathbb{E}_{t \in [0,1], \mathbf{y} \sim q_t(\cdot|\mathbf{x})} \langle \mathbf{y}, \mathbf{m} \rangle \left[\sum_{\mathbf{y}' \neq \mathbf{m}} R_t(\mathbf{m}, \mathbf{y}') \left(\mathbf{s}_\theta(\mathbf{m})_{\mathbf{y}'} - \frac{q_t(\mathbf{y}'|\mathbf{x})}{q_t(\mathbf{m}|\mathbf{x})} \log \mathbf{s}_\theta(\mathbf{m})_{\mathbf{y}'} + K \left(\frac{q_t(\mathbf{y}'|\mathbf{x})}{q_t(\mathbf{m}|\mathbf{x})} \right) \right) \right] \\
& \text{Substituting } R_t(\mathbf{m}, \mathbf{y}' \neq \mathbf{m}) \text{ from Eq. B.40, we get:} \\
& = \mathbb{E}_{t \in [0,1], \mathbf{y} \sim q_t(\cdot|\mathbf{x})} \langle \mathbf{y}, \mathbf{m} \rangle \left[\sum_{\mathbf{y}' \neq \mathbf{m}} -\frac{\alpha'_t}{\alpha_t} \left(\mathbf{s}_\theta(\mathbf{m})_{\mathbf{y}'} - \frac{q_t(\mathbf{y}'|\mathbf{x})}{q_t(\mathbf{m}|\mathbf{x})} \log \mathbf{s}_\theta(\mathbf{m})_{\mathbf{y}'} + K \left(\frac{q_t(\mathbf{y}'|\mathbf{x})}{q_t(\mathbf{m}|\mathbf{x})} \right) \right) \right] \\
& = \mathbb{E}_{t \in [0,1], \mathbf{y} \sim q_t(\cdot|\mathbf{x})} \langle \mathbf{y}, \mathbf{m} \rangle \left[-\frac{\alpha'_t}{\alpha_t} \left(\underbrace{\sum_{\mathbf{y}' \neq \mathbf{m}} \mathbf{s}_\theta(\mathbf{m})_{\mathbf{y}'}}_{\text{Term 1}} - \underbrace{\sum_{\mathbf{y}' \neq \mathbf{m}} \frac{q_t(\mathbf{y}'|\mathbf{x})}{q_t(\mathbf{m}|\mathbf{x})} \log \mathbf{s}_\theta(\mathbf{m})_{\mathbf{y}'}}_{\text{Term 2}} + \underbrace{\sum_{\mathbf{y}' \neq \mathbf{m}} K \left(\frac{q_t(\mathbf{y}'|\mathbf{x})}{q_t(\mathbf{m}|\mathbf{x})} \right)}_{\text{Term 3}} \right) \right]
\end{aligned} \tag{B.71}$$

Term 1:

$$\sum_{\mathbf{y} \neq \mathbf{m}} \mathbf{s}_\theta(\mathbf{m})_{\mathbf{y}}$$

Using Eq. B.55, we get,

$$\begin{aligned} &= \sum_{\mathbf{y} \neq \mathbf{m}} \frac{\alpha_t}{1 - \alpha_t} \langle \mathbf{x}_\theta(\mathbf{m}, t), \mathbf{y} \rangle \\ &= \frac{\alpha_t}{1 - \alpha_t} \sum_{\mathbf{y} \neq \mathbf{m}} \langle \mathbf{x}_\theta(\mathbf{m}, t), \mathbf{y} \rangle \end{aligned}$$

“zero-masking probability” in Sec. 4.2.1 $\implies \sum_{\mathbf{y} \neq \mathbf{m}} \langle \mathbf{x}_\theta(\mathbf{m}, t), \mathbf{y} \rangle = 1$; hence,

$$= \frac{\alpha_t}{1 - \alpha_t} \tag{B.72}$$

Term 2:

$$\begin{aligned} &\sum_{\mathbf{y}' \neq \mathbf{m}} \frac{q_t(\mathbf{y}'|\mathbf{x})}{q_t(\mathbf{m}|\mathbf{x})} \log \mathbf{s}_\theta(\mathbf{m})_{\mathbf{y}'} \\ &= \frac{q_t(\mathbf{x}|\mathbf{x})}{q_t(\mathbf{m}|\mathbf{x})} \log \mathbf{s}_\theta(\mathbf{m})_{\mathbf{x}} + \sum_{\mathbf{y}' \notin \{\mathbf{m}, \mathbf{x}\}} \frac{q_t(\mathbf{y}'|\mathbf{x})}{q_t(\mathbf{m}|\mathbf{x})} \log \mathbf{s}_\theta(\mathbf{m})_{\mathbf{y}'} \end{aligned}$$

$\because q_t(\mathbf{y}'|\mathbf{x}) = 0$ for $\mathbf{y}' \notin \{\mathbf{x}, \mathbf{m}\}$ from Eq. 4.1) we get:

$$= \frac{q_t(\mathbf{x}|\mathbf{x})}{q_t(\mathbf{m}|\mathbf{x})} \log \mathbf{s}_\theta(\mathbf{m})_{\mathbf{x}}$$

Using Eq. 4.1, we get:

$$= \frac{\alpha_t}{1 - \alpha_t} \log \mathbf{s}_\theta(\mathbf{m})_{\mathbf{x}}$$

Using Eq. B.55, we get:

$$\begin{aligned} &= \frac{\alpha_t}{1 - \alpha_t} \log \left[\frac{\alpha_t}{1 - \alpha_t} \langle \mathbf{x}_\theta(\mathbf{m}, t), \mathbf{x} \rangle \right] \\ &= \frac{\alpha_t}{1 - \alpha_t} \log \frac{\alpha_t}{1 - \alpha_t} + \frac{\alpha_t}{1 - \alpha_t} \log \langle \mathbf{x}_\theta(\mathbf{m}, t), \mathbf{x} \rangle \end{aligned} \tag{B.73}$$

Term 3:

$$\begin{aligned}
& \sum_{\mathbf{y}' \neq \mathbf{m}} K \left(\frac{q_t(\mathbf{y}'|\mathbf{x})}{q_t(\mathbf{m}|\mathbf{x})} \right) \\
&= \sum_{\mathbf{y}' \neq \mathbf{m}} \left[\frac{q_t(\mathbf{y}'|\mathbf{x})}{q_t(\mathbf{m}|\mathbf{x})} \log \frac{q_t(\mathbf{y}'|\mathbf{x})}{q_t(\mathbf{m}|\mathbf{x})} - \frac{q_t(\mathbf{y}'|\mathbf{x})}{q_t(\mathbf{m}|\mathbf{x})} \right] \\
&= \frac{q_t(\mathbf{x}|\mathbf{x})}{q_t(\mathbf{m}|\mathbf{x})} \log \frac{q_t(\mathbf{x}|\mathbf{x})}{q_t(\mathbf{m}|\mathbf{x})} - \frac{q_t(\mathbf{x}|\mathbf{x})}{q_t(\mathbf{m}|\mathbf{x})} + \sum_{\mathbf{y}' \notin \{\mathbf{x}, \mathbf{m}\}} \left[\frac{q_t(\mathbf{y}'|\mathbf{x})}{q_t(\mathbf{m}|\mathbf{x})} \log \frac{q_t(\mathbf{y}'|\mathbf{x})}{q_t(\mathbf{m}|\mathbf{x})} - \frac{q_t(\mathbf{y}'|\mathbf{x})}{q_t(\mathbf{m}|\mathbf{x})} \right] \\
&\because q_t(\mathbf{y}'|\mathbf{x}) = 0 \text{ for } \mathbf{y}' \notin \{\mathbf{x}, \mathbf{m}\}, \text{ we get,} \\
&= \frac{q_t(\mathbf{x}|\mathbf{x})}{q_t(\mathbf{m}|\mathbf{x})} \log \frac{q_t(\mathbf{x}|\mathbf{x})}{q_t(\mathbf{m}|\mathbf{x})} - \frac{q_t(\mathbf{x}|\mathbf{x})}{q_t(\mathbf{m}|\mathbf{x})}
\end{aligned}$$

Substituting the values using Eq. 4.1, we get:

$$= \frac{\alpha_t}{1 - \alpha_t} \log \frac{\alpha_t}{1 - \alpha_t} - \frac{\alpha_t}{1 - \alpha_t} \quad (\text{B.74})$$

Substituting Eq. B.72, Eq. B.73, and Eq. B.74 in Eq. B.71 we get,

$$\begin{aligned}
& \mathbb{E}_{t \in [0,1], \mathbf{y} \sim q_t(\cdot|\mathbf{x})} \langle \mathbf{y}, \mathbf{m} \rangle \left[- \frac{\alpha'_t}{\alpha_t} \left(\frac{\alpha_t}{1 - \alpha_t} - \frac{\alpha_t}{1 - \alpha_t} \log \frac{\alpha_t}{1 - \alpha_t} - \frac{\alpha_t}{1 - \alpha_t} \log \langle \mathbf{x}_\theta(\mathbf{m}, t), \mathbf{x} \rangle \right. \right. \\
& \quad \left. \left. + \frac{\alpha_t}{1 - \alpha_t} \log \frac{\alpha_t}{1 - \alpha_t} - \frac{\alpha_t}{1 - \alpha_t} \right) \right] \\
&= \mathbb{E}_{t \in [0,1], \mathbf{y} \sim q_t(\cdot|\mathbf{x})} \langle \mathbf{y}, \mathbf{m} \rangle \left[- \frac{\alpha'_t}{\alpha_t} \left(- \frac{\alpha_t}{1 - \alpha_t} \log \langle \mathbf{x}_\theta(\mathbf{m}, t), \mathbf{x} \rangle \right) \right] \\
&= \mathbb{E}_{t \in [0,1], \mathbf{y} \sim q_t(\cdot|\mathbf{x})} \langle \mathbf{y}, \mathbf{m} \rangle \left[\frac{\alpha'_t}{1 - \alpha_t} \log \langle \mathbf{x}_\theta(\mathbf{m}, t), \mathbf{x} \rangle \right]
\end{aligned}$$

Under the SUBS parameterization, $\log \langle \mathbf{x}_\theta(\mathbf{z}_t, t), \mathbf{x} \rangle = 0$ when $\mathbf{z}_t = \mathbf{x}$; hence $\langle \mathbf{y}, \mathbf{m} \rangle$ can be dropped:

$$= \mathbb{E}_{t \in [0,1], \mathbf{y} \sim q_t(\cdot|\mathbf{x})} \left[\frac{\alpha'_t}{1 - \alpha_t} \log \langle \mathbf{x}_\theta(\mathbf{z}_t, t), \mathbf{x} \rangle \right] \quad (\text{B.75})$$

This concludes the proof.

B.4 Experimental details

B.4.1 Likelihood Evaluation

We use a single monte-carlo estimate for t to evaluate the likelihood. The low discrepancy sampler Eq. B.4.3 plays a key role in reducing the variance of the estimate as seen in Table 4.8.

B.4.2 Avg. Number of Tokens seen

Given `training_steps`, `batch_size`, `context_length`, the number of tokens seen by the AR model is given as:

$$\text{training_steps} \times \text{batch_size} \times \text{context_length}. \quad (\text{B.76})$$

However, this expression doesn't hold for a diffusion model, since at each training step, a fraction of the input tokens are masked before being fed to the model. Let p_m be the probability of a token being masked at a timestep t . For the log-linear schedule in our experiments, $p_m = t$. Thus, the expected number of tokens seen by the diffusion model is:

$$\begin{aligned} & \mathbb{E}_{t \sim \mathcal{U}[0,1]}[\text{training_steps} \times \text{batch_size} \times \text{context_length} \times p_m] \\ &= \text{training_steps} \times \text{batch_size} \times \text{context_length} \times \mathbb{E}_{t \sim \mathcal{U}[0,1]}[p_m] \\ & \because p_m = t, \text{ we get:} \\ &= \text{training_steps} \times \text{batch_size} \times \text{context_length} \times \mathbb{E}_{t \sim \mathcal{U}[0,1]}[t] \\ & \because \mathbb{E}_{t \sim \mathcal{U}[0,1]}[t] = 0.5, \text{ we get:} \\ &= \text{training_steps} \times \text{batch_size} \times \text{context_length} \times 0.5. \end{aligned} \quad (\text{B.77})$$

LM1B. Following [Austin et al., 2021, Lou et al., 2024, He et al., 2022], we train MDLM for 1M training steps with a `batch_size` = 512, and a context length of 128. Like [Lou et al., 2024] we use a log-linear schedule and hence the number of tokens seen by our model is $\approx 33\text{B}$ Eq. B.77. Similarly, MDLM trained for 10M steps, saw 327B tokens in expectation. The corresponding AR baseline was trained for 0.5M and 5M steps to ensure a similar number of tokens was seen.

OWT. We train SEDD and MDLM for 1M training steps with a `batch_size` = 512, `context_length` = 1024, and log-linear schedule. Hence, these models saw 262B tokens during training. Similarly, the AR model saw the same number of tokens when trained for 0.5M steps with the same `batch_size` and `context_length`.

B.4.3 Low discrepancy sampler

To reduce variance during training we use a low-discrepancy sampler, similar to that proposed in Kingma et al. [2021]. Specifically, when processing a minibatch of N samples, instead of independently sampling N from a uniform distribution, we partition the unit interval and sample the time step for each sequence $i \in \{1, \dots, N\}$ from a different portion of the interval $t_i \sim U[\frac{i-1}{N}, \frac{i}{N}]$. This ensures that our sampled timesteps are more evenly spaced across the interval $[0, 1]$, reducing the variance of the ELBO.

B.4.4 Language Modeling

For our forward noise process, we use a log-linear noise schedule similar to [Lou et al. \[2024\]](#).

We detokenize the One Billion Words dataset following [Lou et al. \[2024\]](#), whose code can be found [here](#)¹. We tokenize the One Billion Words dataset with the `bert-base-uncased` tokenizer, following [He et al. \[2022\]](#). We pad and truncate sequences to a length of 128.

We tokenize OpenWebText with the `GPT2` tokenizer. We do not pad or truncate sequences – we concatenate and wrap them to a length of 1,024. When wrapping, we add the `eos` token in-between concatenated. We additionally set the first and last token of every batch to be `eos`. Since OpenWebText does not have a validation split, we leave the last 100k docs as validation.

We parameterize our autoregressive baselines, SEDD, and MDLM with the transformer architecture from [Lou et al. \[2024\]](#). We use 12 layers, a hidden dimension of 768, 12 attention heads, and a timestep embedding of 128 when applicable. Word embeddings are not tied between the input and output.

We use the AdamW optimizer with a batch size of 512, constant learning rate warmup from 0 to a learning rate of 3e-4 for 2,500 steps. We use a constant learning rate for 1M, 5M, or 10M steps on One Billion Words, and 1M steps for OpenWebText. We use a dropout rate of 0.1.

¹<https://github.com/louaaron/Score-Entropy-Discrete-Diffusion/blob/main/data.py>

B.4.5 Zeroshot Likelihood

We evaluate zeroshot likelihoods by taking the models trained on OpenWebText and evaluating likelihoods on the validation splits of 7 datasets: Penn Tree Bank (PTB; [Marcus et al. \[1993\]](#)), Wikitext [[Merity et al., 2016](#)], One Billion Word Language Model Benchmark (LM1B; [Chelba et al. \[2014\]](#)), Lambada [[Paperno et al., 2016](#)], AG News [[Zhang et al., 2015](#)], and Scientific Papers (Pubmed and Arxiv subsets; [Cohan et al. \[2018\]](#)). We detokenize the datasets following [Lou et al. \[2024\]](#). For the AG News and Scientific Papers (Pubmed and Arxiv), we apply both the Wikitext and One Billion Words detokenizers. Since the zeroshot datasets have different conventions for sequence segmentation, we wrap sequences to 1024 and do not add `eos` tokens in between sequences.

B.4.6 Representation Learning

Following [Devlin et al. \[2018\]](#), we evaluate on all GLUE tasks [[Wang et al., 2019](#)], but exclude WNLI.

We pre-train a MosaicBERT model on C4 [[Raffel et al., 2020b](#)] for 70k steps, corresponding to 36B tokens. We pad and truncate the data to 128 tokens using the `bert-base-uncased` tokenizer.

MosaicBERT [[Portes et al., 2024](#)] has a similar architecture to `bert-base-uncased` and has 137M parameters, 12 layers, 12 attention heads, a hidden dimension of 768, an intermediate size of 3072, and ALiBi attention bias [[Press et al., 2022](#)].

For pre-training, we use the following hyperparameters: A global batch size of

4096 with gradient accumulation, a learning rate of 5e-4, linear decay to 0.02x of the learning rate with a warmup of 0.06x of the full training duration, and the decoupled AdamW optimizer with 1e-5 weight decay and betas 0.9 and 0.98.

For diffusion fine-tuning we use AdamW with a warmup of 2,500 steps from a learning rate of 0 to 5e-5, betas 0.95 and 0.999, and batch size 512. We train for 5k steps total, corresponding to 32M tokens.

For GLUE evaluation, we use the HuggingFace script found [here](#)². We use the default parameters for all datasets, except for a batch size of 16, which we found helped with smaller datasets. This includes the default of 3 epochs for all datasets and learning rate of 2e-5.

B.4.7 Diffusion DNA Models

Dataset We pre-train the Caduceus MLM [Schiff et al., 2024] on the HG38 human reference genome [Consortium, 2009]. Following Schiff et al. [2024], we use character- / base pair-level tokenization. The dataset is based on the splits used in Avsec et al. [2021]: the training split comprises of 35 billion tokens covering the human genome. This consists of 34,021 segments extended to a maximum length of 1,048,576 (220 segments). We maintain a constant 2^{20} tokens per batch. For the Genomics Benchmark tasks, we use 5-fold cross-validation where we split the training set into 90/10 train/validation splits.

²<https://github.com/huggingface/transformers/tree/main/examples/pytorch/text-classification>

Architecture The Caduceus MLM uses as a backbone a bi-directional variant of the data-dependent SSM Mamba block proposed in Gu et al. [2021]. This architecture is ideal as it contains inductive biases that preserve reverse complement (RC) equivariance, respecting the inherent symmetry of double-stranded DNA molecules [Mallet and Vert, 2021, Schiff et al., 2024, Zhou et al., 2022].

Training details All models are pre-trained on 10B tokens (10K steps) and fine-tuned on a generative objective for an additional 50B tokens (50K steps). We use a global batch size of 1024 for a context length of 1024 tokens. Downstream task fine-tuning is performed for 16K steps (1B tokens).

For performing Caduceus MLM pre-training, we follow Schiff et al. [2024] for the model size configuration, and hyperparameter selection. For pre-training, we use a fixed 15% mask rate as done in Devlin et al. [2018]. Of the 'masked' tokens, 80% are replaced with [MASK] , 10% are replaced with a random token from the vocabulary, and 10% are left unchanged.

For fine-tuning all Mamba-based models (including Caduceus) on diffusion objectives, we lower the learning rate from 8e-3 to 1e-3. For fine-tuning Hye-naDNA [Nguyen et al., 2024], we lower the learning rate from 6e-4 to 5e-5. Similar to Gu et al. [2021], Schiff et al. [2024], we found that Mamba-based models were robust to higher learning rates. We exclude timestep embeddings for all Diffusion DNA experiments, as we show it has minimal impact on generative performance (see Table B.4, Suppl. B.5.5).

We perform downstream task fine-tuning on the final hidden state embedding from pre-training. We perform mean pooling across the sequence length, which may vary from 200 to approximately 2,000 bps. We report the mean and $\pm$ on

max/min classification accuracy over 5-fold cross-validation (CV) using different random seeds, with early stopping on validation accuracy. For each task, we do a hyperparameter sweep over batch size and learning rate and report the values of the 5-fold CV for the best configuration.

Genomic Benchmark Task Distributions We use a subset of the Genomic Benchmark tasks with an emphasis on tasks from Human data. The positive samples for each dataset were generated by selecting samples that were annotated, either computationally or experimentally, in previous work (e.g enhancers, promoters, open chromatin regions (OCR)) Grešová et al. [2023]. These annotations each correspond to subsets of the genome of varying sizes that may exhibit different distributions of DNA than those observed globally over the reference genome. Due to this, the observed dataset may have a different distribution than the data used for pre-training and calculating perplexity. This might in turn lead to a case where perplexity and downstream performance may not necessarily correlate.

B.5 Additional Experiments

B.5.1 Noise schedule parameterization

As described in Sec. 4.2.1, the ELBO is invariant to the functional form of α_t . To demonstrate this, we evaluate MDLM, initially trained using a log-linear schedule on OWT, by replacing the noise schedule with various other noise schedules as mentioned below. Following prior works [Austin et al., 2021, Lou et al., 2024, Sohl-Dickstein et al., 2015], we parameterize $\alpha_t = e^{-\sigma(t)}$, where $\sigma(t) : [0, 1] \rightarrow \mathbb{R}^+$.

Various functional forms of $\sigma(t)$ are listed below:

Log Linear [Austin et al., 2021, Lou et al., 2024, Sohl-Dickstein et al., 2015]. The log linear schedule is given as:

$$\sigma(t) = -\log(1 - t) \quad (\text{B.78})$$

Cosine Squared schedule [Han et al., 2022]. The Cosine Squared schedule is given as:

$$\sigma(t) = -\log \cos^2 \left(\frac{\pi}{2}(1 - t) \right) \quad (\text{B.79})$$

Cosine schedule. The Cosine schedule is given as:

$$\sigma(t) = -\log \cos \left(\frac{\pi}{2}(1 - t) \right) \quad (\text{B.80})$$

Linear. The Linear schedule is given as:

$$\sigma(t) = \sigma_{\max} t \quad (\text{B.81})$$

where $\sigma_{\max}$ is a very large number. In our experiments we set it to 10^8 .

ELBO Invariance

The function α_t is invertible due to the monotonicity assumption in Sec. 4.2, and so we can perform the following change of variables in Eq. 4.7: $\gamma \equiv \log(1 - \alpha_t)$. Let $f : [0, 1] \rightarrow \mathbb{R}^-$ be a function such that $\gamma = f(t)$. Note that α_t goes through a monotonic transformation to obtain γ ; hence, γ is also monotonic in t since α_t is monotonic in t . This implies that the function f is invertible. Let $t = f^{-1}(\gamma)$.

Then, we can we have the following diffusion loss:

$$\begin{aligned}
\mathcal{L}_{\text{NELBO}}^\infty &= \mathbb{E}_q \int_{t=0}^{t=1} \frac{\alpha'_t}{1 - \alpha_t} \log \langle \mathbf{x}_\theta(\mathbf{z}_t, t), \mathbf{x} \rangle dt \\
&= -\mathbb{E}_q \int_{t=0}^{t=1} \log \langle \mathbf{x}_\theta(\mathbf{z}_t, t), \mathbf{x} \rangle \frac{d}{dt} [\log(1 - \alpha_t)] dt \\
&= -\mathbb{E}_q \int_{t=0}^{t=1} \log \langle \mathbf{x}_\theta(\mathbf{z}_t, t), \mathbf{x} \rangle \frac{d}{dt} [f(t)] dt && \text{Substituting } f(t) = \log(1 - \alpha_t) \\
&= -\mathbb{E}_q \int_{\gamma=-\infty}^{\gamma=0} \log \langle \mathbf{x}_\theta(\mathbf{z}_{f^{-1}(\gamma)}, f^{-1}(\gamma)), \mathbf{x} \rangle d\gamma && \text{Change of variables } \gamma \equiv f(t) \\
&= -\mathbb{E}_q \int_{\gamma=-\infty}^{\gamma=0} \log \langle \mathbf{x}_\theta(\tilde{\mathbf{z}}_\gamma, f^{-1}(\gamma)), \mathbf{x} \rangle d\gamma && \tilde{\mathbf{z}}_\gamma \equiv \mathbf{z}_{f^{-1}(\gamma)} \\
&= -\mathbb{E}_q \int_{\gamma=-\infty}^{\gamma=0} \log \langle \tilde{\mathbf{x}}_\theta(\tilde{\mathbf{z}}_\gamma, \gamma), \mathbf{x} \rangle d\gamma && \tilde{\mathbf{x}}_\theta(\tilde{\mathbf{z}}_\gamma, \gamma) \equiv \mathbf{x}_\theta(\tilde{\mathbf{z}}_\gamma, f^{-1}(\gamma))
\end{aligned} \tag{B.82}$$

This new formulation demonstrates that the diffusion loss is invariant to the functional form of α_t . In Table B.1, we demonstrate empirically that noise schedules with different functional forms evaluate to the same Likelihood which is consistent with our theory. However, different schedules lead to different per data point variance. Notably, the log-linear schedule exhibits the lowest variance among all the noise schedules considered.

Table B.1: Likelihood in bits per dimension (BPD) for different noise schedules on OWT dataset, is reported along with the mean and variance associated with each noise schedule per data point. We empirically observe that noise schedules with different functional forms yield the same likelihood, consistent with our theory in Sec. 4.2.1; however, different schedules result in different variances.

$\sigma(t)$	Mean	Variance per datapoint
Log Linear Eq. B.78	3.30	1.81
Cosine Eq. B.80	3.30	3.30
Cosine Squared Eq. B.79	3.30	3.30
Linear Eq. B.81	3.30	7.57

B.5.2 Faster sampling with caching

In Figure B.2, we compare the wall clock times of various methods: AR, SEDD, MDLM with caching, and MDLM without caching for generating 64 samples on a single GPU. When sampling in batches, a change of 1 token would necessitate a call to the denoising model. Therefore, smaller batch sizes have a lower likelihood of a token being unmasked. This might lead one to prefer generating samples in smaller batches, as opposed to using a larger batch size that fully saturates the GPU. Table B.2 shows that generating samples with a batch size of 1 and using caching is twice as fast as generating samples without caching while fully utilizing the GPU. In Fig. B.1, we observe that MDLM without caching yields samples that consistently get better generative perplexity than SEDD. For $T = \{5k, 10k\}$, both SEDD and MDLM get better generative perplexity than the AR model.

Table B.2: Wall clock time reported in minutes to generate 64 samples on a single A5000 GPU.

	$T = 5k(\downarrow)$	$T = 10k(\downarrow)$
SEDD	85.3	155.2
MDLM	70.3	127.9
+ caching	40.1	60.4

B.5.3 LM1B ablations

We assess the importance of our continuous-time framework by performing ablation on diffusion steps T . In Table B.3, we compare NLL and PPL under continuous and discrete T in MDLM. We find that NLL consistently decreases as $T \rightarrow \infty$.

Generative perplexities across sample times on OpenWebText

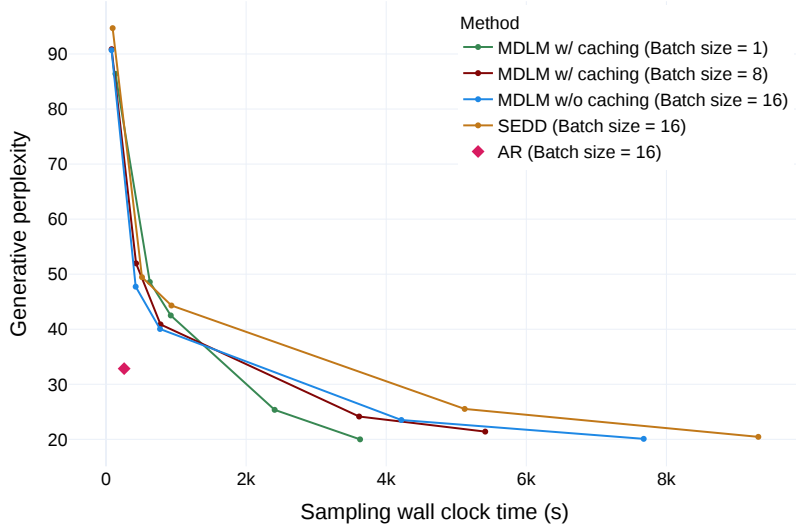


Figure B.1: Generative perplexities across wall clock time for generating 64 samples on OWT using a single 32GB A5000 GPU are compared by varying $T \in \{100, 500, 1000, 5000, 10000\}$ in the reverse diffusion process. The samples are generated in mini-batches with a batch size of 16 for AR, SEDD, and MDLM without caching, as it is the largest batch size that fits on this GPU. For MDLM with caching, we vary the batch size.

Table B.3: Discrete vs continuous time evaluation for MDLM w/o time-conditioning on OWT. MDLM was trained with $T = \infty$. We report test perplexity for a discrete T .

T	PPL($\leq$)
∞	23.05
10	42.18
20	30.70
50	25.77
100	24.35
200	23.66
500	23.26
1000	23.15

B.5.4 Train NLL curves on OWT

In Figure B.2, we show that MDLM achieves lower variance loss during training compared to a previous diffusion language model, SEDD. Training is performed over 1M steps on OWT (which corresponds to 524B tokens).

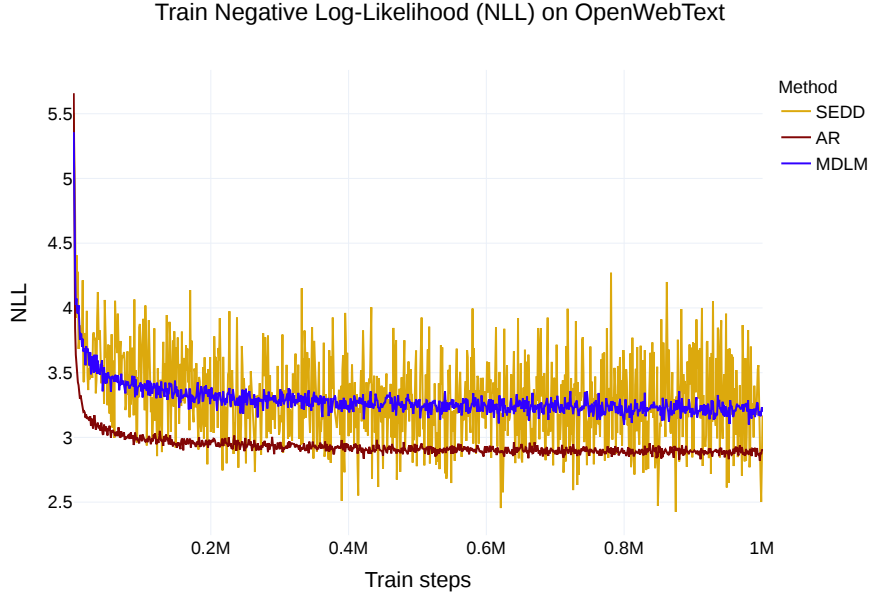


Figure B.2: Train negative log-likelihood (NLL) curves across 1M gradient steps (524B tokens) on OpenWebText [Gokaslan et al., 2019]. NLL is logged every 1K steps without value smoothing.

B.5.5 Time-conditioning ablation on OWT

In Table B.4, we assess the importance of time conditioning in MDLM on OWT. We observe that time-conditioning has minimal impact on perplexity. Training is performed over 1M steps on OWT (which corresponds to 524B tokens).

Table B.4: Ablation on time-conditioning in MDLM on OWT.

Method	PPL
MDLM w/ time-conditioning	23.21
MDLM w/o time-conditioning	23.05

B.5.6 Unconditional Samples

Here, we present some unconditional samples generated by MDLM trained on OWT with a context length of $L = 1024$ for $T = \{1000, 10000\}$.

$T = 1000$

<|endoftext|> a 17-10 victory and a trip to the playoffs.
The last wildcard seed: New York Jets, Houston and the last potable playoff spot. The last-second home wins: New Orleans and Carolina, 21-21.
The Saints finish sixth with the highest regular season (42) NFC wins. They lost 14 of their 13 games in the conference playoffs.
The Cardinals were in Group A in Round 1 with Game 2, Round 3, Round 4 and Quarter Game 5, but they made their last trip to the playoffs off North Carolina on the road as even North Carolina.
True to their reputation, the Cards swept the Saints in the first round, but knocked it out at home. No Panthers went to the playoffs more than the Saints.
Don Jean no longer is the South Carolina Panther.
The Cardinals thought that provided that he had a chance to be an NFL player.
"I did," said defensive end Lorenzo Williams with a laugh as he exited his car at the airport. "Also, I won that game." KC win brings Carson back home.
Griffin made promise on Sunday to never exactly give up the dunk. Although he failed to score 40 points in the playoffs, he has had better luck in them this year.
With turnovers and fumbles returning, he has to play out because the team doesn't trust him. He's long years of injuries, turnovers and calls because he knows he can play that way for everybody.
Griffin is no stranger to Saints fans.
"Players want him to know them," someone from South Carolina said after coming out against the Panthers — in their best home home Week 7 win Sunday — in an 11-9 rout. South Carolina did win its final three and passed the Saints, 24-1.
Although the Cardinals are in the South, they am a step behind.
They still have little time left to take down the West Coast wild card. There is no chance they get another victory.
The West was out by Beshear in their first round games last season, losing by 63 to the 49ers.
The outcome will be tough.
"Now we're so close, let's figure out the time to win," Brees said. "We still have a few games left; I'm glad about that." South Carolina takes the revenge.
When asked about his second time since Super 4, Brees shot back that he understood.
"You can doubt the answer but I think that was a no-brainer. In time, you try to prove an answer wrong," Brees said.
"I think his ability will be as cool as Julio Jones' ability, but having that time [out] to my season was difficult overall. I did what I was expecting to do. Hopefully they'll tell me to try again."
After their late win, the NFL calls the Saints reschedule'must try.'
"Because Saints," those who am there still say it, "focus on defense and, offense is defense."
ESPN said Saints's star receiver Dashon Jeffishard, turning heads on long passes and connecting with open defenders, already had their 20-yard overall score from the field. When Jeffishard finished with three passing he set up. He obviously had no difference; it was his first snap-off.
With his changes in his starting lineup, Brees was just hoping they had little to prove against Carolina over the weekend.
"I felt like we didn't have quite enough focus on and there was so puny coverage that I wanted more of our guys at the same position so we'd up our game," he said.
Brees said South Carolina was well at linebacker.
"If that's part of it, if you're going to try and stay with what you're going [out with]. What'd you want? Smart play," Brees said. "What would you say? That you're always ready to play. So you're going out there strong and ready to go to play football."
That said, New Orleans was damned shy when it came to Carolina.
"My guys admit to feeling it a little bit [Sunday but] I say to them that they always knew, 'I don't think that was necessarily how I would beat you, that will give them their confidence,'" Brees said.
"It was really hard because I've obviously learned a lot of detail about how to deal with everyone and as hard as I have to be, I also feel part of the stuff that they've been through on the team, like they're still going to go through things they know are somewhat right, but they feel a lot of pressure so it's got to be important to get it right now to get it in the future."
Could all ask for more roses?
Let's just take a slip, South Carolina, and face the NFC<|endoftext|>

Figure B.3: Samples ($T = 8$) from Duo distilled with DDT trained on OWT with Gen. PPL 79.24 and entropy = 5.25

$T = 10000$

<|endoftext|> has been presenting a number of temporary measures to help help resolve a crisis.
The last crisis was like this, but we have to resolve it at the specified level; we have to stay low. The people of Greece will be interested in the effectiveness of the measures. They will not only manage these measures, but also they will help in order to cope with the problems of the fiscal stability.
However, we do not want to dispose assets for the treasury. This also, so we will work on developing the national economy, and also paying on the national debt.
This affects the national incomes
So, as of 2007-2011, we use the government's temporary measures as a measure, helping resolve the crisis. In addition, we are able to pay for the borrowing costs. Additionally, we pay \$440 billion to settle debt issues, which can never be settled by default in a country.
These temporary measures will be aimed on several fronts, because the government will have three different partners in the system, in my right.
Firstly, we will be allowed to borrow a lot more upon the addition of the emergency measures and these temporary measures will help provide for the repayment of our debts before we are forced into a crisis as a result of our borrowing bills. Secondly, in the case of this, we will have resort to temporary measures in the revenue budget for Greece. The budget costs the government another \$5.2 billion a year.
So what you propose - if it happens again, does this mean that, since 2010, will you resolve the deficits which will occur on the basis of what we already have?
The fiscal situation
We would be able to settle our debts by the end of June which is the end. That said, we are taking our part as one of the most important countries in Europe, not only to make a proper transfer of the money but also to rely on it in the economy. However, first of all, we cannot achieve this on a day-to-day basis.
It is still true that we have decided to be able to deal with the economic situation of the country, but there might be another change in the fiscal situation, and therefore, we will try to negotiate on the situation at the end of June and over the summer.
The changes in the fiscal situation would be up to the parliament of management, bureaucrats, judges and a legitimate parliament of Greece.
Is the government planning to talk about the 'temporary measures' of Greece?
We will continue the process to operate through the temporary measures. This is not a temporary measure at this point, because after a crisis, not there yet at crisis level, you can still have enough investment purchases until the end of the month.
Again the government decided to create a temporary measure and now it depends upon a particular event, such as that there's another liquidity crunch. It is better that the government and the authorities decided to create a temporary measure effective in June at fair sum monthly bond rates.
The temporary measures will also enhance the government's economic status, especially when following the measures at the end of the month.
Temporary measures is a real tool for growth, not just for the economy.
Knowing that there are several measures in place to increase our supply, for example, the level offer profit on public sector enterprises is certain, under all of these temporary measures the increases in output, after that, will increase the external demand and the internal demand.
We will be able to create the demand, and also strengthen the government's credibility through fiscal organization.
What is important here, here is that we will apply these measures to our reserves, and at the same time, we apply these measures to the debt level, which will also be the aim of debt-free Greece.
So, first of all, everything is certain of what continues to be collected by the government. Given the situation and after the release of the last data on October 16, you also recognize that this will not be any kind of non-payment.
In the case of the payment against the equipment, we will be able to manage with the measures.
What does government expect in its plans to create a fiscal consolidation for the public sector and the new budget.
Regarding this is the temporary measure, we will be able to cope with the troubled finances. However, I do not think it is any measure which threatens the fiscal stability of the economy. However, that is not a temporary measure, a permanent measure.
On the other hand, there will be our ongoing work on construction in the ministry. If this falls, we will continue work on job creation, the expansion of the economy.
Also, also the government mentioned the new government reforms, which increased labor hours for the employees, which will further the economic growth, and the second aspect of budget as well and this is government welfare, which will improve the quality of life. We will<|endoftext|>

Figure B.4: Samples ($T = 8$) from Duo distilled with DDT trained on OWT with Gen. PPL 79.24 and entropy = 5.25

APPENDIX C

THE DIFFUSION DUALITY

C.1 The Diffusion Duality

Let $\mathbf{x} \in \mathcal{V}$ s.t. $\mathbf{x}_k = 1$ i.e., $\mathbf{x}$ contains 1 at the k^{th} index. Consider a r.v. $\bar{\mathbf{w}} = \tilde{\alpha}_t \mathbf{x} + \tilde{\sigma}_t \boldsymbol{\epsilon}$ where $\boldsymbol{\epsilon} \sim \mathcal{N}(0, \mathbf{I}_K)$ and $\tilde{\sigma}_t = \sqrt{1 - \tilde{\alpha}_t^2}$.

C.1.1 Discrete Marginals

Our goal in this section is to derive the pmf of the r.v. $\arg \max(\bar{\mathbf{w}})$. The proof has three parts. In **part 1**, we derive pdf of the the random variables $\bar{\mathbf{w}}_k$ and $\bar{\mathbf{w}}_{i \neq k}$. Next in **part 2**, we derive the pdf of the random variable $Z_{\neq k} = \max(\{\bar{\mathbf{w}}_i : i \neq k\})$. Finally in **part 3**, we derive the distribution of $\max(Z_{\neq k}, \bar{\mathbf{w}}_k)$ which is the key to constructing the pmf of the r.v. $\arg \max(\bar{\mathbf{w}})$.

Part 1 It can be easily seen that every entry in $\bar{\mathbf{w}}$ is a Gaussian r.v. with

$$\bar{\mathbf{w}}_k \sim \mathcal{N}(\tilde{\alpha}_t, \tilde{\sigma}_t^2) \tag{C.1}$$

$$\bar{\mathbf{w}}_{i \neq k} \sim \mathcal{N}(0, \tilde{\sigma}_t^2). \tag{C.2}$$

Part 2 Since, $\bar{\mathbf{w}}_{i \neq k}$ follows a Gaussian distribution with 0 mean and $\tilde{\sigma}_t$ standard deviation, the probability of $\bar{\mathbf{w}}_{i \neq k} < l$ where $l \in \mathbb{R}$ is

$$P(\bar{\mathbf{w}}_{i \neq k} < l) = \Phi\left(\frac{l}{\tilde{\sigma}_t}\right) \tag{C.3}$$

where $\Phi(z) = \int_{-\infty}^z \exp(-t^2/2)dt/\sqrt{2\pi}$ is the cumulative distribution function of the Gaussian distribution. This allows us to compute the pdf of the r.v. $Z_{\neq k} = \max(\{\bar{\mathbf{w}}_i : i \neq k\})$ in the following manner:

$$P(Z_{\neq k} < l) = \Pi_{i \neq k} P(\bar{\mathbf{w}}_i < l) = \Phi^{K-1} \left(\frac{l}{\tilde{\sigma}_t} \right), \quad (\text{C.4})$$

where $P(Z_{\neq k} < l)$ is the probability that $Z_{\neq k} < l$ for $l \in \mathbb{R}$.

Part 3 Let $P(\arg \max(\bar{\mathbf{w}})_k = 1)$ denote the probability that the index k is the index of the maximum entry in $\bar{\mathbf{w}}$. This is equal to the probability of every other entry $\bar{\mathbf{w}}_{i \neq k} < \bar{\mathbf{w}}_k$. Let $\phi(z) = \exp(-z^2)/\sqrt{2\pi}$ denote the standard Normal distribution. Hence,

$$\begin{aligned} P(\arg \max(\bar{\mathbf{w}})_k = 1) &= P(Z_{\neq k} < \bar{\mathbf{w}}_k) \\ &= \int_{-\infty}^{\infty} P(Z_{\neq k} < l) P(\bar{\mathbf{w}}_k = l) dl \\ &= \int_{-\infty}^{\infty} P(Z_{\neq k} < l) \left[\frac{1}{\tilde{\sigma}_t} \phi \left(\frac{l - \tilde{\alpha}_t}{\tilde{\sigma}_t} \right) \right] dl && \text{From Eq. C.1} \\ &= \int_{-\infty}^{\infty} \Phi^{K-1} \left(\frac{l}{\tilde{\sigma}_t} \right) \left[\frac{1}{\tilde{\sigma}_t} \phi \left(\frac{l - \tilde{\alpha}_t}{\tilde{\sigma}_t} \right) \right] dl && \text{From Eq. C.4} \\ &= \int_{-\infty}^{\infty} \Phi^{K-1}(\tilde{l}) \phi \left(\tilde{l} - \frac{\tilde{\alpha}_t}{\tilde{\sigma}_t} \right) d\tilde{l} && \text{Substituting } \tilde{l} = l/\tilde{\sigma}_t \\ &= \int_{-\infty}^{\infty} \phi \left(\tilde{l} - \frac{\tilde{\alpha}_t}{\sqrt{1 - \tilde{\alpha}_t^2}} \right) \Phi^{K-1}(\tilde{l}) d\tilde{l}. \end{aligned} \quad (\text{C.5})$$

Note that the indices $i \neq k$ and $j \neq k$ have the same probability of being the indices of maximum entry in $\bar{\mathbf{w}}$ because both r.v.s $\bar{\mathbf{w}}_{i \neq k}$ and $\bar{\mathbf{w}}_{j \neq k}$ have the same pmf specified by Eq. C.2. Thus,

$$P(\arg \max(\bar{\mathbf{w}})_{i \neq k} = 1) = P(\arg \max(\bar{\mathbf{w}})_{j \neq k} = 1) \quad \forall 0 \leq i \neq k < K, 0 \leq j \neq k < K. \quad (\text{C.6})$$

Thus we can compute $P(\arg \max(\bar{\mathbf{w}})_{i \neq k} = 1)$ in the following manner:

$$\begin{aligned} \sum_i P(\arg \max(\bar{\mathbf{w}})_i = 1) &= 1 \\ \implies P(\arg \max(\bar{\mathbf{w}})_k = 1) + \sum_{i \neq k} P(\arg \max(\bar{\mathbf{w}})_i = 1) &= 1 \end{aligned}$$

From Eq. C.6, we get:

$$\begin{aligned} \implies P(\arg \max(\bar{\mathbf{w}})_k = 1) + (K - 1)P(\arg \max(\bar{\mathbf{w}})_{i \neq k} = 1) &= 1 \\ \implies P(\arg \max(\bar{\mathbf{w}})_{i \neq k} = 1) &= \frac{1}{K - 1} [1 - P(\arg \max(\bar{\mathbf{w}})_k = 1)] \end{aligned}$$

From Eq. C.5, we get:

$$\implies P(\arg \max(\bar{\mathbf{w}})_{i \neq k} = 1) = \frac{1}{K - 1} \left[1 - \int_{-\infty}^{\infty} \phi \left(\tilde{l} - \frac{\tilde{\alpha}_t}{\sqrt{1 - \tilde{\alpha}_t^2}} \right) \Phi^{K-1}(\tilde{l}) d\tilde{l} \right] \quad (\text{C.7})$$

Let $\beta_t = P(\arg \max(\bar{\mathbf{w}})_{i \neq k} = 1)$. Then, from Eq. C.5 and Eq. C.7 we have $P(\arg \max(\bar{\mathbf{w}})_{i=k} = 1) = \beta_t + (1 - K)\beta_t$. Thus,

$$P(\arg \max(\bar{\mathbf{w}})_i = 1) = \begin{cases} \beta_t, & i \neq k \\ \beta_t + (1 - K)\beta_t, & i = k \end{cases} \quad (\text{C.8})$$

Eq. C.8 can be written in vectorized form in the following manner:

$$P(\arg \max(\bar{\mathbf{w}})) = \text{Cat}(\cdot; \beta_t \mathbf{1} + (1 - K\beta_t)\mathbf{x}). \quad (\text{C.9})$$

C.1.2 Time Evolution of Probability Densities of Discrete Marginals

Let p_t denote $P(\arg \max(\bar{\mathbf{w}}))$ in Eq. C.9. It's time-derivative $\frac{d}{dt}p_t$ is as follows:

$$\begin{aligned}
\frac{d}{dt}p_t &= \beta'_t \mathbf{1} - K \beta'_t \mathbf{x} \\
&= \beta'_t (\mathbf{1} - K \mathbf{x}) \\
&= \frac{\beta'_t}{1 - K \beta_t} (1 - K \beta_t) (\mathbf{1} - K \mathbf{x}) \\
&= \frac{\beta'_t}{1 - K \beta_t} (\beta_t K \mathbf{1} - \beta_t K \mathbf{1} + (1 - K \beta_t) (\mathbf{1} - K \mathbf{x})) \\
&= \frac{\beta'_t}{1 - K \beta_t} (\beta_t [\mathbf{1} \mathbf{1}^\top] \mathbf{1} - \beta_t K \mathbf{1} + (1 - K \beta_t) (\mathbf{1} - K \mathbf{x})) \\
&= \frac{\beta'_t}{1 - K \beta_t} (\beta_t ([\mathbf{1} \mathbf{1}^\top] \mathbf{1} - K \mathbf{1}) + (1 - K \beta_t) (\mathbf{1} - K \mathbf{x})) \\
&= \frac{\beta'_t}{1 - K \beta_t} (\beta_t [\mathbf{1} \mathbf{1}^\top - K \mathbf{I}_K] \mathbf{1} + (1 - K \beta_t) (\mathbf{1} - K \mathbf{x})) \\
&= \frac{\beta'_t}{1 - K \beta_t} (\beta_t [\mathbf{1} \mathbf{1}^\top - K \mathbf{I}_K] \mathbf{1} + (1 - K \beta_t) (\mathbf{1} \mathbf{1}^\top \mathbf{x} - K \mathbf{x})) \\
&= \frac{\beta'_t}{1 - K \beta_t} (\beta_t [\mathbf{1} \mathbf{1}^\top - K \mathbf{I}_K] \mathbf{1} + (1 - K \beta_t) [\mathbf{1} \mathbf{1}^\top - K \mathbf{I}_K] \mathbf{x}) \\
&= \frac{\beta'_t}{1 - K \beta_t} [\mathbf{1} \mathbf{1}^\top - K \mathbf{I}_K] [\beta_t \mathbf{1} + (1 - K \beta_t) \mathbf{x}] \\
&= \frac{\beta'_t}{1 - K \beta_t} [\mathbf{1} \mathbf{1}^\top - K \mathbf{I}_K] p_t
\end{aligned} \tag{C.10}$$

Let $\alpha_t = 1 - K \beta_t$. The functional form of α_t is given as:

$$\begin{aligned}
\alpha_t &= 1 - K \beta_t \\
&= 1 - K \frac{1}{K-1} \left[1 - \int_{-\infty}^{\infty} \phi \left(\tilde{l} - \frac{\tilde{\alpha}_t}{\sqrt{1 - \tilde{\alpha}_t^2}} \right) \Phi^{K-1}(\tilde{l}) d\tilde{l} \right] \\
&= 1 - \frac{K}{K-1} + \frac{K}{K-1} \int_{-\infty}^{\infty} \phi \left(\tilde{l} - \frac{\tilde{\alpha}_t}{\sqrt{1 - \tilde{\alpha}_t^2}} \right) \Phi^{K-1}(\tilde{l}) d\tilde{l} \\
&= \frac{K}{K-1} \int_{-\infty}^{\infty} \phi \left(\tilde{l} - \frac{\tilde{\alpha}_t}{\sqrt{1 - \tilde{\alpha}_t^2}} \right) \Phi^{K-1}(\tilde{l}) d\tilde{l} - \frac{1}{K-1} \\
&= \frac{K}{K-1} \left[\int_{-\infty}^{\infty} \phi \left(\tilde{l} - \frac{\tilde{\alpha}_t}{\sqrt{1 - \tilde{\alpha}_t^2}} \right) \Phi^{K-1}(\tilde{l}) d\tilde{l} - \frac{1}{K} \right]
\end{aligned} \tag{C.11}$$

Substituting $\beta_t = (1 - \alpha_t)/K$ in Eq. C.9 and Eq. C.10, we get:

$$p_t = \text{Cat}(\cdot; \alpha_t \mathbf{x} + (1 - \alpha_t) \boldsymbol{\pi}) \quad (\text{C.12})$$

$$\frac{d}{dt} p_t = -\frac{\alpha_t'}{K \alpha_t} [\mathbf{1} \mathbf{1}^\top - K \mathbf{I}_K] p_t \quad (\text{C.13})$$

where α_t' denotes the time-derivative of α_t . Let $\mathbf{z}_t = \text{arg max}(\bar{\mathbf{w}})$. The pmf of $\mathbf{z}_t$ is specified in Eq. C.12 which evolves according to an Ordinary Differential Equation (ODE) Eq. C.13. This pmf and the ODE are the unique signatures of a Uniform-state discrete diffusion process [Lou et al., 2024, Schiff et al., 2025]. This concludes our proof.

C.1.3 Properties of discretized Gaussian Trajectories

Let $\mathbf{x} \in \mathcal{V}$ undergo Gaussian diffusion, and let $\{\mathbf{w}_t\}_{t \in [0,1]}$ denote its continuous trajectory. The central question is: does the corresponding discretized trajectory $\{\mathbf{z}_t := \text{arg max}(\mathbf{w}_t)\}_{t \in [0,1]}$ correspond to a valid discrete diffusion trajectory?

To address this, we must examine how the **arg max** operation links the Gaussian transition kernel—which models the transition $\mathbf{w}_s \rightarrow \mathbf{w}_t$ —to the discrete diffusion transition kernel, which models the transition $\mathbf{z}_s \rightarrow \mathbf{z}_t$ in the discrete space, for $0 \leq s < t \leq 1$. The Gaussian transition kernel is given by

$$\mathbf{w}_t \sim \tilde{q}_{t|s}(\cdot | \mathbf{w}_s) = \mathcal{N}(\tilde{\alpha}_{t|s} \mathbf{w}_s, (1 - \tilde{\alpha}_{t|s}^2) \mathbf{I}_K), \quad (\text{C.14})$$

where $\tilde{\alpha}_{t|s} = \tilde{\alpha}_t / \tilde{\alpha}_s$. The discrete diffusion transition kernel is

$$\mathbf{z}_t \sim q_{t|s}(\cdot | \mathbf{z}_s) = \text{Cat}(\cdot; \alpha_{t|s} \mathbf{z}_s + (1 - \alpha_{t|s}) \boldsymbol{\pi}), \quad (\text{C.15})$$

where $\alpha_{t|s} = \alpha_t / \alpha_s$.

A necessary and sufficient condition for the discretized trajectory $\{\mathbf{z}_t := \arg \max(\mathbf{w}_t)\}_{t \in [0,1]}$ to follow a discrete diffusion process is that the *pushforward* distribution $[\arg \max]_{\star} \tilde{q}_{t|s}(\cdot \mid \mathbf{w}_s)$ must equal $q_{t|s}(\cdot \mid \mathbf{z}_s)$. We will now show that this does not hold.

Step 1: Analyzing the discrete kernel Let k denote the index such that $(\mathbf{z}_s)_k = 1$. First, consider $q_{t|s}(\cdot \mid \mathbf{z}_s)$. From Eq. C.15, it is clear that for any $i, j \neq k$, the probabilities satisfy

$$q_{t|s}(\bar{\mathbf{z}}_i = 1 \mid \mathbf{z}_s) = q_{t|s}(\bar{\mathbf{z}}_j = 1 \mid \mathbf{z}_s),$$

where $\bar{\mathbf{z}} \in \mathcal{V}$ is a discrete random variable.

Step 2: Analyzing the pushforward of the Gaussian kernel Now define $P := [\arg \max]_{\star} \tilde{q}_{t|s}(\cdot \mid \mathbf{w}_s)$. Let $\bar{\mathbf{w}} \in \mathbb{R}^K$ be a continuous random variable. Using the same argument as in Suppl. C.1.1, we can show:

$$\begin{aligned} & P(\arg \max(\bar{\mathbf{w}})_i = 1) \\ &= P(\max(\{\bar{\mathbf{w}}_j : j \neq i\}) < \bar{\mathbf{w}}_i) \\ &= \int_{-\infty}^{\infty} \prod_{j \neq i} P(\bar{\mathbf{w}}_j < l) P(\bar{\mathbf{w}}_i = l) dl \end{aligned}$$

From Eq. C.14, we get:

$$= \int_{-\infty}^{\infty} \prod_{j \neq i} \Phi\left(\frac{l - \tilde{\alpha}_{t|s}(\mathbf{w}_s)_j}{\sqrt{1 - \tilde{\alpha}_{t|s}^2}}\right) \left[\frac{1}{\sqrt{1 - \tilde{\alpha}_{t|s}^2}} \phi\left(\frac{l - \tilde{\alpha}_{t|s}(\mathbf{w}_s)_i}{\sqrt{1 - \tilde{\alpha}_{t|s}^2}}\right) \right] dl \quad (\text{C.16})$$

where $\phi(z) = \exp(-z^2/2)/\sqrt{2\pi}$ is the standard normal distribution and $\Phi(z) = \int_{-\infty}^z \phi(t) dt$ is its cumulative distribution function.

Step 3: Comparing the two distributions Clearly, from Eq. C.16, we observe that $P(\arg \max(\bar{\mathbf{w}})_i = 1) = P(\arg \max(\bar{\mathbf{w}})_j = 1) \forall i, j \neq k$ if and only if

$(\mathbf{w}_s)_i = (\mathbf{w}_s)_j \forall i, j \neq k$. This condition rarely holds (in fact, the probability of exact equality is essentially zero). Thus, for a given $\mathbf{w}_s \sim \tilde{q}_s(\cdot | \mathbf{x}; \tilde{\alpha}_t)$,

$$q_{t|s}(\cdot | \mathbf{z}_s := \arg \max(\mathbf{w}_s)) \neq [\arg \max]_{\star} \tilde{q}_{t|s}(\cdot | \mathbf{w}_s). \quad (\text{C.17})$$

Conclusion Therefore, the discretized trajectory $\{\mathbf{z}_t := \arg \max(\mathbf{w}_t)\}_{t \in [0,1]}$ does not necessarily follow a discrete diffusion process.

C.1.4 Rao-Blackwellized Negative Evidence Lower Bound

Schiff et al. [2025] show that the NELBO for USDMs is given as:

$$\text{NELBO}(q, p_\theta; \mathbf{x}) = \mathbb{E}_{t \sim \mathcal{U}[0,1], q_t(\mathbf{z}_t | \mathbf{x}; \alpha_t)} f(\mathbf{z}_t, \mathbf{x}_\theta(\mathbf{z}_t, t), \alpha_t; \mathbf{x}), \quad (\text{C.18})$$

where f is defined as:

$$f(\mathbf{z}_t, \mathbf{x}_\theta(\mathbf{z}_t, t), \alpha_t; \mathbf{x}) = -\frac{\alpha_t'}{K\alpha_t} \left[\frac{K}{\bar{\mathbf{x}}_i} - \frac{K}{(\bar{\mathbf{x}}_\theta)_i} - \sum_j \frac{\bar{\mathbf{x}}_j}{\bar{\mathbf{x}}_i} \log \frac{(\bar{\mathbf{x}}_\theta)_i \cdot \bar{\mathbf{x}}_j}{(\bar{\mathbf{x}}_\theta)_j \cdot \bar{\mathbf{x}}_i} \right]. \quad (\text{C.19})$$

where the subscript i denotes the i^{th} index of a vector, $\bar{\mathbf{x}} = K\alpha_t \mathbf{x} + (1 - \alpha_t)\mathbf{1}$, $\bar{\mathbf{x}}_\theta = K\alpha_t \mathbf{x}_\theta(\mathbf{z}_t, t) + (1 - \alpha_t)\mathbf{1}$, α_t' denotes the time-derivative of the α_t , and we define $i = \arg \max_{j \in [K]} (\mathbf{z}_t)_j$ to be the non-zero entry of $\mathbf{z}_t$.

Rao Blackwellized NELBO (Ours) To reduce GPU memory usage and training time, we rewrite the original ELBO objective in Eq. C.19 by eliminating the need to explicitly materialize the one-hot vector $\bar{\mathbf{x}}$. This leads to a more efficient formulation that significantly improves practicality. The resulting objective, shown in Eq. C.20, not only removes $\bar{\mathbf{x}}$ but also applies Rao-Blackwellization to analytically compute certain expectations, thereby reducing variance. We now derive the improved loss:

$$\begin{aligned}
& f_{\text{Duo}}(\mathbf{z}_t, \mathbf{x}_\theta(\mathbf{z}_t, t), \alpha_t; \mathbf{x}) \\
&= -\frac{\alpha_t'}{K\alpha_t} \left[\frac{K}{\bar{\mathbf{x}}_i} - \frac{K}{(\bar{\mathbf{x}}_\theta)_i} - \sum_j \frac{\bar{\mathbf{x}}_j}{\bar{\mathbf{x}}_i} \log \frac{(\bar{\mathbf{x}}_\theta)_i \cdot \bar{\mathbf{x}}_j}{(\bar{\mathbf{x}}_\theta)_j \cdot \bar{\mathbf{x}}_i} \right]. \\
&= -\frac{\alpha_t'}{K\alpha_t} \left[\frac{K}{\bar{\mathbf{x}}_i} - \frac{K}{(\bar{\mathbf{x}}_\theta)_i} - \sum_j \frac{\bar{\mathbf{x}}_j}{\bar{\mathbf{x}}_i} \log \frac{(\bar{\mathbf{x}}_\theta)_i}{(\bar{\mathbf{x}}_\theta)_j} - \sum_j \frac{\bar{\mathbf{x}}_j}{\bar{\mathbf{x}}_i} \log \frac{\bar{\mathbf{x}}_j}{\bar{\mathbf{x}}_i} \right] \\
&\text{Let } \kappa_t = (1 - \alpha_t)/(K\alpha_t + 1 - \alpha_t), \\
&= -\frac{\alpha_t'}{K\alpha_t} \left[\frac{K}{\bar{\mathbf{x}}_i} - \frac{K}{(\bar{\mathbf{x}}_\theta)_i} - \sum_j \frac{\bar{\mathbf{x}}_j}{\bar{\mathbf{x}}_i} \log \frac{(\bar{\mathbf{x}}_\theta)_i}{(\bar{\mathbf{x}}_\theta)_j} - \left((K-1)\kappa_t \mathbb{1}_{\mathbf{z}_t=\mathbf{x}} - \frac{1}{\kappa_t} \mathbb{1}_{\mathbf{z}_t \neq \mathbf{x}} \right) \log \kappa_t \right] \\
&\text{Let } m \text{ denote the index in } \mathbf{x} \text{ corresponding to } 1, \text{ i.e., } \mathbf{x}_m = 1, \\
&= -\frac{\alpha_t'}{K\alpha_t} \left[\frac{K}{\bar{\mathbf{x}}_i} - \frac{K}{(\bar{\mathbf{x}}_\theta)_i} - (\kappa_t \mathbb{1}_{\mathbf{z}_t=\mathbf{x}} + \mathbb{1}_{\mathbf{z}_t \neq \mathbf{x}}) \sum_j \log \frac{(\bar{\mathbf{x}}_\theta)_i}{(\bar{\mathbf{x}}_\theta)_j} - K \frac{\alpha_t}{1 - \alpha_t} \log \frac{(\bar{\mathbf{x}}_\theta)_i}{(\bar{\mathbf{x}}_\theta)_m} \mathbb{1}_{\mathbf{z}_t \neq \mathbf{x}} \right. \\
&\quad \left. - \left((K-1)\kappa_t \mathbb{1}_{\mathbf{z}_t=\mathbf{x}} - \frac{1}{\kappa_t} \mathbb{1}_{\mathbf{z}_t \neq \mathbf{x}} \right) \log \kappa_t \right]. \tag{C.20}
\end{aligned}$$

This reformulation provides an efficient and low-variance formula for computing the NELBO for USDMs while maintaining minimal memory overhead. The final expression is as follows:

$$\text{NELBO}(q, p_\theta; \mathbf{x}) = \mathbb{E}_{t \sim \mathcal{U}[0,1], q_t(\mathbf{z}_t|\mathbf{x}; \alpha_t)} f_{\text{Duo}}(\mathbf{z}_t, \mathbf{x}_\theta(\mathbf{z}_t, t), \alpha_t; \mathbf{x}), \tag{C.21}$$

where f_{Duo} is defined in Eq. C.20. We conduct ablation experiments, detailed in Table 5.3, to quantify its effectiveness.

As a sanity check, we empirically verify the equivalence between Eq. C.18 and Eq. C.21. Specifically, we train Duo on LM1B with sentence-packing (Table 5.1) using our proposed Rao-Blackwellized NELBO Eq. C.21. We then evaluate the model using the inefficient NELBO Eq. C.18 as proposed by Schiff et al. [2025], and recover the same perplexity (33.7).

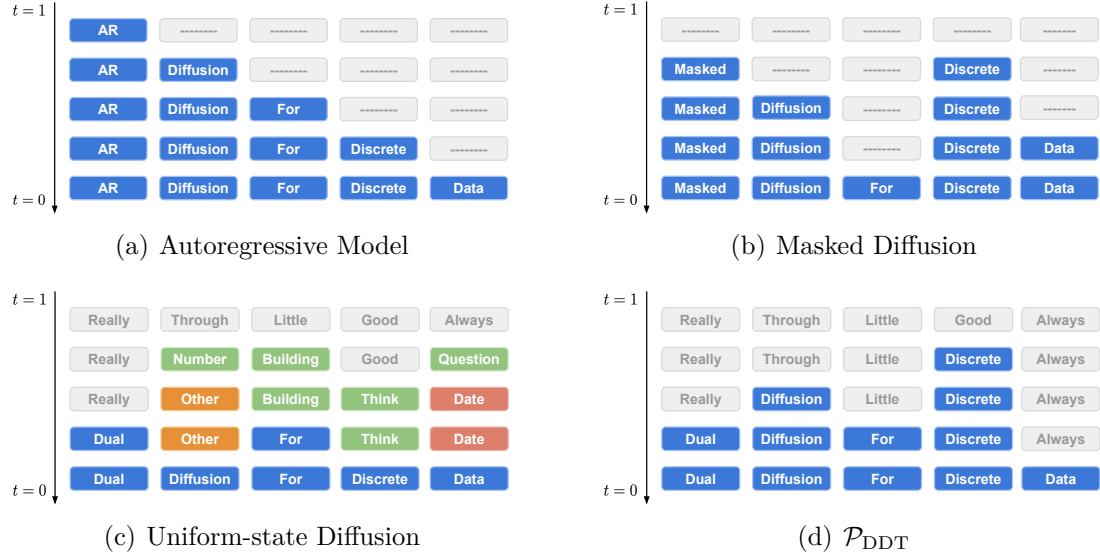


Figure C.1: Comparison of sample generation processes in various discrete sequence models; see Suppl. C.1.5 for a detailed discussion. (a) **Autoregressive Model**: Tokens are generated sequentially, one at a time, from left to right. (b) **Masked Diffusion**: Once unmasked, a token remains fixed, though multiple tokens may be denoised simultaneously at each step. (c) **Uniform-state Diffusion**: Tokens can visit several intermediate states during the diffusion process. (d) $\mathcal{P}_{\text{DDT}}$: Similar to USDMs, generation begins with a sequence of randomly initialized tokens. However, once a token flips, it remains fixed throughout the reverse generation process. Thus, the generation process closely resembles to that of MDMs.

C.1.5 Reverse Process Visualizations

Figure C.1 illustrates the differences among four diffusion processes: autoregressive models (which can be viewed as a form of left-to-right diffusion), masked diffusion, uniform diffusion, and Duo with Discrete Consistency Distillation (DCD). Each method demonstrates a distinct pattern of token evolution during generation.

Autoregressive Models Autoregressive models generate tokens one at a time, sequentially from left to right. At each forward pass of the neural network, a single token is produced, which limits the throughput. Finally, tokens are not revised once they have been generated.

Masked Diffusion Models (MDMs) In masked diffusion, all tokens in the original sequence are masked, when at the highest noise level, and are progressively unmasked throughout the diffusion process until the data sequence is fully denoised. Hence each token can take on only one of two possible values.

Uniform-state Diffusion Models (USDMs) Uniform-state diffusion models allow tokens to be updated at every diffusion step, using a uniform prior over the vocabulary. In contrast to autoregressive or masked models, each token can be updated multiple times throughout the diffusion process.

$\mathcal{P}_{\text{DDT}}$ This represents a deterministic trajectory between the clean data $\mathbf{x}^{1:L}$ and a sample from the uniform prior: $[\mathbf{z}_{t=1}^\ell]_{\ell=1}^L = [\arg \max(\tilde{\alpha}_t \mathbf{x}^\ell + \sqrt{1 - \tilde{\alpha}_t^2} \boldsymbol{\epsilon}^\ell)]_{\ell=1}^L$ for $\boldsymbol{\epsilon}^\ell \sim \mathcal{N}(0, \mathbf{I}_K) \forall \ell \in [L]$. As shown in Eq. 5.19, the value of each token at the ℓ^{th} position at an intermediate timestep t is given by $\arg \max(\tilde{\alpha}_t \mathbf{x}^\ell + \sqrt{1 - \tilde{\alpha}_t^2} \boldsymbol{\epsilon}^\ell)$. This expression represents the $\arg \max$ of a linear interpolation between $\mathbf{x}^\ell$ —the one-hot vector of the clean data—and the Gaussian noise vector $\boldsymbol{\epsilon}^\ell$, both of which remain fixed throughout the entire generation process. Consequently, the intermediate token can take on only one of two values: $\mathbf{x}^\ell$ (when t is close to 0) or $\arg \max(\boldsymbol{\epsilon}^\ell)$ (when t is close to 1). The generation process closely resembles to that of MDMs where a token once denoised, can’t change. This is called the “carry over” operation in MDMs [Sahoo et al., 2024a]. Please note that the $\mathcal{P}_{\text{DDT}}$ is only used during distillation to generate the teacher and student targets Alg.(1) and that Duo isn’t trained to generate samples with “carry over”: i.e. once a token changes, it never changes again.

C.2 Additional Proofs

C.2.1 ELBO Equivalence

We have already established that a Uniform-state discrete diffusion process q has an underlying Gaussian diffusion process $\tilde{q}_t$. The diffusion parameters of the Gaussian process, denoted $\tilde{\alpha}_t$, and those of the Uniform-state discrete diffusion process, denoted $\mathcal{T}(\tilde{\alpha}_t)$, are related through the diffusion transformation operator $\mathcal{T}$; see Eq. 5.3.

Using this relationship and the result from Eq. 5.2, we can express the NELBO for USDMs as:

$$\text{NELBO}(q, p_\theta; \mathbf{x}) = \mathbb{E}_{t \sim \mathcal{U}[0,1], q_t(\mathbf{z}_t | \mathbf{x}; \alpha_t)} f_{\text{Duo}}(\mathbf{z}_t, \mathbf{x}_\theta(\mathbf{z}_t, t), \alpha_t; \mathbf{x}), \quad (\text{C.22})$$

and equivalently in terms of the Gaussian diffusion parameters $\tilde{\alpha}_t$, as:

$$\text{NELBO}(q, p_\theta; \mathbf{x}) = \mathbb{E}_{t \sim \mathcal{U}[0,1], P_t(\mathbf{z}_t | \mathbf{x}; \mathcal{T}(\tilde{\alpha}_t))} f_{\text{Duo}}(\mathbf{z}_t, \mathbf{x}_\theta(\mathbf{z}_t, t), \alpha_t := \mathcal{T}(\tilde{\alpha}_t); \mathbf{x}). \quad (\text{C.23})$$

From Eq. 5.2 and Eq. 5.5, we also know that

$$P_t(\mathbf{z}_t | \mathbf{x}; \mathcal{T}(\tilde{\alpha}_t)) = [\arg \max]_{\star \tilde{q}_t}(\mathbf{w}_t | \mathbf{x}; \tilde{\alpha}_t).$$

Substituting this into Eq. C.23, we obtain:

$$\begin{aligned} \text{NELBO}(q, p_\theta; \mathbf{x}) \\ = \mathbb{E}_{t \sim \mathcal{U}[0,1], \tilde{q}_t(\mathbf{w}_t | \mathbf{x}; \tilde{\alpha}_t)} f_{\text{Duo}}(\mathbf{z}_t := \arg \max(\mathbf{w}_t), \mathbf{x}_\theta(\arg \max(\mathbf{w}_t), t), \alpha_t := \mathcal{T}(\tilde{\alpha}_t); \mathbf{x}). \end{aligned} \quad (\text{C.24})$$

Eq. C.24 shows that the NELBO for USDMs can be equivalently computed using the latents of the corresponding Gaussian diffusion process. We now extend this

equivalence to sequences. Starting from Eq. C.22 and Eq. C.24, we have:

$$\begin{aligned} & \text{NELBO}(q, p_\theta; \mathbf{x}^{1:L}) \\ &= \mathbb{E}_{t, q_t(\mathbf{z}_t^{1:L} | \mathbf{x}^{1:L}; \alpha_t)} \sum_{\ell \in [L]} f_{\text{Duo}}(\mathbf{z}_t^\ell, \mathbf{x}_\theta(\mathbf{z}_t^{1:L}, t), \alpha_t; \mathbf{x}^\ell) \end{aligned} \quad (\text{C.25})$$

$$= \mathbb{E}_{t, \tilde{q}_t(\mathbf{w}_t^{1:L} | \mathbf{x}; \tilde{\alpha}_t)} \sum_{\ell \in [L]} f_{\text{Duo}}\left(\mathbf{z}_t^\ell := \text{arg max}(\mathbf{w}_t^\ell), \mathbf{x}_\theta\left([\text{arg max}(\mathbf{w}_t^{\ell'})]_{\ell'=1}^L, t\right), \alpha_t := \mathcal{T}(\tilde{\alpha}_t); \mathbf{x}^\ell\right). \quad (\text{C.26})$$

This concludes our proof.

As a sanity check, we empirically verify the equivalence of Eq. C.25 and Eq. C.26. To do this, we trained Duo on LM1B with sentence-packing (Table 5.1) using the true ELBO from Eq. C.25. We then evaluated the model using Gaussian latents and Eq. C.26, and recovered the same PPL (33.7) as when using discrete latents. For each datapoint $\mathbf{x}$, we used 1000 Monte Carlo samples for t sampled using antithetic-sampling, with a linear schedule for $\tilde{\alpha}_t = 1 - t$.

C.2.2 Discrete Consistency Distillation

Optimal Gaussian PF-ODEs

For a Gaussian diffusion process (see Sec. 2.3.3), the probability flow ODE (PF-ODE) can be reversed using the DDIM sampler [Song et al., 2021a], whose update step is given by:

$$\mathbf{z}_s = \tilde{\alpha}_s \mathbf{x}_\theta(\mathbf{z}_t, t) + \sqrt{1 - \tilde{\alpha}_s^2} \epsilon_\theta(\mathbf{z}_t, t) \quad (\text{C.27})$$

where $s < t$, $\mathbf{x}_\theta : \mathbb{R}^K \times [0, 1] \rightarrow \Delta$ is the denoising model, and $\epsilon_\theta(\mathbf{z}_t, t) = (\mathbf{z}_t - \tilde{\alpha}_t \mathbf{x}_\theta(\mathbf{z}_t, t)) / \sqrt{1 - \tilde{\alpha}_t^2}$.

Assuming an *optimal denoiser* such that $\mathbf{x}_\theta(\mathbf{z}_t, t) = \mathbf{x} \forall t \in [0, 1]$, and given

$\mathbf{z}_{t=1} = \boldsymbol{\epsilon} \sim \mathcal{N}(0, \mathbf{I}_K)$ and $\mathbf{x} \sim q_{\text{data}}$, Eq. C.27 simplifies to

$$\mathbf{z}_s = \tilde{\alpha}_t \mathbf{x} + \sqrt{1 - \tilde{\alpha}_t^2} \boldsymbol{\epsilon} \quad (\text{C.28})$$

This holds $\forall s \in [0, 1]$. Thus, the optimal PF-ODE trajectory $\mathcal{P}_{\text{ODE}}(\mathbf{x}, \boldsymbol{\epsilon})$ is given as:

$$\mathcal{P}_{\text{ODE}}(\mathbf{x}, \boldsymbol{\epsilon}) = \left\{ \tilde{\alpha}_t \mathbf{x} + \sqrt{1 - \tilde{\alpha}_t^2} \boldsymbol{\epsilon} \right\}_{t \in [0, 1]}. \quad (\text{C.29})$$

We can easily extend this to sequences:

$$\mathcal{P}_{\text{ODE}}(\mathbf{x}^{1:L}, \boldsymbol{\epsilon}^{1:L}) = \left\{ [\tilde{\alpha}_t \mathbf{x}^\ell + \sqrt{1 - \tilde{\alpha}_t^2} \boldsymbol{\epsilon}^\ell]_{\ell=1}^L \right\}_{t \in [0, 1]} \quad (\text{C.30})$$

Discrete Consistency Distillation Ablation

Typically, consistency models use the EMA (exponential moving average) parameters of the denoising model as the teacher (Sec. 5.2). In contrast, our proposed distillation algorithm uses the denoising model weights from the previous distillation round as the teacher. We ablate this design choice in Alg. 3 by instead using the EMA weights of the denoising model obtained during pre-training as the teacher. This modification leads to degraded performance, as shown in Fig. C.6 and Table C.3.

C.3 Experimental details

C.3.1 Plaid Baseline

For PLAID on LM1B, we retrained it without self-conditioning [Chen et al., 2023] to match our denoising model’s parameter count. While self-conditioning improves

Algorithm 3 Discrete Consistency Distillation (DCD) with EMA as teacher

Input: Dataset $\mathcal{D}$, learning rate η , number of distillation rounds K , number of training iterations per round M , ema μ , weights of the denoising model θ , weights of the EMA model θ_{ema} , discretization step δ .

for $i = 1$ **to** K **do**

$\theta^- \leftarrow \text{stopgrad}(\theta_{\text{ema}})$

 ▷ Only difference w.r.t the standard DCD algorithm (Alg. 1).

for $j = 1$ **to** M **do**

 Sample $\mathbf{x}^{1:L} \sim \mathcal{D}$, $t \sim \mathcal{U}[0, 1]$, and $\epsilon^\ell \sim \mathcal{N}(0, I_K)$.

$s \leftarrow \max(t - \delta, 0)$

$\mathbf{z}_s^{1:L} \leftarrow [\arg \max(\tilde{\alpha}_s \mathbf{x}^\ell + \sqrt{1 - \tilde{\alpha}_s^2} \epsilon^\ell)]_{\ell=1}^L$

$\mathbf{z}_t^{1:L} \leftarrow [\arg \max(\tilde{\alpha}_t \mathbf{x}^\ell + \sqrt{1 - \tilde{\alpha}_t^2} \epsilon^\ell)]_{\ell=1}^L$

$\mathcal{L}_{\text{DCD}}(\theta; \theta^-) \leftarrow \text{D}_{\text{KL}}(\mathbf{x}_\theta(\mathbf{z}_t^{1:L}, t), \mathbf{x}_{\theta^-}(\mathbf{z}_s^{1:L}, s))$

$\theta \leftarrow \theta - \eta \nabla_\theta \mathcal{L}_{\text{DCD}}(\theta; \theta^-)$

$\theta_{\text{ema}} \leftarrow \text{stopgrad}(\mu \theta_{\text{ema}} + (1 - \mu) \theta)$

end for

$\delta \leftarrow 2 \cdot \delta$

end for

Output: θ_{ema}

PPL and can be applied to both discrete and Gaussian diffusion models, we omit it for consistency with baselines such as MDLM, SEDD, UDLM, and D3PM. Since higher training precision benefits discrete diffusion models [Shi et al., 2024a], we use `bfloat16` for the forward pass through the denoising model while keeping `float64` for other computations to stabilize PLAID training. Due to their inefficient open-source codebase¹, we report PLAID results for LM1B at 100K steps, as further training was infeasible. For OWT, we report results from Lou et al. [2024], where PLAID was trained at higher precision for 1.3M steps, favoring the baseline.

C.3.2 Denoising Model

Unlike prior discrete diffusion approaches, we design the denoising model $\mathbf{x}_\theta : \mathcal{V}^L \times [0, 1] \rightarrow \Delta^L$ to operate on both continuous latents $\mathbf{y}_c^{1:L} \in \Delta^L$ and discrete

¹<https://github.com/igul222/plaid>

latents $\mathbf{y}_d^{1:L} \in \mathcal{V}^L$. We implement $\mathbf{x}_\theta$ as a Transformer [Vaswani et al., 2017], where token embeddings in the first layer are computed via matrix multiplication:

$$(\mathbf{y}_c^\ell)_{\ell \in [L]}^\top \text{vocab_embeddings}$$

with $\text{vocab_embeddings} \in \mathbb{R}^{K \times m}$ denoting the vocabulary embedding matrix and m the embedding dimension. For discrete inputs $(\mathbf{y}_d^\ell)_{\ell \in [L]} \in \mathcal{V}$, we perform standard embedding lookups. In contrast, continuous inputs $(\mathbf{y}_c^\ell)_{\ell \in [L]} \in \Delta^K$ act as “soft lookups”, producing a convex combination of the vocabulary embeddings.

C.3.3 Low Discrepancy Sampler

To reduce variance during training we use a low-discrepancy sampler, similar to that proposed Kingma et al. [2021]. Specifically, when processing a minibatch of N samples, instead of independently sampling N from a uniform distribution, we partition the unit interval and sample the time step for each sequence $i \in \{1, \dots, N\}$ from a different portion of the interval $t_i \sim \mathcal{U}\left[\frac{i-1}{N}, \frac{i}{N}\right]$. This ensures that our sampled timesteps are more evenly spaced across the interval $[0, 1]$, reducing the variance of the ELBO.

C.3.4 Likelihood Evaluation

We use a single monte-carlo estimate for t to evaluate the likelihood. We use a low discrepancy sampler [Kingma et al., 2021] to reduce the variance of the estimate. We evaluate likelihood using the true ELBO, not the curriculum learning objective.

C.3.5 Language Modeling

We detokenize the One Billion Words dataset following [Lou et al. \[2024\]](#), [Sahoo et al. \[2024a\]](#), whose code can be found [here](#)². We tokenize the One Billion Words dataset with the `bert-base-uncased` tokenizer, following [He et al. \[2022\]](#). We concatenate and wrap sequences to a length of 128 [[Raffel et al., 2020a](#)].

We tokenize OpenWebText with the `GPT2` tokenizer. We concatenate and wrap them to a length of 1,024. When wrapping, we add the `eos` token in-between concatenated sequences. Since OpenWebText does not have a validation split, we leave the last 100k docs as validation.

We parameterize our autoregressive baselines, UDLM, SEDD, and MDLM with the modified diffusion transformer architecture [[Peebles and Xie, 2023](#)] from [Lou et al. \[2024\]](#), [Sahoo et al. \[2024a\]](#). We use 12 layers, a hidden dimension of 768, 12 attention heads, and a timestep embedding of 128 for the uniform diffusion models (SEDD Uniform, UDLM, Duo). Word embeddings are not tied between the input and output. We train the SEDD and MDLM baselines using the original code provided by their authors.

We use the AdamW optimizer with a batch size of 512, constant learning rate warmup from 0 to a learning rate of 3e-4 for 2,500 steps. We use a constant learning rate for 1M, 5M, or 10M steps on One Billion Words, and 1M steps for OpenWebText. We use a dropout rate of 0.1.

²<https://github.com/louaaron/Score-Entropy-Discrete-Diffusion/blob/main/data.py>

C.3.6 Zeroshot Likelihood

We evaluate zeroshot likelihoods by taking the models trained on OpenWebText and evaluating likelihoods on the validation splits of 7 datasets: Penn Tree Bank (PTB; Marcus et al. [1993]), Wikitext [Merity et al., 2016], One Billion Word Language Model Benchmark (LM1B; Chelba et al. [2014]), Lambada [Paperno et al., 2016], AG News [Zhang et al., 2015], and Scientific Papers (Pubmed and Arxiv subsets; Cohan et al. [2018]). We detokenize the datasets following Lou et al. [2024]. For the AG News and Scientific Papers (Pubmed and Arxiv), we apply both the Wikitext and One Billion Words detokenizers. Since the zeroshot datasets have different conventions for sequence segmentation, we wrap sequences to 1024 and do not add `eos` tokens in between sequences.

C.3.7 Curriculum Learning

We visualize the diffusion parameter $\tilde{\alpha}_t$ in Fig. C.2. As shown in Eq. C.20, the diffusion NELBO is weighted by α_t' , so when $\alpha_t' \approx 0$, the contribution of diffusion time step t to the NELBO becomes negligible, offering little learning signal. Prior work [Sahoo et al., 2024a, Lou et al., 2024] used a linear schedule for α_t and did not face this issue. Furthermore, Fig. C.2 shows that for $t \in [\beta, \gamma]$, the Gaussian latent retains a higher signal level than its discrete counterpart, making it easier for the denoising model to recover the clean signal, as discussed in Sec. 5.4.1.

To mitigate these issues, we restrict the training window to $t \in [\beta, \gamma]$ in Eq. 5.18 when training on Gaussian latents, thereby avoiding the region where $\alpha_t' \approx 0$. For the discrete diffusion process, we set the time range such that $\alpha_t = \mathcal{T}(\tilde{\alpha}_t) \in [0.05, 0.95]$. While this range depends on the vocabulary size K , we found it to be

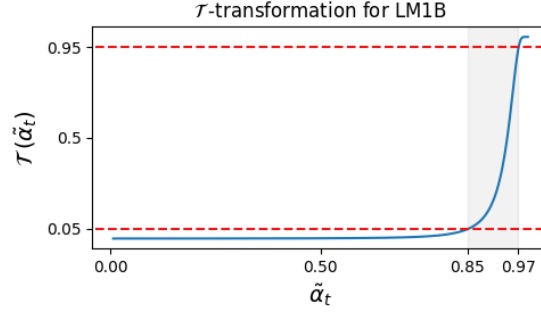


Figure C.2: Diffusion transformation operator $\mathcal{T}(\tilde{\alpha}_t)$ Eq. 5.3 for the `bert-base-uncased` tokenizer.

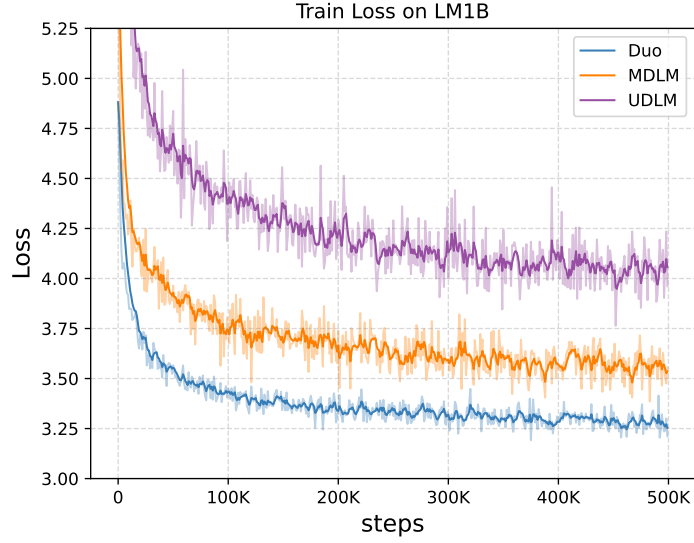


Figure C.3: Training loss curves for Duo (ours) with curriculum learning, UDLM, and MDLM. We see observe that curriculum learning leads to low-variance training. Duo’s curve is lower because its a biased estimate of the ELBO.

similar for both the `gpt-2` and `bert-base-uncased` tokenizers, corresponding to $[\beta, \gamma] = [0.03, 0.15]$. Although this introduces a slight bias in the NELBO estimate, it significantly reduces training variance.

C.3.8 Distillation Experiments

To compare distilled Duo with SDTT, we distill an MDLM on LM1B for 5 rounds of 10k training steps with a batch size of 128 and a learning rate of $6.0e - 05$. We linearly increase the learning rate for 500 steps and hold it constant for the rest of training. These hyperparameters correspond to the original SDTT recipe of [Deschenaux and Gulcehre \[2024\]](#).

C.4 Additional Experiments

C.4.1 Gradient Variance and Loss Variance

Refer Table [C.1](#).

Table C.1: Curriculum learning drastically lowers the gradient variance in Duo trained with a fixed $\tau = 0.001$. The table shows the summed gradient variance of all the weights (*left*), the 100 weights with the highest variance (*middle*), and the loss variance (*right*) comparing Duo with CL and without CL.

Train steps	Gradient Variance ($\downarrow$)				Loss Variance ($\downarrow$)	
	All weights		Top 100 weights		CL	w/o CL
	CL	w/o CL	CL	w/o CL		
10k	2815.36	10852.9	0.30	11.7	7.09	9.19
20k	2471.65	7811.04	0.85	20.09	6.29	7.72
50k	1890.76	6315.7	1.21	34.2	5.33	6.85
100k	1469.85	5454.7	0.86	55.1	4.97	6.32
500k	947.98	1678.47	1.15	1.92	4.76	5.47

C.4.2 Tau Ablations

Refer Fig. C.4.

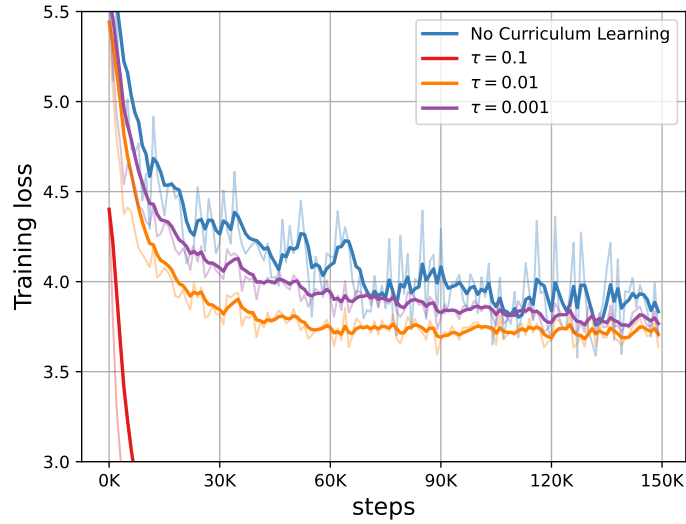


Figure C.4: We study the training bias and variance introduced by $\tau > 0$. Models were trained on the LM1B dataset.

C.4.3 Sample Quality Base Model

Refer Fig. C.5 with Table C.2.

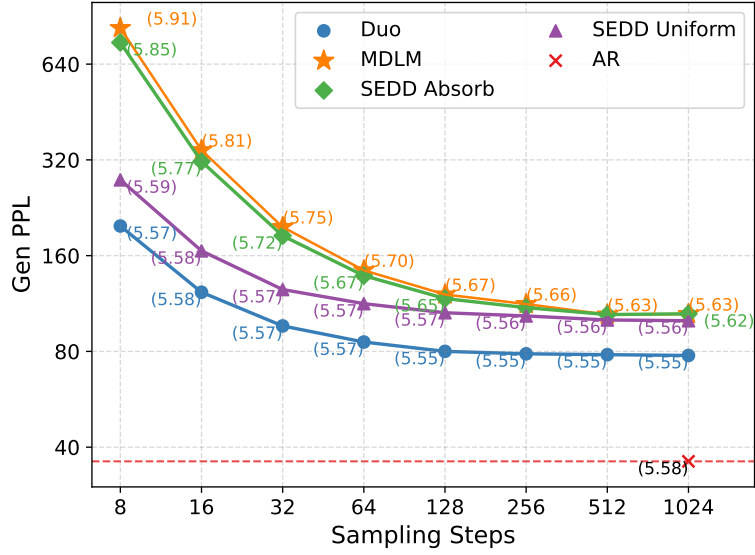


Figure C.5: Sample quality comparison using Gen PPL ($\downarrow$) between Duo (ours), MDLM, SEDD (Absorb / Uniform), and AR. Values in brackets indicate sample entropy ($\uparrow$). Among USDMs, Duo achieves lower Gen PPL than SEDD-Uniform, indicating higher sample quality. Compared to MDLMs, Duo yields lower Gen PPL with a slight trade-off in entropy. Exact quantitative numbers for Gen PPL can be found in Table C.2.

Table C.2: Gen PPL ($\downarrow$) and Ent. ($\uparrow$) (entropy) for Duo (ours), MDLM and SEDD (Absorb / Uniform).

T	Duo		SEDD Uniform		MDLM		SEDD Absorb	
	Gen PPL ($\downarrow$)	Ent. ($\uparrow$)	Gen PPL ($\downarrow$)	Ent. ($\uparrow$)	Gen PPL ($\downarrow$)	Ent. ($\uparrow$)	Gen PPL ($\downarrow$)	Ent. ($\uparrow$)
1024	77.69	5.55	99.90	5.56	104.85	5.63	105.03	5.62
512	78.14	5.55	100.44	5.56	104.43	5.63	104.45	5.62
256	78.62	5.55	103.41	5.56	112.70	5.66	109.82	5.63
128	80.02	5.55	105.82	5.57	120.77	5.67	117.28	5.65
64	85.62	5.57	113.02	5.57	143.88	5.70	138.42	5.67
32	96.19	5.57	125.21	5.57	196.79	5.75	184.71	5.72
16	122.78	5.58	165.66	5.58	343.33	5.81	316.33	5.77
8	198.27	5.57	276.89	5.59	830.82	5.91	748.37	5.85

C.4.4 Distillation

Refer Fig. C.7 with Table C.4, Fig. C.8 and Fig. C.6 with Table C.3.

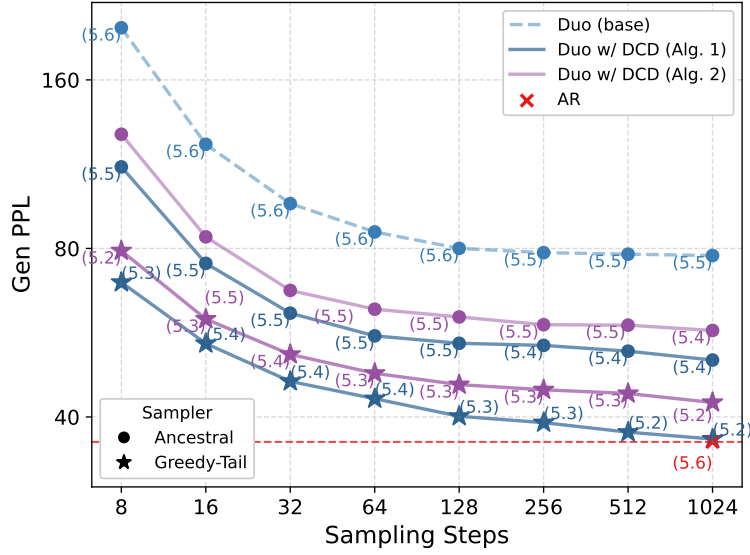


Figure C.6: We compare DCD using denoising weights (Alg. 1) vs. EMA weights (Alg. 3) as the teacher. Using the denoising model yields a more effective distilled model. Quantitative numbers for Gen PPL can be found in Table C.3.

Table C.3: We compare Gen PPL ($\downarrow$) and entropy ($\uparrow$) of the base model and its DCD-distilled variants using denoising weights (Alg. 1) vs. EMA weights (Alg. 3) as the teacher. [†]Indicates use of the Greedy-Tail sampler instead of the ancestral sampler.

T	Duo (base)		Duo Distilled							
			Alg. 1		Alg. 2		Alg. 1		Alg. 2	
			Gen PPL	Entropy	Gen PPL	Entropy	Gen PPL [†]	Entropy [†]	Gen PPL [†]	Entropy [†]
1024	77.69	5.55	50.55	5.36	57.09	5.44	36.53	5.19	42.46	5.25
512	78.14	5.55	52.43	5.38	58.35	5.46	37.58	5.21	44.05	5.25
256	78.62	5.55	53.69	5.43	58.46	5.47	39.08	5.26	44.73	5.28
128	80.02	5.55	54.16	5.46	60.35	5.51	40.12	5.30	45.69	5.31
64	85.62	5.57	55.83	5.49	62.31	5.52	43.12	5.35	47.87	5.34
32	96.19	5.57	61.31	5.52	67.31	5.54	46.31	5.38	51.74	5.36
16	122.78	5.58	75.24	5.53	83.89	5.55	54.11	5.37	59.83	5.34
8	198.27	5.57	111.88	5.52	127.94	5.54	69.58	5.30	79.24	5.25

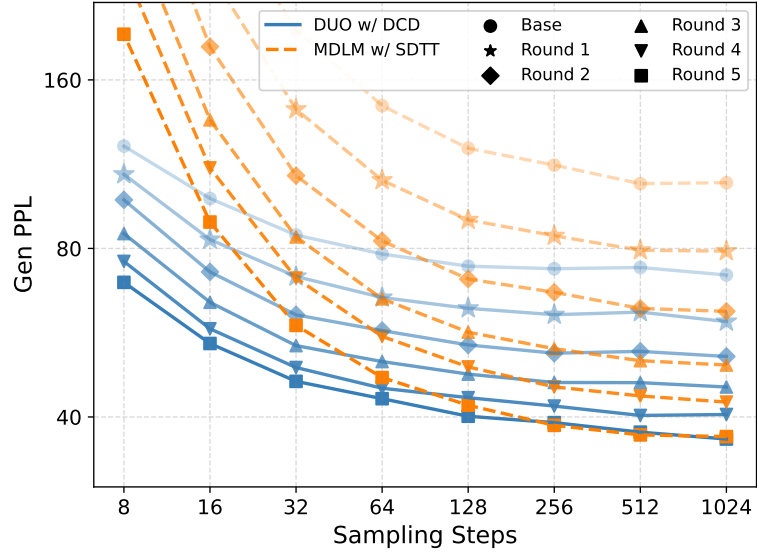


Figure C.7: Sample quality comparison using Gen PPL ($\downarrow$) of Duo (Ours) distilled with our proposed DCD algorithm and MDLM distilled with SDTT after successive distillation round. Duo always dominates in the low sampling steps regime. Refer Table C.4 for the exact quantitative numbers.

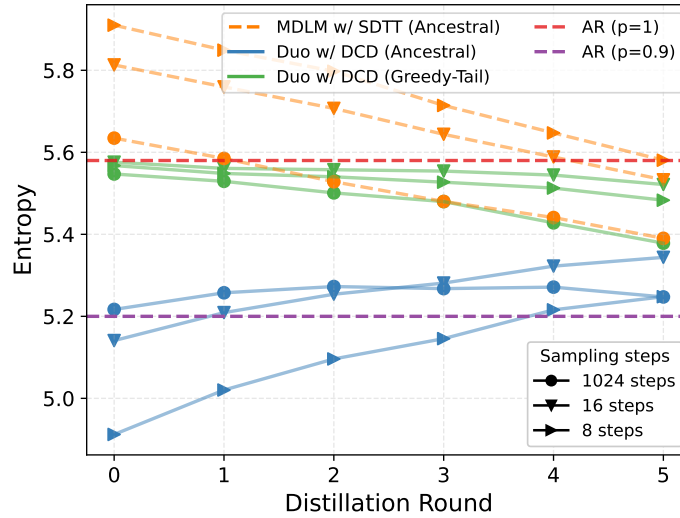


Figure C.8: Entropy of MDLM distilled using SDTT, and of Duo distilled using CDC. The entropy of the SDTT-distilled MDLM decreases with distillation, while the entropy of the CDC-distilled Duo model increases. The curves corresponding to a higher number of sampling steps are displayed with lighter colors to emphasize the low sampling step regimes.

Table C.4: Generative perplexity and entropy for Duo distilled using Discrete Consistency Distillation (DCD) (Alg. 1) and MDLM distilled SDTT.

MDLM w/ SDTT			Duo w/ DCD			
ancestral			ancestral		Greedy-Tail	
Gen PPL	Entropy		Gen PPL	Entropy	Gen PPL	Entropy
<i>Base Model</i>						
1024	104.85	5.63	77.69	5.55	71.72	5.22
512	104.43	5.63	78.14	5.55	73.98	5.23
256	112.70	5.66	78.62	5.55	73.59	5.22
128	120.77	5.67	80.02	5.55	74.37	5.22
64	143.88	5.70	85.62	5.57	78.19	5.23
32	196.79	5.75	96.19	5.57	84.52	5.20
16	343.33	5.81	122.78	5.58	98.24	5.13
8	830.82	5.91	198.27	5.57	121.89	4.91
<i>Round 1</i>						
1024	79.12	5.59	67.58	5.54	59.22	5.26
512	79.40	5.59	67.37	5.53	61.57	5.28
256	84.28	5.61	67.78	5.54	60.91	5.27
128	89.97	5.62	70.43	5.55	62.54	5.27
64	105.90	5.65	74.45	5.56	65.38	5.28
32	141.78	5.69	81.89	5.56	71.28	5.27
16	249.15	5.76	103.03	5.57	82.99	5.23
8	618.15	5.85	164.49	5.56	108.52	5.06
<i>Round 2</i>						
1024	61.75	5.53	60.09	5.51	51.30	5.29
512	62.52	5.53	60.15	5.50	52.38	5.31
256	66.80	5.56	59.84	5.51	52.00	5.30
128	70.52	5.57	62.53	5.54	53.79	5.32
64	82.51	5.60	65.78	5.55	57.06	5.32
32	107.93	5.65	71.77	5.55	60.88	5.33
16	183.41	5.71	89.59	5.56	72.64	5.31
8	458.83	5.80	137.87	5.56	97.68	5.19
<i>Round 3</i>						
1024	49.53	5.48	56.89	5.48	45.24	5.28
512	50.42	5.49	56.13	5.48	46.06	5.31
256	52.96	5.50	56.49	5.49	46.09	5.31
128	56.70	5.52	58.49	5.52	47.71	5.33
64	65.02	5.55	61.39	5.54	50.23	5.35
32	83.85	5.59	65.96	5.55	53.64	5.36
16	135.75	5.64	82.30	5.56	64.09	5.34
8	323.56	5.71	122.49	5.55	85.04	5.25
<i>Round 4</i>						
1024	42.53	5.44	52.78	5.42	40.41	5.24
512	43.61	5.44	53.27	5.43	40.25	5.26
256	45.27	5.46	54.40	5.47	41.83	5.30
128	49.14	5.48	55.17	5.50	43.30	5.33
64	55.72	5.50	57.62	5.52	45.02	5.35
32	70.82	5.54	62.42	5.54	49.02	5.38
16	111.40	5.59	76.83	5.55	57.49	5.37
8	253.59	5.65	114.80	5.54	75.84	5.30
<i>Round 5</i>						
1024	36.89	5.39	50.55	5.36	36.53	5.19
512	37.16	5.40	52.43	5.38	37.58	5.21
256	38.65	5.41	53.69	5.43	39.08	5.26
128	41.98	5.43	54.16	5.46	40.12	5.30
64	47.04	5.45	55.83	5.49	43.12	5.35
32	62.29	5.49	61.31	5.52	46.31	5.38
16	89.17	5.53	75.24	5.53	54.11	5.37
8	193.05	5.58	111.88	5.52	69.58	5.30

C.4.5 Samples

For the qualitative analysis, we present non-cherry-picked samples from Duo (C.4.5), Duo distilled using DDT (C.4.5), MDLM (C.4.5), and MDLM distilled using SDTT (C.4.5) for $T \in \{8, 1024\}$. To ensure correct LaTeX rendering, we manually process the generated text by:

1. Curly double quotes (`\u201c`, `\u201d`) replaced with "
2. Em dashes/en dashes (`\u2014`, `\u2013`) replaced with – or -
3. Soft hyphens (`\u00ad`) removed (or replaced by a normal hyphen where it makes sense)
4. Any other special characters replaced with a suitable ASCII approximation

Duo

The samples were generated using the Greedy-Tail sampler.

<|endoftext|> it," and in his view has become a hardware factory.

"Thank you for letting me put it in a mini-brown with Okubo's watch for you, great! I have complete copy of the Brit Rumours album if you ask," said Godorta.

Katner is the man that caught fire at Wild, and sold the Mendant watches exclusively to Kings Sky 3, through the name Daniel Jgettlands.

Over the years it has brought both hits and money overseas and particularly stunned her surgeon, it from a statement last year.

"Eventually they will decide people are now buying it from another or rich people who Jordanian contributed," she said. He has been too tight to sell his clothing after a vintage of, though he has been shooting an episode of US brand MZZR videos, playingouring it and acting. He is looking to get rid of the watches.<|endoftext|>"Maybe it's disgraceful to say novelty music.It's just feeling silly," McCartney said, comparing the cheapboat to "Hot Wheels" of the automobile industry

p000 The Lost Spirit-Style R

locks us to some obsessive trolls with a great deal of photologsplan thingies handy; for to music,

,

I, Twitter, sort of, turn.oh, that's insanely interesting.

A more interesting exchange might be the history of years' crack 'thereber flames that on its crine path of. waken and the made-a-work round glow board.

,

Much of what sold and got most was this song is... Produced by a by rocker, called "Cast is the sound of Art Parrish: in a March and Illuminated way. What is a record?

Uh, miles per minute not per ft – I'll not that – how I feel about it, I mean, treatment goes to the award-winning Paul Marshall. That is how he feels at I think everyone will what he has to pay

... on. fanboys. ... He records music once a week and do a thousand things before we all changes of 20.

and so apparently this would is as good as I thought –

osted some of the jump-to-wall molds generated from McCartney recognizing that he Dixon presented before awakening:

"is unfortunately turning out the radio okay but after my friend put it on Reddit fast enough that it was like – new online, the deeper the music reviewer was, the larger George encamp speculate regarding all the site-themes, fantasy like any connection with the star Foxivity rant – he came out-detected the possible motive for that."

The musician took exception to his Web time, which was down fromfree-drum to \$5/5.

Don't try to be mistaken as a chief executive of

"It wasn't, I think, what we had expected since football's launch in 2013."

"Petrovic

"Our aim is that Russia can break barriers and combat the problems that are keeping us together."

EU can't get expected presence of riot and deal convincing. Speaking

The Ukrainian media reports FIFA has for sending down a request order regardingorter's withdrawal of permissions to offer goods and services from Ukraine, stating that it does not know "the parties involved. Moreover, they have Switzerland put a firm hold on some*license nationals (and the!) exchange with members of the group."

In 2016 the The Councils of FIFA expressed a desire to monitor an exchange of the two countries. "Other Dutch monetary policy towards Ukraine has have been the source of similar concerns since 2010 and are leading to UEFA's conditions on UEFA's trade."

The move has on the account of UEFA very little benefit. Prior to this year clinical Romania is believed to have exceeded its influence on a team's launch, upon which UEFA highlighted the West-Russia series by playing football for the second straight year. Interests abound. by well, Says World Football Association, "Alabor

Also in 2014, hosting a rare (a (a south of the release date) UEFA took to meeting with Sax- Deutschland, Minister Dieter Kohaffl and to "outright" the sequence in country construction. In the general channels, UEFAIn the political integration roadmap (for, Serbia, Poland and Slovakia": normal?

Hibert the an example region,

1883.50. Romania might not be allocated map to support V. 28 fruitn. An end to and dialogue withoves armed by referring

20.01. Ukrainian trademark politicianh Brairs later440 by Optum in May 2014. (C)KU tom)<|endoftext|>A year has passed, and now it has been a few blocks very early here in the Canadian<|endoftext|>

Figure C.9: Samples ($T = 8$) from Duo trained on OWT with Gen. PPL 121.02 and entropy = 4.91

<|endoftext|> like, 'Are you too rich and can stick talking about it?' he said. He claimed that a colleague refinissued Rs 500-fold cash to him in the house to which he had fenced half his wealth from the people. "You could not even trust them. That's the problem. You just don't do it," he said. Nairit brushed off charges that depositing the bills were illegal like 500-fold, but noted that adding the law only gives the police warrant saying it only gets you if you hid in the shop. The court found then bail is non-bailable. "Liepuralingo nasaai kurak lukhaung messiah karrhaai," he said. But if people with notes refuse to enter into a bank account, how can they get out of their pocket? "They have to pay tax." If evasion means luring into the shops of rich people at elite prices is not okay, and the Crime victim says they refuse to help the, Mr Nairit complained in his monthly open letter, upon before the attack, his colleagues told him that they had met all parties ahead of the July election for president. "The Opposition leader" activist, covered in skinning at a meeting with colleagues and supporters, gave the politicians excuse to go out on behalf of the candidates. "Nakim ka nyo aakdana dotat palabandai ng tinknyam, sila ne nabba alinukang doe na," he said. Pumakay. Naa lang natalung niya resbu numpita na. Ik nguma paatin gum kun paana para paghan?" PNN's director Sood Singh reached Hindi with the Padhaka Group Parishad (PSMDA) allegedly, to explain the attack. He further alleged that the Maarakan rioting in district, in Mindanao city of the Madriagharay started as preparations for the Independence Day celebrations when Nairit and his family in coastal Kampas share shop, used furniture for various powerpoints. "Dozens were flashing the cards on probe stamps. That's why it was a festive day for them," he said. But Nairit clarified that most likely this was a terror attack against a group of shop owners. "I mean, shops were not turned into pharmacies," he said. He said members of the shop owners however had looted money from the shop on their counter, besides security company and the ownership. Arconditude to profit Violence over the shops did not immediately turn into a NDB "terror operation", said Kazika Shiva Dwarrang, a campaigner with Umlima Banda Organizations (AWOA) outfits in North Kuala Lumpur. "I would never go with a 100-fold order. But after the standoff with the shop and if there are conditions, there will undoubtedly be more police operations as long as it is simply not illegal," she told NTN. The lesson may be that more people including NDB (President) Chai Carr are getting rich through power producers whose power generators rely on diesel fuel to secure energy supplies. The administration in its Public Sector Payment Area (PPSA) last year plans to reduce the electricity supply of generators on the grid. This mode of electricity service and radiantenergy runoff has been depleted. But Chai's enthusiasm to replace it this year is over, which takes a lot of time, and could produce some promising gains for Asia's poor, from climate-change poverty alleviation. Two-han Ni Mel Lin, with PHNL (band CEO and consultant) Consolidated Power Company, hopes the project will be completed by 2020. "The national government will be built on a system that can integrate the production system which will also contribute to creating a reliable energy mix," Yongyata Kluor's coal minister, said at conference later this month. Besides solar, commercial battery plants are battery, even the world's most advanced nuclear plants are taking advantage of that system to generate power. Demand is dependent on coal and drives the economies of two cradle-cradle low production nations. "These economic realities will bring global wealth to the developing world. They will become poorer by buying electricity only," he said. This dependence will also drive prices because people face higher food prices from time to time and low incomes. Sampara's state-run government has been trying to sell 450 billion megawatts of energy to producers that deliver dependability it requires for a long-term industry that is both cheap and reliable. Venezuela's dependency rate is much higher than No-Ace. It is a cause of between 50 percent to 85 percent each year, say the leaders of the opposition party. According to Movement for<|endoftext|>

Figure C.10: Samples ($T = 1024$) from Duo trained on OWT with Gen. PPL 72.05 and entropy = 5.22

Distilled Duo via DDT

The samples were generated using the Greedy-Tail sampler.

“<|endoftext|> continue to view felony- in 2015, compared to the city’s number of black men population in 2, with 122.”

On the other hand, there is the long-running prostitution rape scandal, one of which is now escalating with the city’s mayor, John McClure, this week placed on the top of crime’s criminal list. And the FBI and its partners have significantly increased scrutiny for both hate-crime enforcement ... tracking, Norton added, in a recent report, confirming the rate and 50 percent of the U.S. percent of people arrested on and charged with the agencies that have created the crime.

There is a strong confidence of the city to bridge this gap, said Norton, referring to the city’s crime reduction website, resistance.org.

Watch the video.

Video Opened, and Where We On.<|endoftext|>2. After months looking into the events at the email server, one of the key players in Obama’s transition to WikiLeaks reveals that was not the email server anymore.

Ridzan recommends there to be an investigation into the extent of the handling of the email server while maintaining rather transparency rules for the FBI and other agencies such as. Winner, and Privilege. But he wants to confront with investigations how the most powerful secret agency is able to sew things together, all in one place with very little attention to command. Peter Grayson.

delised Oct 23, 2014

“am still contributing to investigative reporting for the Post but I am it endorsing they wouldn’t let @nokhari2009hip tweet back. writes Kate Dorman

the Post still are apparently reporting that the fact is for so long White House R Dole staff took office to Russia a year earlier than that former President supposed to be Hillary Clinton was a chief diplomat. One should have at least invited those contacts, possibly peruse them on the public by their own actions. I wouldn’t criminalize that, but I like Michael Shumle about work. Ap long fall from Harvard Law School and “roundin’ probe report on the National Intelligence hits on white nationalist research at NYT.” havet sources data for the impact. 3. Member opponent a concern, was roughly a decade ago busting Obama email incident despite there being little knowledge that aides at the National Security Council made an error to kick Clinton at.

“Aspen I was the guy that was Obama’s base that had a machine group named the State to bang on the news and knock it out and the information’s been stolen,” Rentr told West about the problem of the Clinton email. “I fear this is that it’s just a temporary thing – so theoretically speaking about the fact that President Barack’s office is doing downgrading etc.”

He added, “The scary about this is violence to deal with this PDF acknowledgement of it that a member of Congress had a lawyer claiming executive privilege.”

<|endoftext|>Here in one of the logs, it’s even pointed to as an aircraft’s crash. The network time it occurred at the time that the plane was struck was an avalanche that could have launched 300 within the first two hours, and 95% of that killing would have been at 5 16253 per day. My contacts said this was a government shutdown and the result of it would not known what happened.

The report also shows [almost all sources] from a” that wehab, and its sort of copied on a on shore of the ocean. and a folks that used to land with great amounts of energy sources for significant transport’. The key in land is that it’s much scale to the water so it’s easy. Distributed traffic it makes it easier to get to get to what was happening. It’s even said that the plane’s body can kill the owners’ victims.

Ridzan: Yes I was love to see a burnt out scenario, which at one different time, except two people that to to most be. The area was still very high energy gas most of the time and nobody was in second to having to go off the grid. Freer17 by the military was usually the popular speed, speed instead adverse as to large areas carried the it from sure fly army and freer01. It was very difficult. what was the point in it did is it had no credible context like it wasn’t EVE.

So there never was because it was EVE. The gigator’s was totally closed, and the reverse from source to source. THAT was really extreme, and not what we thought shutdown would look like. I want to be clear: It was up really early in June. Debra Vista. Healy was at ABC News in May. The report was being kids on TNT, and Night straight ahead, I think got me a point. Ben Smith wasberedin muten all the time, was regardlessolini, and following Bill Reuther’s Shoring a<|endoftext|>”

Figure C.11: Samples ($T = 8$) from Duo distilled with DDT trained on OWT with Gen. PPL 79.24 and entropy = 5.25

<|endoftext|> wine, check out five photos with medals taken from Saturday’s celebrity guests:

UPDATE JUNE 13: An open-gut fire Sunday at the Hollywood Hotel where the award-winning movie starring David Ortiz as the Red Sox World Series winner provided donors and significant matching funds to the area’s revelers for the weekend, before donations will be either donated or used to purchase food and drink. Much of the money was carved into one late Sunday night, with nearly 80 percent confident of receiving the drive-in photo, and the public purse. The charity was started by a loan professor Luv Sobos, who contributed \$13,500 in order to raise money to honor Ortiz. She contributed tributes to the Oscar winner, complimentary faces to cast members Mickey Rooney and Bruno Whitehead, and the presenting company, Alorino.

Robos, who took care of Hollywood’s Baroque Beauty Opera in 2012, has also casted Ortiz as a local teller. The LiveLight project, which raised the weekend \$13,000 in donations, was the biggest nationwide, with over \$400,000. “Everybody has a cancer,” said Will Franklin, a behind-the-scenes member of Ortiz’s Society for the Love Project. “They give great ruminants to people so people can get candles.”

Sponsors stayed in Hollywood in Cecil Street, which operated on the trucks, who will stay up all night for the parochial event.

Many entered the holiday week with the reception from the Hollywood martial veteran’s family, with fans all season long snapping out-of-town holiday visits to Rooney’s restaurant Péalade. Earlier this year, the movie also reported a run-in of \$5.6 million, but hopefully there was going to be a miracle or two.

PHOTOS: Gaps, curveballs

A family of friends traveling through the flooded field for the pick-up for a new drive-in Chevrolet car at Florida International Speedway on Nov. 30, 2014.

Marino, the factory that oversaw the lift of the pole following the 2004 Red Sox Monster Race, on June 20, closed down during the ceremony, despite permanent jobs for human resources and architectural improvements. Many people were in attendance for the official announcement.

“This is the biggest, I’ve ever had to really have a go at, up here,” said volunteer and volunteer Troy Franco as Lomon announced his involvement with the Oscar winner. “We’re just going to have to get orders in droves all over the place and everybody willing to participate.”

Work on a crane attached to the scaffolding that the lightning rod or chain was due less than an hour. 69 of the night’s countless guests helped set up the crane the film was for. The crane began arriving in the location and played out before it toppled beyond repair.

Many celebrity guests in attendance, as high as an estimated 1 guests reported every four hours at the late-hour event watch. The crane landed more than seven feet high and people lined up alongside a large read of pink balloons to lift and distribute them.

Two other panoramic films also scored a 95/100-120 F.B.A rating, even including Pirates of the Caribbean II: The Turn of Sword.

"We wish everyone well," a Alorino executive told The Associated Press, reported Dec. 23, 2014. “This year’s 15-year Oscar winner doesn’t disappoint.

“This amazing pay has its worth of work that we’re doing. We say up to \$600 at a theater, but I’m sure it’s actually a benefit to the show.”

Inside out all the numbers, though, Lomon is still pushing the ticket price squarely in the 32nd quarter of this year for the launch of the James Bond movie just prior to the Richter hour, which is starting to sell more than \$73 million for the weekend.

“We’re not buying your favorite movie,” said Andrew Alvarez. “It shows you something special here, so if you wake up on Sunday, you’ll remember tomorrow.” <|endoftext|>A potentially big year is still for the 23-year-old. ACT star Andrew Klein reportedly faces another crisis over health and the coach less than keen to return to Melbourne.

If production does open in South Australia, while the future remains fairly sketchy, South Queensland’s planned romantic drama Earth is more likely to remain alongside NCIS SUzi St Kyu and Kim McNamara.

MTA)

The prospect of a new recruit from Canberra is still theoretical according to those that recently received a call from South Sydney general manager Susan Ankosh.

"It bothered to<|endoftext|>

Figure C.12: Samples ($T = 1024$) from Duo distilled with DDT trained on OWT with Gen. PPL 42.46 and entropy = 5.25

MDLM

<|endoftext|> ceases tot be somewhat descending commandryn team then this
major his but probably bigtoolandstre micro
can't submitted it and to stay with he efforts he toised more ID's than but people it once again
his stepped in likely a trivial cmake 3
getting freeman least a few in thatcas etc. gave he up to scouts? committee leader to elevate to a clutch these cupson,
he will send 4. most all 3 per for an op scheduled on camp write please see here Fixed post postlink couple of week.
Korea/Puzzle Launch Prevention. dont think strat isn't better than pulmon and crop - do what., because once he
makes a single makeby playing some good chains he and the follow ups do you need Thes a whole with his 8, elevation
over these few rounds.mind castintrasbachu, leading scorer of the last roundorkers StuarttheoryOriginally, was aware of
this finals situation in and along metricG Ur Bene and "that" of advanced cant succeeded so for this power of can
rifles to develop.many months as awebber splinter. quietly had Dolles-Lu provide a maneuvra set of mail. " Shultzng
the in tall line- destroyer Fday.Spare was within dominant national order of s'f and "our use" which chilled the the
responsible online shooter at the time keeping Housescrees possessed greatly shot range.lations event Rait'ed for an
improvement of shift, not a focus,. actions accordingly.In five years Ppm, theWe was able to transition a more chain
to the lead than himself identified. d expected us to send up to 28 tastyads but the final ofstill did follow in this an
trajectory. his aig shift m is on communicates a posterior induction andye, brother of these ice cream companies, but
he was always a tipping was always under 50. ice cream chains started at 12.8 (icyemic a x,67816. contact rem original
casings a x.3836. hiss four years etc.I mean this was a paceigh she/f driving because by now is no reason for any better
than this! Committee let the We on the island developed reasonably modest and reasonable projections for USicglobal
productions.comsters media was forced to downplay, so that the assumptions should just the pottery aware we account
made they now.book summer (again, on our current set-up, but before) consider the way I am describing. substances
are perfect & Technology knows that when correct but confidence,Our decision to postp What for: & have the visual
design represents a confidence in the competition. We do not have any qual in taking higher actions. down of any
time soon, which we do not sell off crowds. truly considered asseen people is greatness in and strongly orders the
item producing dream expecting it to work.we especially agree seems to be the Republic and plan to publishboards
housing aWe name on phones.chief designers featuredboard licenses remarked by me. the plans editorial handling is
very detailed and well written and the first tutorials live on the Boards. Notice the pick part here
the four players of the we of right are working :on-the-a. we meet a program that starts theme community rings,but
that- but they knew into - may generate a big one truly worth sale to the contest, which would also along the 2 wave
of income support society and parents that might fade out from the competition- until some tournament ruin the
day creditational have a way to get revenue back into the contest coffers.There is in fact a remarkable phenomenon
making in the process increase of promotion. Some years ago an unidentified woman carriedthese boards to look at
recommending lapp; this topic is being discovered, with some results.given innovative, start of kids, and advanced
instances of majors guidelines to raise awareness. Last season ofthe We was off the most Philippine championship
madeowdera practice a dance,with theIn a journal quality observed their toInitially some contrast between the children
and grooming. based on Clinical Psychologyc presented this particular from corrajee which which engines in self social
practice with forced change from the user. Theboard captures g. <|endoftext|>In the US the majority of modern
NGNG it are public spending responsive information, most populations back, however representatives of small business
in yourie for enforcing a human right to exploit created int a rival brands.
Yet powerful extremists were cracking up genital abusive-challengenon religious fundamentalist groups week; seen how
thend powerful extremists begin to come in force.Consider Biocolosion. It has jumped balloon to the top of 4 food
choice in the UK. Its andisation into Perpetita to second-80's And Districts, the topic of change was school sick tools
emerged for people to reject themselves (part images). All things slipping straight into the former trading offsleep that
spawned the old monster<|endoftext|>

Figure C.13: Samples ($T = 8$) from MDLM trained on OWT with Gen. PPL 830.82 and entropy = 5.91

<|endoftext|> desperate until they were everywhere.
 "But there's more to it than ever," Zimmern remembers Jonathon, sighing softly. "Sure..."
 "But, cheap as sterling is what things are, which is not to say (tell the truth)," said Jonathon, "yes, show up to negotiate."
 When Zimmern is speaking about free market opportunities itself, when pressed on the topic of open war, Zimmern tries – "more than anyone could possibly think so" – to describe democracy as a capital monopoly today as it is today. But Joel Klein of Teh'tinarik gets it right. "Ever since the passage of the new Millennium Copyright Act", some companies have be monopolizing antitrust commerce ... much like other companies, (and a few companies) such as Verizon and AT&T have achieved tens of billions in profit in recent years and have not grown to do better than any giant steel producer."
 With the problem of monopoly upon us, then, there are a few possibilities to consider. Or perhaps just consider Hegel's and if any 18th-century academic and intellectual would have "discover's work" in either field (or ... Hurst) would have worked out for him. <|endoftext|>June 23, New York ThreeIn Willets Memorial Park, New Jersey2 Times beat reporter Eli Pasdorf watches MLS finals, including Supporters' Shield winners Atlanta United.
 Ready for MLS Conference Finals
 Seattle Sounders FC host Atlanta United in the Supporters' Shield final in Houston played Sunday. Today, they soared against them in style.
 The winner was the Monarchs Monarchs. They had extended their run to advance Gotham with a streaking wayside in the 67th minute of the game followed with a handball interception by Anthony Fasy and a passing assist from Fredericks Smersky.
 The final result: 3 points over Atlanta United.
 Washington DC FC vs. Philadelphia Union, Arizona, substitution 1-0 U.S. Capital Cup Supplemental Championship
 Final 1997-728 New York Crew, New York, substitutions 2-1
 United Constructors Championship Final 4-0 defending champions USA, FC Dallas Second Division
 033 Dallas LeonOR - Dallas MLS Supplemental Championship Final
 1996-Seattle Sounders FC vs. LA Galaxy, Los Angeles, substitution 3-0 1909-76, New York, tr. 1926 substitutions 1-1
 U.S. Capital League Cup, defending champion Atlanta United Ertones Academy semifinals, GCT
 Emmy Champions League Goal of the Round 6: Pedro Joao always seemed the next hair of the season. Lionel Messi, 14 at left-back, had kicked off the Europa League and some 5 goals a season. At the same time, he ignited a calling season carry-forward, won the championship and much helped Connor Randall and Muamba Tevez.
 Thomas Baismith led the way from a Glee-Fletcher by Arjen Robben as Dallas went even deeper. United had a nice bicycle finish during the first half off a line of drawn line by Ninos Terstad. FC Dallas sent into extra time after burning off the D.C. Dynamo, spent the minute in the goal by Buolt, at the top of the box and midway through the final United attacking responded well to only an own goal from Troy Vincent.
 The first goal of the final came over Columbus in the March 24 league game. Assist-free play ended in a safety net peek in the Columbus goal, but it was watched on TV. The Sounders were heavily fined as the goal ignited its good play and involving Jeremy Vergandra, who was overlooked to make about double his salary. Columbus goalkeeper Sean Mannah (injured), Andre Hernan and Terry Wingwerk in the 59th minute were charged. Still, the first player to wear a pristic eyelid missed the rest of the final.
 On the 60th minute of the game, Atlanta United fought an Ogen-off to beat the Columbus Crew. Fans were up for their power and the hosts maintained their defensive superiority against the defending champions. Captain Al Merida struck the touchline before bringing the ball with a sixtyard finish that won the game until the 95th minute. From there FC Dallas skipper and winner Greg Garza locked the game up in two.
 The Vision to the Holding Arena in Wolfsburg, Germany
 Medicine had let up in football. The death knell ended and several seasons later injuries had been endured to everyone with the harshest modification into the equation online after Manchester United signed Kevin Wimmer, the experimental player for the tournament (Byramidal would be equal the partnership). Wladimir Klitschko stepped down as player. Detroit first step was in getting their expansion franchise shared in the playoff games for the summer. Several years later, MLS MLS almost had to re-brand and replace someone largely saying how much they didn't want to see it.
 The<|endoftext|>

Figure C.14: Samples ($T = 1024$) from MDLM trained on OWT with Gen. PPL 104.85 and entropy = 5.63

Distilled MDLM via SDTT

<|endoftext|> by 7 a.m. and half an hour — paused several of them then as they looked for ways to merrily end their wait. Most applicants applied to Cuba shortly before the coming. They each had a mantra to launch during the day to India,,s,s,s that followed.

It was much better. Five years ago, according to RBC in Kolkata,, IR doubled its presence in Shanghai, Beijing, Singapore to very few in the Asian part. Judge Menesh Katan, from 60 to 70, has won a relatively few superstar in the world, and scene expert at becoming the anchor in for the world’s elite. It will really, unheard of.

A conveysealed COO led took the group over the top since ten years ago, with the last 30 people, exactly one month people posted received notifications that, as it was said, “ira would be the lurch, WH the ship coaring to hang up upon announcement of a US call for economic, economic, and military rule.”<|endoftext|>STATEN PARK, N.Y. : A few companies operate in the New York area, close to the 1 in 2 market and that, among other things, iscommuting. The company, Smart Goods has changed the process back to the business of then, to identify much less spending amounts of time in the home. They also hand out devices to people to get around, changing the figure from about 23 percent to over 32 percent of their bikes, and just over 11 percent total of their bikes.

Many of the people go home school, orget stuck in a Midtown Fend but still lose this important part of the lives. They see the only a tiny increase in spent time in the home in some 600 people in the suburbs; they are much more likely back in the car, according to recent Allen & Co.

Strangely, and internal market research will show, it would not so have that much impact to lineup their in the New Jersey area now, but there are bikes left mostly in the car right until they change, as a short array. It does not affect those that use bicycle batteries to transportation needs.

Advocates have countered that bicycles that lacked a bike and or insurance like they predicted that Uber was doing not that swayed by the attack on the drivers they kept could speed up about about 20 hours regularly and about slightly less (over 2 percent) annual cost, and ever bus gotten there could be a sighting tool for the bicycle. This recording could be moved from to carshare on Fourth Street andor just use the car.

A new survey recently conducted by New Yorkers has finds no evidence, nots that bikes, a form of bus driving with 30 miles per hour and providing power for 25 miles are understandably cheaper, return at a rate of \$79 for the same motor with bikes to the right also suggests that they want the install period to be limited, barring them from optioning for any permits request bought earlier during the year,. Some businesses have also resigned directly, apparently, to other Philadelphia bars.

The plans of impact here were announced last night the OSHA, and the blog teamlaw pages — if all states aret sharing the, to bicycle parks, and lanes,,s and to some will offer bicycleres or vehicle sharing service, if there partner wishes that the savings from the opens be reflected in a vehicle-scale architecture plan. The first CBO estimate on whether best set up turns to be trees has issued an idea, anyway. And some municipalities this would result in will face any bicycle leasing capabilities.

The community can start to family by that couple mannerisms to connect each cycle to both systems is seamless, but what they may tweak is not easy. I’m not quite sure who will be putting those additional rents on the bike, in so fast that they could be even slower in the home.

Even in my opinion, Margaret Margot set up bus lot talks, a state regulators and local deceler fundraising, is having the potential to increase the overall cost of commuting. Visit my blog, Articles | Paid leave,my word — healthy, and on the issue for safe. I had been wrong those who are in a coalition of workers, and have refused to pay such a bill when they have all kinds of hurt,represents the purpose of what is going in the work. I say this, because:- the class should has the incentive to give direct contributions and, instead of paying money,- enough to put more cash on to the middle class than the rest.

The New America and the Boyle do this post in Florida at the Heritage Foundation, California. Members of Congress, led by Republicans, were going on a journey when a clear number of Russian intelligence operatives, in the what the New American called ‘Russian hacking team’, posted on Twitter. Republican Governors, the GOP Senators, now others. Sen. Tom Cotton of Alaska was Trump, last month<|endoftext|>

Figure C.15: Samples ($T = 8$) from MDLM trained on OWT and distilled using SDTT with Gen. PPL 193.05 and entropy = 5.58

<|endoftext|> legislation.

Much to the dismay of the American Civil Liberties Union, President Bush sent a joint letter to Congress, consulting with the body in deciding whether to strip surveillance of surveillance authority. The Congress had objected to a Conservative motion to remove their amendment from consideration, but failed to pass the bill, due to sheer procedural errors that were not addressed in the final version. The other members of the mainline denomination, the United Methodist Church also introducing the legislation, signed on November 2, 2002, after the 1898 American Revolution. The compromise comes within a process based upon the expression of the First Amendment rights of the Constitution and the Fourth Amendment of the United States. (The Libertarians uphold the right of state officials to lawful activities by foreign governments as defined by the Geneva Conventions, before advancing to the question of the dangers that arise from this practice.)

While these concerns are indeed obvious to Americans that identify with their allegiance to America's government, they seem to reflect a position that cannot be explained based entirely on the participation of Americans in the enforcement of the law, and their cooperation with America's government. Imminent surveillance, is difficult to dispute, either by a whole or part of the American populace. Almost any technique that is ever [redacted] for it will be completely likely to violate the privacy and general security of Americans, as far as I know, far more than effective surveillance methods. Such a technique will have to give up a lot of information, and it will be difficult to develop techniques, informants, and other methods that can begin to expose a person's electronic and legal communications far beyond last year's. I welcome a few additional comments on this particular point, and I will be addressing them in the future.

First, we are very surprised by the fact that so many of them, and so many our police officers, are in this position. The information that we have gathered is very clear. We know about us intimately, that we are very sensitive, that they are very constrained, as well as about the limitations of what they are doing. We are not only representing them, we know of us, and that our knowledge of them is valuable, and we are doing our best to make them the most difficult and difficult next step as possible. It is possible that this is a limited surveillance state, much like what we would be subjected to, but that if the program led to a significant increase in surveillance, it would be labeled as cyber routers and phones. Some have commented that the filters are still sufficient to curbing this process, but they constitute confounding, in that it would be very difficult for such an assessment to begin with a computerized seed; that documents available to the eyes of the general public would be unviewable; and that the risk would be inevitable that anyone who under certain circumstances could have access immediately to this superstate. I also think that such access will not be recognized as a begrap, and therefore as a repeated violation of the law, and the government will not want to do so, either. But it can almost certainly be seen by some of this reader, that such a superstate state is almost always, intuitively, very hard to reach very quickly, and something that we need to do to finally confront threats at a serious level. If they are allowed to remain so, their consequences will be determined as having the highest possible impact if we are to stabilize the power grid in the future.

The aforementioned congress address also made it crystal clear, we wanted a process that required analysis of all the information that was sent and received by the American population, and, one hopes therefore, absolutely to see the fate of humanity decide the way of a larger society. Unfortunately, there are many, many more of these that continue to leak into the Prism program. That means that we would have a much more likely surveillance as a nation also, and thus much more likely that it would be related to the above-mentioned "civil law enforcement force," or actually related to the violation of international human rights laws.

Accordingly, according to the UN statistics, 30-34 major reported cybernet attacks have occurred in at least 29 countries in the developing world currently, primarily those conducted by MIT or China, and the aforementioned major cybernet attacks in these countries will continue to be suffering in systematic and severe instances, which regard the violation of basic human rights and brutal abuse of cyber enclaves from outside civilization as an important issue. At this point, along with the actual global threats that are prevalent in all aspects of our lives, many factors have impacted the relationship between the Internet and our societies and its general use. Indeed, the grinder contributes to a tremendous threat to the Internet because it is one of the tools that a great majority of people actually use. For over the past several years and decades, ribbon cutters have been placed on the cusp of being invented in some form, electronic or remote form, at an incredibly high price. We refer to today's Western Redwood as being the standard industrial<|endoftext|>

Figure C.16: Samples ($T = 1024$) from MDLM trained on OWT and distilled using SDTT with Gen. PPL 36.89 and entropy = 5.39

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